NEW ISSUE—Book-Entry Only

Ratings:
S&P: "AA"
Moody's: "Aa3"
See "RATINGS"

In the opinion of Stradling, Yocca, Carlson & Rauth, a Professional Corporation, Sacramento, California, and Amira Jackmon, Attorney at Law, Berkeley, California, Co-Bond Counsel, under existing statutes, regulations, rulings and judicial decisions, and assuming the accuracy of certain representations and compliance with certain covenants and requirements pertaining to the 2016 Series AB Bonds, interest on the 2016 Series AB Bonds is excluded from gross income for federal income tax purposes, and is not an item of tax preference for purposes of calculating the federal alternative minimum tax imposed on individuals and corporations. In the further opinion of Co-Bond Counsel, interest on the 2016 Series AB Bonds is exempt from State of California personal income tax. See "TAX MATTERS."

$308,400,000
PUBLIC UTILITIES COMMISSION OF THE CITY AND COUNTY OF SAN FRANCISCO
Wastewater Revenue Bonds

$240,580,000
2016 Series A
(Green Bonds)

$67,820,000
2016 Series B

Dated: Date of Delivery **Due:** October 1, as shown on inside front cover

General. This cover page contains certain information for quick reference only. It is not intended to be a summary of the security or terms of the Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2016 Series A (Green Bonds) (the "2016 Series A Bonds") and 2016 Series B (the "2016 Series B Bonds" and, collectively with the 2016 Series A Bonds, the "2016 Series AB Bonds"). Investors are instructed to read the entire Official Statement to obtain information essential to making an informed investment decision.

Authority for Issuance. The Public Utilities Commission of the City and County of San Francisco (the "SFPUC") is issuing the 2016 Series AB Bonds pursuant to authority granted by the Charter of the City and County of San Francisco (the "City"), through Proposition E approved by voters of the City on November 5, 2002, and under a Fourth Supplemental Indenture dated as of May 1, 2016, by and between the SFPUC and U.S. Bank National Association, as trustee (the "Trustee"), and which supplements the Indenture dated as of January 1, 2003, by and between the SFPUC and the Trustee (as supplemented and amended to date, the "Indenture"). See "SECURITY FOR THE BONDS."

Purposes. The 2016 Series AB Bonds are being issued primarily to finance and refinance (through the retirement of commercial paper notes) certain capital projects benefiting the Wastewater Enterprise (as defined herein). Proceeds of the 2016 Series AB Bonds will also be applied to (a) fund capitalized interest with respect to the 2016 Series AB Bonds for approximately two years, and (b) pay the costs of issuance of the 2016 Series AB Bonds. See "PLAN OF FINANCE," "ESTIMATED SOURCES AND USES OF FUNDS," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION" and "APPENDIX F – ANTICIPATED 2016 SERIES A PROJECTS."

Denominations and Interest. The 2016 Series AB Bonds will be available in denominations of $5,000 or any integral multiple thereof and will mature in the years and amounts and accrue interest from their date of delivery at the rates set forth on the inside cover page of this Official Statement. Interest on the 2016 Series AB Bonds is payable semiannually on April 1 and October 1 of each year, commencing October 1, 2016. See "THE 2016 SERIES AB BONDS."

Book-Entry Only. The 2016 Series AB Bonds will be issued as fully registered bonds, registered in the name of Cede & Co., as nominee of The Depository Trust Company, New York, New York ("DTC"), and will be available to the ultimate purchasers (the "Beneficial Owners"), under the book-entry system maintained by DTC. Beneficial Owners will not receive physical certificates representing their interests in the 2016 Series AB Bonds. The principal of, premium, if any, and interest on the 2016 Series AB Bonds are payable to DTC by the Trustee, and, so long as DTC is acting as securities depository for the 2016 Series AB Bonds, disbursements of such payments to DTC Participants is the responsibility of DTC and disbursements of such payments to the Beneficial Owners is the responsibility of DTC Participants. See "THE 2016 SERIES AB BONDS."

Redemption. The 2016 Series AB Bonds are subject to optional and mandatory sinking fund redemption prior to maturity. See "THE 2016 SERIES AB BONDS – Redemption."

Security. Under the Indenture, the SFPUC has irrevocably pledged the Net Revenues of its Wastewater Enterprise to the punctual payment of principal of, premium, if any, and interest on the 2016 Series AB Bonds and all outstanding parity revenue bonds issued under the Indenture, subject to the allocation of funds provided in the Indenture. The 2016 Series AB Bonds are payable on a parity with certain Outstanding Bonds previously issued by the SFPUC under the Indenture. See "SECURITY FOR THE BONDS."

Limited Obligation. **The SFPUC is not obligated to pay the principal of, premium, if any, or interest on the 2016 Series AB Bonds from any source of funds other than Net Revenues. The SFPUC has no taxing power. The General Fund of the City is not liable for the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds, and neither the credit nor the taxing power of the City is pledged to the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds. The 2016 Series AB Bonds are not secured by a legal or equitable pledge of, or charge, lien, or encumbrance upon, any of the property of the City or of the SFPUC or any of its income or receipts, except Net Revenues. See "SECURITY FOR THE BONDS."**

MATURITY SCHEDULE
(See inside cover)

The 2016 Series A Bonds and the 2016 Series B Bonds were sold through respective competitive sales held on May 10, 2016.

The 2016 Series AB Bonds are offered when, as and if issued by the SFPUC and received by the successful bidder, subject to the approval of validity by Stradling, Yocca, Carlson & Rauth, a Professional Corporation, Sacramento, California, and Amira Jackmon, Attorney at Law, Berkeley, California, Co-Bond Counsel to the SFPUC, and to certain other conditions. Certain matters will be passed upon for the SFPUC and the City by Orrick, Herrington & Sutcliffe LLP, San Francisco, California, Disclosure Counsel, and by the City Attorney of the City and County of San Francisco. It is expected that the 2016 Series AB Bonds in fully registered form will be available for delivery in book-entry form through the facilities of DTC, on or about May 24, 2016.

The date of this Official Statement is May 10 2016.

MATURITY SCHEDULE

2016 Series A Bonds
(Base CUSIP[*] Number: 79768H)

$154,975,000 Serial Bonds

Maturity (October 1)	Principal Amount	Interest Rate	Yield[†]	CUSIP[*]	Maturity (October 1)	Principal Amount	Interest Rate	Yield[†]	CUSIP[*]
2023	$5,475,000	5.00%	1.24%	CH9	2032	$8,590,000	5.00%	2.09%[C]	CS5
2024	5,760,000	5.00	1.36	CJ5	2033	9,030,000	5.00	2.15[C]	CT3
2025	6,055,000	4.00	1.51	CK2	2034	9,495,000	5.00	2.21[C]	CU0
2026	6,365,000	5.00	1.68[C]	CL0	2035	9,980,000	5.00	2.27[C]	CV8
2027	6,690,000	5.00	1.79[C]	CM8	2036	10,495,000	5.00	2.33[C]	CW6
2028	7,035,000	5.00	1.87[C]	CN6	2037	10,975,000	4.00	2.61[C]	CX4
2029	7,395,000	5.00	1.93[C]	CP1	2038	11,425,000	4.00	2.64[C]	CY2
2030	7,775,000	5.00	1.97[C]	CQ9	2039	11,890,000	4.00	2.67[C]	CZ9
2031	8,170,000	5.00	2.03[C]	CR7	2040	12,375,000	4.00	2.70[C]	DA3

$26,285,000	4.00% Term Bonds Due October 1, 2042	Yield: 2.75%[†C]	CUSIP[*] 79768H DC9
$59,320,000	4.00% Term Bonds Due October 1, 2046	Yield: 2.79%[†C]	CUSIP[*] 79768H DB1

2016 Series B Bonds
(Base CUSIP[*] Number: 79768H)

$43,685,000 Serial Bonds

Maturity (October 1)	Principal Amount	Interest Rate	Yield[†]	CUSIP[*]	Maturity (October 1)	Principal Amount	Interest Rate	Yield[†]	CUSIP[*]
2023	$1,545,000	5.00%	1.24%	DD7	2032	$2,420,000	5.00%	2.09%[C]	DN5
2024	1,625,000	5.00	1.36	DE5	2033	2,545,000	5.00	2.15[C]	DP0
2025	1,705,000	4.00	1.51	DF2	2034	2,675,000	5.00	2.21[C]	DQ8
2026	1,795,000	5.00	1.68[C]	DG0	2035	2,815,000	5.00	2.27[C]	DR6
2027	1,885,000	5.00	1.78[C]	DH8	2036	2,960,000	5.00	2.33[C]	DS4
2028	1,980,000	5.00	1.87[C]	DJ4	2037	3,095,000	4.00	2.61[C]	DT2
2029	2,085,000	5.00	1.93[C]	DK1	2038	3,220,000	4.00	2.64[C]	DU9
2030	2,190,000	5.00	1.97[C]	DL9	2039	3,350,000	4.00	2.67[C]	DV7
2031	2,305,000	5.00	2.03[C]	DM7	2040	3,490,000	4.00	2.70[C]	DW5

$7,410,000	4.00% Term Bonds Due October 1, 2042	Yield: 2.75%[†C]	CUSIP[*] 79768H DY1
$16,725,000	4.00% Term Bonds Due October 1, 2046	Yield: 2.79%[†C]	CUSIP[*] 79768H DX3

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[*] CUSIP is a registered trademark of The American Bankers Association. CUSIP data herein is provided by CUSIP Global Services, managed by Standard & Poor's Financial Services LLC on behalf of The American Bankers Association. This data is not intended to create a database and does not serve in any way as a substitute for the CUSIP services. Neither the SFPUC nor the Initial Purchaser of the 2016 Series AB Bonds assumes any responsibility for the accuracy of the CUSIP data.

[†] Reoffering prices and yields have been provided by the Initial Purchaser of the 2016 Series AB Bonds. See "PURCHASE AND REOFFERING."

[C] Yield to par call on October 1, 2025.

WASTEWATER ENTERPRISE MAJOR FACILITIES

Figure 1-1 – SFPUC Wastewater Enterprise – Major Facilities (Not to Scale)

The SFPUC Wastewater Enterprise operates and maintains the City of San Francisco's combined sewer system which collects and treats both sewage and stormwater. This system consists of 993 miles of sewer lines which collect sanitary sewage from homes, businesses, and stormwater runoff; large transport storage box facilities; 27 pump stations that transport sewage and stormwater; and three treatment plants that discharge the treated water into the San Francisco Bay and Pacific Ocean. San Francisco has eight distinct urban watersheds: five on the Bayside (North Shore, Channel, Islais, Sunnydale, and Yosemite) and three on the Westside (Richmond, Sunset, and Lake Merced). The three treatment plants in the sewer system treat sewage and stormwater based on whether the collection point is on the Bayside or the Westside. The Oceanside Treatment Plant is not connected to the Bayside treatment systems, and the Southeast Treatment Plant and North Point Wet Weather Facility are not connected to the Westside treatment system. The map above depicts certain major facilities of the combined sewer system and the ridge separating the Bayside from the Westside facilities. See "THE WASTEWATER ENTERPRISE."

The 2016 Series AB Bonds are not secured by a legal or equitable pledge of, or charge, lien or encumbrance upon, any of the property of the SFPUC or of its income or receipts, except Net Revenues. See "SECURITY FOR THE BONDS."

GENERAL INFORMATION

No dealer, broker, salesperson or other person has been authorized by the SFPUC to give any information or to make any representation other than those contained herein and, if given or made, such other information or representation must not be relied upon as having been authorized by the SFPUC.

This Official Statement does not constitute an offer to sell or the solicitation of an offer to buy, nor there any sale of the 2016 Series AB Bonds, by any person in any jurisdiction in which it is unlawful for such person to make such an offer, solicitation or sale. This Official Statement is not to be construed as a contract with the Initial Purchaser of the 2016 Series AB Bonds. Any statement made in this Official Statement involving any forecast or matter of estimates or opinion, whether or not expressly so stated, is intended solely as such and not as a representation of fact.

The information set forth herein other than that provided by the SFPUC, although obtained from sources which are believed to be reliable, is not guaranteed as to accuracy or completeness. The information and expressions of opinion herein are subject to change without notice and neither delivery of this Official Statement nor any sale made hereunder shall, under any circumstances, create any implication that there has been no change in the affairs of the SFPUC or the City since the date hereof.

The City maintains a website at http://www.sfgov.org and the SFPUC maintains a website at http://www.sfwater.org. In addition, certain information and reports found on other websites, and other information and reports, are referred to in this Official Statement. *The information and reports available on such websites, and the other referenced information and reports, are not incorporated by reference into this Official Statement and should not be relied upon in making an investment in the 2016 Series AB Bonds.*

The issuance and sale of the 2016 Series AB Bonds have not been registered under the Securities Act of 1933 in reliance upon the exemption provided thereunder by Section 3(a)2 for the issuance and sale of municipal securities.

IN CONNECTION WITH THE OFFERING OF THE 2016 SERIES AB BONDS, THE UNDERWRITERS MAY OVERALLOT OR EFFECT TRANSACTIONS WHICH STABILIZE OR MAINTAIN THE MARKET PRICE OF THE 2016 SERIES AB BONDS AT LEVELS ABOVE THAT WHICH MIGHT OTHERWISE PREVAIL IN THE OPEN MARKET. SUCH STABILIZING, IF COMMENCED, MAY BE DISCONTINUED AT ANY TIME.

This Official Statement is delivered for use in connection with the issuance, sale and delivery of the 2016 Series AB Bonds and may not be reproduced or used, in whole or in part, for any other purpose.

FORWARD-LOOKING STATEMENTS

CERTAIN STATEMENTS CONTAINED IN THIS OFFICIAL STATEMENT REFLECT NOT HISTORICAL FACTS BUT FORECASTS AND "FORWARD-LOOKING STATEMENTS." ALL FORWARD-LOOKING STATEMENTS ARE PREDICTIONS AND ARE SUBJECT TO KNOWN AND UNKNOWN RISKS AND UNCERTAINTIES. NO ASSURANCE CAN BE GIVEN THAT THE FUTURE RESULTS DISCUSSED HEREIN WILL BE ACHIEVED, AND ACTUAL RESULTS MAY DIFFER MATERIALLY FROM THE FORECASTS DESCRIBED HEREIN. IN THIS RESPECT, THE WORDS "ESTIMATE," "PROJECT," "ANTICIPATE," "EXPECT," "INTEND," "BELIEVE" AND SIMILAR EXPRESSIONS ARE INTENDED TO IDENTIFY FORWARD-LOOKING STATEMENTS. ALL PROJECTIONS, FORECASTS, ASSUMPTIONS, EXPRESSIONS OF OPINIONS, ESTIMATES AND OTHER FORWARD-LOOKING STATEMENTS ARE EXPRESSLY QUALIFIED IN THEIR ENTIRETY BY THE CAUTIONARY STATEMENTS SET FORTH IN THIS OFFICIAL STATEMENT. GIVEN THEIR UNCERTAINTY, INVESTORS ARE CAUTIONED NOT TO PLACE UNDUE RELIANCE ON SUCH STATEMENTS.

SAN FRANCISCO PUBLIC UTILITIES COMMISSION

Francesca Vietor, Commission President
Anson Moran, Commission Vice President
Ann Moller Caen, Commissioner
Vince Courtney, Commissioner
Ike Kwon, Commissioner

PUBLIC UTILITIES COMMISSION OFFICIALS

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Michael Carlin, Deputy General Manager and Chief Operating Officer
Tommy T. Moala, Assistant General Manager, Wastewater Enterprise
Juliet Ellis, Assistant General Manager, External Affairs
Barbara Hale, Assistant General Manager, Power Enterprise
Kathy How, Assistant General Manager, Infrastructure Division
Steven R. Ritchie, Assistant General Manager, Water Enterprise
Eric L. Sandler, Assistant General Manager, Business Services and Chief Financial Officer

CITY AND COUNTY OF SAN FRANCISCO

MAYOR
Edwin M. Lee

BOARD OF SUPERVISORS
London Breed, Board President, District 5

John Avalos, District 11
David Campos, District 9
Malia Cohen, District 10
Mark Farrell, District 2
Jane Kim, District 6
Eric Mar, District 1
Aaron Peskin, District 3
Katy Tang, District 4
Scott Wiener, District 8
Norman Yee, District 7

CITY ATTORNEY
Dennis J. Herrera

CITY TREASURER
José Cisneros

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Naomi Kelly, City Administrator

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APPENDIX A - SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE
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APPENDIX E - SECURITIES DEPOSITORY AND BOOK-ENTRY SYSTEM
APPENDIX F - ANTICIPATED 2016 SERIES A PROJECTS

OFFICIAL STATEMENT

$308,400,000
PUBLIC UTILITIES COMMISSION
OF THE CITY AND COUNTY OF SAN FRANCISCO
WASTEWATER REVENUE BONDS

$240,580,000	**$67,820,000**
2016 SERIES A (GREEN BONDS)	**2016 SERIES B**

INTRODUCTION

This Introduction is qualified in its entirety by reference to the more detailed information included and referred to elsewhere in this Official Statement. The offering of the 2016 Series AB Bonds to potential investors is made only by means of the entire Official Statement. Terms used in this Introduction and not otherwise defined have the respective meanings assigned to them elsewhere in this Official Statement, including "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE."

General

This Official Statement, including the cover page and Appendices hereto, is provided to furnish certain information in connection with the offering by the Public Utilities Commission of the City and County of San Francisco (the "SFPUC") of its Wastewater Revenue Bonds, 2016 Series A (Green Bonds) (the "2016 Series A Bonds") and 2016 Series B (the "2016 Series B Bonds" and, collectively with the 2016 Series A Bonds, the "2016 Series AB Bonds").

Authority for Issuance

The SFPUC is issuing the 2016 Series AB Bonds pursuant to authority granted by the Charter (the "Charter") of the City and County of San Francisco (the "City"), under Proposition E, approved by the voters of the City on November 5, 2002, and a Fourth Supplemental Indenture, dated as of May 1, 2016 (the "Fourth Supplemental Indenture"), by and between the SFPUC and U.S. Bank National Association, as trustee (the "Trustee"), which supplements an Indenture, dated as of January 1, 2003 ("Original Indenture"), by and between the SFPUC and the Trustee, as amended by a First Supplemental Indenture dated as of May 1, 2010 (the "First Supplemental Indenture"), a First Amendment to Indenture, dated as of May 1, 2010 (the "First Amendment to Indenture"), a Second Supplemental Indenture dated as of January 1, 2013 (the "Second Supplemental Indenture"), and a Third Supplemental Indenture dated as of February 1, 2013 (the "Third Supplemental Indenture"), each between the SFPUC and the Trustee. The Original Indenture, as so amended and supplemented, including and as amended and supplemented by the Fourth Supplemental Indenture, is referred to herein collectively as the "Indenture."

The 2016 Series AB Bonds are being issued pursuant to Ordinance No. 107-14, adopted by the Board of Supervisors of the City (the "Board of Supervisors") on July 2, 2014, and under a resolution adopted by the SFPUC governing body (the "Commission") on March 22, 2016 and a resolution adopted by the Board of Supervisors on April 26, 2016.

See "OBLIGATIONS PAYABLE FROM REVENUES – Authority for Issuance of Revenue Bonds and Other Obligations Payable from Revenues."

Purposes

The 2016 Series A Bonds are being issued primarily to finance and refinance (through the retirement of a portion of the outstanding commercial paper notes) select projects in the SFPUC's Sewer System Improvement Program (as further described herein, the "SSIP"). The SFPUC expects that the projects financed and refinanced from proceeds of the 2016 Series A Bonds will be the projects set forth in Appendix F to this Official Statement.

The 2016 Series B Bonds are being issued primarily to finance and refinance (through the retirement of a portion of the outstanding commercial paper notes) select projects authorized in the capital program for the Wastewater Enterprise (as defined herein). Proceeds of the 2016 Series AB Bonds will also be applied to (a) fund capitalized interest with respect to the 2016 Series AB Bonds for approximately two years, and (b) pay the costs of issuance of the 2016 Series AB Bonds. See "PLAN OF FINANCE," "ESTIMATED SOURCES AND USES OF FUNDS," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION," "APPENDIX F – ANTICIPATED 2016 SERIES A PROJECTS," "WASTEWATER ENTERPRISE CAPITAL PROGRAM" and "SEWER SYSTEM IMPROVEMENT PROGRAM."

The SFPUC and the Wastewater Enterprise

The SFPUC is a department of the City responsible for the maintenance, operation and development of three utility enterprises. See "THE PUBLIC UTILITIES COMMISSION."

The Wastewater Enterprise provides sewage and stormwater collection, treatment and disposal services to residential, commercial and industrial customers in the City, as well as three municipal sewer service providers that serve residents and businesses in northern San Mateo County. The Wastewater Enterprise's services are provided through (i) a combined system that collects sewage and stormwater, (ii) three wastewater treatment plants and (iii) effluent outfalls to the San Francisco Bay and Pacific Ocean. See "THE WASTEWATER ENTERPRISE."

The other two enterprises of the SFPUC deliver retail water services to the City and wholesale water to users in three other Bay Area counties, and power (predominantly hydroelectric) for City government operations and to other users. The revenues of these other two enterprises are not available for, and do not secure, payment of the principal, of premium, if any, or interest on the Bonds. See "THE PUBLIC UTILITIES COMMISSION."

Security for the Bonds

Under the Indenture, the SFPUC has irrevocably pledged the Net Revenues of the Wastewater Enterprise to the punctual payment of principal of, premium, if any, and interest on the Bonds (as defined below), which consist of any parity revenue bonds issued under the Indenture, including the 2016 Series AB Bonds and the outstanding Bonds described below, subject to the flow of funds contained in the Indenture. The 2016 Series AB Bonds and all other Bonds are secured by a parity lien on Net Revenues. See "SECURITY FOR THE BONDS."

The Indenture defines "Net Revenues" as all "Revenues," less all Operation and Maintenance Costs of the Wastewater Enterprise (each as further defined below). Revenues are generated principally from the sewer service charges to customers for the sanitary wastewater and stormwater collection, treatment and disposal services of the Wastewater Enterprise. Wastewater rates are set by the SFPUC, subject to rejection by resolution of the Board of Supervisors. See "FINANCIAL OPERATIONS."

The SFPUC is not obligated to pay the principal of, premium, if any, or interest on the 2016 Series AB Bonds from any source of funds other than Net Revenues. The SFPUC has no taxing power. The General Fund of the City is not liable for the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds, and neither the credit nor the taxing power of the City is pledged to the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds. The 2016 Series AB Bonds are not secured by a legal or equitable pledge of, or charge, lien or encumbrance upon, any of the property of the City or of the SFPUC or any of its income or receipts, except Net Revenues. See "SECURITY FOR THE BONDS."

No Reserve Account will be established for the 2016 Series AB Bonds.

Outstanding and Future Parity Bonds and Other Indebtedness

Parity Bonds. The SFPUC has issued four series of outstanding Bonds:

• wastewater revenue bonds captioned "Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2010 Series A" (the "2010 Series A Bonds"), which were issued in the aggregate principal amount of $47,050,000 on June 8, 2010;

• wastewater revenue bonds captioned "Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2010 Series B (Federally Taxable – Build America Bonds – Direct Payment)" (the "2010 Series B Bonds"), which were issued in the aggregate principal amount of $192,515,000 on June 8, 2010;

• wastewater revenue bonds captioned "Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2013 Series A (Refunding)" (the "2013 Series A Bonds"), which were issued in the aggregate principal amount of $193,400,000 on January 30, 2013; and

• wastewater revenue bonds captioned "Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2013 Series B" (the "2013 Series B Bonds"), which were issued in the aggregate principal amount of $331,585,000 on February 27, 2013.

The Indenture permits, upon the satisfaction of certain conditions, the issuance of additional bonds secured by a pledge of Net Revenues (the "Additional Bonds") on a parity with the 2010 Series A Bonds, the 2010 Series B Bonds, the 2013 Series A Bonds, the 2013 Series B Bonds and the 2016 Series AB Bonds. See "SECURITY FOR THE BONDS – Additional Series of Bonds."

All bonds issued and outstanding at any given time under the Indenture, including the 2016 Series AB Bonds as the context requires, are referred to collectively in this Official Statement as the "Bonds."

State Loans. Under the Indenture, the SFPUC may enter into loan agreements with the State, and any board, department or agency thereof, in order to finance certain categories of projects relating to the facilities of the Wastewater Enterprise. These loans may be payable from Net Revenues on a parity with the Bonds (the "Parity State Loans"), or on a subordinated lien basis relative to the Bonds, as determined by the SFPUC. As of the date of this Official Statement, the SFPUC has entered into one Parity State Loan in the amount of $7,435,000 for the Wastewater Enterprise's Lake Merced Green Infrastructure – Holloway Green Street Project, an SSIP project. However, the Wastewater Enterprise has not yet submitted a request for reimbursement to, nor received reimbursement from, the State, so no repayment obligation has yet been incurred or calculated. See "OBLIGATIONS PAYABLE FROM REVENUES – Parity State Loans."

See "SECURITY FOR THE BONDS – Additional Parity State Loans" and "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE."

Risk Factors

Investment in the 2016 Series AB Bonds is subject to material risks. For a general overview of certain risk factors which should be considered, in addition to other matters set forth in this Official Statement, in evaluating an investment in the 2016 Series AB Bonds, see "RISK FACTORS."

Continuing Disclosure

The SFPUC will covenant in a Continuing Disclosure Certificate, to be executed and delivered by the SFPUC concurrently with the issuance of the 2016 Series AB Bonds, to provide certain financial information and operating data relating to the Wastewater Enterprise and notices of certain enumerated events, in certain cases only if material. Such information and notices will be filed by the SFPUC with the Municipal Securities Rulemaking Board ("MSRB") through its Electronic Municipal Market Access system ("EMMA"). For more information

concerning the SFPUC's continuing disclosure commitment and the form of the Continuing Disclosure Certificate, see "CONTINUING DISCLOSURE" and "APPENDIX D – FORM OF CONTINUING DISCLOSURE CERTIFICATE."

Although in 2013 the SFPUC filed on EMMA materials describing the effectiveness of certain amendments to the Indenture, it failed to file timely a simultaneous and duplicative notice of a material event. The SFPUC has, at least once in the last five years, failed to file in a timely manner notice of a change in the rating of SFPUC bonds resulting from a change in the rating of a bond insurer. The current ratings of the SFPUC's bonds are correct on EMMA.

This Official Statement speaks only as of its date, and the information contained herein is subject to change. Except as required by the Continuing Disclosure Certificate, the SFPUC has no obligation to update the information in this Official Statement. See "CONTINUING DISCLOSURE" and "APPENDIX D – FORM OF CONTINUING DISCLOSURE CERTIFICATE."

Other Matters

Brief descriptions of the 2016 Series AB Bonds, the security and sources of payment for the 2016 Series AB Bonds, the SFPUC, and the Wastewater Enterprise are provided herein. Such descriptions do not purport to be comprehensive or definitive. Definitions of certain capitalized terms used herein may be found in "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE." All references made to various documents herein are qualified in their entirety by reference to the forms thereof, all of which are available for inspection at the office of the SFPUC at:

San Francisco Public Utilities Commission
525 Golden Gate Avenue, 13th Floor
San Francisco, CA 94102
Attention: Assistant General Manager, Business Services and Chief Financial Officer
(415) 554-3155

THE 2016 SERIES AB BONDS

General

The 2016 Series AB Bonds will be dated as of their date of delivery and will accrue interest from the date of delivery at the rates per annum set forth on the inside cover page of this Official Statement. Interest on the 2016 Series AB Bonds is payable on April 1 and October 1 of each year, beginning October 1, 2016. Interest on the 2016 Series AB Bonds will be calculated on the basis of a 360-day year composed of twelve 30-day months.

The 2016 Series AB Bonds will mature on the dates and in the principal amounts set forth on the inside cover page of this Official Statement. The 2016 Series AB Bonds will be issued in fully registered form in denominations of $5,000 or any integral multiple of $5,000.

Under the Indenture, the record date with respect to the payment of principal of and interest on the Bonds is the 15th day of the month immediately preceding an interest payment date, whether or not such day is a Business Day.

Securities Depository and Book-Entry System

The 2016 Series AB Bonds will be issued in fully registered form, registered in the name of Cede & Co., as nominee for The Depository Trust Company ("DTC"), New York, New York, as the Owner of the 2016 Series AB Bonds.

So long as DTC, or its nominee, Cede & Co., is the Owner of the 2016 Series AB Bonds, all payments on the 2016 Series AB Bonds will be made directly to DTC. Disbursement of such payments to the DTC Participants

will be the responsibility of DTC, and disbursement of such payments to the Beneficial Owners of the 2016 Series AB Bonds will be the responsibility of the DTC Participants. See "APPENDIX E–SECURITIES DEPOSITORY AND BOOK-ENTRY SYSTEM."

Redemption

Optional Redemption – 2016 Series A Bonds. The 2016 Series A Bonds maturing prior to October 1, 2026 are not subject to redemption prior to their stated maturity dates.

The 2016 Series A Bonds maturing on and after October 1, 2026 are subject to redemption prior to their stated maturity, at the option of the SFPUC, from and to the extent of any source of available funds, as a whole or in part (and if in part, as described in the sub-section "– Selection of 2016 Series AB Bonds for Redemption" below), on any date on or after October 1, 2025, at a redemption price equal to 100% of the principal amount of the 2016 Series A Bonds to be redeemed, plus accrued interest thereon to the date fixed for redemption, without premium. If 2016 Series A Bonds are to be optionally redeemed in part, the SFPUC shall direct the maturities, or portions thereof, to be redeemed.

Mandatory Sinking Fund Redemption – 2016 Series A Bonds. The 2016 Series A Bonds maturing on October 1, 2042, and October 1, 2046, and payable from the 2016 Series A Sinking Fund Account, are further subject to redemption prior to their stated maturity, from the 2016 Series A Sinking Fund Account, on any October 1 on or after October 1, 2041, and October 1, 2043, respectively, by lot within any such maturity if less than all of the 2016 Series A Bonds of such maturity and tenor be redeemed, upon payment of the principal amount thereof and accrued interest thereon to the date fixed for redemption, without premium.

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2041), will deposit in the 2016 Series A Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series A Term Bonds maturing on October 1, 2042, in the following respective principal amounts on the next succeeding October 1 in each of the following years.

Redemption Date (October 1)	Principal Amount
2041	$12,880,000
2042*	13,405,000

* Maturity

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2043), will deposit in the 2016 Series A Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series A Term Bonds maturing on October 1, 2046, in the following respective principal amounts on the next succeeding October 1 in each of the following years.

Redemption Date (October 1)	Principal Amount
2043	$13,955,000
2044	14,520,000
2045	15,115,000
2046*	15,730,000

* Maturity

Optional Redemption – 2016 Series B Bonds. The 2016 Series B Bonds maturing prior to October 1, 2026 are not subject to redemption prior to their stated maturity dates.

The 2016 Series B Bonds maturing on and after October 1, 2026 are subject to redemption prior to their stated maturity, at the option of the SFPUC, from and to the extent of any source of available funds, as a whole or in part (and if in part, as described in the sub-section "– Selection of 2016 Series AB Bonds for Redemption" below), on any date on or after October 1, 2025, at a redemption price equal to 100% of the principal amount of the 2016 Series B Bonds to be redeemed, plus accrued interest thereon to the date fixed for redemption, without premium. If 2016 Series B Bonds are to be optionally redeemed in part, the SFPUC shall direct the maturities, or portions thereof, to be redeemed.

Mandatory Sinking Fund Redemption – 2016 Series B Bonds. The 2016 Series B Bonds maturing on October 1, 2042, and October 1, 2046, and payable from the 2016 Series B Sinking Fund Account, are further subject to redemption prior to their stated maturity, from the 2016 Series B Sinking Fund Account, on any October 1 on or after October 1, 2041, and October 1, 2043, respectively, by lot within any such maturity if less than all of the 2016 Series B Bonds of such maturity and tenor be redeemed, upon payment of the principal amount thereof and accrued interest thereon to the date fixed for redemption, without premium.

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2041), will deposit in the 2016 Series B Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series B Term Bonds maturing on October 1, 2042, in the following respective principal amounts on the next succeeding October 1 in each of the following years.

Redemption Date (October 1)	Principal Amount
2041	$3,630,000
2042*	3,780,000

* Maturity

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2043), will deposit in the 2016 Series B Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series B Term Bonds maturing on October 1, 2046, in the following respective principal amounts on the next succeeding October 1 in each of the following years.

Redemption Date (October 1)	Principal Amount
2043	$3,935,000
2044	4,095,000
2045	4,260,000
2046*	4,435,000

* Maturity

Selection of 2016 Series AB Bonds for Redemption. Subject to DTC's procedures relating to the selection of bonds for redemption (see "APPENDIX E – SECURITIES DEPOSITORY AND BOOK-ENTRY SYSTEM," whenever less than all of the 2016 Series AB Bond of any one maturity within a Series are called for redemption, the Trustee will select the 2016 Series AB Bonds of such Series and maturity to be redeemed from the Outstanding 2016 Series AB Bonds of that Series and maturity. For purposes of such selection, 2016 Series AB Bonds will be deemed to be made up of $5,000 portions of principal.

Notice of Redemption. Notice of redemption will be mailed by the Trustee at least thirty days but not more than sixty days prior to the redemption date, to DTC (so long as the DTC Book-Entry System is used). The actual receipt by the owner of any 2016 Series AB Bond of notice of such redemption is not a condition precedent to

redemption, and failure to receive a redemption notice or any defect in a redemption notice will not affect the validity of the proceedings for the redemption of such 2016 Series AB Bonds or the cessation of the accrual of interest on the date fixed for such redemption. See "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE – GENERAL REDEMPTION PROVISIONS – Notice of Redemption" and "APPENDIX E – SECURITIES DEPOSITORY AND BOOK-ENTRY SYSTEM."

Rescission of Notice of Redemption. The SFPUC may, at its option, on or prior to the date fixed for redemption in any notice of optional redemption, rescind and cancel such notice of redemption by written request to the Trustee and the Trustee will give notice of such cancellation to the recipients of the notice of redemption being cancelled.

Effect of Redemption. When notice of redemption has been duly given as described above, and moneys for payment of the redemption price are held by the Trustee, the 2016 Series AB Bonds called for redemption will, on the redemption date designated in such notice, become due and payable at the redemption price specified in such notice; and from and after the date so designated interest on the 2016 Series AB Bonds called for redemption will cease to accrue, and such 2016 Series AB Bonds will cease to be entitled to any benefit or security under the Indenture, and the Owners of said 2016 Series AB Bonds will have no rights in respect thereof except to receive payment of the redemption price thereof. The Trustee, upon surrender for payment of any of said 2016 Series AB Bonds, will pay such 2016 Series AB Bonds at the redemption price, together with accrued interest thereon. All 2016 Series AB Bonds redeemed will be cancelled upon surrender and no 2016 Series AB Bonds will be issued in place thereof.

Defeasance

The obligations of the SFPUC and the pledge, lien, covenants and agreements of the SFPUC made or provided for in the Indenture will be fully discharged and satisfied as to any 2016 Series AB Bond and such Bond will no longer be deemed outstanding thereunder if certain conditions set forth in the Indenture are satisfied. See "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE – Defeasance."

PLAN OF FINANCE

The 2016 Series A Bonds are being issued primarily to finance and refinance (through the retirement of a portion of the outstanding commercial paper notes on or prior to May 26, 2016) select SSIP projects. The SFPUC expects that the projects financed and refinanced from proceeds of the 2016 Series A Bonds will be projects set forth in Appendix F to this Official Statement. The 2016 Series B Bonds are being issued primarily to finance and refinance (through the retirement of a portion of the outstanding commercial paper notes on or prior to May 26, 2016) select projects authorized in the Wastewater Enterprise capital program. Proceeds of the 2016 Series AB Bonds will also be applied to (a) fund capitalized interest with respect to the 2016 Series AB Bonds for approximately two years, and (b) pay the costs of issuance of the 2016 Series AB Bonds. The SFPUC expects the financing of projects to include reimbursements to the SFPUC from proceeds of the 2016 Series AB Bonds for certain prior expenditures relating to such projects. See "ESTIMATED SOURCES AND USES OF FUNDS," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION," "APPENDIX F – ANTICIPATED 2016 SERIES A PROJECTS," "WASTEWATER ENTERPRISE CAPITAL PROGRAM" and "SEWER SYSTEM IMPROVEMENT PROGRAM."

ESTIMATED SOURCES AND USES OF FUNDS

The proceeds of the 2016 Series AB Bonds are expected to be applied as follows:

Sources of Funds	
2016 Series A Par Amount	$240,580,000.00
2016 Series A Original Issue Premium	40,801,897.70
2016 Series B Par Amount	67,820,000.00
2016 Series B Original Issue Premium	11,503,617.65
Total Sources	$360,705,515.35
Uses of Funds	
Deposit to 2016 Series A Capital Project Account[1]	$205,123,293.08
Retirement of Commercial Paper Notes—2016 Series A[1]	53,439,773.39
Deposit to 2016 Series B Capital Project Account[2]	52,330,967.86
Retirement of Commercial Paper Notes—2016 Series B[2]	20,560,226.61
Underwriter's Discount	1,072,654.41
Costs of Issuance[3]	885,000.00
Capitalized Interest[4]	27,293,600.00
Total Uses	$360,705,515.35

(1) To finance and refinance a portion of the SSIP. The SFPUC expects the projects financed and refinanced will be the projects set forth in Appendix F to this Official Statement. See "PLAN OF FINANCE," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION," "APPENDIX F – ANTICIPATED 2016 SERIES A PROJECTS," and "SEWER SYSTEM IMPROVEMENT PROGRAM."

(2) To finance and refinance select projects authorized in the Wastewater Enterprise capital program. See "PLAN OF FINANCE," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION" and "WASTEWATER ENTERPRISE CAPITAL PROGRAM."

(3) The costs of issuance include amounts for legal fees, Trustee's fees, financial advisory fees, fees of the Public Utilities Revenue Bond Oversight Committee, rating agency fees, printing costs, and other issuance costs relating to the issuance of the 2016 Series AB Bonds.

(4) Represents capitalized interest with respect to the 2016 Series AB Bonds for approximately two years.

GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION

The 2016 Series A Bonds are being designated by the SFPUC as "Green Bonds." The purpose of designating the offered bonds as "Green Bonds" is to allow investors to invest directly in bonds which finance environmentally beneficial projects ("Green Projects"). For purposes of such determination, the SFPUC considers the projects set forth in Appendix F to this Official Statement to be Green Projects. See "PLAN OF FINANCE," "ESTIMATED SOURCES AND USES OF FUNDS," "APPENDIX F – ANTICIPATED 2016 SERIES A PROJECTS," and "SEWER SYSTEM IMPROVEMENT PROGRAM." The 2016 Series A Bonds will not constitute "exempt facility bonds" issued to finance "qualified green building and sustainable design projects" within the meaning of Section 142(1) of the Code.

In addition, the Climate Bonds Standards Board (the "Climate Board"), an advisory board reporting to the Climate Bonds Initiative, a charity registered in England and Wales, has certified the 2016 Series A Bonds. The certification of the 2016 Series A Bonds reflects only the views of the Climate Board. The explanation of the significance of this certification may be obtained from the Climate Board, which has based its certification upon its own investigations, studies, assumptions and criteria. The SFPUC has provided certain additional information and materials to the Climate Board, including information concerning anticipated capital project expenditures (some of which does not appear in this Official Statement). As part of the certification process, the SFPUC retained Sustainalytics U.S., Inc., a subsidiary of Sustainalytics Holding, B.V, Netherlands (collectively, "Sustainalytics"), to provide third-party analysis and its opinion regarding, among other things, the SFPUC's green bond framework and the compatibility of the SFPUC's financing and refinancing of the projects listed in Appendix F with draft Water Climate Bonds Standards. The SFPUC expects that Sustainalytics will make its report and opinion relating to the

2016 Series A Bonds available on its website at: http://www.sustainalytics.com/sustainability-bond-services. However, neither Sustainalytics's report nor its opinion, nor any other information contained on Sustainalytics's website, are incorporated by reference herein and the SFPUC assumes no responsibility for the accuracy or completeness thereof.

The Climate Board is not a licensed broker-dealer or a nationally recognized statistical ratings organization. Certification by the Climate Board is not a recommendation to buy, sell or hold securities, and such certification may be subject to revision or withdrawal, including, without limitation, in the event that the SFPUC's future capital expenditures from the proceeds of the 2016 Series A Bonds vary from the anticipated expenditures reviewed by the Climate Board. The SFPUC will undertake reasonable efforts to ensure that any adjustment of capital expenditures or other actions taken with respect to the 2016 Series A Bonds will not result in revision or withdrawal of the Climate Board's certification; however, there can be no guarantee that such adjustment or other action or a future revision to the Climate Bonds Initiative's criteria for certifying bonds will not result in a withdrawal or revision of the Climate Board's certification. The SFPUC expects to post information respecting the expenditure of the proceeds of the 2016 Series A Bonds on its website.

The 2016 Series B Bonds are not being designated as "Green Bonds." The Indenture does not restrict the use of proceeds of future issuances of Bonds to the financing of Green Projects and in the future the SFPUC may issue Additional Bonds which are not designated as Green Bonds or certified by the Climate Board.

The repayment obligations of the SFPUC with respect to the 2016 Series AB Bonds are not conditioned on the completion of any particular project or the satisfaction of any condition relating to the status of the 2016 Series A Bonds as Green Bonds or the certification of such bonds by the Climate Board. See "Security for the Bonds."

SECURITY FOR THE BONDS

Pledge of Net Revenues

General. Under the Indenture, the SFPUC has irrevocably pledged the Net Revenues of the Wastewater Enterprise to the punctual payment of principal of, premium, if any, and interest on the Bonds, which consist of any parity revenue bonds issued under the Indenture, and Parity State Loans. This pledge is subject to the flow of funds contained in the Indenture, as described below. See "– Flow of Funds."

The facilities comprising the Wastewater Enterprise have not been pledged or mortgaged and do not otherwise secure payment of the Bonds.

Pursuant to Section 5451 of the California Government Code, the pledge of, lien on and security interest in Net Revenues and certain other funds granted by the Indenture is valid and binding in accordance with the terms thereof from the time of issuance of the 2016 Series AB Bonds; the Net Revenues and such other funds will be immediately subject to such pledge; and such pledge will constitute a lien and security interest which will immediately attach to such Net Revenues and other funds and will be effective, binding and enforceable against the SFPUC, its successors, purchasers of the Net Revenues, creditors, and all others asserting rights therein to the extent set forth and in accordance with the terms of the Indenture irrespective of whether those parties have notice of such pledge and without the need for any physical delivery, recordation, filing or other further act. Such pledge, lien and security interest are not subject to the provisions of Article 9 of the California Uniform Commercial Code.

For definitions of capitalized terms used herein and not otherwise defined, see "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE – Definitions."

Limited Obligation. THE SFPUC IS NOT OBLIGATED TO PAY THE PRINCIPAL OF, PREMIUM, IF ANY, OR INTEREST ON THE 2016 SERIES AB BONDS EXCEPT FROM NET REVENUES. THE SFPUC HAS NO TAXING POWER. THE GENERAL FUND OF THE CITY IS NOT LIABLE FOR THE PAYMENT OF THE PRINCIPAL OF, PREMIUM, IF ANY, OR INTEREST ON THE 2016 SERIES AB BONDS, AND NEITHER THE CREDIT NOR THE TAXING POWER OF THE CITY IS PLEDGED TO THE PAYMENT OF THE PRINCIPAL OF, PREMIUM, IF ANY, AND INTEREST ON THE 2016 SERIES AB BONDS. THE 2016

SERIES AB BONDS ARE NOT SECURED BY A LEGAL OR EQUITABLE PLEDGE OF, OR CHARGE, LIEN, OR ENCUMBRANCE UPON, ANY OF THE PROPERTY OF THE CITY OR OF THE SFPUC OR ANY OF ITS INCOME OR RECEIPTS, EXCEPT NET REVENUES.

Wastewater Enterprise. The Indenture defines "Enterprise" (referred to in this Official Statement as the "Wastewater Enterprise") as meaning the whole and each and every part of the municipal sanitary waste and storm water collection, treatment and disposal system and auxiliary or related facilities of the SFPUC, including all of the presently existing system of the SFPUC for the collection, treatment and disposal of sanitary waste and storm water and all future additions, betterments, and extensions to the system or any part thereof.

Net Revenues. The Indenture defines "Net Revenues" as all of the Revenues, less all Operation and Maintenance Costs of the Enterprise.

The Indenture defines "Revenues" as all gross revenues of the Wastewater Enterprise, including all charges received and all other income and receipts derived by the SFPUC from the operation of the Wastewater Enterprise, or arising from the Wastewater Enterprise, including connection and installation charges, but excluding:

(a) any money received by or for the account of the SFPUC from the levy or collection of taxes;

(b) moneys received from the State of California and the United States of America and required to be deposited in restricted funds;

(c) refundable deposits made to establish credit;

(d) advances and contributions made to the SFPUC to be applied to construction;

(e) moneys received constituting casualty insurance proceeds with respect to all or any part of the Wastewater Enterprise (which shall be received and disposed of pursuant to the Indenture) and moneys received constituting other insurance proceeds;

(f) moneys received from the sale or disposition of all or any part of the Wastewater Enterprise (which shall be received and disposed of pursuant to the Indenture);

(g) moneys received upon the taking by or under the threat of eminent domain of all or any part of the Wastewater Enterprise (which moneys shall be received and disposed of pursuant to the Indenture);

(h) proceeds from Bonds issued by the SFPUC or proceeds from loans or other indebtedness obtained by the SFPUC; and

(i) moneys or securities received by the SFPUC as gifts or grants the use of which is restricted by the donor or grantor.

The term "Revenues" also includes (i) all interest or other income (excluding profits or losses from the sale or disposition of Permitted Investments or other securities owned by or on behalf of the SFPUC) derived from the deposit or investment of any moneys in any fund or account established under the Indenture (excluding any Rebate Fund and any escrow fund pledged for the payment of defeased bonds) or in any fund or account of the Wastewater Enterprise and legally available to pay Debt Service, and (ii) any other moneys, proceeds and other amounts that the SFPUC determines should be "Revenues" under the Indenture.

The Indenture defines "Operation and Maintenance Costs of the Enterprise" as the reasonable and necessary costs of operating and maintaining the Wastewater Enterprise, calculated on the basis of generally accepted accounting principles, including (among other things) salaries and wages, fees for services, costs of materials, supplies and fuel, reasonable expenses of management, legal fees, accounting fees, repairs and other expenses necessary to maintain and preserve the Wastewater Enterprise in good repair and working order, and reasonable amounts for administration, overhead, insurance, taxes (if any), and the payment of pension charges and

proportionate payments to such compensation and other insurance or outside reserve funds as the SFPUC may establish or the Board of Supervisors may require with respect to employees of the SFPUC, as provided in the Charter.

However, the term "Operation and Maintenance Costs of the Enterprise" excludes in all cases (a) depreciation and obsolescence charges or reserves therefor, (b) amortization of intangibles or other bookkeeping entries of a similar nature, (c) costs of capital additions, replacements, betterments, extensions or improvements to the Wastewater Enterprise, which under generally accepted accounting principles are chargeable to a capital account or to a reserve for depreciation, (d) charges for the payment of principal of and interest on any revenue bonds or other indebtedness issued before or after the date of the Indenture for Wastewater Enterprise purposes and (e) such costs as are scheduled to be paid by the SFPUC from moneys other than Revenues, such moneys to be clearly available for such purpose.

Flow of Funds

The Indenture provides that all Revenues must be paid into the Revenue Fund, which must be maintained in the City Treasury. Moneys in the Revenue Fund, including earnings thereon, are required by the Indenture to be applied for the following purposes and only in the following order of priority:

(a) payment of Operation and Maintenance Costs of the Enterprise;

(b) payment of Bonds, Parity State Loans, Policy Costs and amounts due as reimbursement under any Letter of Credit Agreement, as provided in the Indenture; and

(c) any other lawful purpose of the SFPUC.

Net Revenues deposited in the Revenue Fund, as described in (b) above, will be applied to pay interest and principal on the Bonds and to make deposits to the Bond Reserve Fund if the amounts therein are less than the Required Reserve. See "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE."

The Indenture defines "Refundable Credits" as (a) with respect to a Series of Bonds issued as Build America Bonds under Section 54AA of the Code, the amounts which are payable by the Federal government under Section 6431 of the Code, which the SFPUC has elected to receive under Section 54AA(g)(1) of the Code, and (b) with respect to a Series of Bonds issued as Build America Bonds under any other provision of the Code that creates a substantially similar direct-pay subsidy program, the amounts which are payable by the Federal government under the applicable provisions of the Code, which the SFPUC has elected to receive under the applicable provisions of the Code.

The Indenture defines "Build America Bonds" as any bonds or other obligations issued as Build America Bonds under Section 54AA of the Code, or under any other provision of the Code that creates a substantially similar direct-pay subsidy program.

The Indenture provides that all of the Refundable Credits received by the SFPUC shall be deposited immediately upon receipt in the Interest Fund, and such Refundable Credits are irrevocably pledged to the punctual payment of the interest on the Bonds issued as Build America Bonds, and the Refundable Credits shall not be used for any other purpose while any of the Bonds issued as Build America Bonds remain Outstanding.

Rate Covenants

Sufficiency of Revenues. The SFPUC has covenanted in the Indenture that it will, at all times while any of the Bonds remain Outstanding, establish, fix, prescribe and collect rates, fees and charges in connection with the sanitary wastewater and stormwater collection, treatment and disposal services and facilities furnished by the Wastewater Enterprise so as to yield Revenues at least sufficient, after making reasonable allowances for contingencies and error in the estimates, calculated on a cash basis, to pay the following amounts:

(a) the interest on and principal of the Bonds as they become due and payable (but not including any interest for which moneys have been deposited in the Interest Fund from the proceeds of any Series of Bonds or from any other source);

(b) all other payments required for compliance with the terms of the Indenture and of any Supplemental Indenture providing for the issuance of Additional Bonds pursuant to the Indenture;

(c) all other payments to meet any other obligations of the SFPUC which are charges, liens or encumbrances upon, or payable from, Revenues; and

(d) all current Operation and Maintenance Costs of the Enterprise (but not including such Operation and Maintenance Costs of the Enterprise as are scheduled to be paid by the SFPUC from moneys other than Revenues, such moneys to be clearly available for such purpose).

Debt Service Coverage. In addition to the requirements described above, the SFPUC has covenanted that it will, at all times while any of the Bonds remain Outstanding, establish, fix, prescribe and collect rates, fees and charges in connection with the sanitary waste and stormwater collection, treatment and disposal services and facilities furnished by the Wastewater Enterprise so as to yield Net Revenues during the immediately ensuing period of twelve months which (together with any fund balances of the SFPUC or the Wastewater Enterprise which are available for payment of Debt Service and are not budgeted to be expended during such twelve months, but excluding the Bond Reserve Fund), calculated on a cash basis, are at least equal to 1.25 times the Annual Debt Service (but excluding any interest moneys for the payment of which have been deposited in the Interest Fund from the proceeds of any Series of Bonds or from any other source) for said twelve-month period.

The SFPUC may make adjustments from time to time in such rates, fees and charges and may make such classification thereof as it deems necessary, but is obligated not to reduce such rates, fees and charges below those in effect unless the Revenues resulting after such reduced rates are put into effect will at all times be sufficient to meet the requirements described in this section, "Rate Covenants."

"Annual Debt Service" is defined in the Indenture as the sum of principal and interest on all Outstanding Bonds and Parity State Loans as computed for the twelve-month period ending June 30 to which reference is made, and calculated in a manner consistent with the determination of Average Annual Debt Service and Maximum Annual Debt Service as provided in the definitions thereof. In determining the amount of interest coming due during any twelve-month period ending June 30 on any Series of Bonds that were issued as Build America Bonds, including the 2010 Series B Bonds, amounts equal to the Refundable Credits the SFPUC is scheduled to receive during each such twelve-month period ending June 30 will be deducted from such interest. See "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE."

No Reserve Account

The Indenture requires that the Bond Reserve Fund be established with the Trustee and funded in an amount equal to the "Required Reserve," if any, applicable to each series of Bonds. If the Required Reserve for a Series of Bonds is greater than zero, the Indenture requires the establishment of a bond reserve account (each, a "Reserve Account") within the Bond Reserve Fund for such Series of Bonds, and requires the deposit in that bond reserve account of an amount equal to the Required Reserve for the related Series of Bonds.

The Fourth Supplemental Indenture does *not* require the establishment of a Reserve Account for the 2016 Series AB Bonds. The Reserve Accounts established with respect to the 2010 Series A Bonds and the 2010 Series B Bonds do not secure the 2016 Series AB Bonds. No Reserve Account has been established with respect to the 2013 Series A Bonds or the 2013 Series B Bonds.

Additional Series of Bonds

The Charter and the Indenture authorize the issuance of Additional Bonds payable from Net Revenues on a parity with Outstanding Bonds and the 2016 Series AB Bonds, upon satisfaction of the conditions set forth therein.

The SFPUC expects to issue Additional Bonds to finance the costs of additional improvements included in the SSIP and other portions of the Wastewater Enterprise's capital program. See "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM" and "RISK FACTORS – Costs of the SSIP; Timely Completion of the SSIP."

Charter Requirements. Under the Charter, the SFPUC may issue revenue bonds (including Additional Bonds) relating to the Wastewater Enterprise without voter approval in the following circumstances, among others:

(a) to issue revenue bonds (including Additional Bonds) approved by an affirmative vote of two-thirds of the members of the Board of Supervisors for the purpose of reconstructing, replacing, expanding, repairing or improving the Wastewater Enterprise;

(b) to issue bonds (including Additional Bonds) approved by an affirmative vote of three-fourths of the members of the Board of Supervisors if the bonds are to finance buildings, fixtures or equipment which are deemed necessary by the Board of Supervisors to comply with an order of a duly constituted state or federal authority having jurisdiction over the Wastewater Enterprise; and

(c) to issue refunding bonds which are expected to result in net debt service savings to the City on a present value basis, calculated as described in the SFPUC's Debt Management Policies and Procedures.

The Charter also generally authorizes the SFPUC to issue revenue bonds upon the approval of a majority of the voters voting on the proposition at a general or special election.

Notwithstanding any other provision of the Charter or of any ordinance of the City, the Board of Supervisors may, in connection with the issuance of additional Bonds as described under paragraph (a) above, take any and all actions necessary to authorize, issue and repay such Bonds, including, but not limited to, modifying schedules of rates and charges to provide for the payment and retirement of such Bonds, subject to the following conditions:

(a) Certification by an independent engineer retained by the SFPUC that:

(1) the projects to be financed by the additional Bonds, including the prioritization, cost estimates and scheduling, meet utility standards; and

(2) that estimated net revenue after payment of operating and maintenance expenses will be sufficient to meet debt service coverage and other indenture or resolution requirements, including debt service on the bonds to be issued, and estimated renewal and replacement costs.

(b) Certification by the City Planning Department that facilities under the jurisdiction of the SFPUC funded with such Bonds will comply with applicable requirements of the California Environmental Quality Act ("CEQA").

Indenture Requirements. The Indenture provides that Additional Bonds secured on a parity with the Bonds may be issued for any lawful purpose if no Event of Default has occurred and is continuing under the Indenture or any Supplemental Indenture and no event has occurred which, but for the passage of time or the giving of notice, would constitute an Event of Default under the Indenture or any Supplemental Indenture. Prior to the issuance of such Additional Bonds, the SFPUC is required to file with the Trustee, among other documents, the following:

(a) a Certificate of the Commission demonstrating that the SFPUC has complied with the rate covenant under the Indenture and that the requirements for issuing Additional Bonds under the Indenture have been met;

(b) if any portion of the proceeds of such Series of Bonds is to be used to finance construction of a Project, a certificate of the Consulting Engineers setting forth (i) the estimated date of completion for the

portion of the Project for which such Series of Bonds is being issued and for any other uncompleted portion of the Project, and (ii) an estimate of the cost of construction of such portion of the Project and of any other uncompleted portion of the Project;

(c) a Certificate of the Commission setting forth for each of the next three Fiscal Years (and, if any portion of the proceeds of such Series of Bonds is to be used to finance construction, the three Fiscal Years following the Fiscal Year in which the Consulting Engineers estimate such portion of the Project will be completed), estimates of (i) Revenues, (ii) Operation and Maintenance Costs of the Enterprise and (iii) Net Revenues; and

(d) a Certificate of the Commission setting forth (i) the estimates of Net Revenues as set forth in the Certificate of the SFPUC pursuant to paragraph (c) above for each of such three Fiscal Years (and, if any portion of the proceeds of such Series of Bonds is to be used to finance construction, the three Fiscal Years following the Fiscal Year in which the Consulting Engineers estimate such portion of the Project will be completed), (ii) the Annual Debt Service for each of such Fiscal Years, including Annual Debt Service as estimated in such Certificate of the Commission with respect to future Series of Bonds, if any, which such Certificate of the Commission shall estimate will be required to complete payment of the cost of construction of such portion of the Project and any other uncompleted portion of the Project, and (iii) demonstrating that the estimated Net Revenues (together with any fund balances of the SFPUC which are available for the payment of Debt Service but excluding the Bond Reserve Fund) in each of the Fiscal Years set forth in (c) above is at least equal to 1.25 times the Annual Debt Service for such respective Fiscal Years.

See "APPENDIX A – SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE – Definitions" and "– ADDITIONAL SERIES OF BONDS – Issuance of Additional Bonds."

The Indenture provides the SFPUC with flexibility as to the nature and terms of any Additional Bonds issued with a lien and charge on Net Revenues on a parity with the Outstanding Bonds. Such Additional Bonds may: mature over any period of time; bear interest at a fixed, variable or zero rate; be in any denominations; be in any form (including registered, coupon or book-entry); include or exclude redemption provisions; be subject to optional or mandatory tender for purchase; be sold at such price or prices; be further secured by any separate and additional security; and otherwise include such additional terms and provisions as the SFPUC may determine, consistent with the Indenture and applicable provisions of the Charter.

Additional Parity State Loans

Pursuant to the Charter, the SFPUC can incur indebtedness, including additional State loans, without voter approval, but subject to Board of Supervisors' approval. The Indenture permits the SFPUC to enter into Parity State Loans or loans on a subordinated lien basis relative to the Bonds, as determined by the SFPUC. Pursuant to the State's Revolving Fund Loan program (the "SRF Loan Program"), the SFPUC entered into a State Revolving Fund Loan dated as of October 16, 2015 and plans to enter into additional State Revolving Fund Loans (each an "SRF Loan") over the next several years to obtain long-term financing for a portion of the SSIP. See "OBLIGATIONS PAYABLE FROM REVENUES – Parity State Loans."

Under the Indenture, the SFPUC may only enter into additional Parity State Loans if no Event of Default has occurred and is continuing under the Indenture or any Supplemental Indenture (and no event has occurred which, but for the passage of time or the giving of notice, would constitute an Event of Default under the Indenture or any Supplemental Indenture).

In addition, in connection with the execution and delivery of such Parity State Loans, the SFPUC is required to deliver a Certificate to the Trustee setting forth, for each of the next three Fiscal Years after the delivery of the Parity State Loans:

(i) the Revenues and Operation and Maintenance Costs of the Enterprise; and

(ii) the Annual Debt Service assuming delivery of the Parity State Loans;

demonstrating that the estimated Net Revenues (together with any fund balances of the SFPUC which are available for Debt Service, but excluding the Bond Reserve Fund), in each of such three Fiscal Years is at least equal to 1.25 times the Annual Debt Service.

Refunding Bonds

City Charter Requirements. The Charter permits the issuance of refunding bonds without voter approval only if such refunding results in net debt service savings on a present value basis, calculated as described in the SFPUC's Debt Management Policies and Procedures.

Indenture Requirements. The Indenture provides that Additional Bonds may be issued to refund any Bonds without meeting the test for the issuance of Additional Bonds described above, if the SFPUC delivers to the Trustee (among other documents) a certificate of a Qualified Financial Advisor to the effect that the Average Annual Debt Service for the Additional Bonds will be equal to or less than the Average Annual Debt Service on the Bonds to be refunded.

Subordinate Obligations; Obligations Not Payable from Revenues

The Indenture permits the SFPUC to authorize and issue bonds, notes, warrants, certificates or other obligations or evidences of indebtedness, the principal of or interest on which would be payable either (i) from Net Revenues after and subordinate to the payment from Net Revenues of the principal of and interest on the Bonds, or (ii) from moneys which are not Revenues. The SFPUC may issue bonds or incur other indebtedness secured by a pledge of Net Revenues on a basis subordinate to the pledge thereof securing the Bonds without limitation.

Investments

The Indenture provides that moneys in all funds and accounts held by the Trustee under the Indenture shall be invested upon receipt in Permitted Investments as directed by the SFPUC and all accounts funds and accounts held by the Treasurer shall be invested in Legal Investments. "Legal Investments" means any bonds, notes, certificates of indebtedness, bills, acceptances or other securities in which the Treasurer may legally invest the SFPUC's funds. For information regarding the investment of moneys held in the various funds and accounts of the SFPUC, see "FINANCIAL OPERATIONS – Investment of SFPUC Funds."

OBLIGATIONS PAYABLE FROM REVENUES

Authority for Issuance of Revenue Bonds and Other Obligations Payable from Revenues

The Charter authorizes the SFPUC to issue revenue bonds and commercial paper notes and to incur other obligations payable from or secured by a pledge of Revenues. The 2016 Series AB Bonds are being issued under Proposition E, Ordinance 107-14, adopted by the Board of Supervisors on July 2, 2014, a resolution adopted by the Commission on March 22, 2016 and a resolution adopted by the Board of Supervisors on April 26, 2016. See "SECURITY FOR THE BONDS – Additional Series of Bonds," "– Additional Parity State Loans" and "– Refunding Bonds."

Outstanding Parity Revenue Bonds

The following outstanding Bonds (the "Outstanding Bonds") have been issued pursuant to the Indenture and are secured by a parity pledge of Net Revenues.

Series of Bonds	Purpose	Initial Principal Amount	Principal Amount Outstanding as of April 1, 2016
Wastewater Revenue Bonds, 2010 Series A	Refinance capital program costs	$ 47,050,000	$ 47,050,000
Wastewater Revenue Bonds, 2010 Series B	Finance and refinance SSIP and other capital program costs	192,515,000	192,515,000
Wastewater Revenue Bonds, 2013 Series A	Refund 2003 Clean Water Bonds and refund SRF Loans	193,400,000	98,585,000
Wastewater Revenue Bonds, 2013 Series B	Finance and refinance SSIP and other capital program costs	331,585,000	331,585,000
Total:		$764,550,000	$669,735,000

Additional Bonds

The SFPUC intends to issue Additional Bonds under the Indenture to finance select capital program projects, including SSIP projects. See "WASTEWATER ENTERPRISE CAPITAL PROGRAM," "SEWER SYSTEM IMPROVEMENT PROGRAM" and "FINANCING PLAN WASTEWATER ENTERPRISE CAPITAL PROGRAM."

Parity State Loans

The SFPUC plans to enter into Parity State Loans with the State of California over the next several years to obtain long-term financing for a portion of the SSIP. The Parity State Loans are low-cost SRF Loans provided and administered by the State Water Resources Control Board on a project-by-project basis. The State's recently revised SRF Loan Program provides favorable terms to the SFPUC, including up to 30-year amortizing loan terms which commence one year following project completion, an interest rate on each SRF Loan that is equal to one-half of the true interest cost for the State of California's most recent General Obligation Bond issue at the time such loan is entered into, and repayment of the SRF Loans on a parity with the Bonds rather than on a basis senior to the Bonds as had been the case in connection with certain loans entered into under the prior version of the program. The SRF Loan Program offers loans to applicant entities based on available moneys and placement on a statewide priority list. Although placement on the priority list is a necessary condition to receiving SRF Loan Program moneys, placement on the priority list does not create an obligation on the part of the applicant to accept SRF Loan Program moneys.

Pursuant to the State's new SRF Loan Program, the SFPUC entered into an SRF Loan dated as of October 16, 2015 in the amount of $7,435,000 for the Lake Merced Green Infrastructure – Holloway Green Street Project, an SSIP project. However, the Wastewater Enterprise has not yet submitted a request for reimbursement to, nor received reimbursement from, the State, so no repayment obligation has yet been incurred or calculated. Once funds are drawn pursuant to a reimbursement request, the interest rate on the loan will be 1.60%, and principal will amortize over a 30-year term commencing one year following completion of the project. The SFPUC currently expects debt service on the SRF Loan will first become due beginning in Fiscal Year 2018-19. This loan is the first of several SRF Loans the SFPUC anticipates entering into in order to finance SSIP projects. The State estimates that it can enter into approximately $2.24 billion of SRF Loans with the SFPUC over the next five years based on current financial projections of the Wastewater Enterprise.

The SFPUC had previously entered into thirteen SRF Loans totaling an aggregate principal amount of $280 million during the period from 1991 to 2001 under the State's original SRF Loan Program, including certain loans that were payable on a basis senior to repayment of the Bonds. As of February 2013, the SFPUC had paid in full all such prior SRF Loans, including by refunding certain loans with proceeds of the Series 2013 Series A Wastewater

Revenue Bonds. The SFPUC has covenanted in the Fourth Supplemental Indenture not to enter into state loans payable on a basis senior to the Bonds.

Subordinate Debt and Commercial Paper

No Limits on Subordinate Debt. The SFPUC may issue bonds or incur other indebtedness secured by a pledge of Net Revenues on a basis subordinate to the pledge thereof securing the Bonds, without limitation.

Commercial Paper Program. In 2007, the SFPUC established a commercial paper program (authorized by Proposition E) to meet the interim expenditure and encumbrance needs relating to the Wastewater Enterprise's capital program. The commercial paper program has a current authorization of $500 million. As of April 1, 2016, the SFPUC had $135 million aggregate principal amount of Wastewater Commercial Paper Notes outstanding, all of which are secured by and payable from Net Revenues on a basis subordinate to the payment of debt service on the Bonds.

The Wastewater Commercial Paper Notes are supported by respective bank credit facilities, as follows: (a) a Letter of Credit (LOC) with Wells Fargo N.A. in the amount of $75 million and a stated expiration date of February 11, 2017; (b) an LOC with Bank of America, N.A. in the amount of $75 million and a stated expiration date of July 7, 2017; (c) an LOC with Barclays Bank PLC in the amount of $100 million and a stated expiration date of October 19, 2018; (d) a Revolving Credit and Term Loan Agreement ("RCTLA") with The Bank of Tokyo Mitsubishi UFJ, Ltd., acting through its New York Branch in the amount of $75 million and a stated expiration date of July 7, 2018; (e) an RCTLA with U.S. Bank National Association in the amount of $75 million and a stated expiration date of February 8, 2017; and (f) an RCTLA with State Street Bank and Trust Company in the amount of $100 million and a stated expiration date of October 20, 2016.

Contingent Payment Obligations

The Wastewater Enterprise has no interest rate swaps, caps or hedges or other contingent payment obligations payable from Revenues. The Wastewater Enterprise may in the future, however, incur contingent payment obligations payable from Revenues. Such contingent payment obligations may be payable on a parity with the Bonds if the conditions for the issuance of parity debt under the Indenture are met. See "SECURITY FOR THE BONDS – Additional Series of Bonds."

Other Obligations Payable from Revenues

The SFPUC completed the construction of a new, 13-story office building at 525 Golden Gate Avenue in San Francisco to house the administrative offices of the SFPUC's three utility enterprises and moved into the building in July 2012. Total project costs were approximately $202 million and were financed with land sale proceeds, fund balances, grants and the proceeds of certificates of participation (the "2009 Golden Gate COPs"), representing interests in a City General Fund lease, executed and delivered in two series (one of which constitutes Build America Bonds) on October 7, 2009 in the aggregate principal amount of $167,670,000. Pursuant to a Memorandum of Understanding between the City and the SFPUC, the SFPUC will reimburse the City General Fund for all debt service in connection with this City financing (net of Refundable Credits received). The SFPUC allocates such payment obligations internally among its three utility enterprises based on percentage usage. The Wastewater Enterprise has been allocated 18.88% of such obligations, payable from Revenues on a basis subordinate to the payment of principal of and interest on the Bonds and any Parity State Loans.

Revenue Bond Oversight Committee

On November 5, 2002, the voters of the City adopted Proposition P, an ordinance that established the Public Utilities Revenue Bond Oversight Committee ("RBOC") to report publicly to the Mayor, the SFPUC and the Board of Supervisors regarding the expenditure of revenue bond proceeds on the repair, replacement, upgrading and expansion of the Wastewater Enterprise, the Water Enterprise, and the Power Enterprise (each as defined herein).

The RBOC has seven members appointed as follows: two by the Mayor, two by the Board of Supervisors, one by the City Controller, one by the Bay Area Water Users Association under the auspices of the Bay Area Water Supply and Conservation Agency. The seventh member is the City's Budget Analyst or his or her representative. The work of the RBOC is funded by 1/20th of 1% of the gross bond proceeds of revenue bond issuances or sales to the extent permitted by law.

The RBOC's current term expires on January 1, 2019.

The RBOC may, by majority vote of all its members, prohibit the issuance or sale of authorized SFPUC revenue bonds which have yet to be issued or sold if, after reviewing materials provided by the SFPUC and conducting its own independent audit, and after consultation with the City Attorney, the RBOC determines that revenue bond proceeds have been or are being spent on purposes not authorized by the authorizing bond resolution or otherwise in a manner amounting to an illegal expenditure or illegal waste of such revenue bond proceeds. The SFPUC may appeal such a decision to the Board of Supervisors within thirty days. The Board of Supervisors may overturn such a decision by the RBOC by a two-thirds vote of all members of the Board of Supervisors with evidence from the SFPUC of corrective measures satisfactory to the Board of Supervisors or may remand the decision to the RBOC for further consideration.

Debt Service Requirements

Set forth in the following table are debt service requirements for the 2010 Series A Bonds, the 2010 Series B Bonds, the 2013 Series A Bonds, the 2013 Series B Bonds and the 2016 Series AB Bonds.

DEBT SERVICE REQUIREMENTS ON OUTSTANDING BONDS

Fiscal Year (ending June 30)	Outstanding Bonds[1], [2]	2016 Series A Bonds Principal	2016 Series A Bonds Interest	2016 Series B Bonds Principal	2016 Series B Bonds Interest	Capitalized Interest	Total Debt Service[3]
2017	$ 52,253,751	-	$ 9,078,459	-	$ 2,559,229	$(11,637,688)	$ 52,253,751
2018	50,496,301	-	10,645,750	-	3,001,050	(13,646,800)	50,496,301
2019	50,553,826	-	10,645,750	-	3,001,050	(2,009,112)	62,191,513
2020	50,639,601	-	10,645,750	-	3,001,050	-	64,286,401
2021	50,711,476	-	10,645,750	-	3,001,050	-	64,358,276
2022	49,248,476	-	10,645,750	-	3,001,050	-	62,895,276
2023	45,719,966	-	10,645,750	-	3,001,050	-	59,366,766
2024	41,379,962	$ 5,475,000	10,508,875	$ 1,545,000	2,962,425	-	61,871,262
2025	41,252,516	5,760,000	10,228,000	1,625,000	2,883,175	-	61,748,691
2026	41,118,113	6,055,000	9,962,900	1,705,000	2,808,450	-	61,649,463
2027	40,969,768	6,365,000	9,682,675	1,795,000	2,729,475	-	61,541,918
2028	40,818,158	6,690,000	9,356,300	1,885,000	2,637,475	-	61,386,933
2029	40,653,598	7,035,000	9,013,175	1,980,000	2,540,850	-	61,222,623
2030	40,478,303	7,395,000	8,652,425	2,085,000	2,439,225	-	61,049,953
2031	40,299,043	7,775,000	8,273,175	2,190,000	2,332,350	-	60,869,568
2032	40,104,952	8,170,000	7,874,550	2,305,000	2,219,975	-	60,674,477
2033	39,902,608	8,590,000	7,455,550	2,420,000	2,101,850	-	60,470,008
2034	39,695,346	9,030,000	7,015,050	2,545,000	1,977,725	-	60,263,121
2035	39,477,702	9,495,000	6,551,925	2,675,000	1,847,225	-	60,046,852
2036	39,250,596	9,980,000	6,065,050	2,815,000	1,709,975	-	59,820,621
2037	39,012,852	10,495,000	5,553,175	2,960,000	1,565,600	-	59,586,627
2038	38,767,150	10,975,000	5,071,300	3,095,000	1,429,700	-	59,338,150
2039	38,505,339	11,425,000	4,623,300	3,220,000	1,303,400	-	59,077,039
2040	38,240,155	11,890,000	4,157,000	3,350,000	1,172,000	-	58,809,155
2041	37,964,181	12,375,000	3,671,700	3,490,000	1,035,200	-	58,536,081
2042	37,820,700	12,880,000	3,166,600	3,630,000	892,800	-	58,390,100
2043	37,821,600	13,405,000	2,640,900	3,780,000	744,600	-	58,392,100
2044	-	13,955,000	2,093,700	3,935,000	590,300	-	20,574,000
2045	-	14,520,000	1,524,200	4,095,000	429,700	-	20,568,900
2046	-	15,115,000	931,500	4,260,000	262,600	-	20,569,100
2047	-	15,730,000	314,600	4,435,000	88,700	-	20,568,300
TOTAL[3]	$1,143,156,032	$240,580,000	$217,340,584	$67,820,000	$61,270,304	$(27,293,600)	$1,702,873,326

(1) Includes 2010 Series A Bonds, 2010 Series B Bonds, 2013 Series A Bonds and 2013 Series B Bonds. Does not include debt service relating to the Parity State Loan in the amount of $7,435,000 for the Wastewater Enterprise's Lake Merced Green Infrastructure – Holloway Green Street Project. The Wastewater Enterprise has not yet submitted a request for reimbursement to, nor received reimbursement from, the State, so no repayment obligation has yet been incurred or calculated. See "OBLIGATIONS PAYABLE FROM REVENUES – Parity State Loans." Also does not include debt service on subordinate Wastewater Commercial Paper Notes. See "OBLIGATIONS PAYABLE FROM REVENUES – Subordinate Debt and Commercial Paper."

(2) Calculation of interest due on Bonds shown without an offset for Refundable Credits.

(3) Totals may not add due to independent rounding.

THE CITY AND COUNTY OF SAN FRANCISCO

The City is the economic and cultural center of the Bay Area and northern California. The limits of the City encompass over 93 square miles, of which 49 square miles are land, with the balance consisting of tidelands and a portion of the San Francisco Bay (the "Bay"). The City is located at the northern tip of the San Francisco Peninsula, bounded by the Pacific Ocean to the west, the Bay and the San Francisco-Oakland Bay Bridge to the east, the entrance to the Bay and the Golden Gate Bridge to the north, and San Mateo County to the south. Silicon Valley is about a 40-minute drive to the south, and the wine country is about an hour's drive to the north. The City's population in 2014 was 852,469 according to the U.S. Census Bureau.

The San Francisco Bay Area consists of the nine counties contiguous to the Bay: Alameda, Contra Costa, Marin, Napa, San Francisco, San Mateo, Santa Clara, Solano and Sonoma Counties (collectively, the "Bay Area"). The economy of the Bay Area includes a wide range of industries, supplying local needs as well as the needs of national and international markets. Major business sectors in the Bay Area include retail, entertainment and the arts, conventions and tourism, service businesses, banking, professional and financial services, corporate headquarters, international and wholesale trade, multimedia and advertising, biotechnology and higher education.

The City is a major convention and tourist destination. According to the San Francisco Travel Association, a nonprofit membership organization, during the calendar year 2014, approximately 18.01 million people visited the City and spent an estimated $10.67 billion during their stay. The City is also a leading center for financial activity in the State and is the headquarters of the Twelfth Federal Reserve District, the Eleventh District Federal Home Loan Bank, and the San Francisco Regional Office of Thrift Supervision.

The City benefits from a highly skilled, educated and professional labor force. The per-capita personal income of the City for fiscal year 2013-14 was an estimated $75,930 according to the City's CAFR. The San Francisco Unified School District operates 16 transitional kindergarten schools, 72 elementary and K-8 school sites, 12 middle schools, 18 senior high schools (including two continuation schools and an independent study school), and 46 State-funded preschool sites, and sponsors 13 independent charter schools. Higher education institutions located in the City include the University of San Francisco, California State University – San Francisco, University of California – San Francisco (a medical school and health science campus), the University of California Hastings College of the Law, the University of the Pacific's School of Dentistry, Golden Gate University, City College of San Francisco (a public community college), the Art Institute of California – San Francisco, the San Francisco Conservatory of Music, the California Culinary Academy and the Academy of Art University.

San Francisco International Airport ("SFO"), located 14 miles south of downtown San Francisco in an unincorporated area of San Mateo County and owned and operated by the City, is the principal commercial service airport for the Bay Area and one of the nation's principal gateways for Pacific traffic. In fiscal year 2014-15, SFO serviced approximately 48.2 million passengers and handled 441,797 metric tons of cargo. The City is also served by the Bay Area Rapid Transit District ("BART") (electric rail commuter service linking the City with the East Bay and the San Francisco Peninsula, including SFO), Caltrain (a conventional commuter rail line linking the City with the San Francisco Peninsula), and bus and ferry services between the City and residential areas to the north, east and south of the City. San Francisco Municipal Railway ("Muni"), operated by the San Francisco Municipal Transportation Agency (the "SFMTA"), provides bus and streetcar service within the City. The Port of San Francisco (the "Port"), which administers 7.5 miles of Bay waterfront held in "public trust" by the Port on behalf of the people of the State, promotes a balance of maritime-related commerce, fishing, recreational, industrial and commercial activities and natural resource protection.

The City is governed by a Board of Supervisors elected from eleven districts to serve four-year terms, and a Mayor who serves as chief executive officer, elected citywide to a four-year term. Edwin M. Lee is the 43rd and current Mayor of the City, having been elected to his current term by the voters of the City on November 3, 2015. The City's adopted budget for fiscal years 2015-16 and 2016-17 totals $8.94 billion and $8.99 billion, respectively. The City's General Fund portion of each year's adopted budget is $4.59 billion in fiscal year 2015-16 and $4.68 billion in fiscal year 2016-17, with the balance being allocated to all other funds, including enterprise fund departments, such as the SFMTA, SFO, the Port and the SFPUC. The City's CAFR estimates that the City employed approximately 30,156 full-time-equivalent employees at the end of fiscal year 2014-15. According to the

Controller of the City, the fiscal year 2015-16 total net assessed valuation of taxable property in the City is approximately $194.4 billion.

THE GENERAL FUND OF THE CITY IS NOT LIABLE FOR THE PAYMENT OF PRINCIPAL OF, PREMIUM, IF ANY, OR INTEREST ON THE BONDS, AND NEITHER THE CREDIT NOR THE TAXING POWER OF THE CITY IS PLEDGED TO THE PAYMENT OF THE PRINCIPAL OF, PREMIUM, IF ANY, OR INTEREST ON THE 2016 SERIES AB BONDS. THE BONDS ARE NOT SECURED BY A LEGAL OR EQUITABLE PLEDGE, OF, OR CHARGE, LIEN, OR ENCUMBRANCE UPON, ANY OF THE PROPERTY OF THE CITY.

THE PUBLIC UTILITIES COMMISSION

General

The SFPUC is a department of the City responsible for the maintenance, operation and development of three utility enterprises: the Wastewater Enterprise, the Water Enterprise and the Power Enterprise (which is a component of Hetch Hetchy Water and Power), all as further described below.

The revenues of the Water Enterprise and the Power Enterprise are not available for payment of the principal of, premium, if any, or interest on the Bonds. See "SECURITY FOR THE BONDS – Pledge of Net Revenues."

Organization, Purposes and Powers

Wastewater Enterprise. The Wastewater Enterprise's collection and treatment system consists of a combined sewer collection system conveying sewage and stormwater flows within the City to three water pollution control plants, also located within the City. Treated effluent flows are then discharged through deep-water outfalls into the San Francisco Bay and Pacific Ocean. The Wastewater Enterprise also currently provides sewage treatment service on Treasure Island pursuant to contract, and operates an onsite sewage and stormwater reclamation and treatment facility at the SFPUC headquarters at 525 Golden Gate Avenue. See "THE WASTEWATER ENTERPRISE."

Water Enterprise. Nearly 2.6 million people rely on water supplied by the SFPUC to meet their daily water needs through its Water Enterprise. The SFPUC serves as the retail water supplier for the City and is responsible for water deliveries to residents and institutions within the City limits, as well as to a number of retail accounts outside of the City limits. In addition, the SFPUC sells water to 26 Wholesale Customer entities in San Mateo, Alameda and Santa Clara Counties under contractual agreements.

The revenues of the Water Enterprise are not "Revenues" under the Indenture and do not secure the payment of the principal of, premium, if any, or interest on the Bonds. See "SECURITY FOR THE BONDS – Pledge of Net Revenues."

Hetch Hetchy – Water and Power Operations. Hetch Hetchy Water and Power operates dams (among them O'Shaughnessy Dam being the largest), reservoirs (among them Hetch Hetchy Reservoir being the largest), hydroelectric generation and transmission facilities and water transmission facilities from Hetch Hetchy Valley to the connection with the Water Enterprise (collectively, the "Hetch Hetchy Project"), which provides water for distribution through the Water Enterprise and hydroelectric power to municipal and public infrastructure, services and facilities of the City (the "Power Enterprise"). The Power Enterprise, which is a component of the Hetch Hetchy Water and Power Project, was created in February 2005 as a separate system within Hetch Hetchy Water and Power. The Power Enterprise focuses on providing adequate and reliable supplies of electric power to meet the municipal requirements of the City, including power to operate municipal streetcars and electric buses, street and traffic lights, municipal buildings and other City facilities, including SFO. Additionally, the Power Enterprise provides power to the Modesto and Turlock Irrigation Districts (located in the central valley of California) and to other commercial customers consistent with prescribed contractual obligations and federal law.

The revenues of the Power Enterprise are not "Revenues" under the Indenture and do not secure the payment of the principal of, premium, if any, or interest on the Bonds. See "SECURITY FOR THE BONDS – Pledge of Net Revenues."

Commission Members

Under the Charter, the SFPUC is given exclusive charge of the operation and management of all water, wastewater and municipal customers' energy supplies and utilities of the City as well as the real, personal and financial assets under the SFPUC's jurisdiction. The SFPUC is governed by the Commission.

In June 2008, an initiative measure amended the Charter, changing the process for Commission appointments, and establishing qualifications for commissioners, as follows:

- The Commission consists of five members appointed by the Mayor, subject to confirmation by a majority of the Board of Supervisors.

- Seat 1 is designated for a member with experience in environmental policy and an understanding of environmental justice issues.

- Seat 2 is designated for a member with experience in ratepayer or consumer advocacy.

- Seat 3 is designated for a member with experience in project finance.

- Seat 4 is designated for a member with expertise in water systems, power systems, or public utility management.

- Seat 5 is designated for an at-large member.

- Members may be suspended by the Mayor and may be removed by a three-fourths vote of the Board of Supervisors for official misconduct.

The current members of the Commission and the appointment and expiration dates of their terms are:

Name and Title	Seat	Originally Appointed	Term Expires
Francesca Vietor, President	1	September 2008	August 2016
Anson Moran, Vice President	4	July 2009	August 2018
Ann Moller Caen	3	March 1997	August 2016
Vince Courtney	5	January 2011	August 2016
Ike Kwon	2	February 2015	August 2018

Management

Management of the SFPUC is led by the General Manager. The General Manager is appointed by the Mayor from candidates submitted by the Commission. Once appointed by the Mayor, the General Manager serves at the pleasure of the Commission; however, the Commission also has Charter authority to employ the General Manager under an individual contract.

Brief biographies of the General Manager and principal members of the senior management of the SFPUC are set forth below.

Harlan L. Kelly, Jr. Harlan L. Kelly, Jr. became General Manager of the SFPUC in September 2012. He previously served as the SFPUC's Assistant General Manager, Infrastructure, and was responsible for implementing over $10 billion in capital programs for water, sewer and power, including the $4.6 billion Water System Improvement Program, the $6.9 billion Sewer System Improvement Program, and the $202 million SFPUC

Headquarters and Administration Building at 525 Golden Gate Avenue. His civil engineering career spanning nearly three decades includes his tenure as the City Engineer of San Francisco. At San Francisco Department of Public Works, he held functional and project management positions, including Acting General Manager, and Deputy Director of Engineering, during which he managed complex capital improvement programs that included the rebuild and seismic retrofit of City Hall, and expansions of convention, hospital, county jail, and public arts facilities. He is a licensed professional engineer, and a graduate of the University of California at Berkeley. He is the recipient of the Municipal Fiscal Advisory Committee's Public Municipal Excellence Award from the San Francisco Planning and Urban Research Association; the Public Works Leader of the Year Award from the American Public Works Association – Northern California Chapter; the Eminent Engineer Award from the National Engineering Honor Society Tau Beta Pi; and the Heroes and Hearts Award from the San Francisco General Hospital Foundation for exceptional community service. He is a member of the Construction Managers Association of America, the American Society of Civil Engineers, the National Society of Black Engineers, and the American Public Works Association. He is co-founder of the youth internship program Project Pull, which has been in continuous operation since 1995, and he has served on the Board of Directors of the Embarcadero YMCA.

Michael Carlin. Michael Carlin is the SFPUC Deputy General Manager and Chief Operating Officer ("COO"), reporting directly to the General Manager. He was appointed in 2009. As the Deputy General Manager and COO, Mr. Carlin supervises the agency's efforts in capital planning, emergency response, asset management, and other functions across the three business lines—water, power and wastewater. Prior to this position, Mr. Carlin served as the Assistant General Manager for Water where he led the effort to diversify the water supply portfolio. He continues in that role leading many of the environmental initiatives including addressing the impact of climate change on the organization. Mr. Carlin joined the SFPUC in 1996 as the Water Resources Planning Manager where he led the effort to develop comprehensive capital plans. That effort led to the establishment and execution of the Water System Improvement Program. Prior to joining the City, Mr. Carlin worked for more than a decade at the San Francisco Regional Water Board where he was the Planning Chief. Mr. Carlin holds a B.A. in Biology from San Francisco State University and an M.P.A. with an emphasis in Environmental Management from Golden Gate University. Mr. Carlin is a native San Franciscan and continues to live in the City today.

Tommy T. Moala. Tommy T. Moala is the Assistant General Manager of the Wastewater Enterprise which protects public health and safety and the environment through the collection and treatment of wastewater and stormwater. Mr. Moala oversees operations, maintenance, facility improvements and regulatory compliance for the City's three wastewater treatment plants, 993-mile long collection system and Treasure Island facilities. A former Naval Propulsion Engineer, Mr. Moala has more than 15 years' experience in managing wastewater systems. He began his 26-year career with the SFPUC as a Stationary Engineer, moving up steadily through the ranks to Senior Engineer, Chief Stationary Engineer and Operations Manager. A team recipient of the National Protection Agency O & M Award and the National Association of Clean Water Agency Award, Mr. Moala has also received then-Mayor Gavin Newsom's Public Managerial Excellence Award and the SFPUC O'Shaughnessy Award for organizing the SFPUC Emergency Response Team dispatched to Hurricane Katrina. He is a member of the Water Environment Federation, the California Water Environment Federation, the National Association of Clean Water Agencies and the American Water Works Association.

Juliet Ellis. Juliet Ellis is the Assistant General Manager for External Affairs at the SFPUC. Prior to joining the SFPUC as an Assistant General Manager, Juliet served as a Commissioner for two years. Juliet now oversees the implementation of Environmental Justice and Community Benefits policies as Assistant General Manager, along with the Policy and Government Affairs, Communications, and Sustainability Planning teams within the SFPUC. Juliet also oversees the SFPUC's national partnerships with other public utilities with the goal of scaling community benefits programs within the public sector. Before joining the SFPUC, Juliet spent nine years as the Executive Director of Urban Habitat, a regional social and environmental justice organization. She also served as the Associate Program Officer for Neighborhood and Community Development at The San Francisco Foundation. Juliet received her Masters of Science in Business Administration at San Francisco State University with an emphasis in Environmental and Urban Studies.

Barbara Hale. Barbara Hale is Assistant General Manager of the Power Enterprise. Ms. Hale oversees the Power Enterprise, including Power Retail Services, Utilities Services, Regulatory Affairs, Infrastructure Development and Power Purchasing and Scheduling. She is responsible for the development of a strategic business plan for the organization, setting out priorities, objectives, schedules and policy issues. Ms. Hale oversees all

power-related inter-governmental relations, works directly with the Commission on policy and capital matters, and provides direction and leadership to a multi-discipline staff at remote and downtown locations. Ms. Hale provides strategic advice on energy policy matters to the General Manager and manages a staff responsible for developing specific energy efficiency projects and renewable and other advanced sources of electrical generation. Ms. Hale also acts as liaison between the SFPUC and State and federal agencies responsible for energy policy, such as the California Public Utilities Commission, the California Energy Commission, the California Power Authority, the Federal Energy Regulatory Commission, and the United States Department of Energy. Ms. Hale graduated cum laude from San Francisco State University with a B.A. in Economics, receiving special recognition for high achievement with the Department Honors Award. Ms. Hale has pursued extensive graduate coursework in Applied Economics.

Kathy How. Kathy How is Assistant General Manager of Infrastructure, responsible for capital programs and projects implementation for SFPUC facilities, including the Water System Improvement Program, Sewer System Improvement Program and the Hetchy System Improvement Program. Prior to this position, Ms. How was Deputy AGM for Project Delivery, and was responsible for managing engineering design, construction management, and environmental review for all capital projects, whether in-house or consultant designed. Prior to joining the SFPUC in 2003, Ms. How was Assistant City Engineer at the Department of Public Works overseeing project management, architectural and engineering design and construction management for projects in the Seismic Safety bond programs, and Program Director for the Marina Yacht Harbor Renovations at the Recreation and Park Department. She joined the SFPUC to work on initiating program environmental review, development of program goals and objectives for the Water System Improvement Program, and implementation of projects. She is a licensed professional civil engineer in California, and holds a degree in Civil Engineering from the University of California, Berkeley.

Steven R. Ritchie. Steven Ritchie is the Assistant General Manager of the Water Enterprise, responsible for overseeing water system operations and planning from the Hetch Hetchy Project through the Regional Water System to the City Distribution Division. He is also responsible for the management of the SFPUC's lands and natural resources. Mr. Ritchie was the Manager of Planning at the SFPUC from 1995 to 1998. Prior to his current assignment, he managed the South Bay Salt Pond Restoration Project, a multi-agency effort to restore 15,100 acres of valuable habitat in South San Francisco Bay, while providing for flood risk management and public access. In addition, Mr. Ritchie has worked at management positions at the San Francisco Bay Regional Water Quality Control Board (1987-1995), the CalFed Bay-Delta Program (1998-2000), and URS consultants (2000-2004). He has a B.S. and M.S. in Civil Engineering from Stanford University.

Eric L. Sandler. Eric L. Sandler is Assistant General Manager, Business Services and Chief Financial Officer and is responsible for managing a range of internal and external service functions of the Commission including Finance, Customer Service, Information Technology, and Assurance and Internal Controls. Appointed in 2015, he has over 25 years of experience in municipal and infrastructure financing. Prior to joining the Commission, he served as Director of Finance for the East Bay Municipal Utility District, Director of Finance/Treasurer for the San Diego County Water Authority and Director of Financial Planning for the SFPUC. Before joining public service, Mr. Sandler worked in several infrastructure finance positions including renewable energy project finance and municipal investment banking. Mr. Sandler serves on boards and committees of various industry organizations including the Association of California Water Agencies and the National Association of Clean Water Agencies. He has a Bachelor's degree in Biology from Stanford University and a Master's degree in Business Administration from the University of California, Berkeley.

Employee Relations

The wages, hours and working conditions of City employees, including employees of the SFPUC, are determined by collective bargaining pursuant to State law (the Meyers-Milias-Brown Act, California Government Code Sections 3500-3511) and the Charter. Except for nurses and a few hundred unrepresented employees, the Charter requires that bargaining impasses be resolved through final and binding interest arbitration conducted by a panel of three arbitrators. The award of the arbitration panel is final and binding unless legally challenged. Strikes by City employees, including employees of the SFPUC, are prohibited by the Charter. Since 1976, no City employees have participated in a union-authorized strike. See "RISK FACTORS – Risks Related to Wastewater Enterprise Facilities and Operation – Labor Actions."

The City's budget for Fiscal Year 2015-16 includes 29,053 budgeted City positions. City workers are represented by 37 different labor unions. The largest unions in the City are the Service Employees International Union, Local 1021; the International Federation of Professional and Technical Engineers, Local 21; and the unions representing police, fire, deputy sheriffs and transit workers. In May 2014, the City negotiated three-year agreements (for Fiscal Years 2014-15 through 2016-17) with most of its labor unions. In general, the parties agreed to: (1) an annual wage increase schedule of 3% (October 11, 2014), 3.25% (October 10, 2015), and between 2.25% and 3.25% depending on inflation (July 1, 2016); and (2) some structural reforms of the City's healthcare benefit and cost-sharing structures to rebalance required premiums between the two main health plans offered by the City. These changes to health contributions built on reforms agreed to by most unions during earlier negotiations.

The SFPUC employs approximately 2,400 of the City's workers. The Charter governs the SFPUC's employment policies and authorizes the San Francisco Civil Service Commission to establish rules and procedures to implement those policies. Of the 37 labor unions representing City workers more broadly, 14 presently represent SFPUC employees. Most SFPUC employees collectively bargain every three years.

Over the next five years, nearly half of the SFPUC workforce will be eligible for retirement. A new generation of jobs will require workers with specialized training, skills and experience, while local hiring requirements will need to be observed. The SFPUC's Strategic Sustainability Plan includes indicators for employee training and development. The SFPUC also provides ethics training, diversity training, management training, environmental management system training, as well as fraud prevention and awareness training.

The following table summarizes the number of SFPUC and Wastewater Enterprise employees covered by collective bargaining agreements as of February 26, 2016, each of which expire on June 30, 2017.

TABLE 1
COLLECTIVE BARGAINING AGREEMENTS

Employee Bargaining Unit	SFPUC Full-Time Equivalent Employment[1]	Wastewater Enterprise Full-Time Equivalent Employment[1]
International Association of Machinists, Lodge 1414	47	3
Carpenters, Local 22	20	2
International Brotherhood of Electrical Workers, Local 6	127	21
Laborers, Local 261	152	35
San Francisco Association of Personnel Professionals, Local 21	38	2
Municipal Executives Association	157	11
Operating Engineers, Local 3	22	1
Plumbers, Local 38	224	34
International Federation of Professional and Technical Engineers, Local 21	925	163
Service Employees International Union, Local 1021	327	42
San Francisco City Workers United	14	5
Stationary Engineers, Local 39	317	211
Teamsters, Local 856	1	-
Teamsters, Local 853	39	15
Transport Workers Union Local 250-A, Automotive Service Workers	6	1
Unrepresented Employees[2]	1	-
Total	2,417	546

(1) As of February 26, 2016. Actual full-time equivalent employment totals will differ from the number of positions budgeted by the SFPUC for a variety of reasons, including certain requirements in the respective collective bargaining agreements.
(2) Not covered by a collective bargaining agreement.

Source: SFPUC.

Employee Benefit Plans

Retirement System Plan Description. The SFPUC participates in the City's single employer defined benefit retirement plan (the "Plan") which is administered by the San Francisco City and County Employees' Retirement System (the "Retirement System" or "SFERS"). The Plan covers substantially all full time employees of the SFPUC along with other employees of the City. The Plan provides basic service retirement, disability, and death benefits based on specified percentages of final average salary and provides cost-of-living adjustments after retirement. The Plan also provides pension continuation benefits to qualified survivors. The Charter and City Administrative Code are the authorities that establish and amend the benefit provisions and employer obligations of the Plan.

The Retirement System is administered by a Retirement Board consisting of seven members, three appointed by the Mayor, three elected from among the members of the Retirement System, at least two of whom must be actively employed, and a member of the Board of Supervisors appointed by the President of the Board of Supervisors. To aid in the administration of the Retirement System, the Retirement Board appoints an Executive Director and an Actuary. The Executive Director serves as chief executive officer, with responsibility extending to all divisions of the Retirement System. The Actuary's responsibilities include the production of data and a summary

of plan provisions for the independent consulting actuarial firm retained by the Retirement Board to prepare an annual valuation report and other analyses as described below. The independent consulting actuarial firm is currently Cheiron, Inc., a nationally recognized firm selected by the Retirement Board pursuant to a competitive process.

In response to an application filed by the Retirement System, the Internal Revenue Service ("IRS") issued a favorable Determination Letter for SFERS in March 2012. Issuance of a Determination Letter constitutes a finding by the IRS that operation of the defined benefit plan in accordance with the plan provisions and documents disclosed in the application qualifies the plan for federal tax exempt status. The favorable Determination Letter included IRS review of all SFERS provisions, including Proposition C. See "– Pension and Healthcare Costs Reforms – Proposition C."

Plan Financial Reports and Funded Status. The Retirement System issues a publicly available financial report that includes financial statements and required supplementary information for the Plan. That report may be obtained by writing to the San Francisco City and County Employees' Retirement System, 30 Van Ness Avenue, Suite 3000, San Francisco, CA 94102, or by calling (415) 487-7020.

The funded status of the Plan as of July 1, 2014 (the most recent date for which information is available) was as follows:

TABLE 2
RETIREMENT PLAN FUNDED STATUS
(AS OF JULY 1, 2014)

		(000s)
Actuarial Liability	$	21,122,567
Actuarial Value of Assets	$	18,012,088
Unfunded Actuarial Liability	$	3,110,479
Funded Status (assets/liabilities)		85.3%

Source: SFERS.

The Retirement System discloses accounting and financial reporting information under GASB Statement No. 67 (first implemented by the Retirement System in Fiscal Year 2013-14) and the City reports accounting and financial information about the Retirement System under GASB Statement No. 68 (first implemented by the City in Fiscal Year 2014-15). The accounting statements separated retirement system financial reporting from retirement system funding and required certain additional information in the notes to the City's financial statements. Funding requirements relating to the SFPUC are described below in "– Retirement System Funding Policy." In general, the City's funding of its pension obligations is not affected by the changes to its reporting requirements under GASB Statement No. 68, though such changes did result in changes to the SFPUC's reported salary and working capital costs. See "HISTORICAL OPERATING RESULTS – Summary of Historical Operating Results and Debt Service Coverage."

Retirement System Funding Policy. Contributions to the basic Plan are made by both the SFPUC and its employees. Employee contributions are mandatory. Employee contribution rates are approved through collective bargaining and vary by union and employment category. For Fiscal Years 2013-14 and 2014-15, the SFPUC's employee contribution rates varied from 7.5% to 13.0% as a percentage of gross covered salary. For the 2014-15 Fiscal Year, most employee groups agreed through collective bargaining that employees would contribute the full amount of the employee contribution on a pretax basis.

The SFPUC is required to contribute at an actuarially determined rate and allocates the applicable portions of such contribution to the separate enterprises, including the Wastewater Enterprise. For the prior three Fiscal

Years, the Wastewater Enterprise has paid 100% of its required contributions. The contributions by the Wastewater Enterprise required for Fiscal Years 2012-13 through 2015-16 are summarized in the following table.

TABLE 3
WASTEWATER ENTERPRISE RETIREMENT PLAN CONTRIBUTIONS

Fiscal Year	Actuarially Determined Rate as a Percentage of Covered Payroll	Contribution (000s)	Percent of Required Contribution
2012-13	20.70%	$ 9,200	100%
2013-14	24.82	11,290	100
2014-15	26.76	12,288	100
2015-16†	22.80	10,427	100

† Based on Budget for Fiscal Year 2015-16.

Source: SFERS July 1, 2014 Actuarial Valuation Report (produced by Cheiron March 4, 2015); SFPUC.

The annual actuarial valuation of the Retirement System is a joint effort of the Retirement System and its independent consulting actuarial firm. The Charter prescribes certain actuarial methods and amortization periods to be used by the Retirement System in preparing the actuarial valuation. The Retirement Board adopts the economic and demographic assumptions used in the annual valuations. Demographic assumptions such as retirement, termination and disability rates are based upon periodic demographic studies performed by the consulting actuarial firm approximately every five years. Economic assumptions are reviewed each year by the Retirement Board after receiving an economic experience analysis from the consulting actuarial firm.

At the January 2015 Retirement Board meeting, the consulting actuarial firm recommended that the Retirement Board adopt the following economic assumptions for the July 1, 2014 actuarial valuation: long-term investment earnings assumption of 7.50%, long-term wage inflation assumption of 3.75% and long-term consumer price index assumption of 3.25%. After consideration of the analysis and recommendation, the Retirement Board voted to adopt these recommended assumptions. At the November 2015 Retirement Board meeting, the Retirement Board voted to continue these economic assumptions with no changes for the July 1, 2015 actuarial valuation following the recommendation of the consulting actuarial firm. The Retirement Board also voted to update demographic assumptions, including mortality, after review of a new demographic assumptions study by the consulting actuarial firm.

Upon receipt of the consulting actuarial firm's valuation report, Retirement System staff provides a recommendation to the Retirement Board for their acceptance of the consulting actuary's valuation report. In connection with such acceptance, the Retirement Board acts to set the annual employer contribution rates required by the Retirement System as determined by the consulting actuarial firm and approved by the Retirement Board. This process is mandated by the Charter.

Pursuant to the Charter, the consulting actuarial firm and the Retirement Board set the actuarially required employer contribution rate using three related calculations:

First, the normal cost is established for the Retirement System. The normal cost of the Retirement System represents the portion of the actuarial present value of benefits that SFERS will be expected to fund that is attributable to a current year's employment. The Retirement System uses the entry age normal cost method, which is an actuarial method of calculating the anticipated cost of pension liabilities, designed to fund promised benefits over the working careers of the Retirement System members.

Second, the contribution calculation takes account of the amortization of a portion of the amount by which the actuarial accrued liability of the Retirement System exceeds the actuarial value of Retirement System assets, such amount being known as an "unfunded actuarial accrued liability" or "UAAL." The

UAAL can be thought of as a snapshot of the funding of benefits as of the valuation date. There are a number of assumptions and calculation methods that bear on each side of this asset-liability comparison. On the asset side, the actuarial value of Retirement System assets is calculated using a five-year smoothing technique, so that gains or losses in asset value are recognized over that longer period rather than in the immediate time period such gain or loss is identified. On the liability side, assumptions must be made regarding future costs of pension benefits in addition to demographic assumptions regarding the Retirement System members including rates of disability, retirement, and death. When the actual experience of the Retirement System differs from the expected experience, the impacts on UAAL are called actuarial gains or losses. Under the Retirement Board's updated Actuarial Funding Methods Policy any such gain or loss is amortized over a closed 20-year period. Similarly, if the estimated liabilities change due to an update in any of the assumptions, the impact on UAAL is also amortized over a closed 20-year period. Prior to the updated Policy which became effective with the July 1, 2014 actuarial valuation, the amortization period for gains, losses and assumption changes was 15 years at the valuation date.

Third, supplemental costs associated with the various SFERS benefit plans are amortized. Supplemental costs are additional costs resulting from the past service component of SFERS benefit increases. In other words, when the Charter is amended to increase benefits to some or all beneficiaries of the Retirement System, the Retirement System's liability is correspondingly increased in proportion to the amount of the new benefit associated with service time already accrued by the then-current beneficiaries. These supplemental costs are required to be amortized over no more than 20 years according to the Charter. The Retirement Board has adopted a 15-year closed period for changes to active member benefits and a 5-year closed period for changes to inactive or retired members effective for all changes on or after July 1, 2014. The prior Board Retirement Policy specified closed 20-year periods for all benefit changes.

The consulting actuarial firm combines the three calculations described above to arrive at a total contribution requirement for funding the Retirement System in the next Fiscal Year. This total contribution amount is satisfied from a combination of employer and employee contributions. Employee contribution rates are mandated by the Charter. Sources of payment of employee contributions (*i.e.* in the case of the SFPUC, the SFPUC, or its employee) are the subject of collective bargaining agreements with each bargaining unit. As described above, most of the SFPUC's employee groups have agreed through collective bargaining that employees would contribute the full amount of the employee contribution on a pretax basis. The employer contribution rate is established by Retirement Board action each year and is expressed as a percentage of salary applied to all wages covered under the Retirement System.

The assumptions and calculations described above were made as of their respective dates and are subject to change thereafter, including, for example, as a result of a subsequent Retirement Board action to revise the actuarial assumptions applied in the calculations. There is a risk that actual results will differ significantly from such assumptions and calculations.

Projected Future Pension Contributions and Costs. The new funding policies described above, favorable investment returns and the recognition of deferred investment gains following the large investment losses in Fiscal Year 2008-09, which is now fully reflected in the actuarial value of assets after a five-year smoothing period, resulted in a 3.96% decrease in the employer contribution rate for Fiscal Year 2015-16, from 26.76% to 22.80%. The City projects that SFERS employer contribution rates will increase beginning in Fiscal Year 2016-17, however.

Health Care Benefits. Health care benefits of the SFPUC employees, retired employees and surviving spouses are financed by beneficiaries and by the City through the City and County of San Francisco Health Service System (the "Health Service System"). The Wastewater Enterprise's annual contribution for both active and retired employees amounted to approximately $10.1 million and $9.9 million in Fiscal Years 2013-14 and 2014-15, respectively.

Included in these amounts are $2.7 million and $2.8 million for Fiscal Years 2013-14 and 2014-15, respectively, to provide post-retirement benefits for the Wastewater Enterprise's retired employees, on a pay-as-you-go basis.

The City has determined a Citywide Annual Required Contribution ("ARC"), interest on net other post-employment benefits other than pensions ("OPEB") obligation, ARC adjustment, and OPEB cost based upon an actuarial valuation performed in accordance with Governmental Accounting Standards Board Statement ("GASB") No. 45, by the City's actuaries. The ARC represents a level of funding that, if paid on an ongoing basis, is projected to cover the normal cost of each year and any unfunded actuarial liabilities (or funding excess) amortized over 30 years. The City's allocation of the OPEB related costs to the SFPUC for the year ended June 30, 2014 based upon its percentage of Citywide payroll costs is presented below.

The following table shows the components of the City's annual OPEB allocations for the Wastewater Enterprise for the Fiscal Years ending June 30, 2014 and 2015, for the amount contributed to the plan, and changes in the City's net OPEB obligation:

TABLE 4
ANNUAL OPEB OBLIGATION FOR THE WASTEWATER ENTERPRISE
FOR FISCAL YEARS ENDING JUNE 30, 2014 AND JUNE 30, 2015
(IN THOUSANDS)

	2014	2015
Annual required contribution	$ 7,020	$ 7,323
Interest on net OPEB obligation	1,469	1,666
Adjustment to ARC	(1,224)	(1,390)
Annual OPEB cost (expense)	7,265	7,599
Contribution made	(2,678)	(2,771)
Increase in net OPEB obligation	4,587	4,828
Net OPEB obligation – beginning of year	32,565	37,152
Net OPEB obligation – end of year	37,152	41,980

Source: SFPUC, Financial Services.

The City issues a publicly available financial report on a City-wide level that includes the complete note disclosures and required supplementary information related to the City's post-retirement health care obligations. The report may be obtained by writing to the City and County of San Francisco, Office of the Controller, 1 Dr. Carlton B. Goodlett Place, Room 316, San Francisco, CA 94102, or by calling (415) 554-7500.

The City's OPEB Unfunded Actuarial Accrued Liability ("UAAL") was $3.9 billion for Fiscal Year 2014-15. The amount allocable to the Wastewater Enterprise, as of July 1, 2015, was $83.6 million.

Pension and Health Care Costs Reforms

Voters implemented City employee pension and health care cost reforms in recent years to help mitigate future cost increases. These include the following propositions:

Proposition B. Proposition B was a Charter amendment approved by voters in June 2008 that increased the years of service required to qualify for employer-funded retiree health benefits for City employees who retire under SFERS and were hired on or after January 10, 2009. Previously, employees became eligible to participate in the retirement health care system after 5 years of service and the employer paid 100% of the contribution. Beginning with employees hired on or after January 10, 2009, employees remain eligible to participate in the retirement health care system after 5 years of service, however, no employer contributions are required until 10 years of service. From 10 to 15 years of service, employers pay 50% of the contribution, from 15 to 20 years of service 75%, and for employees with 20 years or more of service, 100%.

Proposition B also established a health care trust fund to pay for future costs relating to retiree health care. Employees hired on or after January 10, 2009 contribute up to 2% of their pre-tax pay, with employers contributing an additional 1%, to the health care trust fund. Proposition B also increased maximum pension benefits for employees retiring at and after age 60 and enhances cost of living increases for pensions.

Proposition C. Proposition C was a Charter amendment approved by voters in November 2011 that changed the way the City and current and future employees share in funding SFERS pension and health benefits.

With regard to pension benefits, the base employee contribution rate remains at 7.5% for most employees when the City contribution rate is between 11% and 12% of City payroll. Employees making at least $50,000 will pay an additional amount up to 6% of compensation when the City contribution rate is over 12% of City payroll. When the City contribution rate falls below 11%, employee contributions will be decreased proportionately.

Proposition C creates new retirement plans for employees hired on or after January 7, 2012 that: (1) for miscellaneous employees, increased the minimum retirement age to 53 with 20 years of service or 65 with 10 years; (2) for safety employees, kept the minimum retirement age at 50 with five years of service, but increased the age for maximum benefits to 58; (3) for all employees, limited covered compensation, calculated final compensation from a three-year average, and changed the multipliers used to calculate pension benefits, and (4) for miscellaneous employees, raised the age of eligibility to receive vesting allowance to 53 and reduced by half the City's contribution to vesting allowances.

With regard to health benefits, elected officials and employees hired on or before January 9, 2009, contribute up to 1% of compensation toward their retiree health care, with matching contribution by the City. For employees or elected officials who left the City workforce before June 30, 2001, and retire after January 6, 2012, Proposition C requires that the City contributions toward retiree health benefits remain at the same levels they were when the employee left the City workforce.

Proposition C also limits cost-of-living adjustments for SFERS retirees; however, in 2015, the Court of Appeals held in a suit against the City brought by a retiree organization, *Protect Our Benefits v. San Francisco* (1st DCA Case No. A140095), that certain changes to payment of supplemental cost of living allowances imposed by Proposition C could not be applied to current City employees and those who retired after November 1996 when the supplemental cost of living allowance provisions were originally adopted, but could be applied to SFERS members who retired before November 1996. On June 17, 2015, the California Supreme Court denied review of the Court of Appeals decision. It is estimated that the actuarial liabilities of the Plan will increase by approximately $388 million, or 1.8%, for back payment of supplemental cost of living allowances payable for 2013 and 2014.

THE WASTEWATER ENTERPRISE

Background and History

Initial development of the City's combined system dates back to the second half of the nineteenth century. In accordance with common engineering practice of that period, the combined system collected both sewage and stormwater runoff and transported them to a large number of discharge points on the shoreline of San Francisco Bay. Now all municipal wastewater treatment providers nationally are required to provide some level of urban stormwater treatment.

Today the Wastewater Enterprise provides sewage and stormwater collection, treatment and disposal services through the operation of combined sewage and stormwater system. The Wastewater Enterprise provides such sanitary wastewater and stormwater services across eight distinct urban watersheds, with one all-weather wastewater treatment plant and one wet-weather treatment plant serving the North Shore, Channel, Islais, Sunnydale, and Yosemite urban watersheds with effluent outfalls to the San Francisco Bay (the "Bayside Watersheds"), and a separate all-weather wastewater treatment plant serving the Richmond, Sunset, and Lake Merced urban watersheds with an effluent outfall to the Pacific Ocean (the "Westside Watersheds").

Of an estimated total combined wastewater flow of 40 billion gallons per year throughout the entire system, aggregating both watersheds, approximately 34 billion gallons per year receive full secondary treatment, 4.5 billion gallons per year receive primary or decant treatment and are discharged to deep-water outfalls, and 1.5 billion gallons per year receive the equivalent to wet weather primary treatment and are discharged through nearshore outfalls.

In separate sewer areas of the city, new multi-phase revitalization projects such as Mission Bay, Treasure Island and the Hunter's Point Naval Shipyard are implementing green infrastructure or low-impact design elements that allow for the treatment of stormwater by natural techniques such as vegetated swales, infiltration and other approved methods prior to discharge into the receiving waters.

The Wastewater Enterprise was operated by the City's Department of Public Works until July 1996, when the Mayor transferred its operation to the SFPUC.

Service Area

The service area of the Wastewater Enterprise encompasses approximately 29,773 acres and includes residents of the City and of northern San Mateo County through arrangements with three municipal sewer service providers: North San Mateo County Sanitation District, the Bayshore Sanitary District and the City of Brisbane (collectively, the "Municipal Customers"). The SFPUC also currently provides waste treatment service on Treasure Island pursuant to contract, and expects eventually to expand its service area to include Treasure Island. See "– Wastewater Treatment – Contract Services," "WASTEWATER ENTERPRISE CAPITAL PROGRAM – Treasure Island" and "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM."

The Wastewater Enterprise serves residential, commercial and industrial users, making up a daytime "population equivalent" of approximately 1,069,731 in 2014. United States Census estimates of the total daytime and nighttime populations of the City from 2010 to 2014 are reflected in the table below.

TABLE 5

CITY AND COUNTY OF SAN FRANCISCO POPULATION

	2010	2011	2012	2013	2014
Daytime[1]	950,728	990,449	992,493	1,021,780	1,069,731
Nighttime	805,253	812,826	825,863	841,138	852,469

(1) Includes the estimated number of people who work, but do not live, within the City.

Source: United States Census.

Customer Base

As of January 2016, the SFPUC has 163,742 active retail wastewater accounts. Of these, over 90% are residential accounts, with the remainder being commercial, industrial, or municipal. Major non-residential customer categories include retail, offices, restaurants, and services, which together account for 50% of non-residential customers. The total number of customer accounts has changed very little over the previous five years, increasing by a total of less than 1% since January 2011. Although the City's population is increasing, the SFPUC anticipates similarly modest growth in the total number of customer accounts over the next several years, as the majority of new customer accounts opened in the City are for urban infill construction replacing previously-existing buildings and the SFPUC's multifamily residential accounts comprise multiple dwelling units. In addition to its retail customers, the SFPUC provides wholesale sewage treatment services to three customers located in northern San Mateo County, and operates wastewater treatment facilities on Treasure Island pursuant to a "caretaker" contractual agreement and certain lease agreements between the Treasure Island Development Authority and the U.S. Navy. See "– Wastewater Treatment – Contract Services."

The following table sets forth a five-year history of the number of customer accounts served by the Wastewater Enterprise, grouped by user type.

TABLE 6
SUMMARY OF ACTIVE SEWER ACCOUNTS BY USER TYPE
FOR FISCAL YEARS ENDED JUNE 30

User Type	2011	2012	2013	2014	2015
Multi-Family Residential[1]	36,269	36,275	36,301	36,248	36,313
Single-Family Residential	110,608	110,875	111,007	111,125	111,173
Subtotal Residential	146,877	147,150	147,308	147,373	147,486
Commercial[1]	15,439	15,439	15,400	15,430	15,460
Municipal Customers	716	719	729	725	731
Suburban[2]	9	8	8	9	9
Total	163,041	163,316	163,445	163,537	163,686

(1) Includes Presidio Trust Accounts.
(2) In addition to suburban customers, service is provided to North San Mateo County Sanitation District, Bayshore Sanitary District and the City of Brisbane.

Source: SFPUC Customer Service.

Separate rate schedules apply to single-family residential, multi-family residential, and non-residential customers. See "FINANCIAL OPERATIONS – Wastewater Enterprise Rates and Charges."

Combined Sewage and Stormwater System

History and Background Regarding Combined System. On average, over three quarters of the City's annual rainfall occurs between November and March. Depending on the duration, intensity and storm pattern, combined flows of sewage and stormwater can exceed the collection and treatment system hydraulic capacity, resulting in combined sewer discharges.

In order to reduce the number and volume of combined sewer overflows during wet-weather events, the City built a series of large underground transport/storage structures (box sewers and tunnels) around the perimeter of the City to intercept, temporarily store, and transport the mixture of storm runoff and sewage to new or upgraded treatment facilities. The primary purpose of this system of transport/storage structures is to reduce the incidence and volume and provide treatment of discharges in wet weather. Prior to their construction, untreated combined sewer overflows occurred throughout the City whenever rainfall occurred at a rate of 0.02 inches per hour. By providing both storage volume and detention time, the transport/storage structures allow for treatment of the stored combined sewage and stormwater flows at the treatment plants after storms. In addition, the retention of the combined flows in the transport/storage structures allows solids to settle, and weir and baffle structures retain floatable materials, providing the equivalent of wet-weather primary treatment.

With these improvements, the facilities of the Wastewater Enterprise are designed to minimize the number of wet-weather discharge events. In all methods of discharge, the combined wastewater receives the level of treatment prescribed by federal and state law.

Sewer Lines. The Wastewater Enterprise's collection and transport system currently has a storage capacity of approximately 200 million gallons, and includes approximately 993 miles of sewer lines of which 792 miles is made up of collecting sewers of 36 inches or less in diameter.

TABLE 7

SEWER SYSTEM BY PIPE LENGTH IN MILES AND AGE

Year Built	Gravity Pipe <36" Diameter	All Other Pipe[1]	Total Pipe	Percent of Total
1860 – 1900	70	39	109	11%
1901 – 1940	339	64	403	41
1941 – 1980	178	58	236	24
1981 – present	182	38	220	22
Unknown	23	2	25	3
Total[2]	792	201	993	100%

(1) Includes gravity pipe greater than 36" in diameter, tunnels, force mains, transport/storage, effluent outfall and overflow discharge.
(2) Totals may not add due to independent rounding.
Source: SFPUC, Wastewater Enterprise

Transport/Storage Structures. In addition to sewer lines, the Wastewater Enterprise maintains 24 miles of underground transport/storage structures which are located around the perimeter of the city to intercept, temporarily store, and transport the mix of stormwater runoff and sewage to treatment plants. These structures, which operate in both dry and wet weather, were built between 1979 and 1997. The largest of these structures, the Westside Transport structure, is approximately 2 miles long, 45 feet deep and 25 feet wide. The transport/storage structures were designed with sufficient storage to reduce combined sewer discharges and protect beneficial uses of receiving waters for the San Francisco Bay and the Pacific Ocean. The transport/storage and collection system provides approximately 200 million gallons worth of storage (approximately three days of dry weather flow).

The performance of the transport/storage structures and treatment facilities complies with the requirements of the National Combined Sewer Overflow Control Policy, as implemented by discharge permits issued on behalf of the United States Environmental Protection Agency (the "EPA") by the San Francisco Bay Regional Water Quality Control Board (the "Regional Water Quality Control Board"). See "REGULATORY MATTERS." By providing both storage volume and detention time, the transport/storage structures allow for delayed treatment of combined sewage and stormwater flows at the treatment plants at the conclusion of a wet weather event. They also provide a "buffer" for dry weather flows in the event of an unplanned outage.

Pump Stations. The Wastewater Enterprise has 27 pump stations, which include six major all-weather pump stations, two major wet-weather pump stations, 18 minor pump stations, and one major effluent pump station for bayside effluent discharge.

Outfalls and Nearshore Discharge Structures. The Wastewater Enterprise currently has three offshore outfalls that discharge to deep waters — Southeast Bay Outfall, Northpoint Outfall, and Southwest Ocean Outfall. In addition to these outfalls, 36 combined sewer discharge structures, or near-shore outfalls, serve as relief points of the combined sewer system. These structures operate infrequently, and only during large storm events. Discharges through these permitted sites receive the equivalent of wet weather primary treatment.

Urban Watershed Management. The SFPUC is working to improve the system's stormwater drainage performance and its sewage treatment efficiency. The SFPUC Stormwater Program complies with regulatory requirements and is designed to maximize sewer system performance, engage community members in its work, improve watershed function, enhance the environmental quality of the City's neighborhoods, and protect the water quality of the San Francisco Bay and Pacific Ocean. To achieve these goals, the SFPUC has adopted regulations

that require new and redevelopment projects in San Francisco to install and operate green technologies for managing stormwater runoff.

Useful Life. The average useful life of typical collecting sewers (i.e., gravity pipes 36 inches or less in diameter) is approximately 110 years. The average useful life for other sewer pipes/infrastructure (such as tunnels, force mains, transport/storage boxes) is 50 years, while large outfalls have a typical useful life ranging from 50 to 100 years. However, many conditions, including pipe material and soil conditions, affect actual pipe lifespan. The approximate ages of the various components of the Wastewater Enterprise's sewer lines are set forth in Table 7 above.

The historical rate of replacement for the Wastewater Enterprise's sewer pipes (less than 36-inch diameter) has been at a 200-year replacement cycle. Aging sewers, if left unaddressed, would result in increased leakage, sinkholes, sewer line failures, reduced system reliability and possibly public health and safety risks. The SFPUC has developed its capital program, including SSIP and the Renewal & Replacement program in part to address the aging infrastructure, with the objective of ultimately returning to a 110-year replacement cycle, and in part to upgrade other critical infrastructure. Replacement of specific sewer sections will be prioritized under an asset management approach which considers the current condition of the section in question and applicable risks of failure. The sewers will be inspected through closed-circuit television and the likelihood of failure will be calculated based on a condition scoring algorithm taking into account the severity and quantity of defects identified in each inspection. The consequence of failure is calculated based on level of service, including protection of the public, preservation and improvement of quality of life, cost efficiency, customer service, and environmental responsibility. To implement this Renewal and Replacement program, the SFPUC has divided the city into 182 individual grids for the purpose of scheduling inspections for all gravity sewer mains on a 5-year cycle. These grids are prioritized by predicted total risk, using age of sewer as the main factor. The goal is to inspect the sewers predicted to be in the worst condition and in highly sensitive locations first. As the inspections proceed, each surveyed sewer receives a risk score and sewers identified for replacement are scheduled based on risk scores, proximity, and utility coordination. See "WASTEWATER ENTERPRISE CAPITAL PROGRAM" and "SEWER SYSTEM IMPROVEMENT PROGRAM."

Wastewater Treatment

Wastewater Treatment Plants. The Wastewater Enterprise operates three major wastewater treatment facilities (in addition to the transport/storage structures that provide the equivalent of wet weather primary treatment), two serving the Bayside Watersheds and one serving the Westside Watersheds:

- Southeast Water Pollution Control Plant (the "Southeast Plant"), which treats dry and wet weather flows collected from the Bayside Watersheds (18,597 acres, or 63% of the total city service area) as well as the Municipal Customers;

- North Point Wet-Weather Facility (the "North Point Facility"), which treats a portion of wet-weather flows collected from the north Bayside Watersheds; and

- Oceanside Water Pollution Control Plant (the "Oceanside Plant"), which treats dry and wet weather flows collected from the Westside Watersheds (11,176 acres, or 37% of the total city service area) and minor flows from northern San Mateo County.

The Oceanside Plant and related collection facilities in the Westside Watersheds are not interconnected with the Southeast Plant or the North Point Facility and the collection facilities in the Bayside Watersheds.

Southeast Water Pollution Control Plant. The Southeast Plant is an all-weather secondary wastewater treatment plant serving the sewage and stormwater treatment needs for nearly two-thirds of the City in the Bayside Watersheds, which consists of 18,587 acres including the Marina, Downtown, South of Market Area, Mission, Hunters Point, and Visitacion Valley neighborhoods, plus 1.65 million gallons per day ("mgd") of dry weather flow from North San Mateo City Municipal District, Bayshore Sanitary District and the City of Brisbane. Land uses in the Bayside Watersheds are a mixture of residential, commercial, light industrial and heavy industrial. The

Southeast Plant, as the only Bayside dry-weather facility, is a critical component of the system that must meet high availability requirements. The Bayside Watersheds wastewater flow is collected by the combined sewer system and conveyed to the Southeast Plant via gravity sewers, tunnels, transport/storage structures, pump stations and force mains.

The Southeast Plant provides preliminary, primary and secondary treatment using a high-purity-oxygen activated sludge process prior to effluent disinfection. Sludge treatment consists of gravity-belt thickening, anaerobic digestion, chemical conditioning and dewatering. Plant effluent is discharged during dry weather into the San Francisco Bay through a deep water outfall in the vicinity of Pier 80. During wet weather, treated effluent is discharged through the Pier 80 outfall and through an additional outfall at the shoreline of Islais Creek.

The Southeast Plant was planned and designed in the 1940s and commenced operations in 1951 as a primary treatment facility. In order to meet the mandates of the federal Clean Water Act, the Southeast Plant was expanded in the early 1980s to provide secondary treatment of all Bayside Watersheds dry-weather flows with a daily average design capacity of approximately 85 mgd and peak-hour design flow of 142 mgd. In 1996, the plant's wet weather capacity was increased to 250 mgd, with 150 mgd receiving secondary treatment prior to disinfection and discharge. During wet weather, additional wet-weather facilities are operated at the Southeast Plant to provide primary treatment and disinfection to the remaining 100 mgd of combined wastewater flow.

The Southeast Plant operates every day of the year in both wet weather and dry weather conditions. Many of its critical systems have operated past their expected useful life and replacement facilities are needed to ensure reliability requirements are met, including seismic upgrades. During the last two years, a full condition assessment was conducted of the Southeast Plant and the highest priority needs were identified for both the liquid treatment and solids treatment facilities. The assessment took into account current maintenance records, remaining life, and compliance with levels of service. Projects have been identified for all of the major treatment processes, and for various support, monitoring and control systems. Maintenance and repair projects, including repairs to digester roofs and existing biosolids treatment and storage equipment, have been undertaken to bolster some of the facilities considered to be the most vulnerable so that they function until the new facilities are completes. See "SEWER SYSTEM IMPROVEMENT PROGRAM." Because the Southeast Plant is not interconnected with the other treatment plants, it cannot be shut down during construction of such projects as there would be no substitute wastewater treatment option. Instead, the Southeast Plant will remain in service during construction. See "RISK FACTORS – Risks Related to Wastewater Enterprise Facilities and Operation – Aging Facilities" and "– Limited Redundancy."

North Point Wet-Weather Facility. The North Point Facility is a wet-weather primary wastewater treatment plant serving the northeast quadrant of the Bayside Watersheds, an area comprising predominantly commercial and residential land uses. The North Point Facility was originally constructed in 1951 as an all-weather primary treatment facility, but did not incorporate digesters and so sludge was pumped to the digesters at the newly constructed Southeast Plant for treatment. In response to the mandates of the federal Clean Water Act, the North Point Facility was converted into a strictly wet-weather treatment plant in 1983, providing up to 150 mgd of primary treatment.

The North Point Facility provides preliminary, primary and disinfection treatment. Effluent from the North Point Facility is discharged into the San Francisco Bay through a deep water outfall system at Piers 33 and 35. At the conclusion of each wet-weather event, grit and solids are flushed out of the sedimentation tanks and directed to the Channel Pump Station, which pumps them to the Southeast Plant for treatment.

Oceanside Water Pollution Control Plant. Constructed in 1993, the Oceanside Plant is an all-weather secondary wastewater treatment plant that provides the wastewater treatment and stormwater treatment needs for San Francisco's Westside Watersheds and San Mateo County flows that drain to this basin. Land uses in the Westside Watersheds are primarily residential.

The Oceanside Plant's dry weather capacity is up to 43 mgd of secondary treatment. The Oceanside Plant's wet-weather capacity is up to 65 mgd of primary and 43 mgd of secondary treatment using the high-purity-oxygen activated sludge process. Sludge treatment consists of gravity belt thickening, anaerobic digestion, chemical conditioning and dewatering. The Oceanside Plant began operations in September 1993 and complies with all dry-

and wet-weather discharge requirements. Up to 195 mgd of plant effluent and decanted wet weather flow is discharged approximately 4 miles offshore into the Pacific Ocean through a deep ocean outfall.

Emergency Operations. The Wastewater Enterprise maintains up-to-date contingency plans in the event of an unplanned outage or failure of a treatment facility, process unit, pump station, sewer pipeline or other infrastructure element. The wastewater collection and treatment system is designed with some redundancy and flexibility in order to facilitate responses to emergency events, though the Westside Watershed and Oceanside Watershed are not interconnected. The collection system is equipped with pump stations and isolation valves so that flows can be redirected to help mitigate impacts in the event of a failure.

If an unplanned shutdown of critical treatment facilities were to occur, the transport/storage structures and collection system pipes provide approximately 200 million gallons of storage (providing up to several days of storage in some portions of the system, depending on the inflow). The wastewater treatment facilities and major pump stations are also built with some redundancy, and in some instances may continue operation with select equipment outages. Critical infrastructure elements with higher failure risk or for which adequate redundancy is not available have been identified as projects in the SSIP to increase system flexibility and the ability to respond to unplanned events. See "RISK FACTORS – Risks Related to Wastewater Enterprise Facilities and Operations."

Biosolids Management. The Southeast Plant and the Oceanside Plant produce approximately 75,000 wet tons per year of sewage sludge or biosolids, which are highly treated and anaerobically digested. During wet-weather months, biosolids are trucked to the Potrero Hills Landfill in Solano County and the Altamont and Vasco Road Landfills in Alameda County, where biosolids are used as Alternative Daily Cover. During dry-weather months, biosolids are trucked to various ranches in Solano and Sonoma Counties and applied directly as a soil amendment for farming or to enhance pasture productivity in accordance with federal and local county regulations. Throughout the year, a portion of the SFPUC's biosolids are sent to a ranch in Sacramento County for storage and/or land application in accordance with federal and local county regulations. Additionally, a small portion (about 5%) of the Southeast Plant biosolids are trucked to a Merced County composting site and are processed into a Class A biosolids compost.

The Wastewater Enterprise coordinates the application, testing and sampling procedures required by the applicable regulatory agencies. Because current beneficial use options may not offer a long-term solution (due to more stringent regulations and encroaching suburban development in currently agricultural use areas), the SFPUC is exploring other means of beneficially using biosolids and upgrading the level of treatment. The Oceanside Plant's digesters have been retrofitted for a conversion to a Temperature Phased Anaerobic Digestion ("TPAD") process, which will enable the Oceanside Plant to produce Class A biosolids. The SFPUC anticipates that TPAD processing will commence at the Oceanside Plant in by 2018. The SFPUC also plans to implement technology at the Southeast Plant to produce Class A biosolids as part of an SSIP project.

See "SEWER SYSTEM IMPROVEMENT PROGRAM" and "REGULATORY MATTERS" below.

Contract Services. The Wastewater Enterprise also provides waste treatment service on Treasure Island, by agreement with the Treasure Island Development Authority ("TIDA"), a non-profit public benefit agency vested with the rights to administer municipal services to Treasure Island and Yerba Buena Island during interim reuse of the property. Such waste treatment service is provided at a wastewater treatment plant currently owned by the U.S. Navy, subject to agreements between TIDA and the U.S. Navy for the conveyance of property on Treasure Island to TIDA. The wastewater system serving Treasure Island and Yerba Buena Island, located in San Francisco Bay between San Francisco and Oakland, are served by a wastewater system that currently relies on pumping to convey sewage to a secondary facility for treatment and discharge. The treatment plant is designed to treat an average of 2.0 mgd, with a permitted capacity of up to 4.4 mgd. The wastewater collection system consists of 10 miles of sewers and 29 wastewater pump stations. In addition, a stormwater collection system includes 6 stormwater pump stations and a number of shallow water outfalls.

In connection with ongoing redevelopment activities on Treasure Island, the SFPUC anticipates that the Treasure Island facilities, including collection and other facilities being built by the developer as part of the project, will eventually be transferred to the SFPUC and be integrated into the Wastewater Enterprise's overall system assets. Following such transfer, the SFPUC would provide sewer discharge service and impose rates and charges

directly on customers on Treasure Island, and payments the SFPUC receives from such customers would constitute Revenues as and to the extent provided under the Indenture. The SFPUC would also be responsible for operation and maintenance, and such costs would constitute Operation and Maintenance Costs of the Enterprise as and to the extent provided under the Indenture. Finally, the SFPUC would be responsible for the future repair and replacement of all facilities transferred to it, including collection and treatment facilities. The SFPUC expects that it will need to replace the existing treatment plant on Treasure Island, and has included preliminary projected costs of such project in its Ten-Year Capital Plan. See "–Projected Future Demand," "WASTEWATER ENTERPRISE CAPITAL PROGRAM – Treasure Island" and "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM."

System Capacity

When the three treatment facilities and other elements of the collection system are fully operational, the Wastewater Enterprise can provide up to 575 mgd of combined wastewater and stormwater treatment, including 193 mgd of secondary treatment, 272 mgd of primary treatment and 110 mgd of equivalent-to-primary treatment provided by the transport/storage structures located around the perimeter of the City.

TABLE 8

TREATMENT PLANT MAXIMUM CAPACITY
(MILLION GALLONS PER DAY)

Plant	Dry Weather	Peak Wet Weather
Southeast	105	250
Oceanside	43	65
North Point	N/A	150
Total	148	465

Source: SFPUC, Wastewater Enterprise

The three treatment plants discharge up to 365 mgd through deep-water outfalls. Another 100 mgd of secondary treated effluent with disinfection can be discharged through a shallow water outfall to Islais Creek during peak wet-weather events. The combined sewer system is designed to handle wet weather events and its permits explicitly allows for them, including discharge from the system when a storm exceeds the capacity of the system.

Current System Demands

Wastewater Enterprise facilities collected, treated and discharged an average of approximately 70 mgd of sanitary wastewater during dry-weather periods between 2011 and 2015. Both Bayside and Oceanside have been experiencing declining dry weather flows over the last five years, primarily due to water use conservation efforts.

TABLE 9

AVERAGE DRY WEATHER TREATMENT
IN FISCAL YEARS ENDED JUNE 30
(MILLION GALLONS PER DAY)

Plant	2011	2012	2013	2014	2015
Southeast	57.8	56.6	57.4	57.8	51.9
Oceanside	14.4	15.3	13.8	12.8	12.2
Total	72.2	71.9	71.2	70.6	64.1

Source: SFPUC, Wastewater Enterprise

Projected Future Demand

Results of the SFPUC's water demand forecasts predict that the SFPUC's in-City retail water demand will increase only slightly from 2010 through 2035, even though household population in the City is projected to increase by nearly 12% during the same period. See "– Service Area" and "– Customer Base." This growth differential is attributable to decreases in average per capita water usage due to conservation, including significant use of low-flow plumbing fixtures. For wastewater production, this translates to relatively flat growth of influent flow to the treatment plants, but more solids and biochemical oxygen demand loading.

The Wastewater Enterprise's existing treatment plants have sufficient excess capacity to address the currently anticipated increase in demand, therefore no significant expansion is expected to be required in order to meet the needs of the City based upon these population growth projections alone. However, the SFPUC expects to provide a new treatment plant at Treasure Island with additional capacity and associated infrastructure in order to meet the needs at full build-out of the proposed conversion of the former Treasure Island Naval Station to residential and commercial uses, though agreements for the provision and construction of the treatment plant have not yet been finalized. See "– Contract Services."

Because the SFPUC maintains a combined sewer system, there may be other factors in the future, including environmental changes and regulatory developments, that could require expanded collection or treatment capacity.

WASTEWATER ENTERPRISE CAPITAL PROGRAM

Capital and Financial Planning Process

The SFPUC's long-term capital and financial planning is performed on an annual rolling ten-year forward looking basis. The SFPUC prepares a Ten-Year Capital Plan for each of its enterprises, as required by the Charter of the City and County of San Francisco. The Ten-Year Capital Plan serves as the basis for the development of the annual Ten-Year Financial Plan. Proposed long-term capital programs, projects and investments, and related costs are included in the Ten-Year Financial Plan. Consistent with the Charter, updates to the Ten-Year Capital and Financial Plans are annually reviewed and adopted by the SFPUC Commission each February. The Ten-Year Financial Plan provides estimated rate impacts of projected capital and operating spending and assures compliance with debt service coverage and fund balance reserve policy requirements.

The Ten-Year Capital Plan is not a budget and is not "appropriated" like a budget. The Annual Capital Programs can be revised during the development of the budget and final projects, costs and totals for the Capital Programs can change. Consequently, even though the annual Capital Programs are based on the Ten-Year Capital Plan, they may occasionally differ from it.

The Wastewater Enterprise Ten-Year Capital Plan for Fiscal Years 2016-17 to 2025-26, adopted by the SFPUC Commission in February 2016, totals approximately $5.5 billion, including projected SSIP costs for that

period. See "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM." The Ten-Year Capital Plan includes projects in four categories: (i) the Sewer System Improvement Program; (ii) Renewal and Replacement; (iii) Treasure Island; and (iv) Wastewater Facilities and Infrastructure.

Sewer System Improvement Program

The SFPUC has embarked on a comprehensive $6.9 billion Sewer System Improvement Program ("SSIP") to be implemented over approximately the next twenty (20) years. The SSIP is a series of major capital improvement projects that are intended to bring the City's wastewater and stormwater collection system into a state of good repair, and meet levels of service goals originally endorsed by the Commission in 2012 and revised by the Commission in March 2016.

The SSIP capital projects address specific challenges and deficiencies facing the Wastewater Enterprise, maximize system reliability and flexibility, improve operational and seismic reliability, and ensure present and future regulatory compliance. The SSIP project development will also consider environmental benefits, sustainability, and community benefits in addressing long-term wastewater needs. However, the SFPUC is not subject to any consent decree requiring specific improvements. The implementation of the SSIP projects is being phased over a twenty (20) year period in an effort to maintain ratepayer affordability and minimize impacts throughout the City.

Information on the SSIP may be found in "SEWER SYSTEM IMPROVEMENT PROGRAM".

Renewal and Replacement

The Wastewater Enterprise undertakes renewal and replacement projects to improve performance or extend the service life of an existing asset. These projects are typically annual ongoing projects and fall into two categories: Renewal and Replacement Collection System and Renewal and Replacement Treatment Plants.

The Renewal and Replacement Collection System category includes the following projects: Condition Assessment projects to clean, video inspect, as well as perform condition assessment of sewer facilities; Sewer Improvement projects to maintain the existing functionality of the collection system and repair and replacement of structurally inadequate sewers; Spot Sewer Replacement projects which consist of as-needed replacement of localized failed sections of existing sewer facilities; Hydraulic Improvement projects to upgrade sewers with hydraulic deficiencies; and, Salt Water Intrusion projects to prevent San Francisco Bay water from entering the wastewater collection system, which can be detrimental to the treatment plant's biological system and result in corrosion of exposed metal.

The Renewal and Replacement Treatment Plant Improvements includes projects to maintain the capacity and reliable performance and extend the useful life of aging assets such as transport/storage boxes, discharge structures, pump stations, force mains, tunnels and treatment plants. Priority lists are maintained for both collection and treatment projects, with collection system projects identified using an Asset Management approach which considers the current condition of the section in question and applicable risks of failure. See "THE WASTEWATER ENTERPRISE – Combined Sewage and Stormwater System – Useful Life."

Treasure Island

Treasure Island and Yerba Buena Island, located in San Francisco Bay between San Francisco and Oakland, are served by a wastewater system that currently relies on pumping to convey wastewater to a secondary facility for treatment and discharge. The treatment plant is designed to treat an average flow of 2 mgd, with a permitted capacity of up to 4.4 mgd. These wastewater facilities are not interconnected with the wastewater infrastructure in San Francisco. The Treasure Island treatment plant is currently owned by the United States Navy, but operated and maintained by the Wastewater Enterprise pursuant to a cooperative agreement with the Navy. However, the SFPUC expects eventually to expand its service area to include Treasure Island. See "– Wastewater Treatment – Contract Services." In connection with such expansion, the SFPUC anticipates constructing a new treatment plant with additional capacity and associated infrastructure in order to meet the needs at full build-out of

the proposed conversion of the former Treasure Island Naval Station to residential and commercial uses, though agreements for the provision and construction of the treatment plant have not yet been finalized. The projected costs of such project are included in the SFPUC's Ten-Year Capital Plan, however.

The Treasure Island and Yerba Buena Island wastewater collection system consists of 10 miles of sewers and 29 wastewater pump stations. In addition, a stormwater collection system includes 6 stormwater pump stations and a number of shallow water outfalls. With the exception of any new treatment plant constructed by the SFPUC, the SFPUC expects that the developer undertaking the conversion project will construct all other new wastewater infrastructure relating to the development, including collection facilities, prior to transferring such facilities to the SFPUC.

Wastewater Facilities and Infrastructure

There are three projects in the category of Wastewater Facilities and Infrastructure: (i) the Ocean Beach Protection Project, (ii) the Southeast Community Facility and (iii) the Islais Creek Outfall.

The Ocean Beach Protection Project will facilitate the development of a comprehensive shoreline management and protection plan in partnership with relevant stakeholders and regulatory agencies to provide a long-term solution to the erosion issue along Ocean Beach between Sloat and Skyline Boulevard, in an effort to protect the integrity of critical SFPUC wastewater assets that are jeopardized by the ongoing shoreline erosion and to mitigate potential impacts to the Lake Merced Tunnel and other critical wastewater assets at this location. Such assets include the Lake Merced Transport Tunnel, the Westside Pump Station, the Westside Transport, and the Oceanside Plant.

The Southeast Community Facility was originally built in the 1980s as a community mitigation measure for expansion of the Southeast Plant with the purpose of providing opportunities for residents in the Bayview Hunters Point community to participate in education programs, to establish and expand daycare opportunities and to provide information and resources for the enhancement and growth of the community as a whole. The Southeast Community Facility Project will focus on improving the functional and operational reliability of the facility to ensure it fulfills its goals of providing fir the education and jobs training needs of the community. Infrastructure improvements, such as energy efficiency upgrades, building envelope repairs, tenant space reconfigurations and consolidations, life/safety, and accessibility upgrades, as required by applicable codes will be provided. The project will include evaluation of programming and siting of the community facility and associated green houses.

The Islais Creek Outfall Project will replace the effluent pipelines within Islais Creek and rehabilitate the existing upstream, on-shore pipelines, booster pump station pump manifold and isolation valves. This project was included in the capital program following a determination by the SFPUC in 2015 that one of the two pipelines had extensive corrosion and a risk of pipeline failure would continue to exist until the issue could be addressed.

Environmental Considerations

Projects undertaken by the SFPUC are generally subject to CEQA and certain projects involving the participation of federal agencies, including projects on federal land, are also subject to NEPA.

Under CEQA, a project that may have a significant effect on the environment and is to be carried out or approved by a public agency must comply with a comprehensive environmental review process, including the preparation of an Environmental Impact Report ("EIR"). The EIR reflects not only an independent technical analysis of the project's potential impacts, but also the comments of other agencies with some form of jurisdiction over the project and the comments of interested members of the public. Contents of the EIR include a detailed statement of the project's significant environmental effects; any such effects that cannot be avoided if the project is implemented; mitigation measures proposed to minimize such effects; alternatives to the proposed project; the relationship between local and short-term uses and long-term productivity; any significant irreversible environmental changes that would result from the project; the project's growth-inducing impacts; and a brief statement setting forth the agency's reasons for determining that certain effects are not significant and hence do not require discussion in the EIR. Prior to the sale of bonds, the San Francisco Planning Department Environmental

Review Officer will issue a "Planning Certificate" required under Proposition E approved by the voters of San Francisco at the November 5, 2002 General Election. The Planning Certificate will identify the status of environmental review for each capital project to be funded under the proposed bond sale and the type of CEQA document either completed or to be completed for each project. CEQA compliance must be completed for each project prior to project approval or approval to award a construction contract to implement any project to be funded by the proposed bond sale. The Planning Certificate relating to the projects being financed from proceeds of the 2016 Series AB Bonds was issued on May 19, 2014.

Any action or proceeding challenging the SFPUC's determination must be brought within 30 days following the filing of such notice. Actions have been, and in the future may be, filed against the SFPUC challenging a project's compliance with CEQA, including the adequacy of the EIR and other environmental documents, for particular projects. If an action challenging the SFPUC's compliance with CEQA is successful, the particular project could be delayed, revised, suspended or canceled. CEQA also contains a number of exemptions, which the SFPUC uses for its projects when appropriate.

As part of its regular planning and budgetary process, the San Francisco Planning Department gives careful attention to environmental considerations. All projects are evaluated under the SFPUC's environmental evaluation procedures, developed in compliance with federal and State laws and regulations, and City Ordinances and Administrative Code procedures.

SEWER SYSTEM IMPROVEMENT PROGRAM

Program Summary

The SFPUC has embarked on a comprehensive $6.9 billion Sewer System Improvement Program ("SSIP") to be implemented in 3 phases over approximately the next 20 years. The SSIP is a series of major capital improvement projects that are intended to bring the City's wastewater and stormwater system into a state of good repair, and meet Commission-endorsed levels of service goals. The SSIP has a scope, budget and schedule based on the Commission's endorsed levels of service. The Commission's authorization of the program specified that the program would be implemented in three phases, with the highest priority and best defined projects to be included in Phase 1.

SSIP Program Development and Chronology

In February 2010 the Commission directed staff to proceed with the procurement of a program management consultant to assist City staff with implementation of the SSIP. The AECOM-Parsons Joint Venture was selected and the Program Management Consultant ("PMC") team began work on September 6, 2011 to validate the SSIP scope, schedule, and budget.

On August 28, 2012, the Commission endorsed these SSIP Goals and levels of service, validated the scope and phased implementation approach for the overall $6.9 billion SSIP, and authorized SFPUC staff to move forward with planning and development on a proposed Phase 1 set of SSIP projects representing $2.7 billion of the total.

In October, 2015, the PMC was assigned to work on refining program scope, budget and schedule based on newly available information, and various constraints and challenges. This effort included project re-prioritization, scope refinement, budget re-alignment and schedule re-alignment. The PMC assessment was completed in February 2016 and they provided SFPUC their recommendations for revising program.

On March 22, 2016, the Commission adopted the "March 2016 SSIP Baseline Scope, Budget and Schedule." The overall SSIP budget remained at approximately $6.9 billion, however, the budget for SSIP Phase 1 increased from $2.7 billion to $2.9 billion.

Program Objectives

The SSIP capital improvement projects are designed to address specific infrastructure challenges and deficiencies facing the Wastewater Enterprise, maximize system reliability and flexibility, improve operational and seismic reliability, and promote current and future regulatory compliance. The SSIP project development will also consider environmental benefits, sustainability, and community benefits in addressing the long-term wastewater needs.

Program Goals

The program goals for the SSIP, as endorsed by the Commission in August 2012, include the following:

1. Provide a compliant, reliable, resilient and flexible system that can respond to catastrophic events;

2. Integrate grey and green infrastructure to manage stormwater and minimize flooding;

3. Provide benefits to impacted communities;

4. Modify the system to adapt to climate change;

5. Achieve economic and environmental sustainability; and

6. Maintain ratepayer affordability.

The March 2016 SSIP review did not change these program goals.

Program Scope

The March 2016 SSIP – Baseline divided the overall scope of SSIP projects into three major subprograms: Treatment Plants, Collection System, and Land Reuse.

Treatment Plants. Treatment plant projects are proposed at the Southeast Plant, Oceanside Plant and North Point Facility that would address aging infrastructure and outmoded technologies; increase seismic and operational reliability; and reduce odors, noise, visual and other public impacts. These projects would also address long-term regulatory compliance by ensuring continued performance that is aimed at meeting current and future regulatory mandates. The major treatment project is the Biosolids Digester Project at the Southeast Plant, which will replace the aged existing digester and solids handling treatment processes with a more technologically advanced facility intended to ensure long-term reliable and sustainable treatment, improve the biosolids end product to a Class A level, and to ensure disposal and/or reuse of biosolids.

A majority of the SSIP improvements will occur at the Southeast Plant to upgrade the building and systems which were planned, designed and built in the 1940s and 1950s. See "WASTEWATER ENTERPRISE – Wastewater Treatment – Southeast Plant." The largest project in the SSIP is the Biosolids Digester, a new digester and solids handling facility to replace the aging facility at the Southeast Plant. The new facility will include more modern treatment processes designed to produce Class A biosolids. Biosolids treatment processes are expected to include solids thickening, screening, dewatering, thermal hydrolysis pretreatment, digestion, gas handling, energy generation and reuse, and odor control. Several other SSIP projects are scheduled to be constructed at the Southeast Plant, which will require careful logistical and space planning and scheduling of key project interfaces. These other projects focus on liquid treatment and basic infrastructure, including: construction of a new all-weather 250 million gallon per day headworks facility and other improvements to oxygen generation, clarifiers, gas handling, disinfection, facility-wide distributed control systems and power feed and switchgear. The Southeast Plant will be operational throughout the construction of these capital improvements. See "WASTEWATER ENTERPRISE – Wastewater Treatment – Southeast Plant" and "RISK FACTORS – Risks Related to Wastewater Enterprise Facilities and Operation – Limited Redundancy."

In addition to the work at Southeast Plant, there will be reliability improvements to both Oceanside Plant and North Point Facility, including seismic, electric, and general reliability improvements for these aging facilities.

Collection System. Proposed collection system projects would increase the ability of the sewer system to collect and convey wastewater and stormwater and would address aging infrastructure including large diameter sewers, pump stations, transport/storage boxes, and combined sewer discharge structures. The collection system projects also include interdepartmental projects that capture sewer improvement opportunities arising from capital projects initiated by other City and County agencies. Other collection system improvements are expected to include reliability projects for sewer pipelines and boxes, combined sewer discharge structures, and pump stations. Additionally, the SFPUC anticipates projects focused on stormwater management and control, flood resilience (particularly in low lying areas that are susceptible to flooding) and green infrastructure. The green infrastructure is expected to include eight projects, one in each of the City's eight urban watersheds, designed to assist the SFPUC in monitoring and evaluating the effectiveness of green infrastructure.

The major collection system project is the Central Bayside System Improvement Project. This project would provide redundancy back-up to the existing 66-inch Channel Force Main, which conveys 60% of the flow to the Southeast Plant, and improve stormwater conveyance in the local drainage basins. The Channel Force Main has been subject to long term settlement and previously experienced temporary failures in connection with the Loma Prieta Earthquake in 1989. Phase 1 of the SSIP includes complete planning and design work for this project. Flood Resilience is the next largest project and includes improvements to stormwater management systems intended to reduce flooding during wet weather. Particular improvements have been identified through the SFPUC's urban watershed assessment process. The process involves characterizing each of the eight watersheds in the City through extensive hydraulic modeling, condition assessment and stakeholder input. A suite of projects to meet watershed objectives, including green infrastructure alternatives, have been initiated based on this characterization.

Land Reuse. There are currently projects to acquire 1800 Jerrold and 1801 Jerrold, located adjacent to the Southeast Plant, to serve a variety of functions in support of projects planned at the Southeast Plant. Over the next ten years, multiple construction projects have been planned at the Southeast Plant to upgrade and/or replace aging infrastructure. Each of these potential projects will require laydown, trailers and staging areas, as well as secured space for physical storage of equipment. 1800 Jerrold is a 6.04 acre site which is presently occupied by another city department under the Office of Contract Administration and used as a central shop. 1801 Jerrold, a 1.54 acre site, is under the jurisdiction of the Department of Public Works and is presently used as a dispatch yard for Street Repair. It was formerly an asphalt plant for the city. Subsequent to the relocation of current occupants, the properties would be acquired and the sites cleared and remediated for other potential uses to support the Southeast Plant and SSIP. Following the completion of geotechnical and environmental investigation, plans for demolition, environmental remediation and reuse will be developed.

Program Budget

The Commission has authorized SFPUC staff to proceed with planning and developing projects for Phase 1 of the SSIP. To date, the Board of Supervisors has appropriated $528 million for Phase 1 projects. Subject to Commission and Board of Supervisors consideration of project environmental review and actions to approve project budgets, construction of Phase 1 projects and implementation of Phases 2 and 3 of the SSIP would occur over the next 20 years. The following table summarizes the Approved Budgets for the March 2016 Revised SSIP.

TABLE 10
SEWER SYSTEM IMPROVEMENT PROGRAM
(March 2016 Approved Budget)[1]
(Millions)

Subprograms	Phase 1	Phase 2	Phase 3	Total Project Cost Estimate
Treatment Plants	$ 2,182	$ 1,299	$ 407	$ 3,888
Collection System	505	1,708	476	2,670
Land Reuse	98			98
Program Management (PM)	125	152	43	320
TOTAL SSIP	**$ 2,910**	**$ 3,159**	**$ 926**	**$ 6,976**

(1) The Commission approved what is referred to as the "SSIP – Baseline" on March 22, 2016.

Source: SFPUC, Wastewater Enterprise

Due to the size of the overall program, a phased approach was developed to simplify implementation of the SSIP projects. This was done to manage rate impacts, consider construction sequencing impacts, and maintain existing operations and permit compliance. While all projects are needed to meet the Program's Level of Service, Phase 1 projects focused on ensuring regulatory compliance, enhancing process reliability and redundancy, improving plant odor control, and replacing antiquated biosolids and headworks facilities with state of the art technology. As such, Phase 1 focuses on treatment plant improvements, while Phase 2 emphasizes collection system improvements, and Phase 3 primarily focuses on improvements resulting from condition assessment investigations.

Program Schedule

According to the March 2016 SSIP – Baseline, the SSIP Phase 1 completion date is October 2026.

Management Approach

The development and implementation of the projects in the capital program, including SSIP projects, are led by SFPUC staff, with oversight and ultimate responsibility by the General Manager, Assistant General Manager of the Wastewater Enterprise, and Assistant General Manager of the Infrastructure Division.

Consultants are employed to support a number of programmatic functions, such as strategic program development, risk assessment and mitigation, program controls, various independent technical reviews, construction planning and management, and labor and community relations. The services of consultants are also used on an as-needed or project-specific basis to assist SFPUC staff with functions such as engineering design, environmental review, right-of-way engineering and surveying, and construction management.

Potential Changes to SSIP Projects

Because the SSIP projects are currently in the preliminary and planning stages, uncertainties exist that may result in changes to the scope and phasing of the projects, changes to the project budgets, and delays and cost increases. These factors include, without limitation, the following:

- market conditions and the bidding environment for construction costs, which could worsen over the 20-year life of the program;

- market conditions for financing the various phases of the SSIP, which could worsen over the 20-year life of the program;

• changes in the legal and regulatory requirements affecting the Wastewater Enterprise and the SSIP (see "REGULATORY MATTERS");

• the discovery of unforeseen underground/geotechnical conditions, particularly for projects with tunneling activities and extensive earthwork;

• the discovery of unforeseen site conditions associated with existing infrastructure; many of the SSIP projects involve upgrades to existing structures and facilities that were built a very long time ago and accurate as-built records are not always available;

• the discovery of unforeseen site conditions associated with utility conflicts;

• unexpected failure of aging infrastructure or major equipment that warrant expedited repair or replacement;

• unexpected environmental discoveries that may impact construction activities; those may include the discovery of protected species, archaeological artifacts, contaminated soil or hazardous material at project sites;

• contractor claims, contractor non-performance, failure of contractors to execute within contract price, or failure of contractors to meet schedule terms;

• errors or omissions in contract documents (drawings and specifications) that may result in change orders;

• equipment and material vendors' lack of compliance with quality and schedule requirements;

• inclement weather affecting contractor performance and timeliness of completion;

• labor issues involving work stoppages or slowdowns; or

• the occurrence of a major seismic or other natural or man-made catastrophic event.

• unforeseen public opposition to projects or elements of projects

See "RISK FACTORS."

FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM

Long Term Financing of Capital Program

Pursuant to the Wastewater Enterprise's Ten-Year Financial Plan for Fiscal Years 2016-17 to 2025-26, which is based on the Ten-Year Capital Plan for that period, and which was adopted by the SFPUC Commission in February 2016, it is projected that long-term debt financing will fund approximately $4.6 billion of the Ten-Year Capital Plan. Revenue (pay-as-you-go) funding will provide approximately $819 million of funds for most of the remaining portion of the Ten-Year Capital Plan. Long-term debt financing is expected to be comprised primarily of Additional Bonds and Parity State Loans. See "OBLIGATIONS PAYABLE FROM REVENUES – Parity State Loans."

Interim Funding of Capital Program

The Wastewater Enterprise utilizes its commercial paper program to meet the expenditure and encumbrance needs of capital projects on an interim basis through design and into the early project construction phase. Commercial paper is then refunded and consolidated into either long-term revenue bond issues or Parity State Loans when the outstanding and encumbered amount of commercial paper approaches authorized limits. This

approach allows the SFPUC to take advantage of lower short-term interest rates, and to size and closely time long-term financings with projected need.

The SFPUC has established a commercial paper program authorizing the issuance of up to $500 million in Wastewater Commercial Paper Notes. Of such authorized amount, as of April 1, 2016 the SFPUC had $135 million aggregate principal amount of tax-exempt Wastewater Commercial Paper Notes outstanding, which the SFPUC expects to repay in full using a portion of the proceeds of the 2016 Series AB Bonds. The SFPUC anticipates issuing additional commercial paper to provide interim financing for Wastewater Enterprise capital projects under the $500 million authorization. See "OBLIGATIONS PAYABLE FROM REVENUES – Subordinate Debt and Commercial Paper" and "PLAN OF FINANCE."

Sources and Uses of Funding the Capital Program

The table below sets forth the projected sources and uses of funds for the Wastewater Enterprise's capital program, including the SSIP. Non-SSIP capital program categories include Renewal and Replacement projects, Wastewater Facilities and Infrastructure projects, and Treasure Island projects. The projected repayment of principal and interest on these future debt financings has been incorporated into the SFPUC's approved rates through Fiscal Year 2017-18, and has also been reflected in the remaining projection period set forth in the Ten-Year Financial Plan for Fiscal Years 2018-19 to 2025-26.

If the Wastewater Enterprise's capital program, including the SSIP, is carried out on the currently projected schedule, the SFPUC projects that retail sewer rates will increase annually by an average of approximately 11% from Fiscal Year 2018-19 to Fiscal Year 2025-26, after the current adopted rate period ends in Fiscal Year 2017-18. *However*, no rate increases beyond Fiscal Year 2017-18 have been proposed to, or adopted by, the SFPUC's Commission or submitted to the Board, and any future rate increases are subject to future Commission approval (subject to the Board's ability to reject rate increases). See "FINANCIAL OPERATIONS – Wastewater Enterprise Rates and Charges."

TABLE 11
WASTEWATER ENTERPRISE CAPITAL PROGRAM
FINANCING PLAN FOR FISCAL YEARS ENDED JUNE 30
(IN THOUSANDS)[1]

	2016[2]	2017	2018	2019	2020
APPROPRIATIONS					
SSIP	$422,284	$175,921	$670,560	$1,008,829	$756,833
Renewal and Replacement	91,051	97,315	100,735	104,886	105,644
Wastewater Facilities & Infrastructure	23,300	12,000	19,000	29,000	26,000
Treasure Island	0	0	20,463	22,240	21,090
Total Uses	$536,635	$285,236	$810,758	$1,164,955	$909,567
SOURCES OF FUNDS					
Revenue Bonds / Parity State Loans	$482,635	$230,136	$763,258	$1,114,455	$833,067
Wastewater Revenues	41,000	43,000	45,000	48,000	74,000
Capacity Charge Revenues	13,000	12,100	2,500	2,500	2,500
Total Sources	$536,635	$285,236	$810,758	$1,164,955	$909,567

(1) Amounts set forth are projections. Actual results may differ materially from these projections. See "FORWARD-LOOKING STATEMENTS" above.
(2) Budgeted.
Source: SFPUC, Financial Services

FINANCIAL OPERATIONS

General

The SFPUC is a department of the City and, as such, the financial operations of its three enterprises are included in the Comprehensive Annual Financial Report of the City and shown as enterprise funds. *The City's Comprehensive Annual Financial Report is not incorporated by reference herein.*

The following information is provided with respect to the Wastewater Enterprise only and does not purport to reflect the financial position of the SFPUC or the City as a whole.

Basis of Accounting. The accounts of the Wastewater Enterprise are organized on the basis of a proprietary fund type, specifically an enterprise fund. The financial activities of the Wastewater Enterprise are accounted for on a flow of economic resources measurement focus, using the accrual basis of accounting. Under this method, all assets and liabilities associated with its operations are included on the statement of net assets; revenues are recorded when earned, and expenses are recorded when liabilities are incurred.

The Wastewater Enterprise applies all applicable GASB pronouncements.

City Budget Process. The SFPUC's operating and capital budget preparation and approval is a part of a City-wide process. The SFPUC is one of several departments which prepares biannual budgets. The Commission reviews and approves the SFPUC's two-year budget, which is then submitted to the Mayor's Office for review. The Mayor then incorporates the proposed budget, with amendments, into the City-wide budget that is submitted to the Board of Supervisors for approval. Under the Charter, the Board of Supervisors may increase or decrease any proposed expenditure in the Mayor's budget so long as the aggregate changes do not cause the expenditures to exceed the total amount of expenditures proposed by the Mayor. The Charter further provides that the Mayor may reduce or reject any expenditure authorized by the Board of Supervisors except appropriations for bond interest, redemption or other fixed charges, subject to reinstatement of any such expenditure by a two-thirds vote of the Board of Supervisors.

City Services Auditor. On November 4, 2003, voters in the City adopted Proposition C, an ordinance that established the City Services Auditor ("CSA"), an audit function within the Office of the City Controller. Pursuant to the provisions of this ordinance, which have been incorporated into the Charter, the CSA has broad oversight authority and responsibilities including, but not limited to, (i) to reporting upon level of effectiveness for City public services, (ii) auditing financial and management performance of City departments and functions, (iii) ensuring the financial integrity and improving the overall performance and efficiency of City government, and (iv) maintaining a whistleblower hotline to investigate upon reports of fraud, waste and abuse.

CSA is currently completing a performance audit of the Wastewater Enterprise with a focus upon internal controls surrounding operational functions, procurement and payroll. The audit is scheduled to be published in the summer of 2016.

Wastewater Enterprise Rates and Charges

General. Sewer service charges are the primary funding source for the payment of costs associated with the Wastewater Enterprise's sanitary wastewater and stormwater collection, treatment and disposal services. Pursuant to the Charter, the SFPUC prepares an analysis of projected revenues and revenue requirements of the Wastewater Enterprise at least once every five years and then sets rates projected to be sufficient to fund the proposed budget, to maintain an adequate operating reserve and to comply with Indenture requirements. In addition to meeting funding and reserve requirements, the rates must also comply with regulations or policies promulgated by the EPA, the State Water Resources Control Board and the Board of Supervisors and with the requirements of the City's Charter and the State Constitution; however, the SFPUC's sewer service rates are not subject to any approval proceedings by the California Public Utilities Commission or any other State or federal agencies. See "CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS – State Law Limitations."

Federal and State Requirements. Under federal clean water laws and regulations, entities accepting federal grant funds (such as the SFPUC) must comply with certain requirements, including the requirement that user charges be set to ensure that recipients of sewer services pay their proportionate share of the costs of operation, as well as system maintenance and replacement based on the quantity and characteristics of the users' discharge into the public sewage system. The Federal "Revenue Program Guidelines" are incorporated into the State water quality regulations, which are administered and enforced by the State Water Resources Control Board. See "REGULATORY MATTERS" for a more detailed discussion of State and Federal regulations affecting the Wastewater Enterprise.

Summary of Charter Rate-Setting Requirements. Pursuant to certain provisions of Proposition E, which became effective with respect to the Wastewater Enterprise on January 3, 2003, the Charter was amended to authorize the Commission to set rates, fees and other charges in connection with providing Wastewater Enterprise services. These rate changes are subject to rejection, within 30 days of submission, by the Board of Supervisors. If the Board of Supervisors does not act within 30 days, the rates become effective without further action.

In setting Wastewater Enterprise rates, fees and charges, the Commission must:

(a) Establish rates, fees and charges at levels sufficient (i) to improve or maintain financial condition and bond ratings at or above levels equivalent to highly rated utilities (see "RATINGS"), (ii) to meet requirements and covenants under all bond resolutions and indentures (including the Indenture) and (iii) to provide sufficient resources for the continued financial health (including appropriate reserves), operation, maintenance and repair, consistent with good utility practice;

(b) Retain an independent rate consultant to conduct rate and cost of service studies at least every five years;

(c) Set retail rates, fees and charges based on the cost of service;

(d) Conduct all studies mandated by applicable State or Federal law to consider implementing connection fees servicing new development;

(e) Conduct studies of rate-based conservation incentives and/or lifeline rates and similar rate structures to provide assistance to low income users, and to take the results of each study into account when establishing rates, fees and charges, in accordance with State and federal laws; and

(f) Adopt annually a rolling five-year forecast of rates, fees and charges.

Rate Setting Process. The Wastewater Enterprise periodically adjusts rates based on a comprehensive cost of service analysis and the resulting revenue requirement analysis, as required by the Charter. The biannual budget submitted to the Commission, the Mayor and the Board of Supervisors for approval is used as the basis for determining annual revenue requirements. The Wastewater Enterprise projects revenues under the existing schedule of rates for sufficiency. If additional revenues are required to meet budgetary sufficiency or if the cost structure of the Wastewater Enterprise has changed, recommended rate schedule increases are submitted to the Commission for its consideration. After receiving public comment, the Commission adopts a rate resolution and transmits its recommended rate schedules to the Board of Supervisors. Once submitted, the Board of Supervisors has authority under the Charter to vote to reject the proposed rate schedules within 30 days. If rejected, the existing rate schedules remain in effect until such time as a new rate schedule is resubmitted by the Commission and not rejected by the Board of Supervisors.

Whenever rates are revised, costs are allocated to pollutant parameters, volume, suspended solids, oil and grease, and chemical oxygen demand, and any costs are then allocated to each of the rate categories. See "– Wastewater Enterprise Rates and Charges – Rate Categories."

As part of the annual Ten-Year Financial Plan process, the Wastewater Enterprise develops a ten-year rate forecast using projected revenues under existing rates plus additional revenues from projected rate increases, as

required, to meet the projected revenue requirements during the forecast period. This forecast is updated each year resulting in a "rolling" rate forecast that is intended to moderate the effects of any significant changes in revenue requirements in any year.

In addition to complying with the requirements of the Charter, the rate-setting process must comply with the requirements of the State Constitution. See "CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS – State Law Limitations."

SFPUC Citizens Advisory Committee and Rate Fairness Board The Public Utilities Commission Citizens' Advisory Committee ("CAC"), established by a Charter amendment in 2002, provides recommendations to the SFPUC's General Manager and the Board of Supervisors regarding the SFPUC's long-term strategic, financial and capital improvement plans. The CAC is comprised of seventeen appointees. Each member of the Board of Supervisors may appoint one member who must be a resident of his or her supervisory district. Candidates must demonstrate one or more of the following qualifications: represent a community, business, environmental, or environmental justice organization, or have demonstrated knowledge, skill or experience in a field related to public utilities, environmental justice or environmental science. Two additional members of the CAC are appointed by the President of the Board of Supervisors, one of whom represents a small business and the other of whom represents an environmental justice organization. Four members are appointed by the Mayor and must include one member who represents regional water customers of the SFPUC, one who represents a large City water user, one who has knowledge of engineering or financial management and one who represents a regional or statewide environmental organization.

Proposition E, approved by voters in 2002, directed the establishment of a Rate Fairness Board to advise the SFPUC on water, sewer and power rate matters. The Rate Fairness Board consists of seven members: the City Administrator or his or her designee; the Controller or his or her designee; the Director of the Mayor's Office of Public Finance or his or her designee; two residential City retail customers, consisting of one appointed by the Mayor and one by the Board of Supervisors; and two City retail business customers, consisting of a large business customer appointed by the Mayor and a small business customer appointed by the Board of Supervisors. Specific powers for the Rate Fairness Board include the authority to: (1) annually review the five-year rate forecasts produced by the SFPUC enterprises, including the Wastewater Enterprise; (2) hold one or more public hearings on annual rate recommendations before the SFPUC adopts rates; (3) provide a report and recommendations to the SFPUC on any rate proposal; and, (4) in connection with periodic rate studies, submit to the SFPUC rate policy recommendations for the Commission's consideration, including recommendations to reallocate costs among various retail utility customer classifications, subject to any outstanding bond requirements. The Rate Fairness Board is not authorized, however, to reject proposed rates approved by the Commission.

Currently Approved Four-Year Rate Schedule. In May 2014, the Commission approved four years of annual retail water and sewer service rate increases beginning July 1, 2014 and applicable to Fiscal Years 2014-15 through 2017-18. Separate rate schedules apply to single-family residential, multi-family residential, and non-residential customers. See "THE WASTEWATER ENTERPRISE – Customer Base." Under State law, all sewer rates reflect cost of service by customer class.

Under its current rate structure, 100% of the Wastewater Enterprise's customer bills are calculated based on metered water volumes, with a per-customer "flow factor" adjustment to estimate the percentage of water that does not enter the system as sewage. See "– Residential Users and "– Non-Residential Users." As a result, decreased water sales volumes can reduce Wastewater Enterprise Revenues.

Residential Users. Residential users are charged on the basis of discharge units. Monthly discharge units are determined for residential customer accounts by multiplying an account's total monthly water consumption by the "flow factor" applicable to such account, which calculation is designed to approximate that portion of the account's total water use returned to the sewer system as wastewater. For example, a customer using 10 units of water and having a flow factor of 90% would be billed for 9 discharge units. Each discharge unit represents 100 cubic feet of water discharged as approximated by this calculation. The standard flow factor for single-family residential accounts is 90%. The standard flow factor for multi-family residential accounts is 95%. All residential discharge is assumed to be of a standard strength.

Single and multi-family residential wastewater rates currently have two tiers. For Single-Family Residential customers, tier 1 currently includes the first four hundred cubic feet, representing average water consumption for this class. For multi-family accounts, tier 1 includes three hundred cubic feet, representing average water consumption for this customer class, and that amount is multiplied based on the number of dwelling units at the service address. Both single-family and multi-family residential wastewater tiers will be phased into a uniform, single tier rate by Fiscal Year 2017-18.

TABLE 12

HISTORICAL AND ADOPTED SINGLE-FAMILY RESIDENTIAL USER SEWER RATES

Fiscal Year	Tier One[1]	Additional Discharge Units[1]	Average Monthly Bill
2010-11	$6.91	$9.21	$42.83
2011-12	7.16	9.55	44.40
2012-13	7.52	10.03	46.63
2013-14	7.90	10.53	48.97
2014-15[2]	9.06	11.23	44.89
2015-16[2]	9.82	11.34	45.97
2016-17[2]	10.84	11.66	50.24
2017-18[2]	12.40	12.40	56.92

(1) Until June 30, 2014, Tier One included the first 300 cubic feet of water consumption per dwelling unit per month billed at the lifeline rate and all excess use at the regular residential rate. As of July 1, 2014, first tier includes the first 400 cubic feet. Effective July 1, 2017, residential wastewater billing will be based on a single tier.

(2) Rates approved on May 13, 2014.

Source: SFPUC, Financial Services

TABLE 13
HISTORICAL AND ADOPTED MULTI-FAMILY RESIDENTIAL USER SEWER RATES

Fiscal Year	Tier One[1]	Additional Discharge Units[1]	Average Monthly Bill
2010-11	$6.51	$8.68	$31.68
2011-12	7.49	9.99	36.46
2012-13	7.86	10.49	38.27
2013-14	8.25	11.01	40.16
2014-15[2]	9.24	11.48	36.90
2015-16[2]	9.95	11.51	37.91
2016-17[2]	10.91	11.75	40.96
2017-18[2]	12.40	12.40	45.88

(1) Tier One includes the first 300 cubic feet of water consumption per dwelling unit per month billed at the lifeline rate and all excess use at the regular residential rate. Effective July 1, 2017, residential wastewater billing will be based on a single tier.
(2) Rates approved on May 13, 2014.

Source: SFPUC, Financial Services

Non-Residential Users. Non-residential wastewater rates have a uniform rate for volume plus per pound strength charges. Non-residential users are charged the cost for each parameter according to the schedule of rates in the table below. Customers whose parameter loadings are not based on periodic sampling are charged on the basis of standard parameter loadings established by the General Manager for each Standard Industrial Classification code in accordance with applicable state and federal laws and regulations.

TABLE 14

HISTORICAL AND ADOPTED NON-RESIDENTIAL USER DISCHARGE RATES (PER DISCHARGE UNIT)

Fiscal Year	Volume charge (per 100 cubic feet)	Suspended solids (per pound)	Oil/Grease (per pound)	Chemical Oxygen Demand (per pound)
2010-11	$6.5548	$0.8819	$1.1035	$0.2156
2011-12	6.5548	0.8819	1.1035	0.2156
2012-13	6.5548	0.8819	1.1035	0.2156
2013-14	6.6203	0.8907	1.1145	0.2178
2014-15[1]	6.1450	0.8280	0.8670	0.4400
2015-16[1]	6.4530	0.8700	0.8670	0.4400
2016-17[1]	6.9040	0.9310	0.9740	0.4940
2017-18[1]	7.6640	1.0330	1.0820	0.5480

(1) Rates approved on May 13, 2014.

Source: SFPUC, Financial Services

Appeals. Multi-family residential water rates have a multiplier based on the number of dwelling units at the service address and Sewer service charges are assessed on the basis of water use as billed by the Water Enterprise multiplied by a flow factor. See "– Wastewater Enterprise Rates and Charges." For example, it is assumed that 90% of the volume of water measured at the customer meter for a single-family residential user is discharged to the sewer system as wastewater requiring treatment. Customers who can demonstrate higher rates of consumptive use than that reflected in the applicable flow factor, such as irrigation, can apply to the Residential Users Appeals Board for a lower flow factor (i.e., percentage of metered water returned to the sewage system).

Revenues

Sewer Service Charges. Sewer service charges are imposed based on discharge volume estimates and, in the case of non-residential users, suspended solids, oil and grease and chemical oxygen demand. See "WASTEWATER ENTERPRISE – Rates and Charges." The following tables show billed discharge by category and annual sewer billings by customer class for Fiscal Years 2010-11 to 2014-15.

TABLE 15
ANNUAL BILLED DISCHARGE BY CATEGORY

Fiscal Year	Volume (Hundred Cubic Feet)	Suspended Solids (Thousand Pounds)	Oil and Grease (Thousand Pounds)	Chemical Oxygen Demand (Thousand Pounds)
2010-11	27,063,139	47,101	14,350	115,473
2011-12	26,857,053	46,742	14,240	114,593
2012-13	26,535,868	46,183	14,070	113,223
2013-14	26,796,985	46,638	14,209	114,337
2014-15	24,667,341	42,931	13,079	105,250

Source: SFPUC, Financial Services

TABLE 16
ANNUAL SEWER BILLINGS BY USER TYPE
FOR FISCAL YEARS ENDED JUNE 30
(IN THOUSANDS)

User Type	2011	2012	2013	2014	2015
Multi-Family Residential	$ 82,056	$ 93,264	$ 95,607	$ 99,603	$100,178
Single-Family Residential	53,466	55,776	58,683	64,377	61,048
Subtotal Residential	$135,522	$149,040	$154,290	$163,980	$161,226
Commercial	$ 76,784	$ 75,604	$ 74,978	$ 76,740	$ 76,042
Municipal Customers	6,519	6,316	6,520	6,205	5,534
Suburban	3	2	2	3	2
Total	$218,828	$230,962	$235,790	$246,928	$242,804

Source: SFPUC, Financial Services

Sewer Account Billing and Delinquencies. Sewer service charges are billed on a combined water and sewer utility bill on either a monthly or bi-monthly basis. Payments are due 15 days after the bill date. If payments are not made, late payment charges are assessed 15 days from the due date or 30 days from the bill date. Accounts are considered delinquent 15 days after a second unpaid bill. Water service may be disconnected or a lien may be assessed against the property for non-payment of water and sewer services.

Current accounts receivables are shown in the table below. These amounts exclude receivables from municipal customers.

TABLE 17
ACCOUNTS RECEIVABLES AGING REPORT
AS OF JUNE 30, 2015

Period	Amount	Percent of Total	Percent of Total Fiscal Year 2014-15 Revenues
Current	$ 21,732,839	77.4%	8.9%
31—60 Days	993,988	3.5	0.4
61—90 Days	340,192	1.2	0.1
Over 90 Days	5,017,001	17.9	2.1
Total	$28,084,021	100.00%	
Credit Balances	(660,521)		
Total Aged Receivables	27,423,500		
Less Allowance For Doubtful Accounts	(2,490,390)		
Accounts Receivable, Net of Allowance	$24,933,110		

Source: SFPUC, Financial Services

Once an account is delinquent, a series of notices are sent to the customer and if payment is still not made, action to shut-off the water supply is initiated. However, if the delinquent account is in the name of the property owner and payment is still not made despite such notices, the customer will receive a lien warning notice. After a lien notice is delivered, a lien hearing is held and, if the bill still remains outstanding, the lien is recorded and can only be removed upon full payment of all unpaid charges, plus administrative fees and interest. Liens not paid during the Fiscal Year in which they are recorded are transferred to the City Tax Collector's Office, Bureau of Delinquent Revenues for collection as a lien against the property. Accounts for which property transfers occurred prior to recording the lien and closed accounts with amounts less than $25 are normally written off as uncollectible.

As a result of these collections processes, the SFPUC's annual write-offs for wastewater accounts remain relatively low. The following table shows a five-year history of write-offs for uncollectible accounts, excluding municipal customers:

TABLE 18
WRITE-OFFS FOR UNCOLLECTIBLE ACCOUNTS
(Fiscal Years Ended June 30)

	Amount
2011	$ 37,293
2012	21,297
2013	1,965
2014[1]	771,446
2015[1]	890,010

[1] Write-offs for Fiscal Years 2013-14 and 2014-15 include amounts from Fiscal Years 1999-2000 to 2012-13 initially reported to the SFPUC as uncollectable by the Bureau of Delinquent Revenues in Fiscal Years 2013-14 and 2014-15. In the future, the SFPUC expects that the Bureau of Delinquent Revenues will report such uncollectable amounts to the SFPUC on an annual basis.

Source: SFPUC, Financial Services

Capacity Charges. Effective July 1, 2005, any customer requesting a new connection to the sewer system or requiring additional collection or treatment capacity as a result of any addition, improvement, modification or change in use of an existing connection as determined solely by the General Manager must pay a capacity charge for the new or additional capacity required to serve the customer. The capacity charge is site specific and may not be sold, traded or conveyed in a manner to another site or customer. The capacity charge does not convey or imply ownership in or of any facilities of the Wastewater Enterprise.

Effective July 1, 2014, capacity charges are assessed based on water meter equivalents, which provide a direct estimate of wastewater flow to the system, and Standard Industrial Classification (SIC) code, which accounts for wastewater strength. For a single residential customer with a 5/8" water meter, the wastewater capacity charge fee for the 2015-16 Fiscal Year is $4,343. The capacity charge is adjusted on July 1st of each subsequent year by the annual change in the 20-City Average Construction Cost Index (CCI) published by ENR Magazine.

Capacity charges have averaged approximately 3.24% of revenues over the past five years.

Operating and Maintenance Expenses

"Operating and Maintenance Expenses" cover the general operational expenses of the Wastewater Enterprise. These expenses include labor and employment benefits, contractual services, materials and supplies, depreciation, general and administrative, services from other departments and other miscellaneous costs. See "HISTORICAL OPERATING RESULTS" and "THE PUBLIC UTILITIES COMMISSION – Employee Relations." Services from other departments include payment for services from other City departments, such as the City Attorney's Office and the General Services Agency.

Allocation of Costs. Various common costs incurred by the SFPUC are allocated among the Wastewater Enterprise, the Water Enterprise and the Power Enterprise. The most recent cost allocation review was completed in 2014. Allocations are based on the SFPUC management's best estimate and may change from year to year depending on activities undertaken by each enterprise and information available.

For Fiscal Year 2015-16, the SFPUC budgeted $27.0 million in administrative costs to the Wastewater Enterprise, which is recorded as personal service expenses and also in other various operating expenses in the Wastewater Enterprise financial statements. For Fiscal Year 2014-15, the SFPUC allocated $26.9 million in administrative costs to the Wastewater Enterprise.

Payments to/from the City. The SFPUC receives payments from other agencies of the City for their share of the proportionate cost of the service provided to them. Over the past five years, these service deliveries generated revenues that have averaged approximately 3% of Wastewater Enterprise revenues.

A variety of City departments provide services such as engineering, purchasing, legal, data processing, telecommunications, and human resources to the Wastewater Enterprise and charge amounts designed to recover those costs. These charges totaled approximately $11.9 million and $11.5 million for the years ended June 30, 2015 and 2014, respectively.

On October 7, 2009, the City and County of San Francisco issued $167.67 million in fixed-rate Certificates of Participation, Series 2009 C and D, to fund the construction of the headquarters of the SFPUC at 525 Golden Gate Avenue. Pursuant to a Memorandum of Understanding between the City and the SFPUC, the SFPUC will reimburse the City General Fund for all costs in connection with this City financing. Such obligations are subordinate to debt service on the Bonds and payments related thereto are allocated among the three SFPUC Enterprises. See "OBLIGATIONS PAYABLE FROM REVENUES – Other Obligations Payable from Revenues."

Debt Management and Fund Balance Reserve Policies

The SFPUC has established "Debt Management Policies and Procedures" for debt financing under its jurisdiction. It has also established separate "Bond Disclosure Compliance Policies & Procedures". These policies apply to all SFPUC enterprises, including the Wastewater Enterprise, and are intended to enable the SFPUC to effectively manage its debt issuance and administration practices. The "Debt Management Policies and Procedures" are reviewed bi-annually and revised, as necessary, with Commission approval. The most recent revisions were approved on March 24, 2015.

The SFPUC also established a "Fund Balance Reserve Policy" in 2010. This policy is applied to the long-term financial planning of all SFPUC enterprises, including the Wastewater Enterprise. The Fund Balance Reserve Policy states that operating and capital plans, budgets and rates will be projected and proposed for adoption such that all bond indenture requirements are met or exceeded and that Operating Fund Balance Reserves meet one or more of the following: total at least 15% of annual revenues; total at least 15% of annual expenditures; or result in Debt Service Coverage, on a bond indenture basis, including fund balance reserves available to pay debt service, of at least 1.25 times.

The SFPUC makes no representation that these policies will not be revised or amended and, except to the extent required for compliance with the terms of the Indenture, and makes no representation that these policies will be followed by the SFPUC.

Investment of SFPUC Funds

The SFPUC's pooled deposits and investments are invested pursuant to State law and the investment policy established from time to time by the City Treasurer and overseen by the Treasury Oversight Committee. The current policy seeks the preservation of capital, liquidity and yield, in that order of priority. Under the City Treasurer's current investment procedures, the SFPUC's pooled deposits and investments are invested in the City's larger pooled investment fund (the "City Pool"). Among other purposes, the City Pool serves in effect as a disbursement account for expenditures from the City's various segregated and pooled funds. Investments are generally made so that securities can be held to maturity. The City Treasurer calculated the weighted average maturity of these investments as of December 31, 2015 to be 422 days.

The following table sets forth the approximate book values of the investments held in the City Pool reported by the City Treasurer as of December 31, 2015. The Wastewater Enterprise's pooled deposits and investments accounted for approximately $287.46 million, or approximately 3.8%, of such amounts.

TABLE 19
CITY POOLED INVESTMENT FUND
(AS OF DECEMBER 31, 2015)

Investments	Book Value (millions)
U.S. Treasuries	$ 498.4
Federal Agencies	4,233.3
State & Local Government Agency Obligations	213.7
Public Time Deposits	1.2
Negotiable CDs	1,025.0
Commercial Paper	443.7
Medium Term Notes	722.5
Money Market Funds	185.1
Supranationals	74.9
Total	$7,397.9

Source: Office of the Treasurer & Tax Collector of the City and County of San Francisco.

The SFPUC's non-pooled deposits and investments consist primarily of funds related to the SFPUC's Outstanding Bonds, which are invested pursuant to policy established by the SFPUC, subject to the restrictions contained in the applicable bond documentation.

Risk Management and Insurance

The SFPUC's risk management program encompasses both self-insured and insured coverage. Risk assessments and coverage are coordinated by the SFPUC Enterprise Risk Manager through the City Office of Risk Management. With certain exceptions, the City and SFPUC's general approach is to first evaluate self-insurance for the risk of loss to which it is exposed. Based on this analysis, the SFPUC has determined that mitigating risk through a "self-retention" mechanism is more economical as it manages risks internally and administers, adjusts, settles, defends, and pays claims from budgeted resources (i.e., pay-as-you-go). When economically more viable or when required by debt financing covenants, the SFPUC obtains commercial insurance.

At least annually, the City reviews and actuarially determines general liability and workers' compensation liabilities, which are recorded as "Damages and Claims" and "Accrued Worker's Compensation" in the financial statements.

The SFPUC does not maintain commercial earthquake coverage for the Wastewater Enterprise, with certain minor exceptions, such as a sub-limit for fire-sprinkler leakage due to earthquake under the Property Insurance program.

The following is a summary of the SFPUC's coverage approach to risk:

Primary Risks	Typical Coverage Approach
General Liability	Self-Insured
Property	Purchased Insurance & Self-Insured
Workers' Compensation	Self-Insured through City-Wide Pool

Other Risks	Typical Coverage Approach
Surety Bonds	Purchased and Contractually Transferred
Professional Liability	Combination of Self-Insured, Purchased Insurance and Contractual Risk Transfer
Errors & Omissions	Combination of Self-Insured, Purchased Insurance and Contractual Risk Transfer
Builders Risk	Purchased Insurance & Contractual Risk Transfer
Public Officials Liability	Purchased Insurance

The SFPUC's property risk management approach varies depending on whether the facility is currently under construction, or if the property is part of revenue-generating operations. The majority of purchased insurance is for revenue-generating facilities, debt-financed facilities, and mandated coverage to meet statutory or contractual requirements.

Additionally, the SFPUC acknowledges the importance of aligning strategic planning to the risk management process and has implemented an Enterprise Risk Management ("ERM") program to meet this need. The framework provides a strategic approach to managing operational risks. The ERM program has been implemented thus far at the SFPUC Business & Financial Services Bureau and Power Enterprises and plans are in place to continue implementation across the remainder of the SFPUC.

Capital Project Risk Management. For capital construction projects, the SFPUC has utilized traditional contractual risk transfer, owner-controlled insurance programs or other alternative insurance programs. Under the latter two approaches, the insurance program usually provides coverage for the entire construction project, along with multiple risk coverages, such as general liability and workers compensation. When a contractual risk transfer is used for capital construction risks, the SFPUC requires each contractor to provide its own insurance, while ensuring that the full scope of work be covered with satisfactory levels to limit the SFPUC's risk exposure balanced by that which is commercially available.

Performance bonds are required, and Builder's Risk insurance must be purchased, in most phases of the construction contracting process for such phases, as bid, performance, and payment or maintenance. Additionally, bonds may be required in other contracts where goods or services are provided to ensure compliance with applicable terms and conditions such as warranty.

Professional liability policies are either directly purchased insurance on behalf of the SFPUC, transferred through contract to the contracted professional, or retained through self-insurance on a case by case basis depending on the size, complexity or scope of construction or professional service contracts. Professional liability policies are typically purchased for services provided by engineers, architects, design professionals and other licensed or certified professional service providers.

Builder's Risk policies of insurance are required to be provided either through an owner-controlled insurance program or the contractor on all construction projects for the full value of the construction.

HISTORICAL OPERATING RESULTS

Summary of Historical Operating Results and Debt Service Coverage

The historical results of operations reflected in the table below are based on the tables contained in the Financial Statements entitled "Statements of Revenues, Expenses and Changes in Net Assets" and "Statements of Cash Flows" for the Fiscal Years listed. However, operating and investment income presented in this table differs from the Change in Net Assets. This table presents Debt Service Coverage on a cash basis as prescribed under the Indenture and excludes certain elements of nonoperating revenue and expenses included in the Statements of Revenues, Expenses and Changes in Net Assets. The calculation of debt service coverage includes net operating income and funds not budgeted to be spent in the next twelve months and legally available to pay debt service, as permitted under the Indenture. The audited financial statements of the Wastewater Enterprise for Fiscal Years 2013-14 and 2014-15, prepared by the SFPUC and audited by KPMG LLP, independent certified public accountants, are attached as APPENDIX B to this Official Statement. The following table should be read in conjunction with such financial statements. KPMG LLP has not reviewed the following table. See "APPENDIX B – SFPUC WASTEWATER ENTERPRISE FINANCIAL STATEMENTS."

TABLE 20
HISTORICAL REVENUE, OPERATING & MAINTENANCE EXPENSE
AND DEBT SERVICE COVERAGE FOR FISCAL YEARS ENDED JUNE 30
(IN THOUSANDS)[1]

	2011	2012	2013	2014	2015
OPERATING & INVESTMENT REVENUE					
Sewer Service Charges	$220,586	$233,628	$235,479	$244,705	$244,604
Other Revenues	4,875	4,983	5,738	5,869	5,041
Investing Activities	1,927	2,975	524	2,400	1,207
Capacity Charges[2]	3,755	5,544	11,337	9,523	6,357
Total Revenues	$231,143	$247,130	$253,078	$262,497	$257,209
OPERATING & MAINTENANCE EXPENSE					
Salary and Fringe Benefits[3]	$ 73,630	$ 82,709	$ 84,155	$ 85,114	$ 76,396
Contractual Services	12,577	13,257	13,418	14,314	13,841
Materials and Supplies	8,338	8,921	10,481	10,830	9,815
Depreciation	42,217	44,799	46,347	48,402	50,254
Services of Other Departments	32,689	33,292	34,141	35,274	36,212
General and Administrative[4]	507	2,815	3,760	396	10,232
Other[5]	9,126	10,064	15,958	22,010	19,735
Total Operating Expenses	$179,084	$195,857	$208,260	$216,340	$216,485
OPERATING AND INVESTMENT INCOME	$ 52,059	$ 51,273	$ 44,818	$ 46,157	$ 40,724
COVERAGE CALCULATION[6]					
Operating and Investment Income	52,059	51,273	44,818	46,157	40,724
+ Adjustment to Investing Activities[7]	108	(299)	337	(601)	127
+ Depreciation & Non-Cash Expenses	44,232	46,271	52,108	50,717	51,773
+ Changes in Working Capital[8]	5,633	21,375	(13,795)	12,908	3,923
+ SRF Loan Payments	(16,503)	(10,983)	(9,421)	0	0
= "**Net Revenues**"	$ 85,529	$107,637	$ 74,047	$109,181	$ 96,547
+ Other Available Funds[9]	22,769	50,761	80,094	109,807	134,413
Funds Available for Bond Debt Service	$108,298	$158,398	$154,141	$218,988	$230,960
Bond Debt Service	$ 41,838	$ 42,190	$ 38,750	$ 64,852	$ 60,901
DEBT SERVICE COVERAGE[10]					
Indenture Basis[11]	2.59x	3.75x	3.98x	3.38x	3.79x
Current Basis[12]	2.04x	2.55x	1.91x	1.68x	1.59x

(1) Operating and Investment Income presented in this table differs from the Change in Net Assets presented in the Statement of Revenues, Expenses and Changes in Net Assets in the Audited Financial Statements. See "APPENDIX B – SFPUC WASTEWATER ENTERPRISE FINANCIAL STATEMENTS." This table presents Debt Service Coverage as defined under the Indenture and excludes certain elements of non-operating revenue and expenses included in the Statements of Revenues, Expenses and Changes in Net Assets. Examples of excluded elements are Grant Revenue, Interest Expense and Gains from Sale of Assets. Totals may not add due to independent rounding.

(2) Decrease in Fiscal Year 2014-15 is due to rate structure change effective July 2014.

(3) Decrease in Fiscal Year 2014-15 due to pension cost reduction resulting from implementation of GASB Statement No. 68. See "THE SAN FRANCISCO PUBLIC UTILITIES COMMISSION – Employee Benefit Plans – Plan Financial Reports and Funded Status."

(4) Increase in Fiscal Year 2014-15 due to net change of $9.1 million in general liabilities accrual.

(5) Decrease in Fiscal Year 2014-15 due primarily to lower Capital Project funds expense.

(6) The Indenture defines "Net Revenues" on a cash basis.

(continued)

(7) Represents adjustments to show investing activities on a cash basis.

(8) In accordance with GASB Statement No. 68, roughly $10 million related to pensions was recognized as deferred inflows of resources in Fiscal Year 2014-15, driving the decrease in working capital. See "THE SAN FRANCISCO PUBLIC UTILITIES COMMISSION – Employee Benefit Plans – Plan Financial Reports and Funded Status."

(9) Per the Indenture, includes any fund balances of the SFPUC or the Wastewater Enterprise available for payment of debt service and not budgeted to be expended during the 12 months following a calculation date, excluding monies held in any Reserve Account established under the Indenture.

(10) Coverage does not include debt service on subordinate obligations, including the Wastewater Enterprise's share of lease payments associated with the 2009 Golden Gate COPs and debt service on Commercial Paper Notes.

(11) Calculated as the sum of Net Revenues plus "Other Available Funds," divided by debt service on Bonds and Parity State Loans. The Indenture includes a rate covenant of 1.25x "Debt Service Coverage." See "SECURITY FOR THE BONDS – Rate Covenants – Debt Service Coverage."

(12) Unaudited. Calculated as Net Revenues divided by debt service on Bonds and SRF Loans.

Source: SFPUC, Financial Services

Management Discussion

Certain Impacts of the Current Drought. In Fiscal Year 2014-15 and Fiscal Year 2015-16, budgeted sewer charge revenue was determined assuming a 52.4 mgd and 51.1 mgd billed sewer discharge volume, respectively. Actual billed sewer discharge volumes in Fiscal Year 2014-15 were 50.6 mgd, a 3.5% decrease from budgeted volumes, while Fiscal Year 2015-16 projected billed volumes are 48.5 mgd, a 5.4% decrease from budgeted amounts. Actual Wastewater Enterprise sewer charge revenues (including City Departments) for Fiscal Year 2014-15 were $248.2 million, as compared to budgeted sewer charge revenues of $264.5 million for that Fiscal Year, representing a 6.1% decrease. As of February 9, 2016, projected sewer charge revenues for Fiscal Year 2015-16 were $250 million, as opposed to budgeted sewer charge revenues of $262.2 million, representing a projected 4.6% decrease in Fiscal Year 2015-16. Reduced water sales stemming from drought-reduction efforts contributed to such decrease in Revenues. See "FINANCIAL OPERATIONS – Wastewater Enterprise Rates and Charges."

The Ten-Year Financial Plan adopted by the Commission in February 2016 for Fiscal Year 2016-17 to Fiscal Year 2025-26, adjusted projected sewer discharge volumes down to 48.5 mgd throughout the ten-year period. The SFPUC anticipates that the potential financial impacts of such decreases in sewer discharge volumes will be offset by both cost reductions and increased rates. See "THE WASTEWATER ENTERPRISE – Current System Demand," "HISTORICAL OPERATING RESULTS," "PROJECTED OPERATING RESULTS" and "RISK FACTORS – Risks Related to Wastewater Enterprise Facilities and Operation – Unanticipated Decreases in Water Sales Volumes."

PROJECTED OPERATING RESULTS

The following table presents projected operating results for the Wastewater Enterprise. These projections are based on an analysis of historical trends, adjusted where appropriate for known or anticipated changes in operations. The projections are also based on the assumption that all rate increases necessary to finance future expenses, including the SSIP and other capital programs, will be approved and implemented.

THESE PROJECTIONS, ALL OR SOME OF WHICH MAY OR MAY NOT BE REALIZED, ARE BASED ON THE ISSUANCE OF ADDITIONAL BONDS FOR THE CAPITAL PROGRAM, INCLUDING SSIP. CHANGES IN THE CIRCUMSTANCES THAT FORM THE BASES FOR THE ASSUMPTIONS USED IN DEVELOPING THESE PROJECTIONS, AS WELL AS UNANTICIPATED EVENTS, MAY OCCUR SUBSEQUENT TO THE DATE OF THE OFFICIAL STATEMENT. THEREFORE, ACTUAL RESULTS MAY DIFFER MATERIALLY FROM THE PROJECTIONS SHOWN.

TABLE 21
PROJECTED REVENUE, OPERATING AND MAINTENANCE EXPENSE AND DEBT SERVICE COVERAGE FOR FISCAL YEAR ENDING JUNE 30 (IN THOUSANDS) [1]

	2016	2017	2018	2019	2020
REVENUE					
Sewer Service—Base Rates	$238,102	$250,007	$267,508	$296,934	$338,504
Sewer Service—Rate Increases[2]	11,905	17,501	29,426	41,571	47,391
Capacity Charges	7,587	6,434	6,825	6,949	6,736
Interest Income[3]	800	1,326	1,480	1,633	1,773
Other Miscellaneous Income	4,604	3,427	3,474	3,522	3,573
Total Revenues	$262,998	$278,695	$308,713	$350,609	$397,977
OPERATING AND MAINTENANCE EXPENSE[4]					
O&M Expenses	$162,822	$161,410	$165,505	$170,402	$175,473
NET REVENUES[5]	$100,176	$117,285	$143,208	$180,207	$222,504
plus OTHER AVAILABLE FUNDS[6]	$121,543	$119,599	$136,258	$152,536	$162,553
FUNDS AVAILABLE FOR DEBT SERVICE	$221,719	$236,884	$279,466	$332,743	$385,057
DEBT SERVICE[7]					
2010 Series A Bonds	2,276	9,038	9,042	9,042	9,040
2010 Series B Bonds (Net)	7,200	7,200	7,200	7,200	7,200
2013 Series A Bonds	36,109	18,103	16,341	16,398	16,486
2013 Series B Bonds	14,428	14,428	14,428	14,428	14,428
2016 Series A Bonds	-	-	-	9,078	10,646
2016 Series B Bonds	-	-	-	2,559	3,001
Future Parity Obligations[8]	-	-	-	41,975	68,707
TOTAL DEBT SERVICE	$ 60,013	$ 48,768	$ 47,011	$ 100,680	$129,508
DEBT SERVICE COVERAGE[9]					
Indenture Basis[10]	3.69x	4.86x	5.94x	3.30x	2.97x
Current Basis[11]	1.67x	2.40x	3.05x	1.79x	1.72x

(1) Amounts set forth in the table are projections. Actual results may differ materially from these projections. See "FORWARD-LOOKING STATEMENTS" and "– Assumptions Used in Projections." Totals may not add due to independent rounding.
(2) Reflects adopted annual rate increases of 5%, 7%, and 11% approved for Fiscal Years 2015-16 through 2017-18 in May, 2014. Average annual rate increases of 14% are assumed the following two years. See "–Assumptions Used in Projections."
(3) Assumes approximately 1.0% earnings rate on invested cash balance over projection period.
(4) Operating and Maintenance Expenses net of depreciation and other non-cash items per Indenture.
(5) Represents Net Revenues under the Indenture.
(6) Per the Indenture, includes any fund balances of the SFPUC or the Wastewater Enterprise available for payment of debt service and not budgeted to be expended during the 12 months following a calculation date, excluding monies held in any Reserve Account established under the Indenture.
(7) Debt Service on Outstanding Bonds, including the 2016 Series AB Bonds, net of capitalized interest and anticipated Build America Bonds interest subsidy payments.
(8) Net of capitalized interest. Future debt issuances to fund renewal and replacement and SSIP capital projects are expected to total approximately $1.7 billion over the projection period. Standard assumptions for the issuance of Additional Bonds include 30-year term, 5% interest rate, and 2 years of capitalized interest. Projections also assume additional SRF Loan funding (Parity State Loans) for Fiscal Years 2018-19 and 2019-20, however repayment of such loans does not commence within the projection period. The actual size and timing of issuances will depend on market conditions and other factors as determined by the SFPUC.
(9) Coverage does not include debt service on subordinate obligations, including the Wastewater Enterprise's share of lease payments associated with the 2009 Golden Gate COPs and debt service on Commercial Paper Notes.
(10) Calculated as the sum of Net Revenues plus "Other Available Funds," divided by debt service on Bonds and Parity State Loans. The Indenture includes a rate covenant of 1.25x "Debt Service Coverage." See "SECURITY FOR THE BONDS – Rate Covenants – Debt Service Coverage."
(11) Calculated as Net Revenues divided by debt service on Bonds and Parity State Loans.

Source: SFPUC, Financial Services

Assumptions Used in Projections

In the preparation of the projections set forth in Table 21, the SFPUC has made certain assumptions with respect to conditions that may occur in the future. While the SFPUC believes these assumptions are reasonable for the purpose of the projections, they are dependent on future events, and actual conditions are likely to differ, perhaps materially, from those assumed. To the extent actual future conditions differ from those assumed by the SFPUC or provided to the SFPUC by others, actual results will vary from those projected. This projected information has not been compiled, reviewed or examined by the SFPUC's independent auditors.

The assumptions used in the table include the following:

Projected Revenue and Rate Increases. The projected revenues are based on projected wastewater service sales and the schedules of rates to be effective in each year. In May 2014, the SFPUC adopted schedules of rates to be effective in Fiscal Years 2014-15 through 2017-18. The adopted schedules provide for rate increases of 5% in Fiscal Year 2014-15 and Fiscal Year 2015-16, 7% in Fiscal Year 2016-17, and 11% in Fiscal Year 2017-18. Average rate increases for the remainder of the projection period are assumed at 14% for Fiscal Year 2018-19 and Fiscal Year 2019-20. Any such rate increases will be subject to Commission approval.

Projected Operating and Maintenance Expenses. Operating and Maintenance Expense costs are projected to grow by 3% annually for the projection period.

Projected Debt Service. Projected debt service reflects the scheduled debt service on the 2010 Series A Bonds, the 2010 Series B Bonds, the 2013 Series A Bonds, the 2013 Series B Bonds, and the 2016 Series AB Bonds (net of capitalized interest and expected Build America Bonds interest subsidy payments relating to 2010 Series B Build America Bonds) and the estimated debt service on future series of Additional Bonds and Parity State Loans.

Issuances of Additional Bonds and Parity State Loans assume borrowing rates ranging from 3%-5%, no reserve account and, with respect to the Additional Bonds but not the Parity State Loans, up to two years of capitalized interest. Actual issuance dates, borrowing rates and capitalized interest periods for Additional Bonds may vary from these assumptions.

Following the issuance of the 2016 Series AB Bonds, the SFPUC anticipates that it will issue Additional Bonds in Fiscal Year 2016-17 to fund approximately $480 million of SSIP and other Wastewater Enterprise capital program projects, as well as Parity State Loans to fund certain SSIP projects. The SFPUC may also issue refunding bonds from time to time in response to market conditions in order to achieve debt service savings.

RISK FACTORS

This section provides a general overview of certain risk factors which should be considered, in addition to the other matters set forth in this Official Statement, in evaluating an investment in the 2016 Series AB Bonds. This section is provided for convenience and is not meant to be a comprehensive or definitive discussion of the risks associated with an investment in the 2016 Series AB Bonds, and the order in which this information is presented does not necessarily reflect the relative importance of various risks.

Potential investors in the 2016 Series AB Bonds are advised to consider the following factors, among others, and to review this entire Official Statement to obtain information essential to the making of an informed investment decision. Any one or more of the risk factors discussed below, among others, could lead to a decrease in the market value and/or in the marketability of the 2016 Series AB Bonds or adversely affect the ability of the SFPUC to make timely payments of principal of or interest on the 2016 Series AB Bonds. There can be no assurance that other risk factors not discussed herein will not become material in the future, and the SFPUC has not undertaken to update investors about the emergence of the risk factors in the future.

General

The ability of the SFPUC to comply with its covenants under the Indenture and to generate Net Revenues sufficient to pay principal of and interest on the 2016 Series AB Bonds may be adversely affected by actions and events outside of the control of the SFPUC and may be adversely affected by actions taken (or not taken) by voters, property owners, taxpayers or persons obligated to pay fees and charges. Among other matters, general and local economic conditions and changes in law and government regulations could adversely affect the amount of Revenues realized by the SFPUC or significantly raise the cost of operating the Wastewater Enterprise.

In addition, the realization of future Revenues is subject to, among other things, the capabilities of management of the SFPUC, the ability of the SFPUC to provide service to its customers, the ability of the SFPUC to establish, maintain and collect charges from its customers and the ability of the SFPUC to establish, maintain and collect rates and charges sufficient to pay for Operation and Maintenance Costs of the Wastewater Enterprise, the Bonds and other obligations payable from Revenues. See "FINANCIAL OPERATIONS" and "OBLIGATIONS PAYABLE FROM REVENUES."

Limited Obligation

If the SFPUC defaults on its obligations to make debt service payments on the Bonds, the Trustee has the right under the Indenture to accelerate the total unpaid principal amount of the Bonds. However, in the event of a default and such acceleration, there can be no assurance that the SFPUC, and correspondingly the Trustee, will have sufficient moneys available for payment of the Bonds.

The SFPUC is not obligated to pay the principal of, premium, if any, or interest on the 2016 Series AB Bonds except from Net Revenues of the Wastewater Enterprise. The SFPUC has no taxing power. The General Fund of the City is not liable for the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds, and neither the credit nor the taxing power of the City is pledged to the payment of the principal of, premium, if any, or interest on the 2016 Series AB Bonds. The 2016 Series AB Bonds are not secured by a legal or equitable pledge of, or charge, lien or encumbrance upon, any of the property of the City or of the SFPUC or any of its income or receipts, except Net Revenues.

No Reserve Account

No Reserve Account has been established for the 2016 Series AB Bonds. The Reserve Accounts established with respect to other Series of Bonds do not secure the 2016 Series AB Bonds.

Commercial Paper Credit Facilities

The bank credit facilities supporting the Wastewater Commercial Paper Notes are subject to early termination upon the occurrence of certain events, including the failure of the Wastewater Enterprise to make certain payments, the occurrence of certain bankruptcy or insolvency-related events or the reduction below specified levels or the withdrawal or suspension of ratings on certain obligations of the Wastewater Enterprise payable from Net Revenues. Upon the occurrence of such termination, one or more of the following would likely occur: (a) the SFPUC would be prohibited from issuing additional notes supported by such credit facilities; (b) any principal of or interest on Wastewater Commercial Paper Notes paid from a draw on a credit support facility would become a reimbursement obligation of the Wastewater Enterprise to the bank providing such facility, and such reimbursement obligation could bear interest at rates higher than the rates borne by the Wastewater Commercial Paper Notes; and (c) amortization of such reimbursement obligation would be required. The Wastewater Commercial Paper Notes and any reimbursement obligations are payable from Net Revenues on a basis subordinate to the Bonds. See "OBLIGATIONS PAYABLE FROM REVENUES – Subordinate Debt and Commercial Paper" and "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM – Interim Funding of Capital Program."

Limitations on Rate-Setting

The generation of Revenues sufficient to satisfy the requirements of the Indenture and to pay the principal of and interest on the 2016 Series AB Bonds will require the SFPUC to raise wastewater rates payable by its customers. The increase of wastewater rates is subject to various substantive and procedural requirements and limitations. See "CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS."

Initiative, Referendum and Charter Amendments and Future Legislation

Under the State Constitution, the voters of the State have the ability to initiate legislation and require a public vote on legislation passed by the State Legislature through the powers of initiative and referendum, respectively. The SFPUC is unable to predict whether any such initiatives might be submitted to or approved by the voters, the nature of such initiatives, or their potential impact on the SFPUC or the Wastewater Enterprise. See "CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS."

Under the Charter, the voters of the City can restrict or revise the powers of the SFPUC through the approval of a Charter amendment or other initiative. For example, in June 1998 the electorate of the City approved Proposition H which, subject to certain exceptions, including a limited exception to raise rates to pay debt service on voter-approved debt, and froze the SFPUC's water and sewer rates through July 1, 2006. The SFPUC can give no assurance that the electorate will not seek in the future to freeze or limit rate increases. See "CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS – Charter Limitations."

In addition, the SFPUC is subject to various laws, rules and regulations adopted by the local, State and federal governments and their agencies. The SFPUC is unable to predict the adoption or amendment of any such laws, rules or regulations, or their effect on the operations or financial condition of the SFPUC.

Risks Related to Wastewater Enterprise Facilities and Operation

The operation of the Wastewater Enterprise, and the physical condition of the Wastewater Enterprise facilities, are subject to a number of risk factors that could adversely affect the reliability of the SFPUC to provide sewage and stormwater collection and treatment services, or increase the operating expenses of the Wastewater Enterprise. Prolonged damage to the Wastewater Enterprise facilities could interrupt the ability of the SFPUC to realize Revenues sufficient to pay principal of and interest on the Bonds, or require the SFPUC to increase expenditures for repairs significantly enough to adversely impact the SFPUC's ability to pay the principal of or interest on the Bonds. These factors could include, among others, the following.

Aging Facilities. Certain Wastewater Enterprise facilities are near the end of their useful life. Aging assets result in decreased reliability due to sewer line breakage and unplanned facility outages and place a greater maintenance burden on Wastewater Enterprise operations. The average useful life of the sewer system's sewers (i.e., gravity pipes 36 inches or less in diameter) is approximately 110 years, though many conditions, such as pipe material and soil conditions, affect actual pipe lifespan. The recent historical rate of replacement for the Wastewater Enterprise's sewer pipes has been at a 200-year replacement cycle. See "THE WASTEWATER ENTERPRISE – Combined Sewage and Stormwater System." Aging sewers, if left unaddressed, would result in increasing system failures, sinkholes in the street, reduced system reliability and possibly risks to public health and safety and the environment.

The SFPUC is undertaking extensive efforts to increase the replacement rate significantly and to address aging infrastructure, with the objective of ultimately increasing the aggregate miles of sewer replaced each year and achieving a 110-year replacement cycle. The capital program, including the SSIP, is intended to help increase system reliability through equipment and facility improvement including, among other things, increasing sewer inspections and condition assessments in order to more effectively prioritize areas of pipeline replacement. In addition to the collection system, other types of facilities, such as treatment plants and pump stations, also face reliability issues due to age and poor condition. The aging system will be further challenged in connection with the SSIP because it must remain operational during project delivery. Specifically, during construction of SSIP capital

projects, all work, including any required partial, temporary facility shutdowns, will have to be carefully scheduled so that the plants remain functional and in permit compliance with all permits and other applicable requirements.

Limited Redundancy. Many critical Wastewater Enterprise facilities must remain constantly operational to collect and treat sewage and stormwater flows in accordance with Regional Water Quality Control Board requirements. See "REGULATORY MATTERS." Certain of these Wastewater Enterprise facilities and systems are aging and have limited redundancy. See "– Aging Facilities." In addition, the Oceanside Plant and the other facilities in the Westside Watersheds are not interconnected with the Southeast Plant, the North Point Facility or the other facilities in the Bayside Watersheds. See "THE WASTEWATER ENTERPRISE – Wastewater Treatment." This limited redundancy in facilities and systems, including the absence of an interconnection between the treatment plants and other facilities located in the two separate watersheds, constrains the SFPUC's ability to take components of the system out of service for maintenance and repairs, or to provide backup treatment and other facilities in the event of an unplanned outage. For example, work at the Southeast Treatment Plant will be done while the plant is operating, requiring coordination of facility and systems shutdowns with construction plans. During dry weather, however, the collection system does have excess storage capacity due to the transport/storage structures which can provide up to 200 million gallons worth of capacity and help facilitate treatment flexibility when needed.

The SFPUC expects the SSIP to address certain redundancy issues, including: power supply redundancy for the three treatment plants, and rehabilitation and addition of redundant pumps as necessary at major pump stations. Interconnection of the facilities in the Westside Watersheds and Bayside Watersheds is not included in the SSIP, however. See "THE WASTEWATER ENTERPRISE – Wastewater Treatment."

Seismic Hazards. The San Francisco Bay Area is in a seismically active region. The San Andreas Fault lies immediately west of the City, and the Hayward fault is approximately fifteen miles to the east. A third major fault, the Calaveras Fault, is a branch of the Hayward Fault and lies east of the Hayward Fault.

During the past 150 years, the San Francisco Bay Area has experienced several major and numerous minor earthquakes. The largest was the 1906 San Francisco earthquake along the San Andreas Fault with an estimated magnitude of 8.2 on the Richter scale. Another was the 1868 Hayward earthquake along the Hayward Fault. The most recent significant earthquake was the October 1989 Loma Prieta earthquake on the San Andreas Fault, which had a magnitude of 7.1 on the Richter scale and an epicenter near Santa Cruz, approximately fifty-five miles south of the City. According to a recent United States Geological Survey, a significant earthquake along these or other faults has a high probably of occurrence during the repayment period of the 2016 Series AB Bonds. A significant earthquake that impacts the City could adversely affect the ability of Wastewater Enterprise customers to pay for service and the capital and operating expenses of the Wastewater Enterprise.

Older facilities may have an increased risk of failure in the event of an earthquake. SSIP and other portions of the capital program include planned and proposed improvements to such older facilities for the purpose of improving seismic reliability.

With certain minor exceptions, the SFPUC does not maintain commercial earthquake insurance coverage for the facilities. See "FINANCIAL OPERATIONS – Risk Management and Insurance".

Flooding and Other Natural and Man-Made Disasters. Flooding and other natural disasters, including without limitation flooding, landslides, and fire, or man-made disasters or accidents, including without limitation natural gas pipeline failures or explosions, could interrupt operation of the Wastewater Enterprise, result in liability claims against the Wastewater Enterprise, or otherwise adversely impact the Wastewater Enterprise's ability to provide services or collect Revenues. See "– Unanticipated Decreases in Water Sales Volumes" and "– Potential Impacts of Climatic Change and Sea Level Rise." For example, major rainstorms in the City which exceed the capacity of the combined storm drainage/sewer system, result in flooding and property damage, especially in some low-lying areas with structures built on or near land which was originally creeks or bays. Severe flood damage occurred in 2004 due to major rainstorms. Property owners in low-lying areas filed claims against the City, many of which were settled. Some claims proceeded to litigation, and the City settled those suits in 2007.

In 2009 and 2012, rainstorms resulted in additional flooding and property damage. In December 2014, two severe rainstorms caused flooding in many of the same low-lying areas that had flooded in 2004, 2009 and 2012.

The City responded to both December storms in each of the affected neighborhoods, and helped coordinate and pay for initial clean-up and repair of damage to affected properties. In addition, property owners filed numerous claims for property damage. To date, the City's aggregate costs incurred for restoration, damage mitigation, and claims settlements exceeds $5 million. The SFPUC continues to conduct technical analysis, explore infrastructure improvements, and document design, operation and maintenance practices necessary to minimize flooding in these areas.

Statutory and Regulatory Compliance. The operation of the Wastewater Enterprise is subject to a variety of federal and State statutory and regulatory requirements concerning matters such as water quality, discharge requirements, and biosolids management. SFPUC's failure to comply with applicable laws and regulations could result in significant fines and penalties. In addition to claims by private parties, changes in the scope and standards for public agency wastewater systems such as the Wastewater Enterprise may also lead to administrative orders issued by federal or State regulators. Future compliance with increased regulatory requirements or enforcement orders could impose substantial additional operating expenses on the Wastewater Enterprise. See "REGULATORY MATTERS."

Unanticipated Decreases in Water Sales Volumes. Under the SFPUC's current rate structure, Wastewater Enterprise customer bills are calculated primarily based on metered water volumes (with certain "flow factor" adjustments) and rates are established based upon expected metered water volumes. As a result, water deliveries at less than expected levels would result in lower than expected Revenues. Reduced water deliveries could result from, among other circumstances, reduced water supply, conservation measures or damage to storage, transportation, treatment or delivery systems of the SFPUC's Water Enterprise. See "FINANCIAL OPERATIONS – Wastewater Enterprise Rates and Charges" and "– Management Discussion – Certain Impacts of Current Drought" and "Seismic Hazards" and "Other Natural or Man-Made Disasters" above.

Increased Operating and Maintenance Expenses. There can be no assurance that the Operating and Maintenance expenses of the SFPUC, such as wages and salaries, pension and other benefits, and purchased power costs, will not increase, perhaps substantially. See "FINANCIAL OPERATIONS – Operating and Maintenance Expenses."

Labor Actions. The Charter prohibits SFPUC and other City employees from striking. Nonetheless, a work stoppage or other labor action could limit the SFPUC's ability to operate the Wastewater Facilities and adversely impact Revenues.

Casualty Losses. The SFPUC's risk management program includes both self-insured and insured coverages; however, the program does not provide coverage for every conceivable risk of loss. Damage attributable to seismic events and environmental pollution are excluded. In situations where the SFPUC has not purchased commercial coverage, the Wastewater Enterprise has a 'self-retention' program that it administers and retains budgeted resources internally to provide coverage for loss liabilities. See also "FINANCIAL OPERATIONS – Risk Management and Insurance." The SFPUC is not required to either insure against or self-insure against every potential risk of loss, and there is a risk that damage or destruction of property and equipment comprising the Wastewater Enterprise could occur for which no insurance or self-insurance funds will be available. There can be no assurance that insurance providers will pay claims under any policies promptly, or at all, should a claim be made under such policies in connection with property loss or damage. It is possible that an insurance provider will refuse to pay a claim, especially if it is substantial, and force the SFPUC to sue to collect on or settle the insurance claim. Further, there can be no assurances that any insurance proceeds will be sufficient to rebuild or replace any damaged property.

Safety and Security. Military conflicts and terrorist activities may adversely impact the operations of the Wastewater Enterprise or the finances of the SFPUC. The SFPUC continually plans and prepares for emergency situations and immediately responds to ensure services are maintained. However, there can be no assurance that any existing or additional safety and security measures will prove adequate in the event that terrorist activities are directed against the assets of the Wastewater Enterprise or that costs of security measures will not be greater than presently anticipated.

Cybersecurity. The SFPUC has adopted information security policies and maintains an active information security program, which has been reviewed by independent third-party consultants engaged by the SFPUC. The SFPUC has appointed a Chief Information Security Officer who is responsible for annual updates to the SFPUC's information security policies and is charged with identifying and monitoring threats which are typically addressed by the SFPUC's information technology services team, and educating staff concerning vulnerabilities. The SFPUC's information security policies include policies intended to support network, computer and mobile device security (both digital and physical), e-mail security, anti-virus requirements, operating system and application patching, encryption requirements and secure computing asset disposal. The SFPUC's information security policies further include a guideline that, at least every two years, the SFPUC will engage external consultants to audit and assess the internal controls of the SFPUC's information security program.

The SFPUC does not purchase liability insurance covering cyber-losses. The SFPUC does require its vendors to purchase Technology Errors & Omissions coverage.

Cost of the SSIP; Timely Completion of the SSIP

The SSIP is the largest component of the Wastewater Enterprise's capital program. The completion of various SSIP projects could be delayed and the overall cost of such projects could increase for a variety of reasons, including, but not limited to, actions by State or federal regulatory agencies, voter initiatives, legal challenges on environmental or other grounds, prolonged contractor disputes, changes in price of commodities or labor, unanticipated geologic or soil conditions, or the occurrence of an earthquake or other natural disaster. See "SEWER SYSTEM IMPROVEMENT PROGRAM – Potential Changes to SSIP Projects."

The SFPUC intends to finance the development and implementation of SSIP projects through the issuance of Additional Bonds and Parity State Loans. If SSIP projects are completed at the cost and on the schedule presently under consideration by the SFPUC, the cost of such projects will require a significant planned increase in the amount of debt payable from Revenues, which will result in significant planned rate increases. Correspondingly, debt service coverage for the Bonds will also be significantly lower than it is currently or has been historically. Were SSIP projects delayed or the cost of SSIP projects to increase without an offsetting reduction in the program scope, the SFPUC would be required either to incur more debt payable from Revenues or to cash fund those costs from Revenues. Either option would likely increase rates payable by SFPUC customers to levels higher than presently anticipated by the SFPUC, and could result in lower debt service coverage ratios than presently anticipated by the SFPUC. See "FINANCING PLAN FOR WASTEWATER ENTERPRISE CAPITAL PROGRAM."

Over the next several years the SFPUC expects to issue additional revenue bonds and Parity State Loans to fund development and implementation of SSIP projects. The issuance by the SFPUC of such debt is subject to various approval requirements. See "OBLIGATIONS PAYABLE FROM REVENUES."

The ability of the SFPUC to issue additional revenue bonds and Parity State Loans to finance the development and implementation of SSIP projects may also be adversely affected by any adverse change in the financial position of the SFPUC or by general market conditions. There can be no assurance that the SFPUC will be able to issue revenue bonds and Parity State Loans in an aggregate amount sufficient to finance all of the costs of completing the SSIP projects.

Construction Related Risks

General. Construction projects in the capital program, including the SSIP, are subject to ordinary construction risks and delays applicable to projects of their kind, including but not limited to (i) inclement weather affecting contractor performance and timeliness of completion, which could affect the costs and availability of, or delivery schedule for, equipment, components, materials, labor or subcontractors; (ii) contractor claims or nonperformance; (iii) failure of contractors to execute within contract price; (iv) work stoppages or slowdowns; (v) failure of contractors to meet schedule terms; or (vi) unanticipated project site conditions, including the discovery of hazardous materials on the site or other issues regarding compliance with applicable environmental standards, and other natural hazards or seismic events encountered during construction. In addition, given the limited redundancy of certain Wastewater Enterprise facilities and systems, such systems must remain operational during construction,

which could affect construction schedules or budgets. See "– Risks Related to Wastewater Enterprise Facilities and Operation."

The SFPUC expects numerous other large construction projects to be scheduled in the region during the same time period as the SSIP, which could potentially limit contractor resources available to the SFPUC during bidding or construction phases. The SFPUC conducts regular contractor outreach and expects to continue to work closely with potential contractors to attempt to mitigate the impact of such simultaneous scheduling of large projects.

Another SSIP construction related risk is that much of the work will occur at a single physical location, the Southeast Plant. The phasing staging, materials and/or equipment laydown areas, parking, truck routes, security, and operating needs are being clearly defined for the plant and each individual project by the SFPUC, with consideration of multiple project interfaces. The plant will remain in operation while these projects are being implemented. Currently logistics planning is identifying on and off site parking, construction laydown and staging areas, truck routing for regular operations and construction and the overall facility and equipment shut-down schedule required to maintain permit compliance and complete the SSIP. Where land acquisition or agreements are required for siting, staging, laydown area, storage, or parking to support the projects, land acquisition could potentially delay a project. See "– Land Acquisition."

Increased construction costs or delays for any reason in connection with the SFPUC's capital program and the SSIP could impact the Wastewater Enterprise's financial condition in general and the implementation of its capital programs in particular.

Land Acquisition. Certain SSIP projects will require the SFPUC to acquire land in and around its facilities in the City. For example, the Southeast Plant is located in densely urban environment with industrial and residential buildings—so securing space will be critical.

A project manager has been hired to manage land issues because many parcels require long lead times to gain site control. Property is challenging to acquire or lease in the San Francisco Bay Area generally and in the City in particular. Real estate prices in the City and throughout the San Francisco Bay Area can be volatile and have in general been increasing during recent years. The SSIP currently includes an estimated $98.3 million related to future land purchases required for the expansion of the Southeast Plant and an estimated $11.3 million related to potential land leases for construction staging areas. At present, the SFPUC cannot predict the final cost of such acquisitions and leases; but it is possible that such costs will exceed budgeted amounts.

Potential Impacts of Climatic Change and Sea Level Rise

The impacts of climate change that would most affect the Wastewater Enterprise relate to changing rainfall patterns, sea level rise and rising tides. Existing climate change models show varied results in terms of projected rainfall patterns making proactive, long-term planning difficult. If they do occur, significant increases in rainfall (intensity, duration or frequency or any combination thereof) could exceed the sewer system's capacity to collect and store stormwater and wastewater for treatment. The SFPUC is investigating a partnership with Lawrence Berkeley National Laboratory to better predict and characterize expected long-term rainfall patterns.

Projected levels for sea level rise and rising tides could result in a backflow (or inflow) of San Francisco Bay water into the sewer system at the lowest weir elevation and increased infiltration due to higher groundwater in coastal areas. Already during certain high tides, the Wastewater Enterprise has begun to experience occasional inflows and infiltration from the San Francisco Bay into its sewer system. In response, the Wastewater Enterprise is planning to implement, through the SSIP, a monitoring program and backflow mitigation measures at some points where tides may backflow into the system.

Certain low-lying areas of the City are vulnerable to flooding, especially during seasonal high tides coupled with intense rainfall. See "– Risks Related to Wastewater Enterprise Facilities and Operation – Flooding and Other Natural and Man-Made Disasters." This vulnerability may be exacerbated with predicted sea level rise and possible higher-intensity storms. The SFPUC has embarked on a City-wide flood resilience study to characterize the impact

of storms, develop a risk-based framework to identify and prioritize its investments in projects to reduce or mitigate flooding risks, and provide a set of flood risk reduction measures for storms that exceed the capacity of the SFPUC's stormwater collection and treatment systems. In parallel with the flood resilience study, the SFPUC is expediting the development of certain capital projects in key areas subject to flooding risk in order to improve stormwater management.

Economic, Political, Social and Environmental Conditions

Changes in economic political, social, or environmental conditions on a local, state, federal, and/or international level may adversely affect investment risk generally. Such conditional changes may include (but are not limited to) fluctuations in business production, consumer prices, or financial markets, unemployment rates, availability of skilled labor, technological advancements, shortages or surpluses in natural resources or energy supplies, changes in law, social unrest, fluctuations in the crime rate, political conflict, acts of war or terrorism, environmental damage, and natural disasters.

Bankruptcy

The SFPUC, being an enterprise department of the City, likely cannot itself file for bankruptcy. While an involuntary bankruptcy petition cannot be filed against the City, the City is authorized to file for bankruptcy under certain circumstances. Should the City file for bankruptcy, there could be adverse effects on the holders of the 2016 Series AB Bonds.

To the extent that Revenues are "special revenues" under the United States Bankruptcy Code (the "Bankruptcy Code"), then Revenues collected after the date of the bankruptcy filing should be subject to the lien of the Indenture. If any or all of the Revenues are determined not to be "special revenues," then any such amounts collected after the commencement of the bankruptcy case will likely not be subject to the lien of the Indenture. The holders of the 2016 Series AB Bonds may not be able to assert a claim against any property of the City other than the Revenues, and if any or all of the Revenues are no longer subject to the lien of the Indenture, then there may be limited, if any, funds from which the holders of the 2016 Series AB Bonds are entitled to be paid.

The Bankruptcy Code provides that "special revenues" can be applied to necessary operating expenses of the project or system, before they are applied to other obligations. This rule applies regardless of the provisions of the transaction documents. It is not clear precisely which expenses would constitute necessary operating expenses, and any definition in the transaction documents may not be applicable.

If the City is in bankruptcy, the parties (including the Trustee and the holders of the 2016 Series AB Bonds) may be prohibited from taking any action to collect any amount from the City or to enforce any obligation of the City, unless the permission of the bankruptcy court is obtained. These restrictions may also prevent the Trustee from making payments to the holders of the 2016 Series AB Bonds from funds in the Trustee's possession. The rate covenants (see "SECURITY FOR THE BONDS – Rate Covenants") may not be enforceable in bankruptcy by the Trustee or the holders of the 2016 Series AB Bonds.

Revenues are deposited with and held by the Treasurer and may be commingled with other City funds. See "SECURITY FOR THE BONDS – Flow of Funds". If the City goes into bankruptcy, the City may not be required to turn over to the Trustee any Revenues that are in its possession at the time of the bankruptcy filing. In addition, if the City has possession of Revenues (whether collected before or after commencement of the bankruptcy) and if the City does not voluntarily turn over such Revenues to the Trustee, it is not entirely clear what procedures the Trustee and the holders of the 2016 Series AB Bonds would have to follow to attempt to obtain possession of such Revenues, how much time it would take for such procedures to be completed, or whether such procedures would ultimately be successful.

The City may be able to borrow additional money that is secured by a lien on any of its property (including the Revenues), which lien could have priority over the lien of the Indenture, or to cause some of the Revenues to be released to it, free and clear of lien of the Indenture, in each case as long as the bankruptcy court determines that the rights of the Trustee and the holders of the 2016 Series AB Bonds will be adequately protected.

If the City is in bankruptcy it may be able, without the consent and over the objection of the Trustee and the holders of the 2016 Series AB Bonds, to alter the priority, interest rate, principal amount, payment terms, collateral, maturity dates, payment sources, covenants (including tax-related covenants), and other terms or provisions of the Indenture and the 2016 Series AB Bonds, as long as the bankruptcy court determines that the alterations are fair and equitable.

There may be delays in payments on the 2016 Series AB Bonds while the court considers any of these issues. There may be other possible effects of a bankruptcy of the City that could result in delays or reductions in payments on the 2016 Series AB Bonds, or result in losses to the holders of the 2016 Series AB Bonds. Regardless of any specific adverse determinations in a City bankruptcy proceeding, the fact of a City bankruptcy proceeding could have an adverse effect on the liquidity and value of the 2016 Series AB Bonds.

The City invests Revenues in the City's Pooled Fund. See "FINANCIAL OPERATIONS – Investment of SFPUC Funds." Should those investments suffer losses, Revenues may be lower than expected, and there may be delays or reductions in payments on the 2016 Series AB Bonds.

Limitations on Remedies

The remedies available to the owners of the 2016 Series AB Bonds upon the occurrence of an event of default under the Indenture in many respects depend upon judicial actions which are themselves often subject to discretion and delay and could prove both expensive and time consuming to obtain. In addition to the limitations on remedies contained in the Indenture, the rights and obligations under the 2016 Series AB Bonds and the Indenture may be subject to bankruptcy, insolvency, reorganization, arrangement, fraudulent conveyance, moratorium and other laws relating to or affecting creditors' rights, to the application of equitable principles, to the exercise of judicial discretion in appropriate cases and to limitations on legal remedies against charter cities and counties in the State. The opinions to be delivered by Co-Bond Counsel, concurrently with the issuance of the 2016 Series AB Bonds, that the 2016 Series AB Bonds constitute valid and binding, limited obligations of the SFPUC and the Indenture constitutes a valid and binding obligation of the SFPUC will also be subject to such limitations and the various other legal opinions to be delivered concurrently with the issuance of the 2016 Series AB Bonds will be similarly qualified. See "APPENDIX C – PROPOSED FORM OF OPINION OF CO-BOND COUNSEL."

If the SFPUC fails to comply with its covenants under the Indenture or to pay principal of or interest on the 2016 Series AB Bonds, there can be no assurance that the available legal remedies will be adequate to protect the interests of the holders of the 2016 Series AB Bonds.

Loss of Tax Exemption/Risk of Tax Audit of Municipal Issuers

As discussed under "TAX MATTERS", interest on the 2016 Series AB Bonds could fail to be excluded from the gross income of the owners thereof for purposes of federal income taxation retroactive to the date of the issuance of the 2016 Series AB Bonds as a result of future acts or omissions of the SFPUC in violation of its covenants to comply with requirements of the Internal Revenue Code of 1986, as amended. Should such an event of taxability occur, the 2016 Series AB Bonds are not subject to redemption or any increase in interest rate and will remain outstanding until maturity.

In December 1999, as a part of a larger reorganization of the IRS, the IRS commenced operation of its Tax Exempt and Government Entities Division (the "TE/GE Division"), as the successor to its Employee Plans and Exempt Organizations division. The TE/GE Division has a subdivision that is specifically devoted to tax-exempt bond compliance. Public statements by IRS officials indicate that the number of tax-exempt bond examinations is expected to increase significantly under the new TE/GE Division. There is no assurance that, if an IRS examination of the 2016 Series AB Bonds were undertaken, it would not adversely affect the secondary market value of the 2016 Series AB Bonds.

Change in Tax Law

As discussed under "TAX MATTERS," current and future legislative proposals, if enacted into law, clarification of the Code or court decisions may cause interest on the 2016 Series AB Bonds to be subject, directly or indirectly, in whole or in part, to federal income taxation or to be subject to or exempted from state income taxation, or otherwise prevent beneficial owners from realizing the full current benefit of the tax status of such interest.

Failure to Maintain Credit Ratings

Certain rating agencies have assigned ratings to the 2016 Series AB Bonds. The ratings issued reflect only the views of such rating agencies. Any explanation of the significance of these ratings should be obtained from the respective rating agencies. See "RATINGS." There is no assurance current ratings will continue for any given period or that such ratings will not be revised downward or withdrawn entirely by the rating agencies if, in the respective judgment of such rating agencies, circumstances so warrant. Any such downward revision or withdrawal of such ratings could be expected to have an adverse effect on the market price of the 2016 Series AB Bonds. The SFPUC undertakes no obligation to maintain its current credit ratings on the 2016 Series AB Bonds or to oppose any such downward revision, suspension or withdrawal.

Secondary Market

There can be no guarantee that there will be a secondary market for the 2016 Series AB Bonds or, if a secondary market exists, that the 2016 Series AB Bonds can be sold for any particular price. Occasionally, because of general market conditions or because of adverse history or economic prospects connected with a particular issue, secondary marketing practices in connection with a particular issue are suspended or terminated. Additionally, prices of issues for which a market is being made will depend upon then prevailing circumstances. Such prices could be substantially different from the original purchase price.

Uncertainties of Projections, Forecasts and Assumptions

Compliance with certain of the covenants contained in the Indenture is based upon assumptions and projections. Projections and assumptions are inherently subject to significant uncertainties. Inevitably, some assumptions will not be realized and unanticipated events and circumstances may occur and actual results are likely to differ, perhaps materially, from those projected. Accordingly, such projections are not necessarily indicative of future performance, and the SFPUC assumes no responsibility for the accuracy of such projections. See "FORWARD-LOOKING STATEMENTS" on the inside front cover of this Official Statement.

Other Risks

The discussion in this section, "RISK FACTORS", is not meant to be a comprehensive or definitive list of the risks associated with an investment in the 2016 Series AB Bonds. There may be other risks inherent in ownership of the 2016 Series AB Bonds in addition to those described in this section. Investors are advised to read the entire Official Statement in order to obtain information necessary to make an investment in the 2016 Series AB Bonds.

REGULATORY MATTERS

The Wastewater Enterprise meets all known current regulatory permit requirements for its facilities, and is in material compliance with the Porter-Cologne and Clean Water Acts (as described below). This section summarizes the regulatory framework governing the Wastewater Enterprise and its operations.

History and Background

In 1969, the State adopted the Porter-Cologne Water Quality Act (the "Porter-Cologne Act"), creating the State's current legal framework for the protection of water quality. This adoption was followed at the federal level by the Water Pollution Control Act Amendments of 1972 (the "Clean Water Act"). The Clean Water Act provided

an aggressive timetable for eliminating pollution of the nation's waters and established the basic secondary treatment requirement that 85% of pollutants, as defined in administrative regulations, be removed from sanitary wastewater. The Clean Water Act also required the issuance of discharge permits on a nationwide basis and established a federal grant program for construction of publicly owned wastewater facilities, subsequently replaced by the state revolving fund loan program. Although the EPA has ultimate responsibility for administering the Clean Water Act, many functions have been delegated to the State. The administration of the current loan program and enforcement of regulations are a joint undertaking of the State Water Resources Control Board, the Regional Water Quality Control Board, and EPA Region IX.

City Master Plan

In 1971, the City created its Master Plan to address the requirements of the Porter-Cologne Act, which, following the environmental review process, was adopted by the City and the EPA in 1974, and the City commenced implementation of the plan. With subsequent refinements, the Master Plan has been the SFPUC's guide for improving the performance of the Wastewater Enterprise and meeting the requirements of State and federal water quality laws. All projects and improvements described in the Master Plan, as subsequently modified, have been completed and are fully operational.

In the 1970s and 1980s, the Regional Water Quality Control Board issued cease and desist orders to the SFPUC containing project planning, design and construction schedules for Master Plan projects and discharge limit compliance dates. All such cease and desist orders have been satisfied and are no longer in effect.

Federal and State Clean Water Act Permits

Under the Porter-Cologne and Clean Water Acts, the Regional Water Quality Control Board administers water pollution control programs. The Regional Water Quality Control Board issues discharge permits under Section 402 of the Clean Water Act, which establishes the National Pollutant Discharge Elimination System ("NPDES") permit system. These permits, issued for a five-year period, are also wastewater discharge requirements for the purposes of the Porter-Cologne Act and apply to discharges from the SFPUC's treatment plants and combined sewer discharge facilities.

The Wastewater Enterprise combined sewer system operates under two wastewater NPDES permits: the 2013 Bayside Permit (NPDES Permit No. CA0037664), covering the Southeast Plant, North Point Facility and other bayside facilities that discharge into the San Francisco Bay; and the 2009 Oceanside Permit (NPDES Permit No. CA0037681; issued jointly with EPA Region IX), covering the Oceanside Plant discharges and other westside facilities that discharge into the ocean. A small portion of the Wastewater Enterprise system within the City consists of separate storm and sanitary sewers. This portion of the system is regulated under the State General Permit for Small Municipal Separate Storm Sewer Systems (MS4) (NPDES Permit No. CAS000004) and the State Waste Discharge Requirements for Sanitary Sewer Systems (Order No. 2006-0003-DWQ). Currently, the SFPUC's discharges are in material compliance with its permit requirements, though during the past two years certain discharges have varied from such requirements.

Combined Sewer Overflow Control Policy

With the exception of a small portion of the City of Sacramento, the SFPUC is the only wastewater agency in the State that operates a combined sewer system, in which sanitary wastewater and stormwater runoff are conveyed and treated in the same system of pipes and treatment, storage and discharge facilities. In order to address the unique characteristics of combined sewer systems, the EPA adopted the Combined Sewer Overflow Control Policy (59 FR 18688) in 1994 (the "CSO Policy"). This policy established a consistent national approach for controlling discharges from CSOs to the nation's water, and has since been incorporated into the Clean Water Act by The Wet Weather Water Quality Act of 2000.

The CSO Policy created a two-phased process for combined sewer systems. During the first phase, the permittee is required to implement "nine minimum controls" specified in the CSO Policy. In addition, the permittee is required to develop and implement a long-term control plan for the purpose of providing facilities and controls

sufficient to comply with water quality standards. The SFPUC has implemented the "nine minimum controls", and construction of the Master Plan projects constituted implementation of the long-term control plan.

The SFPUC's discharge permits require the preparation of reports analyzing the efficacy of the system's wet weather operations and the attainment of water quality standards. The SFPUC's efficacy report for the Bayside Watersheds submitted in June 2012; the efficacy report for the Westside was submitted in July 2014. These reports document the City's continued compliance with the CSO Policy's nine minimum controls and long-term control plan requirements.

Stormwater Regulations

In 1987, Congress revised the Clean Water Act to more effectively address pollution caused by stormwater runoff. The regulations require stormwater management plans for municipalities and controls on certain construction sites and other industries. Urban areas with combined sewers, such as most of the City, are exempt. Because a small portion of the City is served by separate sewer systems, the implementation of the Municipal Separate Storm Sewer System ("MS4") permit requirements occurred under Phase II of the stormwater program, following the earlier Phase I implementation for cities with a large separate sewer system. The permit for small MS4s (NPDES Permit No. CAS000004) is issued by the State Water Resources Control Board and regulates the stormwater discharge from the SFPUC's separate sewer systems. The SFPUC operates a stormwater management program that complies with the requirements of the MS4 Permit.

Regulatory Trends

Regulatory developments at the State and Federal level, as well as ongoing permit reissuance activities, may increase operations costs and capital needs of the Wastewater Enterprise and may have an effect on the Wastewater Enterprise operations and its revenues. In the future, additional constituents of concern (possibly including pollutants such as ammonia, nutrients, endocrine disrupting chemicals, human-made chemicals/products) will likely be identified, and additional effluent limits may be added for wastewater discharges into the San Francisco Bay and Pacific Ocean, as water quality objectives are developed for new compounds and improved analytical techniques become available. Additional source control measures, public education and outreach, and additional or advanced treatment processes may be necessary to achieve compliance. SFPUC staff is actively engaged with regulatory officials and the public in the development of these regulatory matters.

These topics and their possible effect on the Wastewater Enterprise are briefly described below:

Impaired Water Bodies and Total Maximum Daily Loads. The Clean Water Act requires states to identify all water bodies that do not achieve designated water quality standards or objectives. Such water bodies are designated as "impaired," and states are required to identify all sources contributing to the impairment under the Total Maximum Daily Load ("TMDL") program. States are required to designate wasteload allocations to each contributing point source, such as the SFPUC discharges, in order to promote the recovery of the water body. Central and lower San Francisco Bay are currently listed as impaired for a number of organic and inorganic pollutants, as well as invasive species and trash. The Regional Water Quality Control Board has completed San Francisco Bay TMDLs for mercury and PCBs and is developing a fecal indicator bacteria TMDL for San Francisco Bay beaches.

The San Francisco Bay Mercury and PCBs TMDLs are implemented through a group Watershed Permit (Order No. R2-2012-0096) that applies to all municipal, industrial, and stormwater discharges to San Francisco Bay. The Watershed Permit contains effluent limitations for mercury and PCB discharges from the Southeast Plant with which the SFPUC currently complies.

Contaminated Bay Sediments. California Water Code, Division 7, Chapter 5.6 established a program to assess sediment contamination of the State's enclosed bays and estuaries. Known as the Bay Protection and Toxic Cleanup Program, the focus of this effort was to identify contaminated sediments with elevated levels of toxins.

In 1999, the State-wide plan identified Mission Creek and Islais Creek as high priority toxic hot spots. For Mission Creek, the plan included preliminary estimates of investigation and study costs at $1 million, remediation and follow-up monitoring ranging from $0.8 to $1.8 million, and possible sewer system structural changes up to $75 million. The plan included a preliminary estimate of investigation and study costs for Islais Creek at $1 million. Remediation and monitoring were estimated to range from $0.8 to $5.2 million, and possible sewer system structural changes up to $75 million for Islais Creek. The estimated costs shown above were developed by the Regional Water Quality Control Board and State Water Resources Control Board in 1998 and 1999. The SFPUC provided comments and data studies disputing both the extent of contamination in the creeks and any asserted causal relationship to SFPUC activities. The plan is not self-executing, and requires further regulatory action by the Regional Water Quality Control Board, which has indicated that contaminated sediment issues will be addressed through the TMDL analysis and implementation programs mentioned above. In 2002, both Mission Creek and Islais Creek were added to the Clean Water Act 303(d) list of water quality limited segments (i.e., impaired waterways). A TMDL (or related action) must be completed for the listed waterways.

The 1999 Statewide plan also identified Central Basin (adjacent to the San Francisco Bayside shoreline) as a moderate priority toxic hot spot, but did not identify investigation or remediation costs (costs were only developed for the high priority sites). Central Basin has also been placed on the CWA 303(d) list.

The SFPUC continues to dispute the extent of contamination in the creeks and any causal relationship to SFPUC activities, and will vigorously participate in the development of any TMDLs for such areas.

EPA Region IX has listed Yosemite Creek as a Superfund site. Its sediments, adjacent to Hunters Point and the former Navy Shipyard, must be addressed. The City, along with other potentially responsible parties, is participating in a voluntary nonbinding mediation to allocate any cleanup costs. Because the process is in its early stages, SFPUC cannot reasonably predict whether or to what extent it may be partially responsible for contributing to this contamination remediation project, or whether such cleanup will have any impact on current sewer system operations.

Pursuant to GASB Statement No. 49, *Accounting and Financial Reporting for Pollution Remediation Obligations*, the Wastewater Enterprise reported $571,000 in pollution remediation liability as of June 30, 2015.

Contaminants of Emerging Concern. Contaminants of emerging concern include alkyl phenols, flame retardants, hormones, personal care products, pharmaceuticals, steroids, perfluorinated compounds, and pesticides, which typically enter municipal wastewater through bathing, cleaning, laundry, and the disposal of unused products. Pharmaceuticals typically include prescription and over-the-counter therapeutic drugs for both human and veterinary treatment. Personal care products typically include soaps, fragrances, and cosmetics. Secondary treatment facilities remove some of these contaminants, even though they are not specifically designed for this purpose. Currently no water quality standards exist for most of these compounds, therefore, the SFPUC permits do not contain effluent limitations for them.

Nutrient Control for the Bayside. San Francisco Bay has long been recognized as a nutrient-enriched estuary; however, until recently, it had not experienced negative effects of over-enrichment. Changes recently observed in San Francisco Bay indicate that the Bay's resilience to the effects of nutrient enrichment may be declining. In response to these changes, the Regional Water Quality Control Board issued a permit in 2014 for all municipal discharges into the San Francisco Bay to monitor and report nutrient discharges, NPDES No. CA0038873. The Regional Water Quality Control Board is also developing numeric nutrient endpoints ("NNEs"),the purpose of which are to provide a scientifically defensible framework that can serve as guidance for adopting numeric water quality criteria. Whether NNEs will result in new limits that would apply to SFPUC treatment facilities is unclear at this time. The SFPUC and regional groups like the Bay Area Clean Water Association (of which the SFPUC is a member) are participating in the Regional Water Quality Control Board's consideration of the need for and potential characteristics of any such future regulation.

Other Regulatory Agencies with Jurisdiction Over the Wastewater Enterprise

Other regulatory agencies with approval or oversight responsibilities over the siting, construction or operational impacts of the Wastewater Enterprise on air, water and natural resources include the Bay Area Air

Quality Management District, the Bay Conservation and Development Commission, the California Coastal Commission, the State Lands Commission, the California Department of Public Health, the National Marine Fisheries Service, the United States Fish and Wildlife Service, the California Department of Fish and Wildlife, and the U.S. Army Corps of Engineers.

Other Laws Affecting the Wastewater Enterprise

As a public agency the SFPUC's actions must be consistent with the California Environmental Quality Act and, where federal approvals or funding is involved, the National Environmental Policy Act. The federal Clean Air Act and the California Clean Air Act of 1988 also regulate emissions from the treatment facilities. All of the SFPUC's treatment facilities meet present Bay Area Air Quality Management District standards.

CONSTITUTIONAL, STATUTORY AND CHARTER LIMITATIONS

The activities of the SFPUC and the Wastewater Enterprise, including, without limitation, the establishment of rates for sewer service and the issuance of Bonds, are subject to a number of limitations under both State and local law. Certain of such limitations are summarized below. Additionally, such limitations may be revised, enhanced, expanded, or otherwise altered as provided under State and local law, including in certain instances by legislation adopted by State, regional or local authorities, including the State Legislature or the Board of Supervisors, or by the voters of the State or the City themselves through the power of initiative or referendum, by voting in favor of amendments to the Charter, or in any other lawful manner.

State Law Limitations

Tax and Spending Limitations. The taxing powers of public agencies in the State are limited by Article XIIIA of the California Constitution, added by an initiative amendment approved by the voters on June 6, 1978, and commonly known as Proposition 13.

Article XIIIA limits the maximum ad valorem tax on real property to 1% of "full cash value," which is defined as "the County Assessor's valuation of real property as shown on the Fiscal Year 1975-76 tax bill under 'full cash value' or, thereafter, the appraised value of real property when purchased, newly constructed, or a change in ownership has occurred after the 1975 assessment." The full cash value may be adjusted annually to reflect inflation at a rate not to exceed 2% per year, or reduction in the consumer price index or comparable local data, or declining property value caused by damage, destruction, or other factors.

The tax rate limitation referred to above does not apply to ad valorem taxes to pay the debt service on any indebtedness approved by the voters before July 1, 1978, or on any bonded indebtedness for the acquisition or improvement of real property approved by two-thirds of the votes cast by the voters voting on the proposition.

Under the terms of Article XIIIA and pursuant to an allocation system created by implementing legislation, each county is required to levy the maximum ad valorem tax permitted by Article XIIIA and to distribute the proceeds to local agencies.

Assessed valuation growth allowed under Article XIIIA (new construction, change of ownership and up to 2% annual value growth) is allocated among the jurisdictions that serve the tax rate area within which the growth occurs. Local agencies and schools share the growth of base revenues from the tax rate area. Each year's growth allocation becomes part of each agency's allocation in the following year. The availability of revenues from tax bases to such entities may be affected by the existence of certain successor agencies to former redevelopment agencies that, under certain circumstances, may be entitled to such revenues resulting from the upgrading of certain property values.

Under State law, any fee that exceeds the reasonable cost of providing the service for which the fee is charged may be considered a "special tax" that must be authorized by a two-thirds vote of the electorate. Accordingly, if a portion of the SFPUC's wastewater user rates or capacity charges were determined by a court to exceed the reasonable cost of providing service, the SFPUC might not be permitted to continue to collect that

portion unless it were authorized to do so by a two-thirds majority of the votes cast in an election to authorize the collection of that portion of the rates or fees. If the SFPUC were unable to obtain such a two-thirds majority vote and were unable to reduce costs, such failure could adversely affect the SFPUC's ability to pay the debt service on the Bonds.

The United States Supreme Court has upheld Article XIIIA against a challenge alleging violation of equal protection under the Fourteenth Amendment to the United States Constitution.

Proposition 218. Proposition 218, a State ballot initiative known as the "Right to Vote on Taxes Act," was approved by the voters on November 5, 1996. The initiative added Articles XIIIC and XIIID to the California Constitution, creating additional requirements for the imposition by most local governments of "general taxes," "special taxes," "assessments," "fees," and "charges." Articles XIIIC and XIIID became effective, pursuant to their terms, as of November 6, 1996, although compliance with some of the provisions was deferred until July 1, 1997, and certain of the provisions purport to apply to any tax imposed for general governmental purposes (i.e., "general taxes") imposed, extended or increased on or after January 1, 1995 and prior to November 6, 1996.

Article XIIID imposes substantive and procedural requirements on the imposition, extension or increase of any "fee" or "charge" subject to its provisions. A "fee" or "charge" subject to Article XIIID includes any levy, other than an ad valorem tax, special tax or assessment, imposed by an agency upon a parcel or upon a person as an incident of property ownership. Article XIIID prohibits, among other things, the imposition of any proposed fee or charge, and, possibly, the increase of any existing fee or charge, in the event written protests against the proposed fee or charge are presented at a required public hearing on the fee or charge by a majority of owners of the parcels upon which the fee or charge is to be imposed. Except for fees and charges for water, sewer and refuse collection services, the approval of a majority of the property owners subject to the fee or charge, or at the option of the agency, by a two-thirds vote of the electorate residing in the affected area, is required not less than 45 days following the public hearing on any such proposed new or increased fee or charge.

The California Supreme Court decisions in *Richmond v. Shasta Community Services District*, 32 Cal. 4th 409 (2004) ("Richmond"), and *Bighorn-Desert View Water Agency v. Verjil*, 39 Cal. 4th 206 (2006) ("Bighorn") have clarified uncertainty surrounding the applicability of Section 6 of Article XIIID to service fees and charges. In Richmond, the Shasta Community Services District charged a water connection fee, which included a capacity charge for capital improvements to the water system and a fire suppression charge. The Court held that both the capacity charge and the fire suppression charge were not subject to Article XIIID because a water connection fee is not a property-related fee or charge because it results from the property owner's voluntary decision to apply for the connection. In both Richmond and Bighorn, however, the Court stated that a fee for ongoing water service through an existing connection is imposed "as an incident of property ownership" within the meaning of Article XIIID, rejecting, in Bighorn, the water agency's argument that consumption-based water charges are not imposed "as an incident of property ownership" but as a result of the voluntary decisions of customers as to how much water to use.

The SFPUC provides public notice of proposed wastewater rate increases in accordance with the requirements of Article XIIID through means that include, among others, holding informational presentations at community group meetings, mailings to residential and commercial customers of public hearings on rate increases, and press releases and media campaigns regarding rate increases, followed by public hearings conducted by the SFPUC's Rate Fairness Board and by the SFPUC itself. The SFPUC also develops and adopts retail utility user rates and fees in accordance with the requirements of Article XIIID(6)(b) that limit property-related fees and charges.

Article XIIIC extends the people's initiative power to reduce or repeal previously authorized local taxes, assessments, fees and charges. This extension of the initiative power is not limited by the terms of Article XIIIC to fees, taxes, assessment fees and charges imposed after November 6, 1996 and absent other authority could result in retroactive reduction in any existing taxes, assessments, fees or charges. In Bighorn, the Court concluded that under Article XIIIC local voters by initiative may reduce a public agency's water rates and delivery charges. The Court noted, however, that it was not holding that the authorized initiative power is free of all limitations, stating that it was not determining whether the electorate's initiative power is subject to the public agency's statutory obligation to set water service charges at a level that will "pay the operating expenses of the agency, … provide for repairs and depreciation of works, provide a reasonable surplus for improvements, extensions, and enlargements, pay the

interest on any bonded debt, and provide a sinking or other fund for the payment of the principal of such debt as it may become due."

Article XIIID imposes the substantive requirement that any wastewater charges may not exceed the proportional cost of the providing service to customers. The Fourth District of the California Court of Appeal decision in *Capistrano Taxpayers Association, Inc. v. City of San Juan Capistrano*, 235 Cal.App.4th 1493 (2015), clarified that tiered rate structures are compatible with the cost of service limitations of Article XIIID when each tier structure is supported by cost of service calculations. The court held that San Juan Capistrano's water rates violated Article XIIID because no evidence in the record adequately showed that each tier corresponded to the cost of providing service at a given level of usage.

The courts have not fully interpreted the provisions of Proposition 218. The SFPUC is unable to predict how courts will further interpret Article XIIIC and Article XIIID, and what, if any, further implementing legislation will be enacted. Under the Bighorn case, City voters could adopt an initiative measure that reduces or repeals the SFPUC's wastewater rates and charges, though it is not clear whether (and courts have not decided whether) any such reduction or repeal by initiative would be enforceable in a situation in which such rates and charges are pledged to the repayment of bonded indebtedness. There can be no assurance that the courts will not further interpret, or the voters will not amend, Article XIIIC and Article XIIID to limit the ability of the SFPUC to impose, levy, charge and collect increased fees and charges for the Wastewater Enterprise, or to call into question wastewater rate increases previously adopted by the SFPUC. No assurance may be given that Articles XIIIC and XIIID will not have a material adverse impact on Revenues.

Proposition 26. Proposition 26, which amended Article XIIIA and XIIIC of the California Constitution, was approved by the electorate at the November 2, 2010 election. Proposition 26 imposes a two-thirds voter approval requirement for the imposition of fees and charges by the State. It also imposes a majority voter approval requirement on local governments with respect to fees and charges for general purposes, and a two-thirds voter approval requirement with respect to fees and charges for special purposes. According to its supporters, Proposition 26 was designed to prevent the circumvention of tax limitations imposed by the voters pursuant to Proposition 13, approved in 1978, Proposition 218, and other measures through the use of non-tax fees and charges.

Proposition 26 expressly excludes from its scope "a charge imposed for a specific government service or product provided directly to the payor that is not provided to those not charged, and which does not exceed the reasonable cost to the governmental entity of providing the service or product to the payor" and "assessments and property-related fees imposed in accordance with the provisions of Article XIIID." The SFPUC believes that the initiative is not intended to, and would not, apply to fees for wastewater services charged by the SFPUC. The SFPUC, however, is unable to predict how Proposition 26 will be interpreted by the courts to apply to the provision of wastewater services by local governments such as the SFPUC.

Initiative and Referendum

Article XIIIA and Articles XIIIC and XIIID of the California Constitution were adopted pursuant to the State's constitutional initiative process. From time to time other initiative measures could be adopted by State voters, or by voters of the City, placing additional limitations on the ability of the SFPUC to increase revenues.

Charter Limitations

The Charter requires that bonds (such as the Bonds) secured by revenues, other than refunding bonds, may be issued only with the assent of a majority of voters. However, under the Charter amendments enacted by the voters in November 2002 (Proposition E), the SFPUC may issue revenue bonds, including notes, commercial paper or other forms of indebtedness, when authorized by ordinance approved by a two-thirds vote of the Board of Supervisors, for the purpose of reconstructing, replacing, expanding, repairing or improving water facilities or wastewater facilities or combinations of water and wastewater facilities under the jurisdiction of the SFPUC (and subject to the further conditions contained in Proposition E). See "OBLIGATIONS PAYABLE FROM REVENUES – Authority for Issuance of Revenue Bonds and Other Obligations Payable from Revenues" and "SECURITY FOR THE BONDS – Additional Series of Bonds," "– Additional Parity State Loans" and "– Refunding Bonds."

Future Charter Amendments

Voters in the City could adopt additional Charter amendments in the future which would limit the ability of the SFPUC to issue debt or to enact rate increases, affect the operation of the Wastewater Enterprise or implement other changes affecting the SFPUC and the Wastewater Enterprise. See "RISK FACTORS – Initiative, Referendum and Charter Amendments and Future Legislation."

LITIGATION

The SFPUC is not aware of any litigation pending or threatened questioning the political existence of the City or the SFPUC or contesting the SFPUC's power to fix water rates and charges, or in any way questioning or affecting:

(i) the proceedings under which the 2016 Series AB Bonds are to be issued;

(ii) the validity of any provision of the 2016 Series AB Bonds or the Indenture;

(iii) the pledge of Net Revenues by the SFPUC under the Indenture; or

(iv) the titles to office of the present members of the Board of Supervisors and the Commission.

There are a number of suits and claims pending against the City and the SFPUC, which may include personal injury, wrongful death and other suits and claims against which the City may self-insure. The aggregate amount of the self-insured liabilities of the City and the SFPUC which may result from such suits and claims will not, in the opinion of the City Attorney, materially impair the ability of the SFPUC to pay principal of or interest on the Bonds as they become due. There is no litigation pending, with service of process having been accomplished, against the City or the SFPUC which if determined adversely to the City or the SFPUC would, in the opinion of the City Attorney, materially impair the ability of the SFPUC to pay principal of and interest on the Bonds as they become due.

TAX MATTERS

In the opinion of Co-Bond Counsel, under existing statutes, regulations, rulings and judicial decisions, and assuming the accuracy of certain representations and compliance with certain covenants and requirements described herein, interest on the 2016 Series AB Bonds is excluded from gross income for federal income tax purposes, and is not an item of tax preference for purposes of calculating the federal alternative minimum tax imposed on individuals and corporations. In the further opinion of Co-Bond Counsel, interest on the 2016 Series AB Bonds is exempt from State of California personal income tax. Co-Bond Counsel notes that, with respect to corporations, interest on the 2016 Series AB Bonds may be included as an adjustment in the calculation of alternative minimum taxable income, which may affect the alternative minimum tax liability of such corporations.

The amount by which a 2016 Series AB Bond owner's original basis for determining loss on sale or exchange, in the applicable 2016 Series AB Bond (generally, the purchase price) exceeds the amount payable on maturity (or on an earlier call date) constitutes amortizable 2016 Series AB Bond premium, which must be amortized under Section 171 of the Code; such amortizable Bond premium reduces the 2016 Series AB Bond owner's basis in the applicable Bond (and the amount of tax-exempt interest received with respect to the 2016 Series AB Bonds), and is not deductible for federal income tax purposes. The basis reduction as a result of the amortization of 2016 Series AB Bond premium may result in a 2016 Series AB Bond owner realizing a taxable gain when a 2016 Series AB Bond is sold by the owner for an amount equal to or less (under certain circumstances) than the original cost of the 2016 Series AB Bonds to the owner. Purchasers of the 2016 Series AB Bonds should consult their own tax advisors as to the treatment, computation and collateral consequences of amortizable 2016 Series AB Bond premium.

The Internal Revenue Service (the "IRS") has initiated an expanded program for the auditing of tax-exempt bond issues, including both random and targeted audits. It is possible that the 2016 Series AB Bonds will be selected for audit by the IRS. It is also possible that the market value of the 2016 Series AB Bonds might be affected as a result of such an audit of the 2016 Series AB Bonds (or by an audit of similar municipal obligations).

No assurance can be given that in the course of the audit, or otherwise, Congress or the IRS might not change the Code (or interpretation thereof) subsequent to the issuance of the 2016 Series AB Bonds to the extent that it adversely affects the exclusion from gross income of interest on the 2016 Series AB Bonds or their market value.

SUBSEQUENT TO THE DELIVERY OF THE 2016 SERIES AB BONDS, THERE MIGHT BE FEDERAL, STATE OR LOCAL STATUTORY CHANGES (OR JUDICIAL OR REGULATORY INTERPRETATIONS OF FEDERAL, STATE OR LOCAL LAW) THAT AFFECT THE FEDERAL, STATE OR LOCAL TAX TREATMENT OF THE 2016 SERIES AB BONDS OR THE MARKET VALUE OF THE 2016 SERIES AB BONDS. LEGISLATIVE CHANGES HAVE BEEN PROPOSED IN CONGRESS, WHICH, IF ENACTED, WOULD RESULT IN ADDITIONAL FEDERAL INCOME TAX BEING IMPOSED ON CERTAIN OWNERS OF TAX-EXEMPT STATE OR LOCAL OBLIGATIONS, SUCH AS THE 2016 SERIES AB BONDS. THE INTRODUCTION OR ENACTMENT OF ANY OF SUCH CHANGES COULD ADVERSELY AFFECT THE MARKET VALUE OR LIQUIDITY OF THE 2016 SERIES AB BONDS. NO ASSURANCE CAN BE GIVEN THAT, SUBSEQUENT TO THE DELIVERY OF THE 2016 SERIES AB BONDS, SUCH CHANGES (OR OTHER CHANGES) WILL NOT BE INTRODUCED OR ENACTED OR INTERPRETATIONS WILL NOT OCCUR. BEFORE PURCHASING ANY OF THE 2016 SERIES AB BONDS, ALL POTENTIAL PURCHASERS SHOULD CONSULT THEIR TAX ADVISORS REGARDING POSSIBLE STATUTORY CHANGES OR JUDICIAL OR REGULATORY CHANGES OR INTERPRETATIONS, AND THEIR COLLATERAL TAX CONSEQUENCES RELATING TO THE 2016 SERIES AB BONDS.

Co-Bond Counsel's opinions may be affected by actions taken (or not taken) or events occurring (or not occurring) after the date hereof. Co-Bond Counsel has not undertaken to determine, or to inform any person, whether any such actions or events are taken or do occur. Co-Bond's engagement with respect to the 2016 Series AB Bonds terminates upon their delivery and Co-Bond Counsel disclaims any obligation to update the matters set forth in its opinion. The Indenture and the Tax Certificate relating to the 2016 Series AB Bonds permit certain actions to be taken or to be omitted if a favorable opinion of Co-Bond Counsel is provided with respect thereto. Co-Bond Counsel expresses no opinion as to the effect on the exclusion from gross income of interest for federal income tax purposes with respect to any 2016 Series AB Bond if any such action is taken or omitted based upon the advice of counsel other than Co-Bond Counsel.

Although Co-Bond Counsel has rendered an opinion that interest (and original issue discount) is excluded from gross income for federal income tax purposes on the 2016 Series AB Bonds provided that the SFPUC continues to comply with certain requirements of the Code, the ownership of the 2016 Series AB Bonds and the accrual or receipt of interest with respect to the 2016 Series AB Bonds may otherwise affect the tax liability of certain persons. Co-Bond Counsel expresses no opinion regarding any such tax consequences. Accordingly, before purchasing any of the 2016 Series AB Bonds, all potential purchasers should consult their tax advisors with respect to collateral tax consequences relating to the 2016 Series AB Bonds.

A copy of the proposed form of opinion of Co-Bond Counsel is attached hereto as Appendix C.

CERTAIN LEGAL MATTERS

Certain legal matters incident to the authorization, sale and delivery of the 2016 Series AB Bonds are subject to the approval of Stradling, Yocca, Carlson & Rauth, a Professional Corporation, Sacramento, California, and Amira Jackmon, Attorney at Law, Berkeley, California, Co-Bond Counsel to the SFPUC. Certain legal matters are being passed upon for the SFPUC by the City Attorney and by Orrick, Herrington & Sutcliffe LLP, San Francisco, California, Disclosure Counsel. Co-Bond Counsel and Disclosure Counsel will receive compensation that is contingent upon the sale and delivery of the 2016 Series AB Bonds.

The form of approving opinion of Co-Bond Counsel is set forth in Appendix C, and will be available at the time of delivery of the 2016 Series AB Bonds. Co-Bond Counsel is not passing upon and undertakes no responsibility for the accuracy, completeness or fairness of the information contained in this Official Statement.

Orrick, Herrington & Sutcliffe LLP, has served as Disclosure Counsel to the SFPUC and in such capacity has advised the SFPUC with respect to the requirements of applicable securities laws and participated with responsible SFPUC officials and staff in conferences and meetings where information contained in this Official Statement was reviewed for accuracy and completeness. Disclosure Counsel is not responsible for the accuracy or

completeness of the information presented in this Official Statement and has not undertaken to independently verify any of such information. Rather, the SFPUC is solely responsible for the accuracy and completeness of the information contained in this Official Statement. Upon the issuance of the 2016 Series AB Bonds, Orrick, Herrington & Sutcliffe LLP will deliver a letter to the SFPUC concerning certain matters with respect to the Official Statement. No purchaser or holder of the 2016 Series AB Bonds, or other person or party other than the SFPUC, will be entitled to rely on such letter or on the fact that Orrick, Herrington & Sutcliffe LLP has acted as Disclosure Counsel to the SFPUC.

RATINGS

Standard & Poor's Ratings Services, a Standard & Poor's Financial Services LLC business ("S&P"), has assigned its municipal bond rating of "AA" to the 2016 Series AB Bonds, and Moody's Investors Service ("Moody's") has assigned its municipal bond rating of "Aa3" to the 2016 Series AB Bonds.

The ratings assigned by S&P and Moody's express only the views of the respective rating agencies. The explanation of the significance of these ratings, and any outlook associated with these ratings, may be obtained from S&P and Moody's, respectively. Each rating agency generally bases its rating on its own investigations, studies, and assumptions. The SFPUC has provided certain additional information and materials to the rating agencies (some of which does not appear in this Official Statement).

A securities rating is not a recommendation to buy, sell or hold securities and may be subject to revision or withdrawal at any time. There is no assurance such ratings will continue for any given period of time or that such ratings will not be revised downward or withdrawn entirely by the rating agencies, if in the judgment of such rating agencies, circumstances so warrant. Any such downward revision or withdrawal of such ratings may have an adverse effect on the market price of the 2016 Series AB Bonds. The SFPUC undertakes no responsibility to maintain its current ratings on the 2016 Series AB Bonds or to oppose any such downward revision, suspension or withdrawal.

PURCHASE AND REOFFERING

The 2016 Series AB Bonds are being purchased by J.P.Morgan Securities LLC (the "Initial Purchaser") as the winner of respective competitive bids conducted on May 10, 2016.

The Initial Purchaser has agreed to purchase the 2016 Series A Bonds from the SFPUC at a purchase price of $280,544,946.82 (consisting of $240,580,000 aggregate principal amount of the 2016 Series A Bonds, plus original issue premium of $40,801,897.70, less an underwriter's discount of $836,950.88). Under the terms of its bid, the Initial Purchaser will be obligated to purchase all of the 2016 Series A Bonds if any are purchased, the obligation to make such purchase being subject to certain terms and conditions to be satisfied by the SFPUC.

The Initial Purchaser has agreed to purchase the 2016 Series B Bonds from the SFPUC at a purchase price of $79,087,914.12 (consisting of $67,820,000 aggregate principal amount of the 2016 Series B Bonds, plus original issue premium of $11,503,617.65, less an underwriter's discount of $235,703.53). Under the terms of its bid, the Initial Purchaser will be obligated to purchase all of the 2016 Series B Bonds if any are purchased, the obligation to make such purchase being subject to certain terms and conditions to be satisfied by the SFPUC.

The Initial Purchaser has certified the reoffering prices or yields set forth on the inside cover hereof with respect to the 2016 Series AB Bonds. The SFPUC takes no responsibility for the accuracy of these prices or yields. The Initial Purchaser may offer and sell the 2016 Series AB Bonds to certain dealers and others at prices lower than the offering prices stated on the inside cover page. The offering prices of the 2016 Series AB Bonds may be changed from time to time by the Initial Purchaser.

FINANCIAL STATEMENTS

Attached as Appendix B are the audited financial statements of the Wastewater Enterprise (the "Financial Statements") for Fiscal Years ended June 30, 2014 and June 30, 2015, prepared by the SFPUC and audited by

KPMG LLP, independent certified public accountants, San Francisco, California (the "Auditor"). The financial statements are included for convenience.

The SFPUC has not requested nor did the SFPUC obtain permission from the Auditor to include the audited financial statements as an Appendix to this Official Statement. Accordingly, the Auditor has made no representation in connection with inclusion of the audits herein that there has been no material change in the financial condition of the SFPUC since the most recent audit was concluded. The Auditor has not been engaged to perform and has not performed, since the date of its report included herein, any procedures on the financial statements addressed in that report. The Auditor also has not performed any procedures relating to this Official Statement.

CONTINUING DISCLOSURE

The SFPUC has covenanted for the benefit of the Owners of the 2016 Series AB Bonds to provide certain financial information and operating data not later than nine months after the end of the SFPUC's Fiscal Year (which currently ends on June 30), commencing with the report for Fiscal Year 2015-16 (the "Annual Report") and to provide notices of the occurrence of certain enumerated events. The Annual Report will be filed by the SFPUC with the MSRB through EMMA.

The specific nature of the information to be contained in the Annual Report or the notices of material events is summarized in "APPENDIX D – FORM OF CONTINUING DISCLOSURE CERTIFICATE." These covenants have been made in order to assist the Initial Purchaser in complying with Securities and Exchange Commission Rule 15c2-12(b)(5) (the "Rule").

Although in 2013 the SFPUC filed on EMMA materials describing the effectiveness of certain amendments to the Indenture, the SFPUC failed to file timely a simultaneous and duplicative notice of material event.

The SFPUC has, at least once in the last five years, failed to file in a timely manner notice of a change in the rating of SFPUC bonds resulting from a change in the rating of a bond insurer. As of the date of this Official Statement, the current ratings of the SFPUC's bonds are correct on EMMA.

CO-FINANCIAL ADVISORS

Public Resources Advisory Group, Los Angeles, California, and FirstSouthwest, a Division of Hilltop Securities Inc., Oakland, California (the "Co-Financial Advisors"), have served as Co-Financial Advisors to the SFPUC in connection with the structuring and delivery of the 2016 Series AB Bonds. The Co-Financial Advisors participated with responsible SFPUC officials and staff in conferences and meetings where information contained in this Official Statement was reviewed for accuracy and completeness, but are not obligated to undertake, and have not undertaken to make, an independent verification or to assume responsibility for the accuracy, completeness or fairness of the information contained in this Official Statement. The Co-Financial Advisors will receive compensation that is contingent upon the sale and delivery of the 2016 Series AB Bonds.

APPROVAL AND EXECUTION

This Official Statement has been duly approved, executed and delivered by the SFPUC.

PUBLIC UTILITIES COMMISSION OF THE
CITY AND COUNTY OF SAN FRANCISCO

By: /s/ Harlan L. Kelly, Jr.
Harlan L. Kelly, Jr.
General Manager

APPENDIX A

SUMMARY OF CERTAIN PROVISIONS OF THE INDENTURE

This Appendix includes summaries of certain provisions of the Indenture, which are in addition and complementary to the summaries found under "INTRODUCTION," "THE 2016 SERIES AB BONDS" and "SECURITY FOR THE BONDS" in the Official Statement. The following summaries are qualified in their entirety by reference to the Indenture, a copy of which can be obtained from the SFPUC.

The 2016 Series AB Bonds are issued under the Indenture, dated as of January 1, 2003, as amended and supplemented (the "Indenture"), including as supplemented by the Fourth Supplemental Indenture dated as of May 1, 2016 (the "Fourth Supplemental Indenture"). The Indenture, which is summarized under the caption "THE INDENTURE," includes definitions of terms and other provisions specific to the 2016 Series AB Bonds and the provisions applicable to all of the Bonds. The terms of the Fourth Supplemental Indenture are summarized under the caption "THE FOURTH SUPPLEMENTAL INDENTURE" below and include provisions specific to the 2016 Series AB Bonds.

THE INDENTURE

DEFINITIONS; EQUAL SECURITY: CERTIFICATES AND OPINIONS

Unless the context otherwise requires, the terms defined in the Indenture will, for all purposes of the Indenture and of any Supplemental Indenture and of any certificate, opinion or other document mentioned in the Indenture, have the meanings specified in the Indenture. Unless otherwise defined in the Indenture, all terms used in the Indenture will have the meanings assigned to such terms in the Law.

"Accreted Value" means, with respect to any Capital Appreciation Bond, the principal amount thereof plus the interest accrued thereon from its date, compounded at the approximate interest rate thereof on each date specified in the Supplemental Indenture pursuant to which such Capital Appreciation Bonds are issued. The Accreted Value on any such date of compounding will be the amount set forth in the Accreted Value Table and, with respect to any date other than a date on which compounding occurs, will be determined by straight-line interpolation (based on a year consisting of 12 30-day months), as calculated by the Trustee. The calculation of Accreted Value by the Trustee will be binding and conclusive as to the Accreted Value of Capital Appreciation Bonds.

"Accreted Value Table" means, with respect to any Capital Appreciation Bonds, the corresponding table attached as an Exhibit to a Supplemental Indenture pursuant to which Additional Bonds constituting Capital Appreciation Bonds are issued.

"Additional Bonds" means bonds, notes or other obligations of the Commission (other than Parity State Loans) payable from Net Revenues and ranking on a parity with the Bonds and issued pursuant to a Supplemental Indenture in compliance with the Indenture, as applicable.

"Annual Debt Service" means the sum of principal and interest on all Outstanding Bonds and Parity State Loans as computed for the twelve-month period ending June 30 to which reference is made, and calculated in a manner consistent with the determination of Average Annual Debt Service and Maximum Annual Debt Service as provided in the definitions thereof. In determining the amount of interest coming due during any twelve-month period ending June 30 on any Series of Bonds that were issued as Build America Bonds, amounts equal to the Refundable Credits the Commission is scheduled to receive during each such twelve-month period ending June 30 will be deducted from such interest.

"Arbitrage Certificate" means the arbitrate certificate or similar tax certificate delivered or to be delivered by the Commission at the time of issuance and delivery of a Series of Bonds, as the same may be amended or supplemented in a accordance with its terms.

"Authorized Officer" of the Trustee means and includes the chairman of the board of directors, the president, every vice president, every assistant vice president, the cashier, every assistant cashier, every trust officer, and every other officer and assistant officer of the Trustee to whom any trust matter is referred because his or her knowledge of, and familiarity with, a particular subject.

"Average Annual Debt Service" means, as of the date of calculation, total remaining Debt Service divided by the number of twelve-month periods ending on June 30 (including any fractional periods) remaining until the last maturity date of any Outstanding Bond, calculated by the Commission using the following assumptions:

(a) In determining the principal amount due in each year, payment will (unless a different subsection of the definition applies for purposes of determining principal maturities or amortization) be assumed to be made in accordance with any amortization schedule established for such debt, including any Minimum Sinking Fund Account Payments or any scheduled redemption or payment of Bonds on the basis of Accreted Value, and for such purpose, the redemption payment or payment of Accreted Value will be deemed a principal payment and interest that is compounded and paid as Accreted Value will be deemed due on the scheduled redemption or Payment Date of such Capital Appreciation Bond.

(b) If any of the Outstanding Series of Bonds constitute Balloon Indebtedness or Balloon Indebtedness and Variable Rate Indebtedness or if Bonds then proposed to be issued would constitute Balloon Indebtedness or Balloon Indebtedness and Variable Rate Indebtedness, then, for purposes of determining Average Annual Debt Service, such amounts as constitute Balloon Indebtedness will be treated as if the principal amount of such Bonds were to be amortized from the date of their original issuance in substantially equal annual installments of principal and interest over a term of 25 years, and the interest rate used for such computation will, in the case of Variable Rate Indebtedness, be the average rate quoted over the 10 years preceding the date of calculation as the BMA Municipal Swap Index of Municipal Market Data or, if such index is no longer available, the J.J. Kenny Index in each case as such term is defined in the 1992 ISDA U.S. Municipal Counterparty definitions.

(c) If any Outstanding Bonds constitute Tender Indebtedness or if Bonds then proposed to be issued would constitute Tender Indebtedness, then for purposes of determining the amounts of principal and interest due in any Fiscal Year on such Bonds, the options or obligations of the owners of such Bonds to tender the same for purchase or payment prior to their stated maturity or maturities will be treated as a principal maturity occurring on the first date on which owners of such Bonds may or are required to tender such Bonds except that any such option or obligation to tender Bonds will be ignored and not treated as a principal maturity, if (i) such Bonds are rated in one of the two highest long-term rating categories (without reference to gradations such as "plus" or "minus") by Moody's and by S&P, or such Bonds are rated in the highest short-term, note or commercial paper rating categories by Moody's and by S&P, and (ii) the obligation, if any, the Commission may have under a Letter of Credit Agreement with respect to such Bonds, other than its obligations on such Bonds, is either subordinated to the obligation of the Commission on the Bonds or is incurred under the conditions and meeting the tests for the issuance of Additional Bonds set forth in the Indenture.

(d) If any Outstanding Bonds constitute Variable Rate Indebtedness, or if Bonds proposed to be issued will be Variable Rate Indebtedness, the average rate quoted over the 10 years preceding the date of calculation as the BMA Municipal Swap Index of Municipal Market Data or, if such index is no longer available, the J.J. Kenny Index, in each case as such term is defined in the 1992 ISDA U.S. Municipal Counterparty definitions.

(e) If Defeasance Obligations have been deposited with and are held by the Trustee or another fiduciary to be used to pay principal or interest on specified Bonds, then the principal or interest to be paid from such Defeasance Obligations or from the earnings thereon will be disregarded and not included in calculating Average Annual Debt Service.

(f) In determining the amount of interest coming due during any twelve-month period ending June 30 on any Series of Bonds that are issued as Build America Bonds, amounts equal to the Refundable Credits the Commission is scheduled to receive during each such twelve-month period ending June 30 will be deducted from such interest.

"Balloon Indebtedness" means a Series of Bonds 25% or more of the principal of which matures on the same date and is not required by the documents governing such Bonds to be amortized by payment or redemption prior to such date. For purposes of this definition, an optional or mandatory tender of Bonds for purchase as described within the definition of Tender Indebtedness will not be treated as a maturity.

"Board of Supervisors" means the Board of Supervisors of the City from time to time or any other governing board of the City provided for by law.

"Bond Obligation" means, as of any given date of calculation, (1) with respect to any Outstanding Current Interest Bond, the principal amount of such Bond, and (2) with respect to any Outstanding Capital Appreciation Bond, the Accreted Value thereof as of the most recent date of the compounding of interest thereon preceding such date of calculation (unless such date of calculation is a date on which interest thereon is compounded, in which case as of such date).

"Bondowner" or "Owner" means any person who is the registered owner of any Outstanding Bond, or the bearer of any Outstanding Bond that has a maturity of one year or less and is issued in bearer form.

"Bond Reserve Fund" means the fund by that name established under the Indenture.

"Bond Reserve Fund Policy" means a financial guaranty issued to satisfy all or a portion of the Required Reserve for a Series of Bonds, which may be (a) a policy of insurance or surety bond issued by a Bond Reserve Fund Policy Provider, obligations insured by which have, at the time of the issuance of such financial guaranty, a rating by Moody's and S&P which is at least as high as the underlying rating on the related Series of Bonds (i.e., the rating given without regard to any municipal bond or financial guaranty insurance, letter of credit, or similar guaranty or credit enhancement on that Series of Bonds), or (b) a Letter of Credit issued by a Qualified Bank.

"Bond Reserve Fund Policy Provider" means a municipal bond insurance company or other insurance company that is the issuer of a Bond Reserve Fund Policy.

"Bonds" means the 2003 Refunding Series A Bonds and any bonds authorized by, and at any time Outstanding under, the Indenture or any Supplemental Indenture, including any Additional Bonds authorized by, and at any time Outstanding under, the Indenture and any Supplemental Indenture. The Commission may assign such name to any Series of Additional Bonds as it determines appropriate, and such name need not include "Clean Water Revenue Bonds."

"Build America Bonds" means any bonds or other obligations issued as Build America Bonds under Section 54AA of the Code, or under any other provision of the Code that creates a substantially similar direct-pay subsidy program.

"Business Day" means any day other than (a) a Saturday, Sunday or day upon which commercial banks in San Francisco, California, or New York, New York are authorized or required to be closed and (b) for purposes of payments and other actions relating to Bonds secured by a Letter of Credit, a day upon which

commercial banks are authorized to be closed in the city in which is located the office of the Qualified Bank at which demands for payment under the Letter of Credit are to be presented.

"Capital Appreciation Bonds" means all or any portion of a Series of Bonds designated as Capital Appreciation Bonds and on which interest is compounded and paid either at maturity or on prior redemption.

"Capital Project Account" means each account by that name established within the Capital Project Fund.

"Capital Project Fund" means the fund by that name established under the Indenture.

"Certificate of the Commission" means an instrument in writing signed by the President or by the General Manager or by any other officer of the Commission duly authorized by the Commission for that purpose, and by the Secretary. Any such instrument and supporting opinions or representations, if any, may, but need not, be combined in a single instrument with any other instrument, opinion or representation, and the two or more so combined will be read and construed as a single instrument. If and to the extent required by the provisions of the Indenture, each Certificate of the Commission will include the statements provided for in the Indenture.

"Charter" means the Charter of the City as it now exists or as it may be later amended, and any new or successor Charter.

"City" means the existing political subdivision known as the City and County of San Francisco, in the State of California, as the same is organized and existing under and by virtue of the Constitution and laws of the State of California and the Charter, and any public body later created as a successor thereto.

"Closing Date" means May 24, 2016, the date of the original issuance and delivery of the 2016 Bonds.

"Code" means the Internal Revenue Code of 1986, as amended.

"Commercial Paper Notes" means the notes in the aggregate principal amount of $300 million, captioned: (i) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $75,000,000 Series A-1 (Tax-Exempt) issued by the Commission under that certain Issuing and Paying Agent Agreement, dated as of February 11, 2014, by and between the Commission and U.S. Bank National Association, as issuing and paying agent; (ii) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $75,000,000 Series A-2 (Tax-Exempt)" issued by the Commission under that certain Issuing and Paying Agent Agreement, dated as of February 10, 2014, by and between the Commission and U.S. Bank National Association, as issuing and paying agent; (iii) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $75,000,000 Series A-3 (Tax-Exempt) issued by the Commission under that certain Issuing and Paying Agent Agreement, dated as of July 1, 2012, as amended and restated by that certain Amended and Restated Issuing and Paying Agent Agreement, dated as of July 10, 2015, each by and between the Commission and U.S. Bank National Association, as issuing and paying agent; (iv) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $75,000,000 Series A-4 (Tax-Exempt) issued by the Commission under that certain Amended and Restated Issuing and Paying Agent Agreement, dated as of July 10, 2015, by and between the Commission and U.S. Bank National Association, as issuing and paying agent; (v) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $100,000,000 Subseries A-5 and Taxable Subseries A-5-T issued by the Commission under that certain Issuing and Paying Agent Agreement, dated as of October 1, 2014, by and between the Commission and U.S. Bank National Association, as issuing and paying agent; and (vi) "Public Utilities Commission of the City and County of San Francisco Commercial Paper Notes (Wastewater Series, Proposition E) $100,000,000 Tax-Exempt Subseries A-6 and Taxable Subseries A-6-T issued by the

Commission under that certain Issuing and Paying Agent Agreement, dated as of October 1, 2014, by and between the Commission and U.S. Bank National Association, as issuing and paying agent.

"Commission" means the Public Utilities Commission of the City, and all commissions, agencies or public bodies later created which succeed to or take over the powers and duties of the Commission with respect to the Enterprise.

"Consulting Engineers" means any engineer or firm of engineers retained by the Commission having a wide and favorable reputation for skill and experience in evaluating the construction and operation of public utilities, including public sanitary waste and storm water collection, treatment and disposal systems, or in Other revenue producing publicly-owned enterprises, to perform the acts and carry out the duties provided for such consulting engineers in the Indenture.

"Controller" means the Controller of the City from time to time, and includes any deputy acting for the Controller.

"Credit Provider" means a Municipal Bond Insurer that has issued an outstanding policy of municipal bond or financial guaranty insurance or a Qualified Bank that has issued an outstanding Letter of Credit which, in each case, secures payment of principal of, and interest on, or tender price of, all or a portion of a Series of Bonds; provided that "Credit Provider" will not refer to a Bond Reserve Fund Policy Provider.

"Current Interest Bonds" means all or any portion of a Series of Bonds designated as Current Interest Bonds and which pay interest at least semiannually to the Owners thereof excluding the first payment of interest thereon.

"Debt Service" means the sum of all principal and interest due on all Outstanding Bonds and Parity State Loans as of the date of calculation.

"Defeasance Obligations" means:

(a) Cash;

(b) Federal Securities;

(c) The interest component of Resolution Funding Corporation strips which have been stripped by request to the Federal Reserve Bank of New York in book entry form;

(d) Pre-refunded municipal bonds rated "Aaa" by Moody's and "AAA" by S&P, provided that, if the issue is rated only by S&P (i.e., there is no Moody's rating), then the pre-refunded municipal bonds must have been pre-refunded with cash, direct U.S. or U.S. guaranteed obligations, or AAA rated pre-refunded municipals; and

(e) Bonds, debentures, notes or other evidence of indebtedness issued or guaranteed by any of the following federal agencies and provided such obligations are backed by the full faith and credit of the United States of America (stripped securities are only permitted if they have been stripped by the agency itself): (i) direct obligations or fully guaranteed certificates of beneficial ownership of the U.S. Export-Import Bank; (ii) certificates of beneficial ownership of the Farmers Home Administration; (iii) participation certificates of the General Services Administration; (iv) Federal Financing Bank bonds and debentures; (v) guaranteed Title XI financings of the U.S. Maritime Administration; and (vi) project notes, local authority bonds, new communities debentures and U.S. public housing notes and bonds of the U.S. Department of Housing and Urban Development.

"Enterprise" means the whole and each and every part of the municipal sanitary waste and storm water collection, treatment and disposal system and auxiliary or related facilities of the Commission, including all of the presently existing system of the Commission for the collection, treatment and disposal of sanitary waste and storm water, and all future additions, betterments, and extensions to that system or any part thereof.

"Event of Default" means an event of that name described in the Indenture.

"Federal Securities" means United States treasury notes, bonds, bills or certificates of indebtedness, or obligations for which the faith and credit of the United States of America are pledged for the payment of principal and interest (including obligations issued or held in book-entry form and securities which represent an undivided interest in such direct obligations and CATs and TGRS), and also any securities now or later authorized, both the principal of and interest on which is guaranteed directly by the full faith and credit of the United States of America.

"Financial Newspaper or Journal" means *The Wall Street Journal* or *The Bond Buyer*, or any other newspaper Or journal publishing financial news and selected by the Trustee, whose decision will be final and conclusive, printed in the English language, customarily published on each Business Day and circulated in San Francisco, California.

"First Amendment" means that certain First Amendment to Indenture dated as of May 1, 2010, between the Commission and the Trustee.

"Fiscal Year" means the period beginning on July 1 of each year and ending on the next succeeding June 30, or such other fiscal year as may be adopted by the Commission for its general accounting purposes or the then current accounting period of the City if the Commission has no separate accounting period.

"Fitch" means Fitch, Inc., doing business as Fitch Ratings, a corporation duly organized and existing under and by virtue of the laws of the State of Delaware, and its successors and assigns, except that if such corporation is dissolved or liquidated or no longer performs the functions of a securities rating agency, then the term "Fitch" will be deemed to refer to any other nationally recognized securities rating agency selected by the Commission.

"General Manager" means the general manager of the Commission appointed by the Mayor from time to time pursuant to the Charter or any other applicable provision of law, and includes any other person acting on behalf of the General Manager.

"Indenture" means the Indenture, dated as of January 1, 2003, by and between the Commission and the Trustee, as originally executed or as it may from time to time be supplemented or amended by any Supplemental Indenture delivered under the provisions of the Indenture.

"Independent Certified Public Accountant" means any certified public accountant or firm of such accountants appointed and paid by the Commission, and who, or each of whom --

(a) is in fact independent and not under control of the City or the Commission;

(b) does not have any substantial interest, direct or indirect, with the City or the Commission; and

(c) is not connected with the City or the Commission as an officer or employee of the City or the Commission, but who may be regularly retained to make annual or other audits of the books of or reports to the City or the Commission.

"Information Services" means:

(a) Financial Information, Inc.'s "Daily Called Bond Service," 30 Montgomery Street, 10th Floor, Jersey City, New Jersey 07302, Attention: Editor;

(b) Mergent/FIS, 5250 77 Center Drive, Suite 150, Charlotte, North Carolina, 28217, Attn: Called Bond Dept.; and

(c) Kenny S&P, 55 Water Street, 45th Floor, New York, New York 10041, Attention: Notification Department;

or, in accordance with then-current guidelines of the Securities and Exchange Commission, such other addresses or such other services providing information with respect to called bonds, or no such services, as the Commission may designate in a Written Request of the Commission delivered to the Trustee.

"Interest Payment Date" means those interest payment dates set forth for the 2003 Refunding Series A Bonds in the Indenture, and those interest payment dates set forth for any Additional Bonds in a Supplemental Indenture in accordance with the Indenture, as appropriate. With respect to the 2016 Bonds, Interest Payment Date means April 1 and October 1 of each year, commencing October 1, 2016.

"Law" means the Charter, the San Francisco Administrative Code, and all laws of the State of California supplemental thereto, including the Revenue Bond Law of 1941 to the extent made applicable by the Charter or by the San Francisco Administrative Code. Whenever reference is made in the Indenture to the "Law," reference is made to the Law as in force on the date of the Indenture or any Supplemental Indenture, unless the context otherwise requires.

"Legal Investments" means bonds, notes, certificates of indebtedness, bills, acceptances or other securities in which funds of the Commission may now or later be legally invested as provided by the law in effect at the time of such investment.

"Letter of Credit" means an irrevocable and unconditional letter of credit, a standby purchase agreement, a line of credit or other similar credit arrangement issued by a Qualified Bank to secure payment of Balloon Indebtedness, Variable Rate Indebtedness, Tender Indebtedness or a Series of Bonds, or to satisfy all or a portion of the Required Reserve.

"Letter of Credit Agreement" means an agreement between the Commission and a Qualified Bank pursuant to which the Qualified Bank agrees to issue a Letter of Credit and which sets forth the repayment obligation of the Commission to the Qualified Bank on account of any advances under the Letter of Credit.

"Letter of Representations" means the letter or letters of representation of the Commission delivered to and accepted by The Depository Trust Company setting forth the basis on which The Depository Trust Company serves as depository for the Bonds, as originally executed or as it may be supplemented or revised or replaced by a letter to a substitute depository.

"Maturity Date" means the maturity dates set forth for the 2003 Refunding Series A Bonds in the Indenture, and each maturity date set forth for any Additional Bonds in a Supplemental Indenture in accordance with the Indenture, as appropriate.

"Maximum Annual Debt Service" means, as, of the date of calculation, the largest Annual Debt Service in the then current or any future Fiscal Year, calculated by the Commission using the following assumptions:

(a) In determining the principal amount due in each year, payment will (unless a different subsection of the definition applies for purposes of determining principal maturities or amortization) be assumed to be made in accordance with any amortization schedule established for such debt, including any Minimum Sinking Fund Account Payments or any scheduled redemption or payment of Bonds on the basis of Accreted Value, and for such purpose, the redemption payment or payment of Accreted Value will be deemed a principal payment and interest that is compounded and paid as Accreted Value will be deemed due on the scheduled redemption or Payment Date of such Capital Appreciation Bond.

(b) If any of the Outstanding Series of Bonds constitute Balloon Indebtedness or Balloon Indebtedness and Variable Rate Indebtedness or if Bonds then proposed to be issued would constitute Balloon Indebtedness or Balloon Indebtedness and Variable Rate Indebtedness, then, for purposes of determining Maximum Annual Debt Service, such amounts as constitute Balloon Indebtedness will be treated as if the principal amount of such Bonds were to be amortized from the date of their original issuance in substantially equal annual installments of principal and interest over a term of 25 years, and the interest rate used for such computation will, in the case of Variable Rate Indebtedness, be the average rate quoted over the 10 years preceding the date of calculation as the BMA Municipal Swap Index of Municipal Market Data or, if such index is no longer available, the J.J. Kenny Index in each case as such term is defined in the 1992 ISDA U.S. Municipal Counterparty definitions.

(c) If any Outstanding Bonds constitute Tender Indebtedness or if Bonds then proposed to be issued would constitute Tender Indebtedness, then for purposes of determining the amounts of principal and interest due in any Fiscal Year on such Bonds, the options or obligations of the owners of such Bonds to tender the same for purchase or payment prior to their stated maturity or maturities will be treated as a principal maturity occurring on the first date on which owners of such Bonds may or are required to tender such Bonds except that any such option or obligation to tender Bonds will be ignored and not treated as a principal maturity, if (i) such Bonds are rated in one of the two highest long-term rating categories (without reference to gradations such as "plus" or "minus") by Moody's and by S&P, or such Bonds are rated in the highest short-term, note or commercial paper rating categories by Moody's and by S&P, and (ii) the obligation, if any, the Commission may have under a Letter of Credit Agreement with respect to such Bonds, other than its obligations on such Bonds, is either subordinated to the obligation of the Commission on the Bonds or is incurred under the conditions and meeting the tests for the issuance of Additional Bonds set forth in the Indenture.

(d) If any Outstanding Bonds constitute Variable Rate Indebtedness, or Bonds proposed to be issued will be Variable Rate Indebtedness, the average rate quoted over the 10 years preceding the date of calculation as the BMA Municipal Swap Index of Municipal Market Data or, if such index is no longer available, the J.J. Kenny Index, in each case as such term is defined in the 1992 ISDA U.S. Municipal Counterparty definitions.

(e) If Defeasance Obligations have been deposited with, and are held by the Trustee or another fiduciary to be used to pay principal or interest on specified Bonds, then the principal or interest to be paid from, such Defeasance Obligations or from the earnings thereon will be disregarded and not included in calculating Maximum Annual Debt Service.

(f) In determining the amount of interest coming due during any twelve-month period ending June 30 on any Series of Bonds that are issued as Build America Bonds, amounts equal to the Refundable Credits the Commission is scheduled to receive during such twelve-month period ending June 30 will be deducted from such interest.

"Mayor" means the Mayor of the City from time to time.

"Minimum Sinking Fund Account Payments" means the aggregate amounts required by the Indenture and any subsequent Supplemental Indenture to be deposited in Sinking Fund Accounts for the payment of Term Bonds.

"Moody's" means Moody's Investors Service, Inc., a corporation duly organized and existing under and by virtue of the laws of the State of Delaware, and its successors and assigns, except that if such corporation is dissolved or liquidated or no longer performs the functions of a securities rating agency, then the term "Moody's" will be deemed to refer to any other nationally recognized securities rating agency selected by the Commission.

"Municipal Bond Insurer" means any insurance company or companies which is or are designated as such in the Indenture or a Supplemental Indenture, and which has or have issued a policy of municipal bond insurance or a financial guaranty insurance policy insuring payment of the principal of and interest on any of the Bonds of any Series of Bonds.

"Net Revenues" means all of the Revenues less all Operation and Maintenance Costs of the Enterprise.

"Operation and Maintenance Costs of the Enterprise" means the reasonable and necessary costs of operating and maintaining the Enterprise, calculated on generally accepted accounting principles, including (among other things) salaries and wages, fees for services, costs of materials, supplies and fuel, reasonable expenses of management, legal fees, accounting fees, repairs and other expenses necessary to maintain and preserve the Enterprise in good repair and working order, and reasonable amounts for administration, overhead, insurance, taxes (if any), and the payment of pension charges and proportionate payments to such compensation and other insurance or outside reserve funds as the Commission may establish or the Board of Supervisors may require with respect to employees of the Commission, as provided in the Charter. However, the term "Operation and Maintenance Costs of the Enterprise" excludes in all cases (a) depreciation and obsolescence charges or reserves therefor, (b) amortization of intangibles or other bookkeeping entries of a similar nature, (c) costs of capital additions, replacements, betterments, extensions or improvements to the Enterprise, which under generally accepted accounting principles are chargeable to a capital account or to a reserve for depreciation, (d) charges for the payment of principal and interest on any revenue bonds or other indebtedness previously or later issued for Enterprise purposes and (e) such costs as are scheduled to be paid by the Commission from moneys other than Revenues, such moneys to be clearly available for such purpose.

"Opinion of Counsel" means a written opinion of counsel (who may be counsel for the City or the Commission) retained by the Commission. If and to the extent required by the provisions of the Indenture, each Opinion of Counsel will include the statements provided for in the Indenture.

"Outstanding," when used as of any particular time with reference to Bonds, means (subject to the provisions of the Indenture) all Bonds theretofore executed, issued and delivered by the Commission under the Indenture except --

(a) Bonds theretofore cancelled by the Trustee or surrendered to the Trustee for cancellation;

(b) Bonds for the payment or redemption of which funds or securities in the necessary amount (as set forth in the Indenture) have theretofore been deposited with a fiduciary (whether upon or prior to the maturity or redemption date of such Bonds), provided that, if such Bonds are to be redeemed prior to the maturity thereof, notice of such redemption is given as in provided in the Indenture, or provision satisfactory to the Trustee is made for the giving of such notice; and

(c) Bonds in lieu of or in substitution for which other Bonds have been executed, issued and delivered by the Commission pursuant to the Indenture.

For purposes of this definition and within the meaning of the Indenture, any Bonds, the principal of or interest on which has been paid by a Credit Provider, will not be deemed paid by or on behalf of the Commission, will not be defeased and will remain Outstanding under the Indenture until the Credit Provider has been paid or reimbursed for such payment by the Commission.

"Parity State Loans" means those loan agreements entered into between the Commission and the State of California (or any board, department or agency thereof) to finance additions, betterments, extensions, repairs, renewals or replacements to the Enterprise, which are entered into after the issuance of the 2003 Refunding Series A Bonds and which, by their terms, are payable from Net Revenues on a parity basis with debt service on the Bonds.

"Payment Date" means any date on which payment of the principal of or interest on the Bonds is due, or on which any Term Bonds are required to be redeemed from any Minimum Sinking Fund Account Payments.

"Permitted Investments" means arty of the following, to the extent permitted by law and by any policy guidelines promulgated by the Commission or the City:

(a) Cash;

(b) Federal Securities;

(c) the interest component of Resolution Funding Corporation strips which have been stripped by request to the Federal Reserve Bank of New York in book entry form;

(d) bonds, debentures, notes or other evidence of indebtedness issued or guaranteed by any of the following federal agencies and provided such obligations are backed by the full faith and credit of the United States of America (stripped securities are only permitted if they have been stripped by the agency itself): (i) direct obligations or fully guaranteed certificates of beneficial ownership of the U.S. Export-Import Bank; (ii) certificates of beneficial ownership of the Farmers Home Administration; (iii) Federal Housing Administration debentures; (iv) participation certificates of the General Services Administration; (v) Federal Financing Bank bonds and debentures; (vi) guaranteed mortgage-backed bonds or guaranteed pass-through obligations of the Government National Mortgage Association; (vii) guaranteed Title XI financings of the U.S. Maritime Administration; and (viii) project notes, local authority bonds, new communities debentures and U.S. public housing notes and bonds of the U.S. Department of Housing and Urban Development;

(e) bonds, debentures, notes or other evidence of indebtedness issued or guaranteed by any of the following non-full faith and credit U.S. government agencies (stripped securities only as stripped by the agency itself): (i) senior debt obligations of the Federal Home Loan Bank System; (ii) participation certificates and senior debt obligations of the Federal Home Loan Mortgage Corporation; (iii) mortgaged-backed securities and senior debt obligations of the Fannie Mae; (iv) senior debt obligations of the Student Loan Marketing Association; (v) obligations of the Resolution Funding Corporation; and (vi) consolidated system-wide bonds and notes of the Farm Credit System;

(f) money market funds registered under the Federal Investment Company Act of 1940, whose shares are registered under the Federal Securities Act of 1933, and having a rating by S&P of at least AAAm-G, AAAm or AAm, and a rating by Moody's of Aaa, Aa1 or Aa2 (such funds may include funds for which the Trustee, its affiliates, parent or subsidiaries provide investment advisory or, other management services);

(g) certificates of deposit (including those of the Trustee, its parent and its affiliates) secured at all times by collateral described in (a) or (b) above, which have a maturity not greater than one year from the date of investment and which are issued by commercial banks, savings and loan associations or mutual savings banks whose short-term obligations are rated "A-1+" or better by S&P and "Prime-1" by Moody's, which collateral must be held by a third party and provided that the Trustee must have a perfected first security interest in such collateral;

(h) certificates of deposit, savings accounts, deposit accounts or money market deposits (including those of the Trustee and its affiliates) which are fully insured by FDIC, including BIF and SAIF;

(i) investment agreements, including guaranteed investment contracts, forward purchase agreements, reserve fund put agreements and collateralized investment agreements acceptable to the Credit Provider;

(j) commercial paper rated "Prime-1" by Moody's and "A-1+" or better by S&P;

(k) bonds or notes issued by any state or municipality which are rated by Moody's and S&P in one of the two highest rating categories assigned by such agencies;

(1) federal funds or bankers acceptances with a maximum term of one year of any bank which an unsecured, uninsured and unguaranteed obligation rating of "Prime-1" or "A3" or better by Moody's, and "A-1+" by S&P;

(m) the Local Agency Investment Fund which is administered by the California Treasurer for the investment of funds belonging to local agencies within the State of California, provided for investment of funds held by the Trustee, the Trustee is entitled to make investments and withdrawals in its own name as Trustee; and

(n) Any other investment approved in writing by the Credit Provider.

"Policy Costs" means the amounts owing to a Bond Reserve Fund Policy Provider, including the principal amount of any draw on a Bond Reserve Fund Policy, interest thereon and reasonable expenses incurred by the Bond Reserve Fund Policy Provider in enforcing payment of Policy Costs, as more fully set forth in the agreement pursuant to which such Bond Reserve Fund Policy is issued.

"President" means the President of the Commission from time to time, or any other person acting on behalf of the President.

"Principal Payment Date" means the principal payment date set forth for the 2003 Refunding Series A Bonds in the Indenture, and the principal payment date set forth for any Additional Bonds in a Supplemental Indenture in accordance with the Indenture, as appropriate. With respect to the 2016 Bonds, Principal Payment Date means October 1 of each year, commencing October 1, 2016.

"Prior Resolution" means, collectively, Resolution No. 656-92 adopted by the Board of Supervisors on August 3, 1992, Resolution No. 58-94 adopted by the Board of Supervisors on January 24, 1994 and Resolution No. 385-95 adopted by the Board of Supervisors on May 8, 1995.

"Project" means any repairs, replacements, additions, enlargements, betterments, extensions and other improvements to or benefiting, and the equipping of, the Enterprise, including, without limitation, the acquisition of land therefor.

"Proportionate Basis," when used with respect to the redemption of Bonds, means that the amount of Bonds of each maturity to be redeemed will be determined as nearly as practicable by multiplying the total amount of funds available for redemption by the ratio which the amount of Bond Obligation of Bonds of such maturity bears to the amount of all Bond Obligation of Bonds to be redeemed, provided that if the amount available for redemption of Bonds of any maturity is insufficient to redeem a multiple of $5,000 principal amount or Accreted Value payable at maturity, such amount will be applied to the redemption of the highest possible integral multiple (if any) of $5,000 principal amount or Accreted Value payable at maturity. For purposes of the foregoing, Term Bonds will be deemed to mature in the years and in the amounts of the Minimum Sinking Fund Account Payments and Capital Appreciation Bonds and Current Interest Bonds maturing or subject to Minimum Sinking Fund Account Payments in the same year will be treated as separate maturities.

When used with respect to the payment or purchase of Bonds, "Proportionate Basis" will have the same meaning set forth above except that "pay" or "purchase" will be substituted for "redeem" or "redemption" and "paid" or "purchased" will be substituted for "redeemed."

"Qualified Bank" means a state or national bank or trust company or savings and loan association or a foreign bank with a domestic branch or agency which is organized and in good standing under the laws of the United States or any state thereof or any foreign country, which has a debt rating at least as high as the underlying rating on the related Series of Bonds at the time such Qualified Bank delivers a Letter of Credit or a Bond Reserve Fund Policy (i.e., the rating given without regard to any municipal bond or financial guaranty insurance, letter of credit, or similar guaranty or credit enhancement on that Series of Bonds) as provided by Moody's, by S&P or Fitch.

"Qualified Financial Advisor" means a person or a firm selected by the Commission who or which engages in the business of advising the management of public agencies similar to the Commission concerning the issuance of debt.

"Qualified Independent Consultant" means a person or a firm who or which engages in the business of advising the management of public agencies concerning the operation and financing of public utilities, including public municipal sanitary waste and storm water collection, treatment and disposal systems, and also including advice and consultation generally concerning the use and operation of public utilities, including public municipal sanitary waste and storm water collection, treatment and disposal systems, and which person or firm, by reason of his or its knowledge and experience, has acquired a reputation as a recognized consultant. Such Qualified Independent Consultant may include a person or firm rendering professional engineering or accounting services in addition to his or its occupation as a public utility consultant and may include any person or firm regularly employed by the City or the Commission as a consultant to the City or the Commission.

"Rebate Fund" means the fund established and so designated for a Series of Bonds.

"Refundable Credits" means (a) with respect to a Series of Bonds issued as Build America Bonds under Section 54AA of the Code, the amounts which are payable by the Federal government under Section 6431 of the Code, which the Commission has elected to receive under Section 54AA(g)(1) of the Code, and (b) with respect to a Series of Bonds issued as Build America Bonds under any other provision of the Code that creates a substantially similar direct-pay subsidy program, the amounts which are payable by the Federal government under the applicable provisions of the Code, which the Commission has elected to receive under the applicable provisions of the Code.

"Required Reserve" means, with respect to Series of Bonds issued prior to the effective date of the amendments set forth in the First Amendment to Indenture dated as of May 1, 2010 (the "First Amendment"), between the Commission and the Trustee, but only until the effective date of the amendments set forth in the First Amendment, for any Series of Bonds, as of any date of calculation, an amount equal to the lesser of: (i)

Maximum Annual Debt Service on all such Series of Bonds (excluding from such calculation Parity State Loans, if any) then Outstanding; or (ii) 125% of Average Annual Debt Service on all such Series of Bonds (excluding from such calculation Parity State Loans, if any) then Outstanding; provided, that (i) on and after the effective date of the amendments set forth in the First Amendment, "Required Reserve will mean, with respect to a Series of Bonds issued prior to the effective date of the amendments set forth in the First Amendment, 50% of Maximum Annual Debt Service on all such Series of Bonds (excluding from such calculation Parity State Loans, if any) then Outstanding, and (ii) in no event will the Commission, in connection with issuance of a Series of Additional Bonds, be obligated to deposit an amount in the Bond Reserve Fund which is in excess of the amount permitted by the applicable provisions of the Code to be so deposited from the proceeds of tax-exempt bonds without having to restrict the yield of any investment purchased with any portion of such deposit.

"Required Reserve" means, with respect to Series of Bonds- issued on or after the effective date of the amendments set forth in the First Amendment, for any Series of Bonds, as of any date of calculation, the amount, if any, required to be deposited into a Reserve Account for that Series of Bonds, as defined in and provided by the Supplemental Indenture pursuant to which such Series of Bonds is issued; provided, however, that in no event will the Commission, in connection with issuance of a Series of Additional Bonds, be obligated to deposit an amount in the Bond Reserve Fund which is in excess of the amount permitted by the applicable provisions of the Code to be so deposited from the proceeds of tax-exempt bonds without having to restrict the yield of any investment purchased with any portion of such deposit.

"Reserve Account" means each account established in the Bond Reserve Fund with respect to each Series of Bonds issued under the Indenture.

"Revenue Fund" means the fund by that name established pursuant to the Indenture.

"Revenues" means all gross revenues of the Enterprise, including all charges received for and all other income and receipts derived by, the Commission from the operation of the Enterprise, or arising from the Enterprise, including connection and installation charges, but excluding – (a) any money received by or for the account the Commission from the levy or collection of taxes, (b) moneys received from the State of California and the United States of America and required to be deposited in restricted funds, (c) refundable deposits made to establish credit, (d) advances and contributions made to the Commission to be applied to construction, (e) moneys received constituting casualty insurance proceeds with respect to all or any part of the Enterprise (which will be received and disposed of pursuant to the Indenture) and moneys received constituting other insurance proceeds, (f) moneys received from the sale or disposition of all or any part of the Enterprise (which will be received and disposed of pursuant to the Indenture), (g) moneys received upon the taking by or under the threat of eminent domain of all or any part of the Enterprise (which moneys will be received and disposed of pursuant to the Indenture), (h) proceeds from Bonds issued by the Commission or proceeds from loans or other indebtedness obtained by the Commission, and (i) moneys or securities received by the Commission as gifts or grants, the use of which is restricted by the donor or grantor.

The term "Revenues" also includes (i) all interest or other income (excluding profits or losses from the sale or disposition of Permitted Investments or other securities owned by or on behalf of the Commission) derived from the deposit or investment of any moneys in any fund or account established under the Indenture (excluding any Rebate Fund and any escrow fund pledged for the payment of defeased bonds) or in any fund or account of the Enterprise and legally available to pay Debt Service, and (ii) any other moneys, proceeds and other amounts that the Commission determines should be "Revenues" under the Indenture.

"Secretary" means the Secretary of the Commission from time to time.

"Securities Depositories" means The Depository Trust Company, 711 Stewart Avenue, Garden City, New York 11530, Fax-(516) 227-4039 or 4190; or, in accordance with then-current guidelines of the Securities and Exchange Commission, to such other addresses or such other securities depositories, or no such

depositories, as the Commission may designate in a Written Request of the Commission delivered to the Trustee.

"Senior State Loans" means those loan agreements entered into between the Commission and the State of California (or any board, department or agency thereof) to finance additions, betterments, extensions, repairs, renewals or replacements to the Enterprise, which (i) have been entered into and are in effect prior to the issuance of the 2003 Refunding Series A Bonds, or (ii) which are entered into after the issuance of the 2003 Refunding Series A Bonds and which, by their terms, are payable from Revenues on a basis senior to Debt Service. The Commission acknowledges that the payment of the Senior State Loans described in (i) above is not, by the terms of such Senior State Loans, senior to the payment of Debt Service, but has, for purposes of the Indenture, elected to treat such payment of such Senior State Loans as senior to the payment of Debt Service.

"Serial Bonds" means all or any portion of a Series of Bonds designated as Serial Bonds and for which no Minimum Sinking Fund Account Payments are provided.

"Series" means any series of Bonds executed, authenticated and delivered pursuant to the Indenture and identified as a separate Series of Bonds, including any Additional Bonds issued pursuant to a Supplemental Indenture and the Indenture.

"Sinking Fund Accounts" means any special account or accounts established by the Indenture or any Supplemental Indenture or Indentures in the Principal Fund (established pursuant to the Indenture) for the payment of Term Bonds.

"S&P" means Standard & Poor's Ratings Services, a corporation duly organized and existing under and by virtue of the laws of the State of New York, and its successors and assigns, except that if such corporation is dissolved or liquidated or no longer performs the functions of a securities rating agency, then the term "S&P" will be deemed to refer to any other nationally recognized securities rating agency selected by the Commission.

"Supplemental Indenture" means any indenture amendatory of or supplemental to the Indenture that complies with the provisions of the Indenture for amendments and supplements, and includes any amended and restated indenture that complies with the provisions of the Indenture for amendments and supplements.

"Tender Indebtedness" means any Bonds or portions of Bonds a feature of which is an option, on the part of the Bondowners, or an obligation, under the terms of such Bonds, to tender all or a portion of such Bonds to the Commission, the Trustee or other fiduciary or agent for payment or purchase and requiring that such Bonds or portions of Bonds be purchased if properly presented.

"Term Bonds" means all or any portion of a Series of Bonds designated as Term Bonds and which are payable at or before their specified maturity date or dates from Minimum Sinking Fund Account Payments established for that purpose and calculated to retire such Bonds on or before their specified maturity date or dates.

"Treasurer" means the Treasurer of the City and includes any deputy acting for the Treasurer.

"Trustee" means U.S. Bank, National Association, acting as an independent trustee with the duties and powers provided in the Indenture, its successors and assigns, and any other corporation or association which may at any time be substituted in its place, as provided in the Indenture.

"2016 Bonds" means the 2016 Series A Bonds and the 2016 Series B Bonds. As used in the Official Statement, the term 2016 Series AB Bonds means the 2016 Series A Bonds and the 2016 Series B Bonds.

"2016 Continuing Disclosure Certificate" means the Continuing Disclosure Certificate, dated the date of initial issuance of the 2016 Bonds, executed and delivered by the Commission, as it may be supplemented and amended in accordance with its terms.

"2016 Series A Bonds" means the Public Utilities Community of the City and County of San Francisco Wastewater Revenue Bonds 2016 Series A (Green Bonds).

"2016 Series A Capital Project Account" means the account by that name established within the Capital Project Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series A Capitalized Interest Account" means the account by that name established within the Interest Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series A Costs of Issuance Fund" means the fund by that name established pursuant to the Fourth Supplemental Indenture.

"2016 Series A Project" means financing, from amounts on deposit in the 2016 Series A Capital Project Account, the reconstructing, replacing, expanding, repairing or improving facilities that are part of, or of benefit to the Enterprise pursuant to the Law, including Section 8B.124 of the Charter.

"2016 Series A Rebate Fund" means the fund by that name established pursuant to the Fourth Supplemental Indenture.

"2016 Series A Reimbursement Account" means the account by that name established within the Principal Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series A Sinking Fund Account" means the account by that name established within the Principal Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series B Bonds" means the Public Utilities Community of the City and County of San Francisco Wastewater Revenue Bonds 2016 Series B.

"2016 Series B Capital Project Account" means the account by that name established within the Capital Project Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series B Capitalized Interest Account" means the account by that name established within the Interest Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series B Costs of Issuance Fund" means the fund by that name established pursuant to the Fourth Supplemental Indenture.

"2016 Series B Project" means financing, from amounts on deposit in the 2016 Series B Capital Project Account, the reconstructing, replacing, expanding, repairing or improving facilities that are part of, or of benefit to the Enterprise pursuant to the Law, including Section 8B.124 of the Charter.

"2016 Series B Rebate Fund" means the fund by that name established pursuant to the Fourth Supplemental Indenture.

"2016 Series B Reimbursement Account" means the account by that name established within the Principal Fund pursuant to the Fourth Supplemental Indenture.

"2016 Series B Sinking Fund Account" means the account by that name established within the Principal Fund pursuant to the Fourth Supplemental Indenture.

"2003 Refunding Series A Bonds" has the meaning set forth in the Indenture.

"2003 Refunding Series A Reserve Account" means the account by that name established within the Bond Reserve Fund pursuant to the Indenture.

"Variable Rate Indebtedness" means any portion of indebtedness, the interest rate on which is not fixed at the time such indebtedness is incurred, and has not at some subsequent date been fixed for the entire term of the indebtedness.

"Written Request of the Commission," "Written Requisition of the Commission" and "Written Statement of the Commission" mean, respectively, a written request, requisition or statement signed by or on behalf of the Commission by the President or the General Manager or the Secretary or by any person (whether or not an officer of the Commission) who is authorized by resolution of the Commission (which resolution will be provided to the Trustee) or otherwise to sign or execute such a document on its behalf.

Equal Security. In consideration of the acceptance of the Bonds by those who will hold the same from time to time, the Indenture will be deemed to be and will constitute a contract between the Commission, the Trustee and the Owners from time to time of the Bonds to secure the full and final payment of the interest and principal on the Bonds, subject to the agreements, conditions, covenants and terms contained therein; and the covenants and agreements therein set forth to be performed on behalf of the Commission or the Trustee will be for the equal and proportionate benefit, security and protection of all Owners of the Bonds without preference, priority or distinction as to security or otherwise of any of the Bonds over any of the others by reason of the Series, number or date thereof or the time of issue, sale, execution or delivery thereof, or otherwise for any cause whatsoever, except as expressly provided therein or therein.

Content of Certificates and Opinions. (a) Every certificate or opinion with respect to compliance with a condition or covenant provided for in the Indenture, including each Certificate of the Commission, will include (i) a statement that the person or persons making or giving such certificate or opinion have read such covenant or condition and the definitions in the Indenture relating thereto; (ii) a brief statement as to the nature and scope of the examination or investigation upon which the statements or opinions contained in such certificate or opinion are based; (iii) a statement that, in the opinion of the signers, they have made or caused to be made such examination or investigation as is necessary to enable them to express an informed opinion as to whether or not such covenant or condition has been complied with; and (iv) a statement as to whether, in the opinion of the signers, such condition or covenant has been complied with.

(b) Any such certificate or opinion made or given by an officer of the Commission may be based, insofar as it relates to legal, accounting or Enterprise matters, upon a certificate or opinion of or representations by counsel, accountants or consultants, unless such officer knows, or in the exercise of reasonable care should have known, that the certificate or opinion or representations with respect to the matters upon which his certificate or opinion may be based, as aforesaid, are erroneous. Any such certificate or opinion made or given by counsel, accountants or consultants may be based, insofar as it relates to factual matters, information with respect to which is in the possession of the Commission, upon the certificate or opinion of or representations by an officer or officers of the Commission, unless such counsel, accountant or consultant knows, or in the exercise of reasonable care should have known, that the certificate or opinion or representations with respect to the matters upon which his opinion may be based as aforesaid are erroneous.

THE BONDS

Authorization. (a) Issuance of Bonds. (i) Bonds may be issued under the Indenture from time to time in order to obtain funds for the purposes authorized therein. The Bonds will be issued under the Charter and the Law for the purpose of financing or refinancing the acquisition, construction, replacement, reconstruction, extension, improvement and development of the Enterprise.

(ii) The aggregate principal amount of Bonds which may be issued under the Indenture is not limited (subject, however, to the right of the Commission and the Board of Supervisors of the City, which is reserved by the Indenture, to limit or restrict the aggregate principal amount of Bonds which may at any time be issued and Outstanding under the Indenture) and consists or may consist of one or more Series of varying denominations, dates, maturities, interest rates and other provisions, all issued and to be issued pursuant to the Indenture and the Law, subject to the limitations contained in the Indenture.

(iii) The Indenture constitutes a continuing agreement with the Owners of all of the Bonds issued or to be issued under the Indenture and then Outstanding to secure the full and final payment of the principal of and the premiums, if any, and the interest on all Bonds which may from time to time be executed and delivered under the Indenture, subject to the covenants, agreements, provisions and conditions contained in the Indenture.

(iv) The Bonds are designated generally as the "Clean Water Revenue Bonds," each Series thereof to bear such additional designation as may be necessary or appropriate to distinguish such Series from every other Series of Bonds. The Bonds may be issued in such Series as from time to time will be established and authorized by the Commission, subject to the provisions and conditions contained in the Indenture.

Execution of Bonds. (a) The Bonds will be executed on behalf of the Commission by the manual or facsimile signatures of its President or General Manager and the Controller and under the seal of the Commission attested by the manual or facsimile signature of the Secretary. Such seal may be in the form of a facsimile of the Commission's seal and may be imprinted or impressed upon the Bonds. The Bonds will then be delivered to the Trustee for authentication by it. In case any of the officers who have signed or attested any of the Bonds cease to be such officer or officers before the Bonds so signed or attested have been authenticated or delivered by the Trustee or issued by the Commission, such Bonds may nevertheless be authenticated, delivered and issued and, upon such authentication, delivery and issue, will be as binding upon the Commission as though those who signed and attested the same had continued to be such officers, and also any Bond may be signed and attested on behalf of the Commission by such persons as on the actual date of the execution of such Bond will be the President or General Manager or the Secretary or the Controller although at the nominal date of such Bond any such person did not hold such title.

(b) Except as may be provided in any Supplemental Indenture, only such of the Bonds as will bear thereon a certificate of authentication and registration in the form recited in the Indenture, executed by the Trustee, will be valid or obligatory for any purpose or entitled to the benefits of the Indenture, and such certificate of the Trustee will be conclusive evidence that the Bonds so authenticated have been duly authenticated and delivered under the Indenture and are entitled to the benefits of the Indenture.

Transfer of Bonds. (a) Any Bond may, in accordance with its terms, be transferred, upon the books required to be kept pursuant to the provisions of the Indenture, by the person in whose name it is registered, in person or by his or her duly authorized attorney, upon surrender of such Bond for cancellation, accompanied by delivery of a written instrument of transfer in a form approved by the Trustee duly executed.

(b) Whenever any Bond or Bonds are surrendered for transfer, the Commission will execute and the Trustee will authenticate and deliver a new Bond or Bonds of the same Series, tenor and maturity, for a like aggregate principal amount. The Trustee will require the payment by any Bondowner requesting any such transfer of any tax or other governmental Charge required to be paid with respect to such transfer.

(c) No transfer of Current Interest Bonds will be required to be made by the Trustee after the fifteenth day of the month next preceding each interest payment date, or, as, to any Bonds called for redemption, within 30 days of the date fixed for redemption.

Exchange of Bonds. (a) Bonds may be exchanged at the principal corporate trust office of the Trustee in San Francisco, California, for a like aggregate principal amount of Bonds of other authorized denominations of the same Series, tenor and maturity. The Trustee will require the payment by the Bondowner requesting such exchange of any tax or other governmental charge required to be paid with respect to such exchange.

(b) No exchange of Current Interest Bonds will be required to be made by the Trustee after the fifteenth day of the month next preceding each interest payment date, or, as to any Bonds called for redemption, within thirty days of the date fixed for redemption.

Bond Register. The Trustee will keep or cause to be kept, at the principal corporate trust office of the Trustee in St. Paul, Minnesota, sufficient books for the registration and transfer of the Bonds, which will during regular business hours of the Trustee be open to inspection by the Commission; and, upon presentation for such purpose, the Trustee will, under such reasonable regulations as the Trustee may prescribe, register or transfer or cause to be registered or transferred, on said books, Bonds as provided in the Indenture.

Temporary Bonds. The Bonds may be initially issued in temporary form exchangeable for definitive Bonds when ready for delivery. The temporary Bonds may be printed, lithographed or typewritten, will be of such denominations as may be determined by the Commission, will be in registered, form without coupons and may contain such reference to any of the provisions of the Indenture as may be appropriate. Every temporary Bond will be executed by the Commission and be authenticated by the Trustee upon the same conditions and in substantially the same manner as the definitive fully registered Bonds. If the Commission issues temporary Bonds it will execute and furnish definitive Bonds without delay, and thereupon the temporary Bonds may be surrendered, for cancellation, in exchange therefor at the principal corporate trust office of the Trustee in St. Paul, Minnesota, and the Trustee will deliver in exchange for such temporary Bonds definitive Bonds of an equal aggregate principal amount of Bonds of the same Series, tenor and maturity or maturities. Until so exchanged, the temporary Bonds will be entitled to the same benefits under the Indenture as definitive Bonds executed and delivered under the Indenture.

Bonds Mutilated, Lost, Destroyed or Stolen. If any Bond becomes mutilated, the Commission, at the expense of the Owner of said Bond, will execute, and the Trustee will thereupon authenticate and deliver, a new Bond of like tenor and number in exchange and substitution for the Bond so mutilated (except that such number may be preceded by a distinguishing prefix), but only upon surrender to the Trustee of the Bond so mutilated. Every mutilated Bond so surrendered to the Trustee will be cancelled by it and delivered to, or upon the order of, the Commission. If any Bond becomes lost, destroyed or stolen, evidence of such loss, destruction or theft may be submitted to the Commission and the Trustee and, if such evidence is satisfactory to both and indemnity satisfactory to them will be given, the Commission, at the expense of the Owner, will execute, and the Trustee will thereupon authenticate and deliver a new Bond in lieu of and in substitution for the Bond so lost, destroyed or stolen (except that such number may be preceded by a distinguishing prefix). The Commission may require payment of a sum not exceeding the actual cost of preparing each new Bond issued under the Indenture and of the expenses which may be incurred by the Commission and, the Trustee in the premises. Any Bond issued under the provisions of the Indenture in exchange for any Bond mutilated or in lieu of any Bond alleged to be lost, destroyed or stolen will constitute an original additional contractual obligation on the part of the Commission, whether or not the Bond so mutilated or so alleged to be lost, destroyed or stolen be at any time enforceable by anyone, and will be equally and proportionately entitled to the benefits of the Indenture with all other Bonds secured by the Indenture. Neither the Commission nor the Trustee will be required to treat both the original Bond and any duplicate Bond as being Outstanding for the purpose of determining the principal amount of Bonds which may be issued under the Indenture or for the

purpose of determining any percentage of Bonds Outstanding thereunder, but both the original and duplicate Bond will be treated as one and the same.

ISSUANCE OF ADDITIONAL SERIES OF BONDS

Capital Project Fund. (a) The Commission has covenanted and agreed to maintain under the Indenture the separate fund known as the "Clean Water Revenue Bond Capital Project Fund" (called the "Capital Project Fund" in the Indenture). The Treasurer will hold the amounts on deposit in the Capital Project Fund. The Controller will maintain and account for the Capital Project Fund so long as any moneys are on deposit therein. The Commission will establish within the Capital Project Fund separate Capital Project Accounts relating to separate Series of Bonds, to the extent needed for a Series of Bonds.

(b) Upon completion of the acquisition and construction of the Project, the Commission may direct the transfer of any remaining balance in the Capital Project Fund to the Interest Fund. Upon completion of acquisition or construction of the Project or any portion thereof, the Commission will file with the Trustee a Certificate or Written Statement of the Commission stating the fact and date of such completion of construction.

(c) The moneys in the Capital Project Fund will be held by the Treasurer in trust and applied to the costs of acquisition, construction, expansion, improvement, financing and refinancing of the Project and the expenses incident thereto or connected therewith, including, if necessary, interest to the extent permitted by law, reimbursement to the Commission for expenses incurred prior to the issuance of the applicable Series of Bonds to the extent permitted by law, and the costs incurred in connection with the issuance of the applicable Series of Bonds to the extent not provided for.

(d) The Treasurer will pay out moneys from the Capital Project Fund only upon warrants drawn by the Controller in the mariner provided by law. No withdrawals will be made from the Capital Project Fund for any purpose not authorized by law.

Issuance of Additional Bonds. In addition to the 2003 Refunding Series A Bonds, the Commission may, subject to the requirements of the Law, by Supplemental Indenture establish one or more other Series of Bonds payable from Net Revenues on a parity with the 2003 Refunding Series A Bonds and secured by a lien upon and pledge of Net Revenues equal to the lien and pledge securing the 2003 Refunding Series A Bonds, and the Commission may issue and the Trustee may authenticate and deliver Bonds of any Series so established, in such principal amount and for such lawful purpose or purposes (including refunding of any Bonds issued under the Indenture and then Outstanding) as will be determined by the Commission in said Supplemental Indenture, but only upon compliance by the Commission with the provisions of the Indenture, and subject to the following specific conditions, which have been made conditions precedent to the issuance of any such additional Series of Bonds by the Indenture:

(a) No Event of Default has occurred and is continuing under the Indenture or any Supplemental Indenture and no event has occurred which, but for the passage of time or the giving of notice, would constitute an Event of Default under the Indenture or any Supplemental Indenture.

(b) The Supplemental Indenture providing for the issuance of such additional Series of Bonds requires that the amount on deposit in the Bond Reserve Fund to be established pursuant to the Indenture be increased, if and to the extent necessary, immediately upon the receipt of the proceeds of the sale of such additional Series of Bonds, to an amount equal to the Required Reserve. This deposit may be made from such proceeds or any other source, as provided in the Supplemental Indenture.

(c) The Supplemental Indenture providing for the issuance of such additional Series of Bonds provides for the payment of interest and principal as follows:

(i) Principal on such Additional Bonds is payable either semiannually on April 1 and October 1 of each year in which principal falls due or annually on October 1 of each year in which principal falls due, and Term Bonds of any Series have a principal maturity date of October 1. Interest on such Additional Bonds that are Current Interest Bonds is payable semiannually on April 1 and October 1 of each year excepting the first year, provided that the first installment of interest is payable on either April 1 or October 1 and is for a period of not longer than twelve months and that the interest is payable thereafter semiannually on April 1 and October 1.

(ii) Notwithstanding the foregoing, the Supplemental Indenture authorizing the issuance of such Additional Bonds may provide for the payment of principal and interest on dates other than those specified above if the Supplemental Indenture provides for the monthly payment of a portion of interest and principal becoming due and payable on the succeeding Interest Payment Date and Principal Payment Date, as applicable, as set forth in detail in the Supplemental Indenture.

(iii) Interest on any Bonds constituting Variable Rate Indebtedness or Tender Indebtedness may be payable on such Payment Dates as will be specified in the Supplemental Indenture.

(d) Fixed serial maturities or mandatory Minimum Sinking Fund Account Payments, or any combination thereof, is established in amounts sufficient to provide for the retirement of all of the Bonds of such additional Series on or before their respective maturity dates, unless such Bonds are Balloon Indebtedness.

(e) The aggregate principal amount of Bonds issued under the Indenture does not exceed any limitation imposed by law or by any Supplemental Indenture.

(f) After the sale of the Series of Additional Bonds proposed to be issued (but prior to the issuance and delivery thereof and receipt of payment therefor), the Commission will file the following documents with the Trustee; these documents will, with respect to such Series of Additional Bonds, be based upon the actual interest rate or rates determined at the time of sale thereof (except that, with respect to Variable Rate Indebtedness, the interest rate for the Series of Additional Bonds will be calculated in accordance with the provisions of the definitions of Average Annual Debt Service and Maximum Annual Debt Service).

(i) If any portion of the proceeds of such Series of Bonds is to be used to finance construction of a Project, a certificate of the Consulting Engineers setting forth (A) the estimated date of completion for the portion of the Project for which such Series of Bonds is being issued and for any other uncompleted portion of the Project, and (B) an estimate of the cost of construction of such portion of the Project and of any other uncompleted portion of the Project.

(ii) A Certificate of the Commission setting forth for each of the next three Fiscal Years (and if any portion of the proceeds of such Series of Bonds is to be used to finance construction, the three Fiscal Years following the Fiscal Year in which the Consulting Engineers estimate such portion of the Project will be completed) estimates of (A) Revenues, (B) Operation and Maintenance Costs of the Enterprise and (3) Net Revenues.

(iii) A Certificate of the Commission setting forth (A) the estimates of Net Revenues, as set forth in the Certificate of the Commission pursuant to paragraph (ii) above, for each of such three Fiscal Years (and, if any portion of the proceeds of such Series of Bonds is to be used to finance construction, the three Fiscal Years following the Fiscal Year in which the Consulting Engineers estimate such portion of the Project will be completed), (B) the Annual Debt Service for each of such Fiscal Years, including Annual Debt Service as estimated in such Certificate of the Commission with respect to future Series of Bonds, if any, which such Certificate of the Commission will estimate will be required to complete payment of the cost of construction of such portion of the Project and any other uncompleted portion of the Project, and (C) demonstrating that the estimated Net Revenues (together with any fund balances of the Commission which

are available for the payment of Debt Service but excluding the Bond Reserve Fund) in each of the Fiscal Years set forth in (ii) above is at least equal to 1.25 times the Annual Debt Service for such respective Fiscal Years.

(iv) A Certificate of the Commission that all of the requirements of the Indenture have been met.

Issuance of Additional Bonds for Refunding. In addition to the 2003 Refunding Series A Bonds, the Commission may, subject to the requirements of the Law, by Supplemental Indenture establish one or more other Series of Bonds payable from Net Revenues on a parity with the 2003 Refunding Series A Bonds and secured by a lien upon and pledge of Net Revenues equal to the lien and pledge securing the 2003 Refunding Series A Bonds, and the Commission may issue, and the Trustee may authenticate and deliver, Bonds of any Series so established, for the purpose of refunding any Bonds issued under the Indenture and then Outstanding, but only upon compliance by the Commission with the provisions of the Indenture, and subject to the following specific conditions, which are made conditions precedent to the issuance of any such additional Series of Bonds by the Indenture:

(a) No Event of Default has occurred and is continuing under the Indenture or any Supplemental Indenture and no event has occurred which, but for the passage of time or the giving of notice would constitute an Event of Default under the Indenture or any Supplemental Indenture.

(b) The Supplemental Indenture providing for the issuance of such additional Series of Bonds requires that the amounts on deposit in the Bond Reserve Fund to be established pursuant to the Indenture be increased, if necessary, upon the receipt of the proceeds of the sale of such additional Series of Bonds to an amount equal to the Required Reserve. This deposit may be made from such proceeds or any other source, as provided in the Supplemental Indenture.

(c) The Supplemental Indenture providing for the issuance of such additional Series of Bonds provides for the payment of interest and principal as follows:

(i) Principal on such Additional Bonds is payable either semiannually on April 1 and October 1 of each year in which principal falls due or annually on October 1 of each year in which principal falls due, and Term Bonds of any Series will have a principal maturity date of October 1. Interest on such Additional Bonds that are Current Interest Bonds is payable semiannually on April 1 and October 1 of each year excepting the first year, provided that the first installment of interest is payable on either April 1 or October 1 and is for a period of not longer than twelve months and that the interest will be payable thereafter semiannually on April 1 and October 1.

(ii) Notwithstanding the foregoing, the Supplemental Indenture authorizing the issuance of such Additional Bonds may provide for the payment of principal, and interest on dates other than those specified above if the Supplemental Indenture provides for the monthly payment of a portion of interest and principal becoming due and payable on the succeeding Interest Payment Date and Principal Payment Date, as applicable, as set forth in detail in the Supplemental Indenture.

(iii) Interest on any Bonds constituting Variable Rate Indebtedness or Tender Indebtedness may be payable on such Payment Dates as specified in the Supplemental Indenture providing for the issuance of such Bonds.

(d) Fixed serial maturities or mandatory Minimum Sinking Fund Account Payments, or any combination thereof, are established in amounts sufficient to provide for the retirement of all of the Bonds of such additional Series on or before their respective maturity dates, unless such Bonds are Balloon Indebtedness.

(e) The aggregate principal amount of Bonds issued under the Indenture does not exceed any limitation imposed by law or by any Supplemental Indenture.

(f) The proceeds of the Bonds of such additional Series will be used, together with any other available moneys, to refund (by defeasance, current refunding or crossover refunding) all or a portion of the Bonds then Outstanding, and the Average Annual Debt Service for the Bonds of such additional Series (during the period from their issuance to their last maturity date) will be equal to or less than the Average Annual Debt Service on the Bonds to be refunded (during the period from the issuance of the additional Series to the last maturity date of the Bonds to be refunded).

(g) After the sale of the additional Series of Bonds proposed to be issued (but prior to the issuance and delivery thereof and receipt of payment therefor), the Commission will file the following documents with the Trustee; these documents will, with respect to such additional Series of Bonds, be based upon the actual interest rate or rates determined at the time of sale thereof.

(i) A Certificate of the Commission that all of the requirements of the Indenture have been met.

(ii) A certificate of one or more Qualified Financial Advisors that the requirements of clause (f) above have been met.

Proceedings for the Issuance of Additional Series of Bonds. (a) Whenever the Commission determines to issue an additional Series of Bonds pursuant to the Indenture, as the case may be, the Commission will execute or adopt a Supplemental Indenture providing for the issuance of such additional Series of Bonds.

(b) Such Supplemental Indenture will specify the maximum principal amount of Bonds of such Series, provide for the distinctive designation of Bonds of such Series, and prescribe the other terms and conditions of such additional Series of Bonds in accordance with the Indenture and subject to the provisions of the Indenture, as the case may be. The Commission may by such Supplemental Indenture prescribe any provisions respecting the Bonds of such Series not inconsistent with the terms of the Indenture, including registration, transfer and exchange provisions, provisions for the payment of principal and interest and sinking fund provisions.

(c) Before such additional Series of Bonds may be issued and delivered, the Commission will file the following documents with the Trustee:

(i) An Opinion of Counsel setting forth (1) that such counsel has examined the Supplemental Indenture and found it to be in compliance with the requirements of the Indenture; (2) that the execution and delivery of the additional Series of Bonds have been sufficiently and duly authorized by the Commission; (3) that said additional Series of Bonds, when duly executed by the Commission and, if required, authenticated and delivered by the Trustee, will be valid and binding special obligations of the Commission, payable from Net Revenues as provided in the Indenture; and (4) that the issuance of the additional Series of Bonds will not adversely affect the exclusion from federal income taxation of interest on any Bonds then Outstanding.

(ii) The certificates and reports required by the Indenture if the Additional Bonds constitute an additional lien on the Net Revenues or if the Additional Bonds are issued to refund any Outstanding Bonds, as appropriate.

(iii) The Supplemental Indenture, duly executed or certified and approved by the Trustee.

Upon the delivery to the Trustee of the foregoing instruments, the Trustee will authenticate and deliver said additional Series of Bonds, in the aggregate principal amount specified in such Supplemental Indenture, to, or upon the Written Request of, the Commission, when such additional Series of Bonds is presented to it for that purpose.

No Issuance of Additional Bonds or Other Obligations Except as Permitted in the Indenture; Exceptions. So long as any of the Bonds remain Outstanding, the Commission may issue any Additional Bonds or obligations payable from Net Revenues on a parity with the Bonds only pursuant to the Indenture, *except* under any of the following conditions, in which case none of the limitations or restrictions on the issuance of additional Series of Bonds set forth in the Indenture will be applicable: (a) if the Owners of a majority in aggregate amount of the Bond Obligation and any Credit Provider consent in writing to the issuance of such Additional Bonds or obligations, or (b) the obligation constitutes debt of the Commission (including without limitation loan agreements entered into between the Commission and the State of California (or any board, department or agency thereof) to finance or refinance additions, betterments, extensions, repairs, renewals or replacements to the Enterprise) payable by its terms from Net Revenues on a subordinate basis to the payment of Debt Service on the Bonds.

In addition, the Commission may enter into additional Senior State Loans and into Parity State Loans if no Event of Default has occurred and is continuing under the Indenture or any Supplemental Indenture (and no event has occurred which, but for the passage of time or the giving of notice, would constitute an Event of Default under the Indenture or any Supplemental Indenture) and, in connection with the execution and delivery of such Senior State Loans or Parity State Loans, as applicable, the Commission delivers a Certificate to the Trustee setting forth, for each of the next three Fiscal Yeats after the delivery of the Senior State Loans or Parity State Loans, as applicable, (i) the Revenues, Operation and Maintenance Costs of the Enterprise and Net Revenues (assuming the delivery of the Senior State Loans, as applicable) and (ii) the Annual Debt Service (assuming the delivery of the Parity State Loans, as applicable), and demonstrating that the estimated Net Revenues (together with any fund balances of the Commission, which are available for Debt Service, but excluding the Bond Reserve Fund), in each of such Fiscal Years is at least equal to 1.25 times the Annual Debt Service.

Validity of Bonds. The validity of the authorization and issuance of the Bonds will not be dependent on or affected in any way by any proceedings taken by the Commission for the improvement of the Enterprise, or by any contracts made by the Commission in connection therewith, or the failure to construct the Project, the Enterprise or any part thereof. The recital contained in the Bonds that they are regularly issued pursuant to the Law will be conclusive evidence of their validity and of compliance with the provisions of law in their issuance.

GENERAL REDEMPTION PROVISIONS

Selection of Bonds for Redemption. Whenever less than all of the Bonds of any one maturity and tenor of any Series are called for redemption and such Bonds are redeemable by lot, the Trustee will select the Bonds of such maturity and tenor to be redeemed, from the Outstanding Bonds of such maturity and tenor, by lot or by any other manner which the Trustee deems fair and equitable. For purposes of such selection, Bonds will be deemed to be composed of $5,000 portions (of principal in the case of Current Interest Bonds or of Accreted Value at maturity in the case of Capital Appreciation Bonds) and any such portion may be redeemed separately. The Trustee will promptly notify the Commission in writing of the numbers of the Bonds so selected for redemption.

Notice of Redemption. (a) The Trustee will mail notice of redemption, not less than 30 nor more than 60 days prior to the redemption date, to (i) the respective Owners of any Bonds designated for redemption at their addresses appearing on the bond registration books of the Trustee, (ii) the Securities Depositories and (iii) one or more Information Services.

(b) Each notice of redemption will state the date of such notice, the Series of Bonds to be redeemed, the date of issue of such Series of Bonds, the redemption date, the redemption price including, in the case of Capital Appreciation Bonds, the Accreted Value thereof, the place or places of redemption (including the name and appropriate address or addresses of the Trustee), the CUSIP number (if any) of the maturity or maturities, and, if less than all of any such maturity is to be redeemed, the distinctive certificate numbers of the Bonds of such maturity, to be redeemed, and, in the case of Bonds to be redeemed in part only, the respective portions of the principal amount thereof to be redeemed. Each such notice will also state that on said date there will become due and payable on each of said Bonds the principal amount thereof or of said specified portion of the principal amount thereof in the case of a Bond to be redeemed in part only, together with interest accrued thereon to the redemption date, and the premium, if any, thereon (such premium to be specified) and that from and after such redemption date interest thereon will cease to accrue, and will require that such Bonds be then surrendered at the address or addresses of the Trustee specified in the redemption notice.

(c) Failure by the Trustee to give notice pursuant to the Indenture to any one or more of the Information Services or Securities Depositories, or the insufficiency of any such notices will not affect the sufficiency of the proceedings for redemption. Neither failure by the Trustee to mail notice of redemption pursuant to the Indenture to any one or more of the respective Owners of any Bonds designated for redemption nor any defect in such notice will affect the sufficiency of the proceedings for redemption with respect to the Owners to whom such notice was mailed.

(d) Notice of redemption of Bonds will be given by the Commission or, at the request of the Commission, by the Trustee for and on behalf and at the expense of the Commission.

Partial Redemption of Bond. Upon surrender of any Bond redeemed in part only (except as otherwise provided in the Indenture) the Commission will execute and the Trustee will authenticate and deliver to the Owner thereof, at the expense of the Commission, a new Bond or Bonds of authorized denominations equal in aggregate principal amount to the unredeemed portion of the Bond surrendered and of the same Series, tenor, interest rate and maturity.

Effect of Redemption. (a) When notice of redemption has been duly given under the Indenture, and moneys for payment of the redemption price are held by the Trustee, the Bonds so called for redemption will, on the redemption date designated in such notice, become due and payable at the redemption price specified in such notice; and from and after the date so designated, interest on the Bonds so called for redemption will cease to accrue, those Bonds will cease to be entitled to any benefit or security under the Indenture, and the Owners of those Bonds will have no rights in respect thereof except to receive payment of the redemption price thereof. The Trustee will, upon surrender for payment of any of said Bonds (except as otherwise provided in the Indenture), pay such Bonds at the redemption price as aforesaid, together with accrued interest thereon.

(b) All Bonds redeemed pursuant to the provisions of the Indenture will be cancelled upon surrender, and no Bonds will be issued in place thereof.

Rescission of Notice of Redemption. The Commission may, at its option, prior to the date fixed for redemption in any notice of redemption rescind and cancel such notice of redemption by Written Request to the Trustee and the Trustee will mail notice of such cancellation to the recipients of the notice of redemption being cancelled.

REVENUES AND FUNDS

Pledge and Assignment of Net Revenues; Revenue Fund.

(a) Establishment of Revenue Fund. In accordance with the Charter, but subject to the budget and fiscal provisions of the Charter, whenever revenue bonds issued by the Commission pursuant to the Charter or refunding bonds of such revenue bonds are Outstanding, all of the Revenues of the Enterprise will be set aside and deposited into a fund in the City treasury previously established and known as the "Enterprise Revenue Fund" (called the "Revenue Fund" in the Indenture). All amounts paid into the Revenue Fund will be maintained by the Treasurer separate and apart from all other City funds. Separate accounts will be kept of the Revenue Fund with respect to receipts and disbursements. The Revenue Fund will be exempted from the requirements of the Charter.

(b) Priority of Disbursements from Revenue Fund. Moneys in the Revenue Fund, including earnings thereon, will be appropriated, transferred, expended or used for the following purposes, and only in accordance with the following priority: (i) the payment of the Operation and Maintenance Costs of the Enterprise; (ii) the payment of Senior State Loans; (iii) the payment of Bonds, Parity State Loans, Policy Costs and amounts due as reimbursement under any Letter of Credit Agreement, as provided in the Indenture and, as applicable, any Supplemental Indenture; and (iv) any other lawful purpose of the Commission.

(c) Pledge of Net Revenues; Perfection of Lien. (i) Subject to the provisions above, all of the Net Revenues (except amounts on deposit in the various Rebate Funds) are irrevocably pledged by the Indenture to the punctual payment of the principal of and interest and redemption premium, if any, on the Bonds and the Policy Costs, and the Net Revenues will not be used for any other purpose while any of the Bonds remain Outstanding or Policy Costs remain unpaid; except that the Net Revenues may be used for such purposes as are expressly permitted in the Charter and the Indenture. Pursuant to Section 5451 of the California Government Code, said pledge will constitute a lien on and security interest in the Net Revenues for the payment of the Bonds and the Policy Costs in accordance with the terms thereof and of the Indenture, and will immediately attach to the collateral and be effective, binding, and enforceable against the Commission, its successors, purchasers of the Net Revenues, creditors and all others asserting any rights thereto, irrespective of whether such parties have notice of such pledge and without the need for any physical delivery, recordation, filing or further act.

(ii) The Commission will not take any action which alters the pledge of Net Revenues or the order of priority of payment of the Net Revenues used for the payment of principal of and interest on the Bonds.

(d) Disposition of Excess Amounts in Revenue Fund. Except as otherwise provided in a Supplemental Indenture, all moneys remaining in the Revenue Fund on each October 5 (or on such earlier day as the amounts required for the transfers set forth in the Indenture are on deposit in the Revenue Fund), after the setting aside and transferring of all of the amounts required to be set aside or transferred by the Treasurer under the Indenture, will be applied for any lawful purpose of the Commission.

Establishment and Maintenance of Funds for Net Revenues; Use and Withdrawal of Revenues. (a) The Trustee will establish and maintain, in trust, so long as any Bonds are Outstanding, the Interest Fund, Principal Fund, and Bond Reserve Fund. The Trustee will hold all amounts deposited in each of these funds in trust and apply, use and withdraw those funds only for the purposes authorized in the Indenture.

(b) After the making of the payments required by the Indenture the Commission will transfer all Net Revenues in the Revenue Fund, on a parity basis, as follows: (i) to the Trustee, at the times and in the amounts set forth in the Indenture, for deposit in the following funds, to the extent necessary, in the following order of priority: the Interest Fund, Principal Fund, and Bond Reserve Fund, and (ii) as needed for the payment of any Parity State Loans. The requirements of the Interest Fund, the Principal Fund and the Reserve

Fund at the time of deposit will be satisfied before any transfer is made to any fund subsequent in priority. The Trustee will disburse amounts on deposit in these funds in accordance with the Indenture.

Interest Fund. (a) On or before 5 Business Days before each Interest Payment Date, the Treasurer will pay to the Trustee for deposit in the Interest Fund an amount equal to the interest becoming due and payable on the Outstanding Current Interest Bonds on that Interest Payment Date (taking into account amounts on deposit in the Interest Fund and available for the payment of interest on such Interest Payment Date and excluding any interest for which there is moneys deposited in the Interest Fund or the Capital Project Fund from the proceeds of any Series of Bonds, or other source and reserved as capitalized interest to pay such interest on such Interest Payment Date).

(b) The Trustee will use and withdraw moneys in the Interest Fund solely for the purpose of paying the interest on the Bonds as it becomes due and payable (including accrued interest on any Bonds purchased or redeemed prior to maturity under the Indenture).

All of the Refundable Credits received by the Commission will be deposited immediately upon receipt in the Interest Fund, and such Refundable Credits are irrevocably pledged by the Indenture to the punctual payment of the interest on the Bonds issued as Build America Bonds, and the Refundable Credits will not be used for any other purpose while any of the Bonds issued as Build America Bonds remain Outstanding. Pursuant to Section 5451 of the California Government Code, this pledge constitutes a lien on and security interest in the Refundable Credits for the payment of interest on the Bonds issued as Build America Bonds in accordance with the terms thereof and of the Indenture, and will immediately attach and be effective, binding, and enforceable against the Commission, its successors, purchasers of the Refundable Credits, creditors and all others asserting any rights thereto, irrespective of whether such parties have notice of such pledge and without the need for any physical delivery, recordation, filing or further act. Notwithstanding the foregoing deposit and pledge, the Refundable Credits are not considered Revenues or included in the calculation of Revenues under the Indenture. Additionally, in calculating the amount that the Treasurer pays to the Trustee for deposit in the Interest Fund as provided above, the Treasurer and the Trustee will take into account Refundable Credits only if they have been deposited in the Interest Fund on or prior to the fifth Business Day prior to the applicable Interest Payment Date and have not been previously expended to pay Debt Service on the Bonds issued as Build America Bonds or otherwise transferred out of the Interest Fund.

Principal Fund; Sinking Fund Accounts. (a) On or before 5 Business Days before each Principal Payment Date, the Treasurer will pay to the Trustee for deposit in the Principal Fund an amount equal to at least the following:

(i) the aggregate amount of Bond Obligation becoming due and payable on the Outstanding Serial Bonds of all Series having a maturity date on the next succeeding Principal Payment Date, plus

(ii) the aggregate of the Minimum Sinking Fund Account Payments to be paid on the next succeeding Principal Payment Date into the respective Sinking Fund Accounts for the Term Bonds of all Series for which Sinking Fund Accounts have been created (all such Minimum Sinking Rind Account Payments will be made without priority of any payment into any one such Sinking Fund Account over any other such payment), plus

(iii) if any Balloon Indebtedness is Outstanding and principal is due on such Balloon Indebtedness on or prior to the next succeeding Principal Payment Date, sufficient amounts to pay when due the Balloon Indebtedness, plus

(iv) if any Letter of Credit Agreement has been entered into on a parity with the Bonds, sufficient amounts to pay when due the obligations of the Commission under such Letter of Credit Agreement due on the next succeeding Maturity Date.

(b) If the moneys in the Principal Fund on any Principal Payment Date are less than the amount of Bond Obligation and redemption premium on the Outstanding Term Bonds required to be redeemed or paid at maturity on such Principal Payment Date, plus the Balloon Indebtedness and the obligations of the Commission under any Letter of Credit Agreements due on such Principal Payment Date, then the moneys in the Principal Fund will be applied on a Proportionate Basis and in such proportion as the Serial Bonds, Term Bonds, Balloon Indebtedness and Letter of Credit Agreement obligations bear to each other.

(c) No deposit need be made into the Principal Fund so long as it contains an amount equal to an amount sufficient to make the payment required above.

(d) The Trustee will use and withdraw all moneys in the Principal Fund solely for the purpose of paying the Bond Obligation of the Bonds, any Balloon Indebtedness and any Letter of Credit Agreement obligations when due and payable, except that the Trustee will use and withdraw all moneys in any Sinking Fund Account (except as otherwise provided in the Indenture) only to purchase or to redeem or to pay at maturity Term Bonds of the Series for which such Sinking Fund Account was created, as provided in the Indenture or in any Supplemental Indenture.

Bond Reserve Fund; Reserve Accounts. (a) On or before 5 Business Days before each Interest Payment Date, the Treasurer will pay to the Trustee for deposit in the Bond Reserve Fund the aggregate amount of each unreplenished prior withdrawal from the Bond Reserve Fund until there is on deposit in the Bond Reserve Fund a balance equal to the Required Reserve.

(b) If a Bond Reserve Fund Policy satisfies all or a portion of the Required Reserve and a drawing is made on the Bond Reserve Fund Policy, on or before 5 Business Days before each Interest Payment Date, the Treasurer will pay to the Trustee or to the Bond Reserve Fund Policy Provider, with notice to the Trustee, an amount at least equal to the aggregate amount of Policy Costs owing with respect to such Bond Reserve Fund Policy. If the Trustee receives such payment, it will immediately remit the same to the Bond Reserve Fund Policy Provider.

(c) No deposit need be made into the Bond Reserve Fund so long as there exists in the Bond Reserve Fund an amount equal to the Required Reserve, or when and if the sum of the amounts contained (excluding all Bond Reserve Fund Policies) therein and in the Interest Fund and in the Principal Fund is at least equal to the sum of the aggregate principal amount of all of the Bonds then Outstanding and all of the interest then due or thereafter to become due on all such Bonds.

(d) The Trustee will establish and hold within the Bond Reserve Fund a 2003 Refunding Series A Reserve Account with respect to the 2003 Refunding Series A Bonds. In addition, the Trustee will establish and hold a Reserve Account for each Series of Additional Bonds issued under the Indenture, if and to the extent required by the Supplemental Indenture pursuant to which that Series of Bonds is issued. Upon the issuance of a Series of Additional Bonds, there will be deposited into the Reserve Account for that Series an amount equal to the Required Reserve, if any, established for that Series of Bonds under the Supplemental Indenture pursuant to which that Series of Bonds is issued. Upon the issuance of a Series of Additional Bonds, the Commission will advise the Trustee of the Required Reserve to be maintained in the Reserve Account for that Series. Unless otherwise provided in the Supplemental Indenture pursuant to which a Series of Bonds is issued, the Reserve Account established with respect to a Series of Bonds will be available only to pay Debt Service on such Series of Bonds, and will not be available to pay Debt Service on any other Series of Bonds, provided that amounts in the Reserve Accounts (the "Prior Reserve Accounts") with respect to the Series of Bonds (the "Prior Series of Bonds") issued prior to the effective date of the amendments set forth in the First Amendment, will be available to pay the Debt Service on all the Prior Series of Bonds, except that the Trustee will apply any Bond Reserve Fund Policy on deposit in a Prior Reserve Account solely to the payment of Debt Service on the Prior Series of Bonds to which such Prior Reserve Account relates, and such Bond Reserve Fund Policy will not be available for payment of any other Series of Bonds.

(e) Except with respect to the Prior Reserve Accounts and the Prior Series of Bonds, the Trustee will use and withdraw cash amounts on deposit in each respective Reserve Account within the Bond Reserve Fund solely to pay the principal of, Minimum Sinking Fund Account Payments with respect to, and interest on, the Series of Bonds with respect to which that Reserve Account was established (unless otherwise provided in the Supplemental Indenture pursuant to which a Series of Bonds was issued), in the event that no other moneys are available therefor, or for payment or redemption of such Series of Bonds. Additionally, unless otherwise provided in the Supplemental Indenture pursuant to which a Series of Bonds was issued, the Trustee will apply any Bond Reserve Fund Policy on deposit in a Reserve Account solely to the payment of the Series of Bonds to which such Reserve Account relates, and such Bond Reserve Fund Policy will not be available to pay Debt Service on any other Series of Bonds. With respect to the Prior Reserve Accounts, the Trustee will use and withdraw cash amounts on deposit in the respective Prior Reserve Accounts within the Bond Reserve Fund on a proportionate basis solely to pay the principal of, Minimum Sinking Fund Account Payments with respect to and interest on any of the Outstanding Prior Series of Bonds in the event that no other moneys are available therefor, or for payment or redemption of Outstanding Bonds. However, as stated above, the Trustee will apply any Bond Reserve Fund Policy on deposit in a Prior Reserve Account solely to the payment of the Prior Series of Bonds to which such Prior Reserve Account relates, and such Bond Reserve Fund Policy will not be available for payment of any other Series of Bonds.

(f) Following application of all other funds held in the Reserve Account relating to a Series of Bonds, the Trustee will draw under any Bond Reserve Fund Policy issued with respect to such Series of Bonds, in a timely manner and pursuant to the terms of such Bond Reserve Fund Policy, to the extent necessary in order to obtain sufficient funds on or prior to the date such funds are needed to pay the Bond Obligation of, Minimum Sinking Fund Account Payments with respect to, and interest on such Series of Bonds when due.

(g) If the Trustee has notice, that any payment of principal of or interest on a Bond has been recovered from its Bondowner under the United States Bankruptcy Code by a trustee in bankruptcy in accordance with the final, nonappealable order of a court having competent jurisdiction, the Trustee, pursuant to the terms of the Bond Reserve Fund Policy, if any, securing the Series of Bonds of which such Bond is a part, will so notify the Bond Reserve Fund Policy Provider and draw on such policy to the lesser of the extent required or the maximum amount of such policy in order to pay to such Bondowners the principal of and interest so recovered.

(h) If and to the extent that more than one Bond Reserve Fund Policy satisfies the portion of the Required Reserve relating to a Series of Bonds, drawings under such Bond Reserve Fund Policies and payment of Policy Costs with respect to such Bond Reserve Fund Policies will be made on a pro rata basis (calculated by reference to the maximum amounts of such Bond Reserve Fund Policies).

(i) If a Bond Reserve Fund Policy is deposited in a Reserve Account in which cash has been previously deposited in satisfaction of the Requited Reserve for the applicable Series of Bonds, the Trustee will release cash from that Reserve Account in an amount equal to the Bond Reserve Fund Policy being deposited, and will transfer the cash so released to the Commission to be used for any lawful purpose, *provided, however*, that the Commission will ensure that the use of any cash so released will not adversely affect the exclusion from gross income of the interest on the Bonds under Section 103 of the Code.

(j) Deposits to the Bond Reserve Fund will be applied on a pro rata basis to the respective Reserve Accounts, calculated by reference to the amounts initially deposited in such Reserve Accounts, and within each Reserve Account first to satisfying any portion of the Required Reserve to be maintained within such Reserve Account not covered by a Bond Reserve Fund Policy, and second to the pro rata payment of Policy Costs until satisfied.

(k) So long as the Commission is not in default under the Indenture, and in each Reserve Account there is a balance equal to the Required Reserve for the Series of Bonds relating to that Reserve Account, the

Trustee will withdraw any amount in the Bond Reserve Fund in excess of the Required Reserve, semiannually on March 15 and September 15 of each year, and transfer that amount to the Treasurer for deposit in the Revenue Fund or, during the period of construction of the Project or any portion thereof, the Capital Project Account for the related Series of Bonds. Notwithstanding the foregoing, the Commission will have the right to request the Trustee to withdraw excess amounts on deposit in the Bond Reserve Fund at any time.

(l) Except as provided in the Indenture with respect to the Prior Reserve Accounts and the Prior Series of Bonds, unless otherwise provided in a Supplemental Indenture, amounts on deposit in any Reserve Account will be available for the payment of Debt Service only with respect to the Series of Bonds for which that Reserve Account was established.

Deposit and Investment of Moneys in Funds. (a) All moneys held by the Treasurer in the Revenue Fund or the Capital Project Fund may be invested in Legal Investments, maturing not later than the date on which such moneys are required for payment by the Treasurer.

(b) All moneys held by the Trustee and allocated to any of the funds held by it, subject to the restrictions set forth in the Arbitrage Certificate, will be invested in Permitted Investments, as directed by the Commission, maturing not later than the date on which such moneys are required for payment by the Trustee, except that moneys in the Bond Reserve Fund will be deposited or invested in Permitted Investments which mature not more than seven years from the date of investment or the final date of maturity of the Outstanding Bonds, whichever is earlier.

(c) If at any time any of the investments stated to be Permitted Investments under the Indenture cease to be a Legal Investment for the funds held under the Indenture, the Commission will so advise the Trustee by a Written Statement. The Trustee will not be responsible for making any investment which is not a Legal Investment if the Commission has not previously delivered a Written Request or Statement correctly advising the Trustee that such investment was no longer a Legal Investment.

(d) Permitted Investments on deposit in the Bond Reserve Fund or in any account therein will not have a maturity extending beyond five years from the date of acquisition thereof unless otherwise approved by the Credit Provider or unless such Permitted Investment is described in clause (i) of the definition thereof.

(e) For the purpose of determining the amount of money in the Bond Reserve Fund, all investments of moneys therein will be valued at least annually at the market value of such investments.

(f) All interest received on any moneys held and invested by the Treasurer or the Trustee under the Indenture will be deposited in the Revenue Fund, except: (i) all interest received on any moneys invested in the Principal Fund, Interest Fund or Rebate Fund will remain in the Principal Fund, Interest Fund or Rebate Fund, respectively, and (ii) prior to receipt by the Trustee of notice of completion of construction of the Project or any portion thereof, all interest received on any moneys invested in the Capital Project Fund will remain in the Capital Project Fund held by the Treasurer; and (iii) all interest on any amounts on deposit in the Bond Reserve Fund to the extent that amounts on deposit in the Bond Reserve Fund exceed the Required Reserve will be deposited in the Interest Fund.

(g) The Trustee may sell or present for redemption any obligations so purchased by it whenever it is necessary in order to provide moneys to meet any payment, and the Trustee will not be liable or responsible for any loss resulting from such investment. The Trustee may act as principal or agent in the acquisition or disposition of any investment. The Trustee may commingle any of the moneys held by it under the Indenture for investment purposes only; provided, however, that the Trustee will account separately for the moneys belonging to each fund or account established pursuant to the Indenture and held by it.

Interest Rate Swaps. (a) The Commission may and the Trustee will, upon the Written Request or Statement of the Commission, enter into an interest rate swap agreement corresponding to the interest rate or rates payable on a Series of Bonds or any portion thereof, provided that the Trustee is supplied with an Opinion of Counsel to the effect that (i) such action is permitted under the laws of the State of California, (ii) entering into the interest rate swap agreement will not adversely affect the tax-exempt status of interest on the bonds, and (iii) entering into the interest rate swap agreement complies with the terms of the Indenture. The amounts received by the Commission or the Trustee, if any, under such a swap agreement may be applied to the deposits required under the Indenture. The entity with which the Commission or the Trustee may contract for an interest rate swap is limited to entities that are rated in one of the two highest short-term or long-term debt rating categories by Moody's and S&P. If the Commission so designates, amounts payable under the interest rate swap agreement will be made on a parity basis with payments on the Bonds and, in such event, the Commission will pay to the Trustee for deposit in the Interest Fund, at the times and in the manner provided by the Indenture, the amounts to be paid under such interest rate swap agreement, as if such amounts were additional interest due on the Bonds to which such interest rate swap relates.

(b) Notwithstanding the foregoing, the Commission will not enter into an interest rate swap agreement without first making the determination required pursuant to Section 5922 of the California Government Code.

COVENANTS OF THE COMMISSION

Payment of Principal and Interest. The Commission will punctually pay or cause to be paid the principal and interest (and premium, if any) to become due in respect of every Bond issued under the Indenture at the times and places and in the manner provided therein and in the Bonds, in strict conformity with the terms of the Bonds and of the Indenture, but solely from Net Revenues, as provided therein.

Against Encumbrances. (a) Subject to any rights of the United States of America or the State of California, and subject to the "Sale or Other Disposition of Property" provisions below, the Commission will not mortgage or otherwise encumber, pledge or place any charge upon the Enterprise or any part thereof, or upon any of the Net Revenues, prior to or on a parity with the Bonds, provided that Letter of Credit Agreements entered into in connection with Balloon Indebtedness, Variable Rate Indebtedness or Tender Indebtedness may be payable on a parity with the Bonds.

(b) So long as any Bonds are Outstanding, the Commission will not issue any bonds or obligations payable from Net Revenues or secured by a pledge, lien or charge upon Net Revenues prior to or on a parity with the Bonds, other than the Bonds, provided that Letter of Credit Agreements entered into in connection with Balloon Indebtedness, Variable Rate Indebtedness, or Tender Indebtedness may be payable on a parity with the Bonds.

(c) Nothing in the Indenture, and particularly nothing in the two foregoing clauses, will prevent the Commission from authorizing and issuing bonds, notes, warrants, certificates or other obligations or evidences of indebtedness which as to principal or interest, or both, (i) are payable from Net Revenues after and subordinate to the payment from Net Revenues of the principal of and interest on the Bonds, or (ii) are payable from moneys which are not Revenues as such term is defined in the Indenture.

Sale or Other Disposition of Property. (a) The Commission will not sell or otherwise dispose of the Enterprise or any part thereof essential to the proper operation of the Enterprise or to the maintenance of the Revenues except as expressly permitted in the Indenture. The Commission will not enter into any lease or agreement which impairs the operation of the Enterprise or impedes the rights of the Owners of the Bonds with respect to the Net Revenues or the operation of the Enterprise, but the Commission may enter into any lease or agreement concerning all or any part of the Enterprise if such lease or agreement will not impair the operation of the Enterprise or impede the rights of the Owners of the Bonds with respect to the Net Revenues or the operation of the Enterprise.

(b) Any real or personal property which has become nonoperative or which is not needed for the efficient and proper operation of the Enterprise, or any material or equipment which has worn out, may be sold if all of the net proceeds of such sale (less any amounts payable to the United States of America or the State of California or required by the United States of America or the State of California to be deposited in a restricted fund) are deposited in the Revenue Fund.

(c) The Commission expressly reserves by the Indenture the right to sell all or a portion of the Enterprise, and to enter into and execute agreements for and to complete such sale, but subject to the following specific conditions, which have been made conditions precedent to such sale by the Indenture:

(i) The Commission is in compliance with all covenants set forth in the Indenture and in all Supplemental Indentures theretofore adopted by the Commission, and no Event of Default has occurred and is continuing under the Indenture or any Supplemental Indentures theretofore adopted by the Commission (and no event has occurred which but for the passage of time would constitute an Event of Default under the Indenture or any Supplemental Indenture). The Commission will file a Certificate of the Commission to that effect with the Trustee.

(ii) The Commission determines by resolution whether the net proceeds of the sale (less any amounts payable to the United States of America or the State of California or required to be deposited in a restricted fund) are to be used for the redemption of Bonds or for the making of additions or improvements to or extensions of the Enterprise.

(iii) If the Commission determines that the net proceeds of the sale (less any amounts payable to the United States of America or the State of California or required to be deposited in a restricted fund) are to be used for the redemption of Bonds, such proceeds of the sale are deposited with the Trustee, and the following conditions are satisfied:

(A) The Commission adopts a resolution providing for the redemption of the maximum principal amount of Bonds which can be redeemed from such proceeds of such sale, or, if no Bonds are subject to redemption on the next succeeding Interest Payment Date, directing the Trustee (1) to hold such proceeds in trust, (2) to invest such proceeds in the investments permitted in the Indenture until any Bonds become redeemable, subject to any restrictions imposed by the Indenture, (3) to deposit the interest and income on such proceeds in the Revenue Fund as such interest and income is received, and (D) to use such proceeds to redeem Bonds in the amount and manner specified in the Indenture and any Supplemental Indenture on the first Interest Payment Date on which the Bonds can be redeemed; and a certified copy of such resolution has been filed with the Trustee along with a Written Request or Certificate of the Commission containing such direction.

(B) If such proceeds are not to be immediately used for the redemption of Bonds but instead are to be held by the Trustee until Bonds become redeemable, the Commission will file with the Trustee a written report of an Independent Certified Public Accountant stating (1) the amount of proceeds to be deposited with the Trustee from such sale, (2) an estimate of the total amount of Bond Obligation and the amount of Bonds of each maturity which could be redeemed from such proceeds on the first Interest Payment Date on which Bonds are redeemable, and (3) the estimated annual interest and income to be earned on such proceeds while held and invested by the Trustee. Such interest and income on such proceeds upon receipt by the Trustee will be deposited in the Revenue Fund and will be treated as Revenues for all purposes of the Indenture, including determining whether the Commission is in compliance with the rate covenant contained in the Indenture.

(C) If such proceeds of such sale are to be immediately used to redeem Bonds, the Net Revenues for the last Fiscal Year or last recorded twelve-month period preceding the date of the adoption by the Commission of the resolution authorizing such sale, less a deduction for the portion of such Net Revenues attributable to the portion of the Enterprise to be sold, all as shown by a certificate or opinion of

an Independent Certified Public Accountant or a written report of a Qualified Independent Consultant, have produced a sum equal to at least 1.25 times Maximum Annual Debt Service on the Bonds to be Outstanding following the redemption of Bonds from the proceeds of such sale.

(D) If such proceeds are not to be immediately used for the redemption of Bonds but instead are to be held by the Trustee until Bonds become redeemable, the Net Revenues for the last Fiscal Year or last recorded twelve-month period preceding the date of adoption by the Commission of the resolution authorizing such sale, less a deduction for the portion of such Net Revenues attributable to the portion Of the Enterprise to be sold, plus an allowance for the estimated annual interest or income to be earned on the invested proceeds of such sale while held and invested by the Trustee, all as shown by a certificate or opinion of an Independent Certified Public Accountant or a written report of a Qualified Independent Consultant, have produced a sum equal to at least 125 times Maximum Annual Debt Service.

(iv) If the Commission determines that the net proceeds of the sale (less any amounts payable to the United States of America or the State of California or required to be deposited in a restricted fund) are to be used for the making of additions or improvements to or extensions of the Enterprise, such proceeds of the sale will be deposited by the Treasurer in a special fund in trust to be held by the Treasurer to be used for the making of additions or improvements to or extensions of the Enterprise, and the condition set forth in the following sentence will have been satisfied. The Net Revenues for the last Fiscal Year or last recorded twelve-month period preceding the date of the adoption by the Commission of the resolution authorizing such sale, less a deduction for the portion of such Net Revenues attributable to the portion of the Enterprise to be sold, all as shown by a written report of an Independent Certified Public Accountant, plus

(A) An allowance for Net Revenues from any additions or improvements to or extensions of the Enterprise to be made with the proceeds of such sale or with the proceeds of Bonds previously issued, and also for Net Revenues from any such additions, improvements or extensions which have been made from moneys from any source but which, during all or any part of such Fiscal Year or recorded twelve-month period, were not in service, all in an amount equal to 100% of the estimated additional average annual Net Revenues to be derived from such additions, improvements and extensions for the first twenty-four months in which each addition, improvement or extension is respectively to be in operation, all as shown by the certificate or opinion of a Qualified Independent Consultant; and

(B) An allowance for earnings arising from any increase in the charges made for the use of the Enterprise which has become effective prior to such sale, but which, during all or any part of :such Fiscal Year or recorded twelve-month period, was not in effect, in an amount equal to 100% of the amount by which the Net Revenues would have been increased if such increase in charges had been in effect during the whole of such Fiscal Year or recorded twelve-month period, as shown by the certificate or opinion of a Qualified Independent Consultant;

have produced a sum equal to at least 1.25 times the Maximum Annual Debt Service on the Bonds then Outstanding. Any balance of such proceeds from any such sale not required by the Commission for the purposes aforesaid will be deposited in the Revenue Fund established pursuant to the Indenture and applied as provided in the Indenture.

Operation and Maintenance of Enterprise. The Commission will maintain and preserve the Enterprise in good repair and working order at all times from the Revenues available for such purposes, in conformity with standards customarily followed for municipal sanitary waste and storm water collection, treatment and disposal systems of like size and character. The Commission will from time to time make all necessary and proper repairs, renewals, replacements and substitutions to the properties of the Enterprise, so that at all times business carried on in connection with the Enterprise will and can be properly and advantageously conducted in an efficient manner and at reasonable cost, and will operate the Enterprise in an efficient and economical manner, consistent with the protection of the Owners of the Bonds, and will not commit or allow any waste with respect to the Enterprise.

Liens and Claims. Subject to any rights of the United States of America or the State of California, the Commission will keep the Enterprise and all parts thereof free from judgments, from mechanics' and materialmen's liens and from all liens and claims of whatsoever nature or character, to the end that the security provided pursuant to the Indenture may at all times be maintained and preserved, and the Commission will keep the Enterprise and the Revenues free from any liability which might hamper the Commission in conducting its business, or operating the Enterprise. Subject to the provisions of the Indenture, the Trustee at its option (after first giving the Commission thirty days' written notice to comply therewith and failure of the Commission to so comply within said thirty-day period) may defend against any and all actions or proceedings in which the validity of the Indenture is or might be questioned, or may pay or compromise any claim or demand asserted in any such actions or proceedings; provided, however, that, in defending against such actions or proceedings or in paying or compromising such claims or demands, the Trustee will not in any event be deemed to have waived or released the Commission from liability for or on account of any of its covenants and warranties contained in the Indenture, or from its liability under the Indenture to defend the validity of the Indenture and the pledge made in the Indenture and to perform such covenants and warranties.

Insurance. The Commission will procure, and maintain at all times while any of the Bonds will be Outstanding, adequate fidelity insurance or bonds on all officers and employees handling or responsible for any Revenues or funds of the Enterprise, such insurance or bonds to be in an aggregate amount at least equal to the maximum amount of such Revenues or funds at any one time in the custody of all such officers and employees or in the amount of $1,000,000, whichever is less.

The insurance described above may be provided as a part of any comprehensive fidelity and other insurance and not separately for the Enterprise.

The Commission may purchase, on all or any of the Bonds of any Series, insurance assuring the Bondowners that the principal of and interest on the insured Bonds will be paid when due and payable. The purchase of any such insurance will not constitute a preference or priority of the insured Bonds over any Bonds not so insured, and all Bonds Outstanding, irrespective of the providing of such insurance on some of the Bonds, will be equally and proportionately secured by the Indenture.

Books and Accounts; Financial Statements. (a) The Commission will keep proper books of record and accounts of the Enterprise, separate from all other records and accounts of the Commission, in which complete and correct entries will be made of all transactions relating to the Enterprise. Such books of record and accounts will at all times during business hours be subject to the inspection of the Trustee (who will have no duty to inspect) or of any Owner of Bonds then Outstanding or their representatives authorized in writing, at reasonable hours, upon reasonable prior notice and under reasonable conditions.

(b) So long as any of the Bonds are Outstanding, the Commission will prepare and file with the Trustee annually, within seven months after the close of each Fiscal Year, financial statements of the Enterprise for the preceding Fiscal Year, prepared in, accordance with generally accepted accounting principles applied on a consistent basis from year to year ("Enterprise Financial Statements"), which will include a statement of net assets, statement of revenues, expenses and changes in net assets, and statement of cash flows. The Enterprise Financial Statements will be examined by and include the certificate or opinion of an Independent Certified Public Accountant. The Trustee will not be required to review any such statement.

(c) The Commission will furnish a copy of the Enterprise Financial Statements to any Bondowner upon request, and will furnish to the Trustee such reasonable number of copies thereof (not exceeding 100 copies) as may be required by the Trustee for distribution to investment bankers, security dealers and others interested in the Bonds and to the Owners of Bonds requesting copies thereof. The Trustee will not be required to incur any nonreimbursable expenses in making such distribution.

Enterprise Budgets. The Commission will prepare and submit to the Mayor for review and submission to the Board of Supervisors for approval an annual budget for the Enterprise for each Fiscal Year.

Such budget will set forth in reasonable detail the Revenues anticipated to be derived in such Fiscal Year, the expenditures anticipated to be paid or provided for therefrom in such Fiscal Year, and the amounts required to provide for the payment of the principal of and interest and redemption premium, if any, on the Bonds during such Fiscal Year, to pay or provide for Operation and Maintenance Costs of the Enterprise for such Fiscal Year, to make up any deficiencies in any fund or account anticipated for the then current Fiscal Year, and to pay or provide for the payment of all other claims or obligations required to be paid from Revenues in such Fiscal Year, and will show that Net Revenues will be at least adequate to satisfy the requirements of the Indenture. Such budget will comply with any conditions or restrictions set forth in any agreements between the Commission and users of the Enterprise. The Commission will take all action available and necessary to obtain approval or acceptance of the budget by the Mayor and the Board of Supervisors. The Commission will supply to the Trustee and to any Bondowners who will so request in writing a copy of the annual budget for the Fiscal Year covered by such budget. Such budget will be open for inspection by any Owner at the principal corporate trust office of the Trustee during normal business hours.

Maintenance of Revenues. The City will not acquire, construct, operate or maintain, and will not within the scope of its powers permit any other public or private corporation, political subdivision, district or agency or any person whatsoever to acquire, construct, operate or maintain, within the City or any part thereof, any system or utility competitive with the Enterprise. The Commission will have in effect, or cause to have in effect, at all times an ordinance or resolution requiring all customers of the Enterprise to pay the fees, rates and charges applicable to the municipal sanitary waste and storm water collection, treatment and disposal services and facilities furnished by the Enterprise. The Commission will not provide any service of the Enterprise free of charge to any person, firm or corporation, or to any public agency (including the United States of America, the State of California and any public corporation, political subdivision, city, county, district or agency of any thereof), except (i) for free use by the City and its agencies, (ii) to the extent that any such free use is required by the terms of any existing contract or agreement and (iii) for incidental insignificant free use so long as such free use does not prevent the Commission from satisfying the other covenants of the Indenture, including, without limitation, the rate covenant set forth in the Indenture.

Payment of Taxes, Etc. The Commission will pay and discharge, or cause to be paid and discharged, all taxes, assessments and other governmental charges which may later be lawfully imposed upon the Commission on account of the Enterprise or any portion thereof or upon any Revenues and which, if unpaid, might impair the security of the Bonds, when the same becomes due, but nothing contained in the Indenture will require the Commission to pay any such tax, assessment or charge so long as it will in good faith contest the validity thereof. The Commission will duly observe and conform with all valid requirements of any governmental authority relative to the Enterprise or any part thereof.

Acquisition and Construction of Improvements. The Commission will commence and will continue to completion the acquisition and construction of the improvements to the Enterprise proposed to be financed from any Series of Bonds, in a timely manner in accordance with sound engineering practice, and said improvements will be acquired, constructed and completed in a sound and economical manner and in conformity with law.

Eminent Domain Proceeds. If all or any part of the Enterprise is taken by or under threat of eminent domain proceedings, the net proceeds realized by the Commission or the City therefrom (excluding any portion thereof payable to the United States of America or the State of California or required by the United States of America or the State of California to be deposited in a restricted fund) will be deposited by the Treasurer in a special fund in trust and applied and disbursed by the Treasurer subject to the following conditions:

(a) If such eminent domain proceedings have had a material adverse effect upon the Revenues and the security of the Bonds, the Commission will by resolution determine to apply such proceeds for one of the following purposes:

(i) The Commission may determine to apply such proceeds to the purchase, defeasance or redemption of Bonds then Outstanding. In that event, the Treasurer will transfer such proceeds to the Trustee who will apply such proceeds on a Proportionate Basis to the redemption of Bonds of each Series then Outstanding in the proportion which the Bond Obligation amount of each Series bears to the aggregate Bond Obligation amount of all Bonds then Outstanding. If no Bonds are subject to redemption on the next succeeding Interest Payment Date, the Commission will direct the Trustee (A) to hold such proceeds in trust, (B) to invest such proceeds in the investments permitted in the Indenture until any Bonds become redeemable, subject to any restrictions imposed by the Indenture, (C) to deposit the interest and income on such proceeds in the Revenue Fund as such interest and income is received, and (D) to use such proceeds to redeem Bonds in the amount and manner specified in the Indenture and any Supplemental Indenture on the first Interest Payment Date on which the Bonds can be redeemed. Additionally, in such event, the Commission will file with the Trustee a written report of an Independent Certified Public Accountant stating (A) the amount of proceeds to be deposited with the Trustee from such eminent domain proceedings, (B) an estimate of the total amount of Bond Obligation and the amount of Bonds of each maturity that could be redeemed from such proceeds on the first Interest Payment Date on which Bonds are redeemable, and (C) the estimated annual interest and income to be earned on such proceeds while held and invested by the Trustee. Such interest and income on such proceeds upon receipt by the Trustee will be deposited in the Revenue Fund and will be treated as Revenues for all purposes of the Indenture, including determining whether the Commission is in compliance with the rate covenant contained in the Indenture.

(ii) The Commission may determine to apply such proceeds to the cost of additions or improvements to or extensions of the Enterprise if (A) the Commission first secures and files with the Trustee a written report of a Qualified Independent Consultant showing (1) the loss in annual Revenues, if any, suffered, or to be suffered, by the Commission by reason of such eminent domain proceedings, (2) a general description of the additions, improvements or extensions then proposed to be acquired by the Commission from such proceeds, and (3) an estimate of the additional Revenues to be derived from such additions, improvements or extensions; and (B) such written report states that such additional Revenues will sufficiently offset the loss of Revenues resulting from such eminent domain proceedings so that the ability of the Commission to meet its obligations under the Indenture will not be substantially impaired. The Commission will then promptly proceed with the construction of the additions, improvements or extensions substantially in accordance with such written report. Payments for such construction will be made by the Commission from such proceeds. Any balance of such proceeds not required by the Commission for the purposes aforesaid will be deposited in the Revenue Fund and applied as provided in the Indenture.

(b) If such eminent domain proceedings has had no effect, or at the most a relatively immaterial effect, upon the Revenues and the security of the Bonds, and a Qualified Independent Consultant so concludes in a written report filed with the Trustee, the Commission may determine to apply such proceeds to the costs of additions or improvements to or extensions of the Enterprise or may deposit such proceeds in the Revenue Fund, to be applied as provided in the Indenture.

Compliance with Indenture. The Commission will faithfully observe and perform all the covenants, conditions and requirements of the Indenture, and will not suffer or permit any default to occur under the Indenture, or do or permit to be done, in, upon or about the Enterprise, or any part thereof, anything that might in any way weaken, diminish or impair the security intended to be given pursuant to the Indenture.

Observance of Laws and Regulations. The Commission will comply promptly, fully and faithfully with and abide by any statute, law, ordinance, order, rule or regulation, judgment, decree, direction or requirement now in force or later enacted, adopted or entered by any competent governmental authority or agency applicable or with respect to or affecting the acquisition, construction or reconstruction of the Enterprise or any part thereof or applicable or with respect to or affecting the operation, manner, use or condition of the Enterprise or any part or parcel thereof or adjoining public ways; provided that the Commission need not comply with any such statute, law, ordinance, rule, regulation, judgment, decree, direction or requirement if and so long as the Commission in good faith is contesting or permitting or causing

to be contested the applicability or validity thereof by appropriate proceedings diligently prosecuted, even though such contest may result in the imposition of a lien or charge against the Enterprise or the Revenues, if (1) the Commission will effectively prevent foreclosure or enforcement of any such lien or charge and (2) the foreclosure or enforcement of any such lien or charge will be stayed, and if said stay thereafter expires, the Commission will forthwith discharge such lien or charge or cause the same to be discharged, so that pending such proceedings the Enterprise and the Revenues thereof will not be affected thereby, and the security of the Bonds will not be impaired.

Prosecution and Defense of Suits. The Commission will promptly from time to time take such action as may be necessary or proper to remedy or cure any defect in or cloud upon the title to the Enterprise later developing, and will prosecute all such suits, actions and other proceedings as may be appropriate for such purposes and, to the extent permitted by law, will indemnify and save the Trustee and every Bondowner harmless from all loss, cost, damage and expense, including attorneys' fees, which they or any of them may incur by reason of any such defect, cloud, suit, action or proceeding.

The Commission will defend against every suit, action or proceeding at any time brought against the Trustee or any Bondowner upon any claim arising out of the receipt, application or disbursement of any of the Revenues or involving the rights of the Trustee or any Bondowner under the Indenture; provided, that the Trustee or any Bondowner at its or his election may appear in and defend any such suit, action or proceeding. The Commission will, to the extent permitted by law and without making any representation as to the enforceability of the covenants in this paragraph, indemnify and hold harmless the Trustee and the Bondowners against any and all liability claimed or asserted by any person arising out of such receipt, application or disbursement, and will indemnify and hold harmless the Bondowners against any attorneys' fees or other expenses which any of them may incur in connection with any litigation to which any of them may become a defendant by reason of his ownership of Bonds. The Commission, to the extent permitted by law, will promptly reimburse any Bondowner in the full amount of any attorneys' fees or other expenses which he may incur in litigation or Otherwise in order to enforce his rights under the Indenture or the Bonds, provided that such litigation has concluded favorably to such Bondowner's contentions therein. Notwithstanding any contrary provision of the Indenture, this covenant will remain in full force and effect, even though all indebtedness and obligations issued under the Indenture may have been fully paid and satisfied, until the Commission has been dissolved.

Governmental Approvals. The Commission will perform any construction, reconstructions and restorations of, improvements, betterments and extensions to, and equippings and furnishings of, and will operate and maintain the Enterprise at standards required in order that the same may continue to be approved by the proper and competent authority or authorities of the State of California as a public municipal sanitary waste and storm water collection, treatment and disposal system.

Further Assurances. Whenever and so often as requested so to do by the Trustee or any Bondowner, the Commission will promptly execute and deliver or cause to be executed and delivered all such other and further instruments, documents or assurances, and promptly do or cause to be done all such other and further things, as may be necessary or reasonably required in order to further and more fully vest in the Trustee and the Bondowners all rights, interest, powers, benefits, privileges and advantages conferred or intended to be conferred upon them by the Indenture.

Casualty Insurance; Use of Proceeds. (a) The Commission will at all times maintain such insurance on the Enterprise as is customarily maintained with respect to works and properties of like character against accident to, loss of or damage to such works or properties. If any useful part of the Enterprise is damaged or destroyed, or such damage or destruction have had a material adverse effect upon the Revenues and the security of the Bonds, the Commission will by resolution determine to apply the proceeds of any insurance for such loss or damage for one of the following purposes:

(i) The Commission may determine to apply such proceeds to the purchase, defeasance or redemption of Bonds then Outstanding. In that event, the Treasurer will transfer such proceeds to the Trustee who will apply such proceeds on a Proportionate Basis to the redemption, defeasance or purchase of Bonds of each Series then Outstanding in the proportion which the Bond Obligation amount of each Series bears to the aggregate Bond Obligation amount of all Bonds then Outstanding. If no Bonds are subject to redemption on the next succeeding Interest Payment Date, the Commission will direct the Trustee (A) to hold such insurance proceeds in trust, (B) to invest such insurance proceeds in the investments permitted in the Indenture until any Bonds become redeemable, subject to any restrictions imposed by the Indenture, (C) to deposit the interest and income on such insurance proceeds in the Revenue Fund as such interest and income is received, and (D) to use such insurance proceeds to redeem Bonds in the amount and manner specified in the Indenture and any Supplemental Indenture on the first interest payment date on which the Bonds can be redeemed. Additionally, in such event, the Commission will file with the Trustee a written report of an Independent Certified Public Accountant stating (A) the amount of insurance proceeds to be deposited with the Trustee in connection with any insured loss or damage, (B) an estimate of the total amount of Bond Obligation and the amount of Bonds of each maturity which could be redeemed from such proceeds on the first interest payment date on which Bonds are redeemable, and (C) the estimated annual interest and income to be earned on such insurance proceeds while held and invested by the Trustee. Such interest and income on such insurance proceeds upon receipt by the Trustee will be deposited in the Revenue Fund and will be treated as Revenues for all purposes of the Indenture, including determining whether the Commission is in compliance with the rate covenant contained in the Indenture.

(ii) The Commission may determine to apply such insurance proceeds to the cost of additions or improvements to or extensions of the Enterprise if (A) the Commission first secures and files with the Trustee a written report of a Qualified Independent Consultant showing (A) the loss in annual Revenues, if any, suffered, or to be suffered, by the Commission by reason of the loss or damage to the Enterprise, (B) a general description of the additions, improvements or extensions then proposed to be acquired by the Commission from such proceeds, and (C) an estimate of the additional Revenues to be derived from such additions, improvements or extensions; and (B) such written report states that such additional Revenues will sufficiently offset the loss of Revenues resulting from such insured loss or damage so that the ability of the Commission to meet its obligations under the Indenture will not be substantially impaired. The Commission will then promptly proceed with the construction of the additions, improvements or extensions substantially in accordance with such written report. Payments for such construction will be made by the Commission from such insurance proceeds. Any balance of such proceeds not required by the Commission for the purposes aforesaid will be deposited in the Revenue Fund and applied as provided in the Indenture.

(b) If such insured loss, damage or destruction has had no effect, or at the most a relatively immaterial effect, upon the Revenues and the security of the Bonds, and a Qualified Independent Consultant so concludes in a written report filed with the Trustee, the Commission may determine to apply such proceeds to the costs of additions or improvements to or extensions of the Enterprise, or such proceeds may be used for any other lawful purpose of the Enterprise.

(c) Any such insurance will be in the form of policies or contracts for insurance with insurers of good standing and will be payable to the Commission, or may be in the form of self-insurance by the Commission. The Commission will establish such fund or funds or reserves as it determines, in its sole judgement, are necessary to provide for its share of any such self-insurance.

EVENTS OF DEFAULT AND REMEDIES OF BONDOWNERS

Events of Default; Acceleration; Waiver of Default. If one or more of the following events (called "Events of Default" in the Indenture) happens, that is to say – (a) if default is made in the due and punctual payment of the principal of, or the premium (if any) on, any Bond when and as the same becomes due and payable, whether at maturity as therein expressed, by proceedings for redemption, by declaration or otherwise, or if default is made in the redemption from any Sinking Fund Account of any Term Bonds in the amounts and

at the times provided therefor; (b) if default is made in the due and punctual payment of any installment of interest on any Bond when and as such interest installment becomes due and payable; (c) if default is made by the Commission in the observance of any of the other covenants, agreements or conditions on its part in the Indenture or in the Bonds contained, and such default continues for a period of 30 days after written notice thereof, specifying such default and requiring the same to be remedied, is given to the Commission by the Trustee or by a Credit Provider, or to the Commission and the Trustee by the Owners of not less than 25% of the Bond Obligation; or (d) if the Commission or the City files a petition or answer seeking reorganization or arrangement under the federal bankruptcy laws or any other applicable law of the United States of America, or if a court of competent jurisdiction approves a petition, filed with or without the consent of the Commission or the City, as the case may be, seeking reorganization under the federal bankruptcy laws or any other applicable law of the United States of America, or if, under the provisions of any other law for the relief or aid of debtors, any court of competent jurisdiction assumes custody or control of the Commission or the City or of the whole or any substantial part of the property of either; then and in each and every such case during the continuance of such Event of Default, the Trustee may, and upon the written request of the Owners of not less than a majority in aggregate amount of the Bond Obligation or of a Credit Provider will, upon notice in writing to the Commission, declare the principal of all of the Current Interest Bonds then Outstanding, and the interest accrued thereon, and of all Capital Appreciation Bonds then Outstanding, in the amount of the Accreted Value thereof, to be due and payable immediately, and upon any such declaration the same will become and will be immediately due and payable, anything in the Indenture or in the Bonds contained to the contrary notwithstanding.

This provision, however, is subject to the condition that if, at any time after the principal of the Bonds have been so declared due and payable, and before any judgment or decree for the payment of the moneys due has been obtained or entered as provided in the Indenture, the Commission deposits with the Trustee a sum sufficient to pay all principal and Accreted Value of the Bonds maturing prior to such declaration and all matured installments of interest (if any) upon all the Current Interest Bonds, with interest on such overdue payments of principal and Accreted Value and interest installments at the rate or rates of interest borne by the respective Bonds, and the reasonable expenses of the Trustee, and any and all other defaults known to the Trustee (other than in the payment of principal and Accreted Value of and interest on the Bonds due and payable solely by reason of such declaration) have been made good or cured to the satisfaction of the Trustee, or provision deemed by the Trustee to be adequate has been made therefor, then, and in every such case, (i) if such declaration has been made by the Trustee, the Trustee, or (ii) if such declaration has been made upon the written request of Bondowners, the Owners of not less than a majority in aggregate amount of the Bond Obligation of the Bonds then Outstanding, or (iii) if such declaration has been made upon the written request of a Credit Provider, such Credit Provider, may, by written notice to the Commission and, in cases (ii) and (iii) above, to the Trustee, on behalf of the Owners of all of the Bonds, rescind and annul such declaration and its consequences; but no such rescission and annulment will extend to or will affect any subsequent default, or will impair or exhaust any right or power consequent thereon.

Application of Funds Upon Acceleration. All of the Revenues, including all sums in all of the funds provided for in the Indenture upon the date of the declaration of acceleration as provided in the Indenture and all sums thereafter received by the Commission or the Trustee under the Indenture, will, if received by the Commission, be transmitted to the Trustee and be applied by the Trustee in the order following, upon presentation of the several Bonds, and the stamping thereon of the payment if only partially paid, or upon the surrender thereof if fully paid --

First, to the payment of the costs and expenses of the Bondowners in declaring such Event of Default, including reasonable compensation to their agents, attorneys and counsel; and to the payment of the costs and expenses of the Trustee, including but not limited to reasonable compensation to its agents, attorneys and counsel;

Second, to the payment of the whole amount of Bond Obligation then owing and unpaid upon the Bonds, with interest on, with respect to the Current Interest Bonds, the overdue principal and

installments of interest, and, with respect to the Capital Appreciation Bonds, the Accreted Value thereof, at the rate or rates of interest borne by the respective Bonds, and in case such moneys are insufficient to pay in full the whole amount so owing and unpaid upon the Bonds, then to the payment of such principal and interest and Accreted Value without preference or priority of principal over interest, or of interest over principal, or of any installment of interest over any other installment of interest, ratably to the aggregate of such principal and interest and Accreted Value; and

Third, to the payment of all Policy Costs, and in case such moneys are insufficient to pay in full all Policy Costs owing and unpaid, then to the payment of such Policy Costs pro rata (calculated by reference to the maximum amounts available under the respective Bond Reserve Fund Policies).

Suits at Law or in Equity and Mandamus. In case one or more of the Events of Default happens, then and in every such case the Owner of any Bond at the time Outstanding will be entitled to proceed to protect and enforce the rights vested in such Owner by the Indenture by such appropriate judicial proceeding as such Owner deems most effectual to protect and enforce any such right, either by suit in equity or by action at law, whether for the specific performance of any covenant or agreement contained in the Indenture, or in aid of the exercise of any power granted in the Indenture, or to enforce any other legal or equitable right vested in the Owners of Bonds by the Indenture or by law; provided, however, that no such Bondowner will have the right to institute any such judicial proceeding pursuant to the Indenture unless (a) such Owner has previously given to the Trustee written notice of the occurrence of an Event of Default under the Indenture; (b) the Owners of at least ten percent (10%) in aggregate amount of Bond Obligation of the Bonds then Outstanding have made written request to the Trustee to exercise the powers granted in the Indenture or to institute such action, suit or proceeding in its own name; (c) such Owner or said Owners have tendered to the Trustee reasonable indemnity against the costs, expenses and liabilities to be incurred in compliance with such request; (d) the Trustee has refused or omitted to comply with such request for a period of sixty (60) days after such written request has been received by, and said tender of indemnity has been made to, the Trustee; and (e) the Trustee has not received contrary directions from the Owners of a majority in aggregate amount of Bond Obligation of the Bonds Outstanding. The provisions of the Indenture constitute a contract with the Owners of the Bonds, and such contract and duties of the Commission and of the Commission members and of the officers and employees of the Commission and of the City are enforceable by any Bondowner by mandamus or other appropriate suit, action or proceeding in any court of competent jurisdiction.

Non-waiver. Nothing in the Indenture, or in the Bonds, will affect or impair the obligation of the Commission, which is absolute and unconditional, to pay the principal of and the interest (and premium, if any) on the Bonds to the respective Owners of the Bonds at the respective dates of maturity, or upon call for redemption, as provided in the Indenture, but only out of the Revenues pledged in the Indenture for such payments, or affect or impair the right of action, which is also absolute and unconditional, of such Owners to institute suit to enforce such payment by virtue of the contract embodied in the Bonds.

A waiver of any default or breach of duty or contract by any Bondowner will not affect any subsequent default or breach of duty or contract, or impair any rights or remedies on the subsequent default or breach. No delay or omission of the Trustee or of any Owner of any of the Bonds to exercise any right or power arising upon the happening of any Event of Default will impair any such right or power or will be construed to be a waiver of any such Event of Default or an acquiescence therein, and every power and remedy given by the Law or the Indenture to the Trustee or to the Owners of Bonds may be exercised from time to time and as often as will be deemed expedient by the Trustee or the Owners of Bonds.

If a suit, action or proceeding to enforce any right or exercise any remedy is abandoned or determined adversely to the Bondowners, the Commission and the Bondowners will be restored to their former positions, rights and remedies as if such suit, action or proceeding had not been brought or taken.

Actions by Trustee as Attorney-in-Fact. Any suit, action or proceeding which any Owner of Bonds will have the right to bring to enforce any right or remedy under the Indenture may be brought by the Trustee

for the equal benefit and protection of all Owners of Bonds similarly situated (notwithstanding any conditions upon the bringing of any such action, suit or proceeding set forth in the Indenture) and the Trustee is hereby appointed (and the successive respective Owners of the Bonds issued under the Indenture, by taking and holding the same, will be conclusively deemed so to have appointed it) the true and lawful attorney-in-fact of the respective Owners of the Bonds for the purpose of bringing any such suit, action, or proceeding and to do and perform any and all acts and things for and in behalf of the respective Owners of the Bonds as a class or classes, as may be necessary or advisable in the opinion of the Trustee as such attorney-in-fact.

Remedies Not Exclusive. No remedy conferred in the Indenture upon or reserved to the Trustee or to the Owners of Bonds is intended to be exclusive of any other remedy. Every such remedy will be cumulative and will be in addition to every other remedy given under the Indenture or now or later existing, at law or in equity or by statute or otherwise and may be exercised without exhausting and without regard to any other remedy conferred by the Law or any other law.

Power of Trustee to Control Proceedings. In the event that the Trustee, upon the happening of an Event of Default, has taken any action, by judicial proceedings or otherwise, pursuant to its duties under the Indenture, whether upon its own discretion or upon the request of the Owners of at least ten percent (10%) in aggregate amount of the Bond Obligation, it will have full power, in the exercise of its discretion for the best interests of the Owners of the Bonds, with respect to the continuance, discontinuance, withdrawal, compromise, settlement or other disposal of such action; provided, however, that the Trustee does not, unless there no longer continues an Event of Default under the Indenture, discontinue, withdraw, compromise or settle, or otherwise dispose of any litigation pending at law or in equity, if at the time there has been filed within a written request signed by the Owners of at least a majority in aggregate amount of the Bond Obligation under the Indenture opposing such discontinuance, withdrawal, compromise, settlement or other disposal of such litigation.

Remedies of Bond Reserve Fund Policy Provider. If the Commission fails to pay Policy Costs to a Bond Reserve Fund Policy Provider to the extent and at the times required by the provisions of the Indenture and such failure continues for 30 days after written notice of such default is received by the Commission and the Trustee from such Bond Reserve Fund Policy Provider or if an Event of Default set forth in the Indenture occurs and is continuing, then the Bond Reserve Fund Policy Provider may exercise any remedy provided under the Indenture to the Trustee or available at law or in equity to protect and enforce its right to receive payment of Policy Costs; provided, that, in no event, will the Bond Reserve Fund Policy Provider be able to declare the principal and Accreted Value of the Bonds and the interest accrued thereon to be due and payable immediately or to exercise any remedy that the Trustee, in its sole discretion, determines would adversely affect the Bondowners.

Rights of Credit Provider. Each Credit Provider, during any period in which an Event of Default has occurred and is continuing, will be recognized as the Owner of each Bond which it guarantees or insures for the purposes of exercising all rights and privileges available to Bondowners. Any acceleration of principal payments with respect to Bonds guaranteed or insured by a Credit Provider are subject to such Credit Provider's prior written direction or consent (but only if such Credit Provider is not in default under its guaranty or insurance policy). Each Credit Provider, will have the right to institute any suit, action or proceeding at law or in equity under the same terms as an Owner of the Bonds that such Credit Provider guarantees or insures.

THE TRUSTEE

Appointment and Duties of Trustee. The Trustee accepts by the Indenture the trusts imposed upon it as Trustee under the Indenture for the purpose of receiving all moneys which the Commission is required to deposit with the Trustee under the Indenture and agrees to allocate, use and apply the same as provided in the Indenture and otherwise to hold all the offices and to perform all the functions and, duties provided in the Indenture to be held and performed by the Trustee and agrees to perform such duties and obligations, subject to

the terms and conditions set forth in the Indenture. The Commission agrees that it will maintain a Trustee having an office in San Francisco or Los Angeles, California, so long as any Bonds are Outstanding and unpaid.

The Trustee will, prior to an Event of Default, and after the curing of all Events of Default which may have occurred, perform such duties and only such duties as are specifically set forth in the Indenture. The Trustee will, during the existence of any Event of Default (which has not been cured), exercise such of the rights and powers vested in it by the Indenture, and use the same degree of care and skill in their exercise, as a prudent person would exercise or use under the circumstances in the conduct of his own affairs.

So long as there is no Event of Default under the Indenture, the Commission may remove the Trustee, by giving written notice to such Trustee and by giving Bondowners notice by mail, first class postage prepaid, of such removal, and any successor thereto, and may appoint a successor or successors thereto; provided that any such successor will be a bank or trust company doing business and having an office in San Francisco or Los Angeles, California, having a combined capital and surplus of at least $100,000,000, and subject to supervision or examination by federal or state authority. If such bank or trust company publishes a report of condition at least, annually, pursuant to law or to the requirements of any supervising or examining authority above referred to, then for the purpose of the Indenture the combined capital and surplus of such bank or trust company will be deemed to be its combined capital and surplus as set forth in its most recent report of condition so published.

The Trustee may at any time resign by giving written notice to the Commission and by giving the Bondowners notice by mail, first class postage prepaid, of such resignation. Upon receiving such notice of resignation, the Commission will promptly appoint a successor Trustee by an instrument in writing. Any resignation or removal of the Trustee and appointment of a successor Trustee will become effective upon acceptance of appointment by the successor Trustee. If no successor Trustee has been appointed and has accepted appointment within forty-five days of giving notice of removal or notice of resignation as aforesaid, the resigning Trustee or any Bondowner (on behalf of himself or herself and all other Bondowners) may petition any court of competent jurisdiction for the appointment of a successor Trustee, and such court may thereupon, after such notice (if any) as it may deem proper, appoint such successor Trustee.

Any company into which the Trustee may be merged or converted or with which it may be consolidated or any company resulting from any merger, conversion or consolidation to which it will be a party or any company to which the Trustee may sell or transfer all or substantially all of its corporate trust business, provided such company will be eligible under the Indenture, will be the successor to such Trustee without the execution or filing of any paper or any further act, anything in the Indenture to the contrary notwithstanding.

Authorization of Trustee. The Trustee is authorized by the Indenture to pay the principal of and interest and redemption premium, if any, on the Bonds when due and payable, or on call and redemption or on purchase by the Trustee prior to maturity, and to cancel all Bonds upon payment thereof and to return the same so cancelled to the Commission, subject to the provisions of the Indenture. The Trustee will keep accurate records of all funds administered by it and of all Bonds and interest payments paid and discharged.

The Trustee and its directors, officers, employees or agents may in good faith buy, sell, own, hold and deal in any of the Bonds and may join in any action which any Owner of a Bond may be entitled to take, with like effect as if the Trustee was not the Trustee under the Indenture. The Trustee may in good faith hold any other form of indebtedness of the Commission or of the City, own, accept or negotiate any drafts, bills of exchange, acceptances or obligations of the Commission and make disbursements for the Commission and enter into any commercial or business arrangement therewith, without limitation.

Fees and Expenses. The Commission will compensate the Trustee for its services rendered pursuant to the provisions of the Indenture, and also for all reasonable expenses, charges, counsel fees and other

disbursements, including those of its attorneys, agents and employees, incurred in and about the performance of its powers and duties under the Indenture. The obligations of the Commission under the Indenture to compensate the Trustee for services and to pay or reimburse the Trustee for disbursements, liabilities and advances will constitute additional indebtedness under the Indenture.

Liability of Trustee. (a) The recitals of facts, covenants and agreements in the Indenture and in the Bonds contained will be taken as statements, covenants and agreements of the Commission, and the Trustee assumes no responsibility for the correctness of the same, does not make any representations as to the validity or sufficiency of the Indenture or of the Bonds, and will not incur any responsibility in respect thereof, other than in connection with the duties or obligations in the Indenture or in the Bonds expressly assigned to or imposed upon it. The Trustee will not be liable in connection with the performance of its duties under the Indenture, except for its own negligence or default.

(b) The Trustee will not be liable for any error of judgment made in good faith by an Authorized Officer of the Trustee, unless it is proved that the Trustee was negligent in ascertaining the pertinent facts.

(c) The Trustee will not be liable with respect to any action taken or omitted to be taken by it in good faith in accordance with the direction of the Owners of not less than a majority in aggregate amount of Bond Obligations of the Bonds at the time Outstanding relating to the time, method and place of Conducting any proceeding for any remedy available to the Trustee, or exercising any trust or power conferred upon the Trustee under the Indenture.

(d) The Trustee will be under no obligation to exercise any of the rights or powers vested in it by the Indenture at the request, order or direction of any of the Bondowners pursuant to the provisions of the Indenture unless such Bondowners have offered to the Trustee reasonable security or indemnity against the costs, expenses and liabilities which may be incurred therein or thereby.

(e) The Trustee will not be liable for any action taken by it in good faith and believed by it to be authorized or within the discretion or rights or powers conferred upon it by the Indenture.

(f) The Trustee will not be deemed to have knowledge of any Event of Default under the Indenture unless and until it has actual knowledge thereof, or has received written notice thereof, at its principal corporate trust office in San Francisco, California. Except as otherwise expressly provided in the Indenture, the Trustee will not be bound to ascertain or inquire as to the performance or observance of any of the terms, conditions, covenants or agreements in the Indenture or of any of the documents executed in connection with the Bonds or the delivery of the Bonds, or as to the existence of an Event of Default thereunder. The Trustee will not be responsible for the validity or effectiveness of any collateral given to or held by it The Trustee will not be responsible for the recording or filing of any document relating to the Indenture or of financing statements (or continuation statements in connection therewith) or of any supplemental instruments or documents of further assurance as may be required by law in order to perfect the security interests in any collateral given to or held by it.

(g) No provision of the Indenture will require the Trustee to expend or risk its own funds or otherwise incur any financial liability in the performance of any of its duties under the Indenture, or in the exercise of any of its rights or powers.

(h) The Commission further agrees to indemnify, to the extent permitted by law and without making any representation as to the enforceability of this covenant, and save the Trustee, its directors, officers, employees and agents harmless against any liabilities which it may incur in the exercise and performance of its powers and duties under the Indenture, including but not limited to costs and expenses incurred in defending against any claim or liability, which are not due to its negligence or default.

(i) The provisions above will survive the resignation or removal of the Trustee, payment of the Bonds and discharge of the Indenture.

Rights of Trustee to Rely Upon Documents. The Trustee will be protected in acting upon any notice, resolution, request, consent, order, certificate, report, direction, Bond or other paper or document believed by it to be genuine and to have been signed or presented by the proper party or parties. The Trustee may consult with counsel, who may be counsel to the Commission or to the City, with regard to legal questions, and the opinion of such counsel will be full and complete authorization and protection in respect of any action taken or suffered under the Indenture in good faith and in accordance therewith.

The Trustee will not be bound to recognize any person as the Owner of a Bond unless and until his title thereto is satisfactorily established, if disputed.

Whenever in the administration of its duties under the Indenture the Trustee deems it necessary or desirable that a matter be proved or established prior to taking or suffering any action under the Indenture, such matter (unless other evidence in respect thereof be in the Indenture specifically prescribed) may, in the absence of bad faith on the part of the Trustee, be deemed to be conclusively proved and established by a Certificate of the Commission, and such Certificate will be full warrant to the Trustee for any action taken or suffered under the provisions of the Indenture or any Supplemental Indenture upon the faith thereof, but in its discretion the Trustee may, in lieu thereof, accept other evidence of such matter or may require such additional evidence as to it may seem reasonable.

MODIFICATION OR AMENDMENT OF THE INDENTURE

Amendments Permitted. (a)(i) The Indenture and the rights and obligations of the Commission and of the Owners of the Bonds and of the Trustee may be modified or amended at any time by a Supplemental Indenture which will become binding when the written consents of the Owners of a majority in aggregate amount of the Bond Obligation of the Bonds (or, if such Supplemental Indenture is only applicable to a Series of Bonds, such Series of Bonds) then Outstanding, exclusive of Bonds disqualified as provided in the Indenture, and of each Credit Provider (so long as such Credit Provider is not in default under the policy of municipal bond insurance or Letter of Credit issued by it in connection with any Series of Bonds) have been filed with the Trustee (provided, that no such Credit Provider will unreasonably withhold consent to such modification or amendment). (ii) The Indenture and the rights and obligations of the Commission and of the Owners of the Bonds and of the Trustee may also be modified or amended at any time by a Supplemental Indenture which will become binding when the written consents of each Credit Provider have been filed with the Trustee, provided that at such time the payment of the principal of and interest on all Outstanding Bonds will be insured by a policy or policies of municipal bond insurance or payable under a Letter of Credit issued by a Credit Provider. (iii) No such modification or amendment will (A) extend the fixed maturities of the Bonds, or extend the time for making any Minimum Sinking Fund Account Payments, or reduce the rate of interest thereon, or extend the time of payment of interest, or reduce the amount of principal thereof, or reduce any premium payable on the redemption thereof, without the consent of the Owner of each Bond so affected, or (B) reduce the aforesaid percentage of the Bond Obligation the consent of the Owners of which is required for the execution of any amendment or modification of the Indenture, or (C) modify any of the rights or obligations of the Trustee without its written consent thereto.

(b) The Indenture and the rights and obligations of the Commission and of the Owners of the Bonds may also be modified or amended at any time by a Supplemental Indenture which will become binding upon adoption, without the consent of any Bondowners or any Credit Provider (but with notice to each Credit Provider), but only to the extent permitted by law and only if the Trustee determines, which determination may be based upon a good faith reliance upon an Opinion of Counsel, that the provisions of such Supplemental Indenture will not materially adversely affect the interests of the Owners, including, without limitation, for any one or more of the following purposes – (i) to add to the covenants and agreements of the Commission in the Indenture contained other covenants and agreements thereafter to be observed or to surrender any right or

power in the Indenture reserved to or conferred upon the Commission; (ii) to cure, correct or supplement any ambiguous or defective provision or omission or mistake contained in the Indenture or in regard to questions arising under the Indenture, as the Commission may deem necessary or desirable; (iii) to provide for the issuance of additional Series of Bonds, and to provide the terms and conditions under which such additional Series of Bonds may be issued, subject to and in accordance with the provisions of the Indenture; and (iv) to provide additional security for the Bonds.

Procedure for Amendment with Written Consent of Bondowners. The Commission may at any time adopt a Supplemental Indenture amending the provisions of the Bonds or of the Indenture or any Supplemental Indenture, to the extent that such amendment is permitted by the Indenture, to take effect when and as provided in the Indenture. A copy of or a summary of the provisions of such Supplemental Indenture, together with a request to Bondowners and to each Credit Provider for their consent thereto, will be mailed, first class postage prepaid, by the Commission to each Owner of Bonds and to each Credit Provider, but failure to mail copies of such Supplemental Indenture or summary thereof and request will not affect the validity of the Supplemental Indenture when assented to as provided in the Indenture.

Such Supplemental Indenture will not become effective unless there are filed with the Trustee the written consents of the Owners of a majority in aggregate amount of the Bond Obligation of the Bonds then Outstanding (exclusive of Bonds disqualified as provided in the Indenture) and of each Credit Provider and a notice has been mailed as provided below. Each such consent will be effective only if accompanied by proof of ownership of the Bonds for which such consent is given, which proof will be such as is permitted by the Indenture. Any such consent will be binding upon the Owner of the Bonds giving such consent and on any subsequent Owner (whether or not such subsequent Owner has notice thereof) unless such consent is revoked in writing by the Owner giving such consent or a subsequent Owner by filing such revocation with the Trustee prior to the date when the notice provided below for has been mailed.

After the Owners of the required percentage of Bond Obligation and each Credit Provider has filed their consents to the Supplemental Indenture, the Commission will mail a notice to the Bondowners in the manner provided above for the mailing of the Supplemental Indenture, stating in substance that the Supplemental Indenture has been consented to by the Owners of the required percentage of Bond Obligation and each Credit Provider and will be effective as provided in the Indenture (but failure to mail copies of said notice will not affect the validity of the Supplemental Indenture or consents thereto), and proof of the mailing of such notice will be filed with the Trustee. A record, consisting of the papers required by the Indenture to be filed with the Trustee, will be proof of the matters therein stated until the contrary is proved. The Supplemental Indenture will become effective upon the filing with the Trustee of the proof of the mailing of such last-mentioned notice.

In lieu of obtaining any demand, request, direction, consent or waiver in writing, the Trustee may call and hold a meeting of the Bondowners upon such notice and in accordance with such rules and regulations as the Trustee considers fair and reasonable for the purpose of obtaining any such action.

Disqualified Bonds. Bonds owned or held by or for the account of the Commission or of the City (but excluding Bonds held in any pension or retirement fund) will not be deemed Outstanding for the purpose of any consent or other action or any calculation of Outstanding Bonds provided for in the Indenture, and will not be entitled to consent to or take any other action provided for in the Indenture. For the purpose of the Indenture no bank organized under the laws of the State of California and no national banking association doing business in said State, or elsewhere, will be deemed to be an agency of the Commission.

The Commission may adopt appropriate regulations to require each Bondowner, before his consent provided for in the Indenture will be deemed effective, to reveal if the Bonds as to which such consent is given are disqualified as provided in the Indenture.

Effect of Supplemental Indenture. From and after the time any Supplemental Indenture becomes effective pursuant to the Indenture, the Indenture will be deemed to be modified and amended in accordance therewith, and the respective rights, duties and obligations under the Indenture of the Commission, the Trustee and all Owners of Bonds Outstanding will thereafter be determined, exercised and enforced under the Indenture subject in all respects to such modification and amendment, and all the terms and conditions of any such Supplemental Indenture will be deemed to be part of the terms and conditions of the Indenture for any and all purposes.

Endorsement or Replacement of Bonds Issued Before Amendments. The Commission may determine that Bonds issued and delivered before the effective date of any action taken as provided in the Indenture will bear a notation, by endorsement or otherwise, in form approved by the Commission, as to such action. In that case, upon demand of the Owner of any Bond Outstanding at such effective date and presentation of his Bond for the purpose at the office of the Commission or at such other office as the Commission may select and designate for that purpose, a suitable notation will be made on such Bond. The Commission may determine that new Bonds, so modified as in the opinion of the Commission is necessary to conform to such action, will be prepared, executed and delivered. In that case, upon demand of the Owner of any Bonds then Outstanding, such new Bonds will be exchanged at the principal corporate trust office of the Trustee in San Francisco, California, without cost to such Owner, for Bonds of the same character then Outstanding, upon surrender of such Bonds.

Amendatory Endorsement of Bonds. The provisions of the Indenture will not prevent any Bondowner from accepting any amendment as to the particular Bonds held by such Bondowner, provided that due notation thereof is made on such Bonds.

DEFEASANCE

Discharge of Indenture. (a) If the Commission pays and discharges the entire indebtedness on all Bonds Outstanding in any one or more of the following ways – (1) by well and truly paying or causing to be paid the principal of (including redemption premiums, if any) and interest on all Bonds Outstanding, as and when the same become due and payable (but this clause will not include Bonds the principal of or interest on which has been paid by a Credit Provider until the Credit Provider has been paid or reimbursed for such payment by the Commission); or (2) by depositing with the Trustee, an escrow agent or other fiduciary, in trust, at or before maturity, money which, together with the amounts then on deposit in the Principal Fund, the Interest Fund and the Bond Reserve Fund, is fully sufficient to pay or redeem all Bonds Outstanding, including all principal, interest and redemption premiums, if any; or (3) by delivering to the Trustee, for cancellation by it, all Bonds Outstanding; or (4) by depositing with the Trustee, an escrow agent or other fiduciary, in trust, Defeasance Obligations in such amount which, in the determination of an Independent Certified Public Accountant, who certifies such determination to the Trustee, will, together with the income or increment to accrue thereon and any other moneys of the Commission made available for such purpose, be fully sufficient to pay and discharge the indebtedness on all Bonds (including all principal, interest and redemption premiums, if any) at or before their respective maturity dates; and if the Commission also pays or causes to be paid all other sums payable under the Indenture by the Commission, including all Policy Costs, then and in that case, at the election of the Commission (evidenced by a Certificate of the Commission signifying its intention to pay and discharge all such indebtedness, which will be filed with the Trustee), and notwithstanding that any Bonds have been surrendered for payment, the pledge of the Net Revenues and other funds provided for in the Indenture and all other obligations of the Commission under the Indenture will cease, terminate and be completely discharged, except only as provided in the Indenture, and the Owners of the Bonds not so surrendered and paid will thereafter be entitled to payment only out of the Defeasance Obligations deposited with the Trustee, escrow agent or other fiduciary as aforesaid for their payment; subject, however, to the provisions of the Indenture. The discharge of the obligations of the Commission under the Indenture will be without prejudice to the rights of the Trustee to charge for and be reimbursed by the Commission for any expenditures which it may thereafter incur in connection with the Indenture.

The Commission may at any time surrender to the Trustee for cancellation by it any Bonds previously issued and delivered, which the Commission may have acquired in any manner whatsoever, and such Bonds, upon such surrender and cancellation, will be deemed to be paid and retired.

(b) In the event that any portion or all of the Bonds are to be paid and discharged pursuant to the Indenture, the Credit Provider will be provided with 15 days advance notice and will be provided with a draft copy of any proposed escrow agreement establishing the defeasance trust, the form of the Independent Certified Public Accountants certificate or letter, the preliminary official statement of the refunding issue (if applicable) and the form of Opinion of Counsel to the effect that upon the deposit of funds and investments under the escrow or other applicable agreement, the Bonds being paid and discharged will no longer be deemed to be outstanding under the Indenture. Substitution of securities held in trust under the escrow agreement will not be permitted unless there has first been delivered to the escrow agent (1) a certificate of Independent Certified Public Accountant to the effect that the escrow investments, as substituted, are sufficient to pay debt service on the Bonds being paid and discharged whether to maturity or date of redemption, as applicable, and (2) an Opinion of Counsel to the effect that the substitution is permitted under the escrow agreement and does not, in and of itself, adversely affect the exclusion from gross income of interest on the Bonds being so paid and discharged for purposes of federal income taxation.

A final copy of any such escrow agreement and the Independent Certified Public Accountant's certificate stating that the escrow is sufficient to meet the standards of the Indenture, the final official statement for the refunding issue (if applicable), the approving opinion of bond counsel with respect to the refunding issue, the Trustee's receipt, and the Trustee's certification as to the application of funds will be furnished to the Credit Provider no later than 10 days (or such later date which is the first date practicable to deliver such items in the event that the refunding issue is not sold early enough to provide such items 10 days prior to the date of defeasance) prior to the defeasance of Bonds by the Commission.

If a forward purchase or supply contract is employed in connection with the defeasance of any of the Bonds, the verification report relating to the defeasance of such Bonds will expressly state that the adequacy of the escrow to accomplish the defeasance relies solely on the initial escrowed investments and the maturing principal thereof and interest income thereon and does not assume performance under or compliance with the forward supply contract, and the applicable escrow agreement will provide that in the event of any discrepancy or difference between the terms of the forward supply contract and the escrow agreement, the terms of the escrow agreement will be controlling.

(d) Notwithstanding anything in the Indenture to the contrary, in the event that the principal and/or interest due of the Bonds is paid by any Credit Provider in accordance with the terms of the Indenture, such Bonds will remain Outstanding for all purposes, not be defeased or otherwise satisfied and not be considered paid by the Commission, and the assignment and pledge of the Net Revenues and other assets under the Indenture and all covenants, agreements and other obligations of the Commission to the Owners so paid will continue to exist and will run to the benefit of Credit Provider, and the Credit Provider will be subrogated to the rights of such Owners, as applicable.

Discharge of Liability on Bonds. Upon the deposit with the Trustee, an escrow agent or other fiduciary, in trust, at or before maturity, of Defeasance Obligations in the necessary amount to pay or redeem Outstanding Bonds (whether upon or prior to their maturity or the redemption date of such Bonds), provided that if such Bonds are to be redeemed prior to the maturity thereof, notice of such redemption has been given as provided in the Indenture or provision satisfactory to the Trustee has been made for the giving of such notice, then all liability of the Commission in respect of such Bonds will cease, determine and be completely discharged, except only that thereafter the Owners thereof will be entitled to payment of the principal of and interest on such Bonds by the Commission, and the Commission will remain liable for such payment, but only out of the Defeasance Obligations deposited in an escrow fund established for this purpose and held by the Trustee, an escrow agent, or other fiduciary, as aforesaid for their payment, subject, however, to the provisions of the Indenture.

Payment of Bonds after Discharge of Indenture. Notwithstanding any provisions of the Indenture, any moneys deposited in trust for the payment of the principal of, or interest or premium on, any Bonds and remaining unclaimed for two years after the principal of all the Outstanding Bonds has become due and payable (whether at maturity or upon call for redemption or by declaration as provided in the Indenture) will then be repaid to the Commission upon its Written Request, and the Owners of such Bonds will thereafter be entitled to look only to the Commission for payment thereof, and all liability of the Trustee or any other fiduciary with respect to such moneys will thereupon cease; provided, however, that before the repayment of such moneys to the Commission as aforesaid, the Trustee may (at the cost of the Commission) first publish at least once in a Financial Newspaper or Journal a notice, in such form as may be deemed appropriate by the Trustee, with respect to the Bonds so payable and not presented and with respect to the provisions relating to the repayment to the Commission of the moneys held for the payment thereof. In the event of the repayment of any such moneys to the Commission as aforesaid, the Owners of the Bonds in respect of which such moneys were deposited will thereafter be deemed to be general creditors of the Commission for amounts equivalent to the respective amounts deposited for the payment of such Bonds and so repaid to the Commission (without interest thereon).

MISCELLANEOUS

Liability of Commission Limited to Net Revenues. (a) Notwithstanding anything in the Indenture contained, the Commission will not be required to advance any moneys derived from the proceeds of any taxes levied or collected by the City, or from any source of income other than the Net Revenues, for the payment of the principal of or interest on the Bonds, for the operation and maintenance of the Enterprise, for the performance of any covenants contained in the Indenture or for the payment of any obligations under the Indenture, including indemnification. The Commission may, however, advance funds for any such purpose, provided that such funds are derived from a source legally available for such purpose and may be used by the Commission for such purpose without incurring indebtedness.

(b) The Bonds will be revenue bonds, payable exclusively from the Net Revenues as provided in the Indenture. The general fund of the City is not liable, and the credit or taxing power of the City is not pledged, for the payment of the Bonds or their interest. The Owners of the Bonds will never have the right to compel the exercise of the taxing power of the City or the forfeiture of any property of the Commission or of the City. The principal of and interest on the Bonds and any premiums upon the redemption of any thereof will not be a debt of the Commission or the City, nor a legal or equitable pledge, charge, lien or encumbrance upon any property of the Commission or of the City or upon any of its income, receipts or revenues except the Net Revenues pledged to the payment thereof as provided in the Indenture.

Successor Is Deemed Included in All References to Predecessor. Whenever in the Indenture or any Supplemental Indenture either the Commission or the Trustee is named or referred to, such reference will be deemed to include the successors or assigns thereof, and all the covenants and agreements in the Indenture contained by or on behalf of the Commission or the Trustee will bind and inure to the benefit of the respective successors and assigns thereof whether so expressed or not.

Limitation of Rights to Parties and Bondowners. Nothing in the Indenture or in the Bonds expressed or implied is intended or will be construed to give to any person, other than the Commission, the Trustee, any Bond Reserve Fund Policy Provider, any Credit Provider and the Owners of the Bonds issued under the Indenture, any legal or equitable right, remedy or claim under or in respect of the Indenture or any covenant, condition or provision therein or contained in the Indenture; and all such covenants, conditions and provisions are and will be held to be for the sole and exclusive benefit of the Commission, the Trustee, the Bond Reserve Fund Policy Provider, the Credit Provider and the Owners of the Bonds issued under the Indenture.

Waiver of Notice. Whenever in the Indenture the giving of notice by mail or otherwise is required, the giving of such notice may be waived in writing by the person entitled to receive such notice and in any

such case the giving or receipt of such notice will not be a condition precedent to the validity of any action taken in reliance upon such waiver.

Destruction of Bonds. Whenever provision is made in the Indenture for the cancellation by the Trustee and the delivery to the Commission of any Bonds, the Trustee may destroy such Bonds (in the presence of an officer of the Commission, if the Commission will so require), and, deliver a certificate of such destruction to the Commission, unless the Commission will, by Written Request of the Commission, request the Trustee to instead cancel and deliver said Bonds to the Commission.

Severability of Invalid Provisions. If any one or more of the provisions contained in the Indenture or in the Bonds is for any reason held to be invalid, illegal or unenforceable in any respect, then such provision or provisions will be deemed severable from the remaining provisions contained in the Indenture and such invalidity, illegality or unenforceability will not affect any other provision of the Indenture, and the Indenture will be construed as if such invalid or illegal or unenforceable provision had never been contained in the Indenture. The Commission declares by the Indenture that it would have adopted the Indenture and each and every other Section, paragraph, sentence, clause or phrase of the Indenture and authorized the issuance of the Bonds pursuant thereto irrespective of the fact that any one or more Sections, paragraphs, sentences, clauses or phrases of the Indenture may be held illegal, invalid or unenforceable. If, by reason of the judgment of any court, the Trustee or any successor. Trustee is rendered unable to perform its duties under the Indenture, and if no successor Trustee be then appointed, all such duties and all of the rights and powers of the Trustee under the Indenture will be assumed by and vest in the Treasurer in trust for the benefit of the Bondowners.

Evidence of Rights of Bondowners. Any request, consent or other instrument required by the Indenture to be signed and executed by Bondowners may be in any number of concurrent writings of substantially similar tenor and may be signed or executed by such Bondowners in person or by an agent or agents duly appointed in writing. Proof of the execution of any such request, consent or other instrument or of a writing appointing any such agent, will be sufficient for any purpose of the Indenture and will be conclusive in favor of the Trustee and of the Commission if made in the manner provided in the Indenture.

The fact and date of the execution by any person of any such request, consent or other instrument or writing may be proved in any manner reasonably acceptable to the Trustee.

The ownership of Bonds will be proved by the bond registration books maintained pursuant to the Indenture

Any request, consent, vote or declaration of the Owner of any Bond will bind every future Owner of the same Bond and the Owner of every Bond issued in exchange therefor or in lieu thereof, in respect of anything done or suffered to be done by the Trustee or the Commission in pursuance of such request, consent, vote or declaration.

Credit Provider and Bond Reserve Fund Policy Provider Provisions. Any provisions in the Indenture requiring consent from the Credit Providers will have no force or effect with respect to a Credit Provider or a Bond Reserve Fund Policy Provider during any period in which such Credit Provider or a Bond Reserve Fund Policy Provider is in default in its obligations under the related Letter of Credit, insurance policy or Bond Reserve Fund Policy. The provisions with respect to the Credit Provider and Bond Reserve Fund Policy Provider may be disregarded if no Credit Providers or Bond Reserve Fund Policy Providers are in existence with respect to the Outstanding Bonds.

Funds and Accounts. Any fund required by the Indenture to be established and maintained by the Commission or the Controller or the Treasurer or the City or the Trustee may be established and maintained in the accounting records of the Commission or the Controller or the Treasurer or the City or the Trustee either as a fund or an account, and may, for the purposes of such records, any audits thereof and any reports or statements with respect thereto, be treated either as a fund or as an account; but all such records with respect to

all such funds will at all times be maintained in accordance with generally accepted accounting practices and with due regard for the protection of the security of the Bonds and the rights of every Owner thereof.

Repeal of Inconsistent Resolution. Any resolution of the Commission, and any part of any resolution, inconsistent with the Indenture, is repealed to the extent of such inconsistency by the Indenture.

Waiver of Personal Liability. No Commission member or officer, agent or employee of the Commission or of the City will be individually or personally liable for the payment of the principal of or interest on the Bonds; but nothing contained in the Indenture will relieve any such Commission member or officer, agent or employee from the performance of any official duty provided by law.

Governing Law. The Indenture will be construed and governed in accordance with the laws of the State of California.

Business Day. Except as specifically set forth in a Supplemental Indenture, any payments or transfers which would otherwise become due on any day which is not a Business Day become due or will be made on the next succeeding Business Day.

FOURTH SUPPLEMENTAL INDENTURE

PROVISIONS RELATING TO 2016 BONDS

Authorization and Terms of 2016 Bonds. The principal of and premium, if any, on the 2016 Bonds will be payable by check or wire in lawful money of the United States of America to the Owner thereof, upon the surrender thereof at the corporate trust office of U.S. Bank National Association, in San Francisco, California, or such other office designated by the Trustee.

The interest on the 2016 Bonds will be payable in like lawful money to the person whose name appears on the bond registration books of the Trustee as the Owner thereof as of the close of business on the 15th day of the calendar month immediately preceding an interest payment date, whether or not such day is a Business Day, such interest to be paid by check mailed to such Owner at such address as appears on such registration books or at such address as such Owner may have filed with the Trustee for that purpose, or at the request of an Owner of at least $1,000,000 in aggregate principal amount of 2016 Bonds filed with the Trustee by such 15th day, by wire transfer to such account designated in such request at a financial institution in the United States.

Each 2016 Bond will bear interest from the interest payment date next preceding the date of authentication thereof unless it is authenticated as of a day during the period from the 16th day of the calendar month next preceding any interest payment date to the interest payment date, inclusive, in which event it will bear interest from such interest payment date, or unless it is authenticated on or before September 15, 2016, in which event it will bear interest from the Closing Date; provided, however, that if, at the time of authentication of any Bond, interest is in default on Outstanding Bonds, such Bond will bear interest from the interest payment date to which interest has previously been paid or made available for payment on the Outstanding Bonds and will be payable to the Owners thereof of record as of a special date as will be established by the Trustee following such default.

On January 30, 2013, certain amendments set forth in the First Amendment that govern the sizing of the Required Reserve for each Series of Bonds became effective in accordance with the terms of the Indenture. As a result, the Commission has determined not to fund the Required Reserve for the 2016 Bonds.

The Commission has reviewed all proceedings previously taken relative to the authorization of the 2016 Bonds and has found, as a result of such review, that all conditions, things and acts required by law to exist, happen or be performed precedent to and in the issuance of the 2016 Bonds do exist, have happened and

have been performed in due time, form and manner as required by law, and the Commission is authorized, pursuant to each and every requirement of law, including the Law, to issue the 2016 Bonds in the manner and form provided in the Fourth Supplemental Indenture.

Use of Depository. (a) The 2016 Bonds will be initially registered as provided in the Fourth Supplemental Indenture. Registered ownership of the 2016 Bonds, or any portions thereof, may not thereafter be transferred except: (i) to any successor of The Depository Trust Company or its nominee, or of any substitute depository designated pursuant the Fourth Supplemental Indenture (a "Substitute Depository"); provided that arty successor of The Depository Trust Company or Substitute Depository will be qualified under any applicable laws to provide the service proposed to be provided by it; (ii) to any Substitute Depository not objected to by the Trustee, upon (1) the resignation of The Depository Trust Company or its successor (or any Substitute Depository or its successor) from its functions as depository, or (2) a determination by the Commission that The Depository Trust Company (or its successor) is no longer able to carry out its functions as depository; provided that any such Substitute Depository will be qualified under any applicable laws to provide the services proposed to be provided by it; or (iii) to any person as provided below, upon (1) the resignation of The Depository Trust Company or its successor (or any Substitute Depository or its successor) from its functions as depository, or (2) a determination by the Commission to remove The Depository Trust Company or its successor (or Substitute Depository or its successor) from its functions as depository.

(b) In the case of any transfer pursuant to the foregoing, upon receipt of all Outstanding 2016 Bonds by the Trustee, together with a Written Request of the Commission to the Trustee designating the Substitute Depository, a single new 2016 Bond, which the Commission will prepare or cause to be prepared, will be executed and delivered for each maturity of 2016 Bonds then Outstanding, registered in the name of such successor or such Substitute Depository, or their nominees, as the case may be, all as specified in such Written Request of the Commission.

In the case of any transfer pursuant to clause (iii) above, upon receipt of all Outstanding 2016 Bonds by the Trustee, together with a Written Request of the Commission to the Trustee, new 2016 Bonds, which the Commission will prepare or cause to be prepared in definitive form, will be executed and delivered in such denominations and registered in the names of such persons as are requested in such Written Request of the Commission, subject to the limitations of the Fourth Supplemental Indenture, provided that the Trustee will not be required to deliver such new 2016 Bonds within a period less than 60 days after the date of receipt of such Written Request from the Commission.

(c) In the case of a partial redemption or an advance refunding of any 2016 Bonds evidencing a portion of the principal maturing in a particular year, The Depository Trust Company or its successor (or any Substitute Depository or its successor) will make an appropriate notation on such 2016 Bonds indicating the date and amounts of such reduction in principal, in form acceptable to the Trustee. The Trustee will not be liable for such depository's failure to make such notations or errors in making such notations.

(d) The Commission and the Trustee will be entitled to treat the person in whose name any 2016 Bond is registered as the Owner thereof for all purposes of the Indenture and any applicable laws, notwithstanding any notice to the contrary received by the Trustee or the Commission; and the Commission and the Trustee will not have responsibility for transmitting payments to, communicating with, notifying, or otherwise dealing with any beneficial owners of the 2016 Bonds. Neither the Commission nor the Trustee will have any responsibility or obligation, legal or otherwise, to any such beneficial owners or to any other party, including The Depository Trust Company or its successor (or Substitute Depository or its successor), except to the Owner of any 2016 Bonds, and the Trustee may rely conclusively on its records as to the identity of the Owners of the 2016 Bonds.

(e) Notwithstanding any other provision of the Fourth Supplemental Indenture and so long as all Outstanding 2016 Bonds are registered in the name of Cede & Co. or its registered assigns, the Commission

and the Trustee will cooperate with Cede & Co., as sole registered Bondowner, and its registered assigns, in effecting payment of the principal of and redemption premium, if any, and interest on the 2016 Bonds by arranging for payment in such manner that funds for such payments are properly identified and are made available on the date they are due all in accordance with the· Letter of Representations delivered by the Commission and the Trustee to The Depository Trust Company with respect to the 2016 Bonds, the provisions of which the Trustee may rely upon to implement the foregoing procedures notwithstanding any inconsistent provisions in the Fourth Supplemental Indenture.

Application of Proceeds of 2016 Series A Bonds. Amounts on deposit in the 2016 Series A Capitalized Interest Account will be applied to the payment of interest on the 2016 Series A Bonds on each April l and October 1, commencing on October 1, 2016, in such amounts as specified in a written certificate of the Commission delivered to the Trustee not less than five Business Days prior to each April 1 and October 1, prior to amounts on the Interest Fund being so used. Amounts remaining on deposit in the 2016 Series A Capitalized Interest Account at such time as the Commission has informed the Trustee in a written certificate of the Commission that the 2016 Series A Capitalized Interest Account will be closed will be transferred by the Trustee to the commission for deposit in the 2016 Series A Capital Project Account. All moneys held by the Trustee in the 2016 Series A Capitalized Interest Account may be invested in Legal Investments maturing not later than the date on which such moneys are required for payment by the Trustee.

Application of Proceeds of 2016 Series B Bonds. Amounts on deposit in the 2016 Series B Capitalized Interest Account will be applied to the payment of interest on the 2016 Series B Bonds on each April l and October 1, commencing on October 1, 2016, in such amounts as specified in a written certificate of the Commission delivered to the Trustee not less than five Business Days prior to each April 1 and October 1, prior to amounts on the Interest Fund being so used. Amounts remaining on deposit in the 2016 Series B Capitalized Interest Account at such time as the Commission has informed the Trustee in a written certificate of the Commission that the 2016 Series B Capitalized Interest Account will be closed will be transferred by the Trustee to the commission for deposit in the 2016 Series B Capital Project Account. All moneys held by the Trustee in the 2016 Series B Capitalized Interest Account may be invested in Legal Investments maturing not later than the date on which such moneys are required for payment by the Trustee.

Establishment and Application of the 2016 Series A Rebate Fund. (a) The Trustee will establish and maintain a fund separate from any other fund established and maintained under the Fourth Supplemental Indenture to be known as the "2016 Series A Rebate Fund." Within the 2016 Series A Rebate Fund, the Trustee will maintain such accounts as it is instructed by the Commission as necessary in order to comply with the terms and requirements of the Certificate as to Arbitrage with respect to the 2016 Series A Bonds, dated the date of issuance of the 2016 Series A Bonds (for purposes of the following, the "Tax Certificate").

Subject to the transfer provisions provided in the Fourth Supplemental Indenture, all money at any time deposited in the 2016 Series A Rebate Fund will be held by the Trustee for the account of the Commission in trust, to the extent required to satisfy the requirements for rebate, as set forth in the Tax Certificate (for purposes of this Article, the "Rebate Requirement"), for payment to the federal government of the United States of America, and no other person will have any rights in or claim to such money. All amounts deposited into or on deposit in the 2016 Series A Rebate Fund will be governed by the Fourth Supplemental Indenture and by the Tax Certificate (which is incorporated in the Fourth Supplemental Indenture by reference). The Trustee will be deemed conclusively to have complied with such provisions and fulfilled its obligation with respect to rebate as long as it follows the written directions of the Commission, including supplying all necessary information in the manner provided in the Tax Certificate. The Trustee will not be required to take any actions under the Fourth Supplemental Indenture or the Tax Certificate in the absence of written directions by the Commission, and will have no liability or responsibility to enforce compliance by the Commission with the terms of the Tax Certificate nor make computations in connection therewith.

(b) Upon the Commission's written direction, an amount will be deposited to the 2016 Series A Rebate Fund by the Trustee from deposits by the Commission so that the balance of the amount on deposit

thereto equals the Rebate Requirement Computations of the Rebate Requirement will be furnished by or on behalf of the Commission in accordance with the Tax Certificate.

(c) The Trustee will have no obligation to rebate any amounts required to be rebated pursuant to the Fourth Supplemental Indenture, other than from moneys held in the funds and accounts created under the Indenture or from other moneys provided to it by or on behalf of the Commission.

(d) The Trustee will invest all amounts held in the 2016 Series A Rebate Fund in Permitted Investments as instructed in writing by the Commission, subject to the restrictions set forth in the Tax Certificate. Moneys will not be transferred from the 2016 Series A Rebate Fund except as provided below.

(e) Upon receipt of the Commission's written directions, the Trustee will remit part or all of the balances in the 2016 Series A Rebate Fund to the United States, as so directed. In addition, if the Commission so directs, the Trustee will deposit moneys into or transfer moneys out of the 2016 Series A Rebate Fund from or into such accounts or funds as directed by the Commission's written directions. Any funds remaining in the 2016 Series A Rebate Fund after redemption and payment of all of the 2016 Series A Bonds and payment and satisfaction of any Rebate Requirement will be withdrawn and remitted to the Commission upon the Commission's written request.

(f) Notwithstanding any other provision of the Indenture, the obligation to remit the Rebate Requirement to the United States and to comply with all other requirements of the Fourth Supplemental Indenture and the Tax Certificate will survive the defeasance or payment in full of the 2016 Series A Bonds.

Establishment and Application of the 2016 Series B Rebate Fund. (a) The Trustee will establish and maintain a fund separate from any other fund established and maintained under the Fourth Supplemental Indenture to be known as the "2016 Series B Rebate Fund." Within the 2016 Series B Rebate Fund, the Trustee will maintain such accounts as it is instructed by the Commission as necessary in order to comply with the terms and requirements of the Certificate as to Arbitrage with respect to the 2016 Series B Bonds, dated the date of issuance of the 2016 Series B Bonds (for purposes of the following, the "Tax Certificate").

Subject to the transfer provisions provided in the Fourth Supplemental Indenture, all money at any time deposited in the 2016 Series B Rebate Fund will be held by the Trustee for the account of the Commission in trust, to the extent required to satisfy the requirements for rebate, as set forth in the Tax Certificate (for purposes of this Article, the "Rebate Requirement"), for payment to the federal government of the United States of America, and no other person will have any rights in or claim to such money. All amounts deposited into or on deposit in the 2016 Series B Rebate Fund will be governed by the Fourth Supplemental Indenture and by the Tax Certificate (which is incorporated in the Fourth Supplemental Indenture by reference). The Trustee will be deemed conclusively to have complied with such provisions and fulfilled its obligation with respect to rebate as long as it follows the written directions of the Commission, including supplying all necessary information in the manner provided in the Tax Certificate. The Trustee will not be required to take any actions under the Fourth Supplemental Indenture or the Tax Certificate in the absence of written directions by the Commission, and will have no liability or responsibility to enforce compliance by the Commission with the terms of the Tax Certificate nor make computations in connection therewith.

(b) Upon the Commission's written direction, an amount will be deposited to the 2016 Series B Rebate Fund by the Trustee from deposits by the Commission so that the balance of the amount on deposit thereto equals the Rebate Requirement Computations of the Rebate Requirement will be furnished by or on behalf of the Commission in accordance with the Tax Certificate.

(c) The Trustee will have no obligation to rebate any amounts required to be rebated pursuant to the Fourth Supplemental Indenture, other than from moneys held in the funds and accounts created under the Indenture or from other moneys provided to it by or on behalf of the Commission.

(d) The Trustee will invest all amounts held in the 2016 Series B Rebate Fund in Permitted Investments as instructed in writing by the Commission, subject to the restrictions set forth in the Tax Certificate. Moneys will not be transferred from the 2016 Series B Rebate Fund except as provided below.

(e) Upon receipt of the Commission's written directions, the Trustee will remit part or all of the balances in the 2016 Series B Rebate Fund to the United States, as so directed. In addition, if the Commission so directs, the Trustee will deposit moneys into or transfer moneys out of the 2016 Series B Rebate Fund from or into such accounts or funds as directed by the Commission's written directions. Any funds remaining in the 2016 Series B Rebate Fund after redemption and payment of all of the 2016 Series B Bonds and payment and satisfaction of any Rebate Requirement will be withdrawn and remitted to the Commission upon the Commission's written request.

(f) Notwithstanding any other provision of the Indenture, the obligation to remit the Rebate Requirement to the United States and to comply with all other requirements of the Fourth Supplemental Indenture and the Tax Certificate will survive the defeasance or payment in full of the 2016 Series B Bonds.

Establishment and Application of the 2016 Series A Capital Project Account and the 2016 Series A Reimbursement Account.

(a) *2016 Series A Capital Project Account.* The Commission covenants and agrees by the Fourth Supplemental Indenture to establish, maintain and hold under the Fourth Supplemental Indenture within the Capital Project Fund, established under the Indenture, a separate account known as the "2016 Series A Capital Project Account" (called the "2016 Series A Capital Project Account" in the Fourth Supplemental Indenture). The Treasurer will hold the amounts on deposit in the 2016 Series A Capital Project Account, which will be maintained and accounted for by the Controller so long as any moneys are on deposit therein. Upon completion of the 2016 Series A Project, the Commission may direct the transfer of any remaining balance in the 2016 Series A Capital Project Account to any other lawfully available fund or account of the Commission; provided such transfer is consistent with the Commission's covenants in the Tax Certificate.

(The moneys in the 2016 Series A Capital Project Account will be held by the Treasurer in trust and applied to the costs of the 2016 Series A Project and the expenses incident thereto or connected therewith, including, if necessary, interest to the extent permitted by law, reimbursement to the Commission for expenses incurred prior to the issuance of the 2016 Series A Bonds or in connection with the Enterprise, architectural, engineering and inspection fees and expenses, apparatus, equipment and furnishings for the Enterprise, testing and inspection, surveys, insurance premiums, losses during construction not insured against because of deductible amounts, the fees and expenses of the Trustee, legal accounting and consultant fees and expenses, and similar expenses. All moneys held by the Treasurer in the 2016 Series A Capital Project Account may be invested in Legal Investments maturing not later than the date on which such moneys are required for payment by the Treasurer. The Treasurer will pay out moneys from the 2016 Series A Capital Project Account only upon warrants drawn by the Controller in the manner provided by law. No withdrawals will be made from the 2016 Series A Capital Project Account for any purpose not authorized by law.

(b) *2016 Series A Reimbursement Account.* The Commission covenants and agrees by the Fourth Supplemental Indenture to establish, maintain and hold under the Fourth Supplemental Indenture within the Capital Project Fund, established under the Indenture, a separate account known as the "2016 Series A Reimbursement Account" (called the "2016 Series A Reimbursement Account" in the Fourth Supplemental Indenture), which will be maintained and accounted for by the Trustee so long as any moneys are on deposit therein. The moneys in the 2016 Series A Reimbursement Account will be held by the Trustee in trust and transferred by the Trustee to U.S. Bank National Association, as issuing and paying agent of the Commercial Paper Notes (for purposes of the Fourth Supplemental Indenture, the "Issuing and Paying Agent") on May 26, 2016, in connection with the reimbursement of certain capital costs previously paid with the proceeds of the Commercial Paper Notes.

Any balance remaining in the 2016 Series A Reimbursement Account following such application of moneys will be transferred to the Treasurer for deposit in the 2016 Series A Capital Project Account. All moneys held by the Trustee in the 2016 Series A Reimbursement Account will be invested in Permitted Investments specified by the Commission or, if the Commission does not so specify, then in Permitted Investments of the type described in clause (f) of the definition thereof that are rated AAAm-G by S&P and Aaa by Moody's.

Establishment and Application of the 2016 Series B Capital Project Account and the 2016 Series B Reimbursement Account.

(a) *2016 Series B Capital Project Account.* The Commission covenants and agrees by the Fourth Supplemental Indenture to establish, maintain and hold under the Fourth Supplemental Indenture within the Capital Project Fund, established under the Indenture, a separate account known as the "2016 Series B Capital Project Account" (called the "2016 Series B Capital Project Account" in the Fourth Supplemental Indenture). The Treasurer will hold the amounts on deposit in the 2016 Series B Capital Project Account, which will be maintained and accounted for by the Controller so long as any moneys are on deposit therein. Upon completion of the 2016 Series B Project, the Commission may direct the transfer of any remaining balance in the 2016 Series B Capital Project Account to any other lawfully available fund or account of the Commission; provided such transfer is consistent with the Commission's covenants in the Tax Certificate.

(The moneys in the 2016 Series B Capital Project Account will be held by the Treasurer in trust and applied to the costs of the 2016 Series B Project and the expenses incident thereto or connected therewith, including, if necessary, interest to the extent permitted by law, reimbursement to the Commission for expenses incurred prior to the issuance of the 2016 Series B Bonds or in connection with the Enterprise, architectural, engineering and inspection fees and expenses, apparatus, equipment and furnishings for the Enterprise, testing and inspection, surveys, insurance premiums, losses during construction not insured against because of deductible amounts, the fees and expenses of the Trustee, legal accounting and consultant fees and expenses, and similar expenses. All moneys held by the Treasurer in the 2016 Series B Capital Project Account may be invested in Legal Investments maturing not later than the date on which such moneys are required for payment by the Treasurer. The Treasurer will pay out moneys from the 2016 Series B Capital Project Account only upon warrants drawn by the Controller in the manner provided by law. No withdrawals will be made from the 2016 Series B Capital Project Account for any purpose not authorized by law.

(b) *2016 Series B Reimbursement Account.* The Commission covenants and agrees by the Fourth Supplemental Indenture to establish, maintain and hold under the Fourth Supplemental Indenture within the Capital Project Fund, established under the Indenture, a separate account known as the "2016 Series B Reimbursement Account" (called the "2016 Series B Reimbursement Account" in the Fourth Supplemental Indenture), which will be maintained and accounted for by the Trustee so long as any moneys are on deposit therein. The moneys in the 2016 Series B Reimbursement Account will be held by the Trustee in trust and transferred by the Trustee to the Issuing and Paying Agent on May 26, 2016, in connection with the reimbursement of certain capital costs previously paid with the proceeds of the Commercial Paper Notes.

Any balance remaining in the 2016 Series B Reimbursement Account following such application of moneys will be transferred to the Treasurer for deposit in the 2016 Series B Capital Project Account. All moneys held by the Trustee in the 2016 Series B Reimbursement Account will be invested in Permitted Investments specified by the Commission or, if the Commission does not so specify, then in Permitted Investments of the type described in clause (f) of the definition thereof that are rated AAAm-G by S&P and Aaa by Moody's.

Terms of Redemption – 2016 Series A Bonds.

Rescission of Notice of Redemption. The Commission, may, at its option, on or prior to the date fixed for redemption in any notice of redemption of 2016 Series A Bonds, rescind and cancel such notice of

redemption by Written Request to the Trustee and the Trustee will mail notice of such cancellation to the recipients of the notice of redemption being cancelled.

The provisions of the Indenture relating to redemption of Bonds set forth in the Fourth Supplemental Indenture will apply to the redemption of the 2016 Series A Bonds.

2016 Series A Sinking Fund Account. The Trustee will establish and hold within the Principal Fund established under the Indenture, a 2016 Series A Sinking Fund Account, which the Commission covenants and agrees by the Fourth Supplemental Indenture to cause to be maintained, for payment of the Bond Obligation of the 2016 Series A Term Bonds.

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2041), will deposit in the 2016 Series A Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series A Term Bonds in the respective principal amounts set forth in the Fourth Supplemental Indenture on the next succeeding October 1 in each of the years set forth in the Fourth Supplemental Indenture (each such deposit of moneys being referred to as a "2016 Series A Minimum Sinking Fund Account Payment").

During the 12-month period immediately preceding each redemption from the 2016 Series A Sinking Fund Account, the Commission may satisfy, in whole or in part, the 2016 Series A Minimum Sinking Fund Account Payment and the redemption therefrom by depositing with the Trustee 2016 Series A Term Bonds for cancellation prior to the Trustee's selection of the 2016 Series A Term Bonds for redemption.

All moneys in the 2016 Series A Sinking Fund Account, at the Written Request of the Commission, will be used and withdrawn by the Trustee at any time for the purchase of 2016 Series A Term Bonds (except that no 2016 Series A Term Bonds maturing in any year will be purchased so long as any 2016 Series A Term Bonds maturing in any earlier year and being of like tenor are Outstanding) at public or private sale, as and when and at such prices (including brokerage and other charges, but excluding accrued interest, which is payable from the Interest Fund) as the Commission will direct by Written Request, but not to exceed the principal thereof, and all 2016 Series A Term Bonds so purchased by the Trustee or deposited by the Commission, will be cancelled and delivered to the Commission; provided, however, that:

(i) all moneys in the 2016 Series A Sinking Fund Account on each September 15, beginning on September 15, 2041 and ending on September 15, 2046, together with any additional sums the Trustee expects to receive for deposit in the 2016 Series A Sinking Fund Account after such date and on or before the next succeeding October 1, will be used and withdrawn by the Trustee solely for the purpose of redeeming the 2016 Series A Term Bonds that are subject to redemption under the Fourth Supplemental Indenture; and

(ii) the Trustee will during each 12-month period beginning with the 12-month period ending on October 1, 2041, purchase or call and redeem (as provided in the Fourth Supplemental Indenture) an aggregate amount of 2016 Series A Term Bonds equal to at least the amount of Bond Obligation of the 2016 Series A Term Bonds identified in the Fourth Supplemental Indenture for such 12-month period reduced by the principal amount of 2016 Series A Term Bonds deposited by the Commission with the Trustee, except that if 2016 Series A Term Bonds of any Term Bond maturity have previously been redeemed or purchased by the Trustee or deposited by the Commission in excess of the amount of Bond Obligation of the 2016 Series A Term Bonds identified in the Fourth Supplemental Indenture, there will be deemed to have been a reduction of the remaining amounts stated above on a Proportionate Basis, and further except that moneys in the 2016 Series A Sinking Fund Account will be used, to the extent necessary, to purchase or retire the Outstanding 2016 Series A Term Bonds at the maturity thereof.

The Commission by the Fourth Supplemental Indenture covenants and agrees with the Owners of the 2016 Series A Term Bonds to call and redeem 2016 Series A Term Bonds from the 2016 Series A Sinking Fund Account pursuant to the Fourth Supplemental Indenture on October 1 in each of the years, and in the amounts, stated above.

Terms of Redemption – 2016 Series B Bonds.

Rescission of Notice of Redemption. The Commission, may, at its option, on or prior to the date fixed for redemption in any notice of redemption of 2016 Series B Bonds, rescind and cancel such notice of redemption by Written Request to the Trustee and the Trustee will mail notice of such cancellation to the recipients of the notice of redemption being cancelled.

The provisions of the Indenture relating to redemption of Bonds set forth in the Fourth Supplemental Indenture will apply to the redemption of the 2016 Series B Bonds.

2016 Series B Sinking Fund Account. The Trustee will establish and hold within the Principal Fund established under the Indenture, a 2016 Series B Sinking Fund Account, which the Commission covenants and agrees by the Fourth Supplemental Indenture to cause to be maintained, for payment of the Bond Obligation of the 2016 Series B Term Bonds.

The Trustee, on or before September 30 of each year (commencing on or before September 30, 2041), will deposit in the 2016 Series B Sinking Fund Account from the Principal Fund moneys in an amount sufficient to call and redeem or to pay at maturity, as the case may be, the principal of 2016 Series B Term Bonds in the respective principal amounts set forth in the Fourth Supplemental Indenture on the next succeeding October 1 in each of the years set forth in the Fourth Supplemental Indenture (each such deposit of moneys being referred to as a "2016 Series B Minimum Sinking Fund Account Payment").

During the 12-month period immediately preceding each redemption from the 2016 Series B Sinking Fund Account, the Commission may satisfy, in whole or in part, the 2016 Series B Minimum Sinking Fund Account Payment and the redemption therefrom by depositing with the Trustee 2016 Series B Term Bonds for cancellation prior to the Trustee's selection of the 2016 Series B Term Bonds for redemption.

All moneys in the 2016 Series B Sinking Fund Account, at the Written Request of the Commission, will be used and withdrawn by the Trustee at any time for the purchase of 2016 Series B Term Bonds (except that no 2016 Series B Term Bonds maturing in any year will be purchased so long as any 2016 Series B Term Bonds maturing in any earlier year and being of like tenor are Outstanding) at public or private sale, as and when and at such prices (including brokerage and other charges, but excluding accrued interest, which is payable from the Interest Fund) as the Commission will direct by Written Request, but not to exceed the principal thereof, and all 2016 Series B Term Bonds so purchased by the Trustee or deposited by the Commission, will be cancelled and delivered to the Commission; provided, however, that:

(i) all moneys in the 2016 Series B Sinking Fund Account on each September 15, beginning on September 15, 2041 and ending on September 15, 2046, together with any additional sums the Trustee expects to receive for deposit in the 2016 Series B Sinking Fund Account after such date and on or before the next succeeding October 1, will be used and withdrawn by the Trustee solely for the purpose of redeeming the 2016 Series B Term Bonds that are subject to redemption under the Fourth Supplemental Indenture; and

(ii) the Trustee will during each 12-month period beginning with the 12-month period ending on October 1, 2041, purchase or call and redeem (as provided in the Fourth Supplemental Indenture) an aggregate amount of 2016 Series B Term Bonds equal to at least the amount of Bond Obligation of the 2016 Series B Term Bonds identified in the Fourth Supplemental Indenture for such 12-month period reduced by the principal amount of 2016 Series B Term Bonds deposited by the

Commission with the Trustee, except that if 2016 Series B Term Bonds of any Term Bond maturity have previously been redeemed or purchased by the Trustee or deposited by the Commission in excess of the amount of Bond Obligation of the 2016 Series B Term Bonds identified in the Fourth Supplemental Indenture, there will be deemed to have been a reduction of the remaining amounts stated above on a Proportionate Basis, and further except that moneys in the 2016 Series B Sinking Fund Account will be used, to the extent necessary, to purchase or retire the Outstanding 2016 Series B Term Bonds at the maturity thereof.

The Commission by the Fourth Supplemental Indenture covenants and agrees with the Owners of the 2016 Series B Term Bonds to call and redeem 2016 Series B Term Bonds from the 2016 Series B Sinking Fund Account pursuant to the Fourth Supplemental Indenture on October 1 in each of the years, and in the amounts, stated above.

Tax Covenants. The Commission covenants that it will not take any action, or fail to take any action, if any such action or failure to take action would adversely affect the exclusion from gross income of the interest on the 2016 Bonds under Section 103 of the Code.

The Commission will not directly or indirectly use or permit the use of any proceeds of the 2016 Bonds or any other funds of the Commission, or take or omit to take any action that would cause the 2016 Bonds to be "arbitrage bonds" within the meaning of Section 148(a) of the Code. To that end, the Commission will comply with all requirements of Section 148 of the Code to the extent applicable to the 2016 Bonds. If at any time the Commission is of the opinion that for purposes of the tax covenants it is necessary to restrict or limit the yield on the investment of any moneys held by the Trustee under the Indenture or otherwise, the Commission will so instruct the Trustee in writing, and the Trustee will take such action as required by such instructions. Without limiting the generality of the foregoing, the Commission agrees that there will be paid from time to time all amounts required to be rebated to the United States pursuant to Section 148(f) of the Code and any temporary, proposed or final Treasury Regulations as may be applied to the 2016 Bonds from time to time. This covenant will survive payment in full or defeasance of the 2016 Bonds. The Commission specifically covenants to pay or cause to be paid to the United States, at the times and in the amounts determined, the Rebate Requirement. The Trustee agrees to comply with all written instructions of the Commission given in accordance with the Tax Certificate.

Notwithstanding any provision of the Fourth Supplemental Indenture, if the Commission provides to the Trustee· an opinion of nationally recognized bond counsel to the effect that any action required under the Fourth Supplemental Indenture or under the Tax Certificate is no longer required, or to the effect that some further action. is required, to maintain the exclusion from gross income of the interest on the 2016 Bonds under Section 103 of the Code, the Commission and the Trustee may rely conclusively on such opinion in complying with the provisions of the Fourth Supplemental Indenture, and the covenants under the Fourth Supplemental Indenture will be deemed to be modified to that extent. The Commission will assure that the proceeds of the 2016 Bonds are not so used as to cause the 2016 Bonds to satisfy the private business tests of Section 141(b) of the Code or the private loan financing test of Section 141(c) of the Code. The Commission will not take any action or permit or suffer any action to be taken if the result of the same would be to cause any of the 2016 Bonds to be "federally guaranteed" within the meaning of Section 149(b) of the Code.

Continuing Disclosure. The Commission covenants and agrees that it will comply with and carry out all of the provisions of the 2016 Continuing Disclosure Certificate. Notwithstanding any other provision of the Indenture, failure of the Commission to comply with the 2016 Continuing Disclosure Certificate will not be considered an Event of Default; however, any Participating Underwriter (as such term is defined in the 2016 Continuing Disclosure Certificate) or any Bondowner or beneficial owner may take such actions as may be necessary and appropriate, including seeking specific performance by court order, to cause the Commission to comply with its obligations under the Fourth Supplemental Indenture, and the sole remedy in the event of any failure of the Commission to comply with the 2016 Continuing Disclosure Certificate will be an action to compel performance.

No Additional Senior State Loans. No Senior State Loans are currently Outstanding. Notwithstanding anything contained in the Indenture, the Commission will not issue or enter into any additional Senior State Loans or pay any amounts with respect to any loan agreement with the State of California (or any board, department or agency thereof) prior to the payment of amounts described in the Indenture.

MISCELLANEOUS

Terms of 2016 Bonds Subject to the Indenture. Except as expressly provided in the Fourth Supplemental Indenture, every term and condition contained in the Indenture will apply to the Fourth Supplemental Indenture, and to the 2016 Bonds, with the same force and effect as if the same were set forth at length in the Fourth Supplemental Indenture, with such omissions, variations and modifications thereof as may be appropriate to make the same conform to the Fourth Supplemental Indenture.

The Fourth Supplemental Indenture and all the terms and provisions contained in the Fourth Supplemental Indenture will form part of the Indenture as fully and with the same effect as if all such terms and provisions had been set forth in the Indenture. The Indenture is ratified and confirmed by the Fourth Supplemental Indenture and will continue in full force and effect in accordance with the terms and provisions thereof, as supplemented and amended by the Fourth Supplemental Indenture.

APPENDIX B

SFPUC WASTEWATER ENTERPRISE FINANCIAL STATEMENTS

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SAN FRANCISCO WASTEWATER ENTERPRISE

Financial Statements

June 30, 2015 and 2014

(With Independent Auditors' Report Thereon)

SAN FRANCISCO WASTEWATER ENTERPRISE

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KPMG LLP
Suite 1400
55 Second Street
San Francisco, CA 94105

Independent Auditors' Report

The Honorable Mayor and Board of Supervisors
City and County of San Francisco:

Report on the Financial Statements

We have audited the accompanying financial statements of the San Francisco Wastewater Enterprise (the Enterprise), an enterprise fund of the City and County of San Francisco, California (the City), as of and for the years ended June 30, 2015 and 2014, and the related notes to the financial statements, which collectively comprise the Enterprise's basic financial statements as listed in the table of contents.

Management's Responsibility for the Financial Statements

Management is responsible for the preparation and fair presentation of these financial statements in accordance with U.S. generally accepted accounting principles; this includes the design, implementation, and maintenance of internal control relevant to the preparation and fair presentation of financial statements that are free from material misstatement, whether due to fraud or error.

Auditors' Responsibility

Our responsibility is to express an opinion on these financial statements based on our audits. We conducted our audits in accordance with auditing standards generally accepted in the United States of America and the standards applicable to financial audits contained in *Government Auditing Standards*, issued by the Comptroller General of the United States. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the financial statements are free from material misstatement.

An audit involves performing procedures to obtain audit evidence about the amounts and disclosures in the financial statements. The procedures selected depend on the auditors' judgment, including the assessment of the risks of material misstatement of the financial statements, whether due to fraud or error. In making those risk assessments, the auditor considers internal control relevant to the entity's preparation and fair presentation of the financial statements in order to design audit procedures that are appropriate in the circumstances, but not for the purpose of expressing an opinion on the effectiveness of the entity's internal control. Accordingly, we express no such opinion. An audit also includes evaluating the appropriateness of accounting policies used and the reasonableness of significant accounting estimates made by management, as well as evaluating the overall presentation of the financial statements.

We believe that the audit evidence we have obtained is sufficient and appropriate to provide a basis for our audit opinion.

KPMG LLP is a Delaware limited liability partnership, the U.S. member firm of KPMG International Cooperative ("KPMG International"), a Swiss entity.

Opinion

In our opinion, the financial statements referred to above present fairly, in all material respects, the respective financial position of the San Francisco Wastewater Enterprise, an enterprise fund of the City and County of San Francisco, California, as of June 30, 2015 and 2014, and the respective changes in financial position, and where applicable, cash flows thereof for the years then ended in accordance with U.S. generally accepted accounting principles.

Emphasis of Matter

As discussed in Note 1, the financial statements of the Enterprise are intended to present the financial position, and the changes in the financial position of only that portion of the City that is attributable to the transactions of the Enterprise. They do not purport to, and do not, present fairly the financial position of the City as of June 30, 2015 and 2014, the changes in its financial position, or where applicable, its cash flows for the years then ended in conformity with U.S. generally accepted accounting principles.

As discussed in Note 2(u) to the financial statements, in 2015, the Enterprise adopted Governmental Accounting Standards Board (GASB) Statement No. 68, *Accounting and Financial Reporting for Pensions*. The July 1, 2014 beginning net position has been restated for the retrospective application of this new accounting guidance. Our opinion is not modified with respect to this matter.

Other Matters

Required Supplementary Information

U.S. generally accepted accounting principles require that the management's discussion and analysis on pages 3 through 14 be presented to supplement the basic financial statements. Such information, although not a part of the basic financial statements, is required by the Governmental Accounting Standards Board who considers it to be an essential part of financial reporting for placing the basic financial statements in an appropriate operational, economic, or historical context. We have applied certain limited procedures to the required supplementary information in accordance with auditing standards generally accepted in the United States of America, which consisted of inquiries of management about the methods of preparing the information and comparing the information for consistency with management's responses to our inquiries, the basic financial statements, and other knowledge we obtained during our audits of the basic financial statements. We do not express an opinion or provide any assurance on the information because the limited procedures do not provide us with sufficient evidence to express an opinion or provide any assurance.

Other Reporting Required by *Government Auditing Standards*

In accordance with *Government Auditing Standards*, we have also issued our report dated November 6, 2015 on our consideration of the Enterprise's internal control over financial reporting and on our tests of its compliance with certain provisions of laws, regulations, contracts, and grant agreements and other matters. The purpose of that report is to describe the scope of our testing of internal control over financial reporting and compliance and the results of that testing, and not to provide an opinion on internal control over financial reporting or on compliance. That report is an integral part of an audit performed in accordance with *Government Auditing Standards* in considering the Enterprise's internal control over financial reporting and compliance.

KPMG LLP

San Francisco, California
November 6, 2015

This section presents management's analysis of the San Francisco Wastewater Enterprise's (the Enterprise) financial condition and activities as of and for the fiscal years ended June 30, 2015 and 2014. Management's Discussion and Analysis (MDA) is intended to serve as an introduction to the Enterprise's financial statements. This information should be read in conjunction with the audited financial statements that follow this section. All dollar amounts, unless otherwise noted, are expressed in thousands of dollars.

The information in this MDA is presented under the following headings:

- Organization and Business
- Overview of the Financial Statements
- Financial Analysis
- Capital Assets
- Debt Administration
- Rates and Charges
- Request for Information

Organization and Business

The San Francisco Public Utilities Commission (SFPUC or the Commission) is a department of the City and County of San Francisco (the City) that is responsible for the maintenance, operation, and development of three utility enterprises: Water, Hetch Hetchy Water and Power, and Wastewater (the Enterprise). The primary responsibility of the Enterprise is to protect the public health and the surrounding bay and ocean receiving waters by collecting, transmitting, treating, and discharging storm and sanitary flows generated in the service area. This includes 993 miles of combined storm and sanitary collection system pipes, sewer mains, transport/storage boxes, other storage structures, and tunnels. San Francisco is the only coastal city in California with a combined sewer system that collects both wastewater and stormwater in the same network of pipes and provides treatment to remove harmful pollutants before discharging into the San Francisco Bay and Pacific Ocean. In addition, the Enterprise serves on a contractual basis certain municipal customers located outside of the City limits, including the North San Mateo County Sanitation District No. 3, Bayshore Sanitary District, and the City of Brisbane. The Enterprise recovers costs of service through user fees based on the volume and strength of sanitary flow. As of June 30, 2015, the Enterprise serves 147,486 residential accounts, which discharge about 16.5 million units of sanitary flow per year (measured in hundreds of cubic feet, or ccf) and 16,200 non-residential accounts, which discharge about 8.2 million ccf per year.

The Enterprise also maintains a Water Pollution Prevention Program that works to keep pollutants from entering the City's sewer system and street storm drains. This includes an industrial and commercial pre-treatment program, which monitors individual businesses that have been issued permits to discharge wastewater into the City's sewer system, as well as outreach, education, and a best management practices program for residents, businesses, and governments.

Overview of the Financial Statements

The Enterprise's financial statements include:

Statements of Net Position present information on the Enterprise's assets, deferred outflows, liabilities, and deferred inflows as of year-end, with the difference reported as net position. Over time, increases or decreases in net position may serve as a useful indicator of whether the financial position of the Enterprise is improving or worsening.

While the *Statements of Net Position* provide information about the nature and amount of resources and obligations at year-end, the *Statements of Revenues, Expenses, and Changes in Net Position* present the results of the Enterprise's operations over the course of the fiscal year and information as to how the net position changed during the year. These statements can be used as an indicator of the extent to which the Enterprise has successfully recovered its costs through user fees and other charges. All changes in net position are reported during the period in which the underlying event giving rise to the change occurs, regardless of the timing of the related cash flows. Thus revenues and expenses are reported in these statements from some items that will result in cash flows in future fiscal periods, such as delayed collection of operating revenues and the expenses of employee earned but unused vacation leave.

The *Statements of Cash Flows* present changes in cash and cash equivalents resulting from operational, capital, non-capital, and investing activities. These statements summarize the annual flow of cash receipts and cash payments, without consideration of the timing of the event giving rise to the obligation or receipt and exclude non-cash accounting measures of depreciation or amortization of assets.

The *Notes to Financial Statements* provide information that is essential to a full understanding of the financial statements that is not displayed on the face of the financial statements.

Financial Analysis

Financial Highlights for Fiscal Year 2015

- Total assets of the Enterprise exceeded total liabilities by $1,161,891.
- Net position decreased by $39,815 or 3.4% during the year.
- Capital assets, net of accumulated depreciation, increased by $127,229 or 7.0% to $1,932,788.
- Operating revenues, excluding interest and investment income, and other non-operating revenues, decreased by $4,095 or 1.6% to $256,002.
- Operating expenses, excluding interest expense, amortization of premium, refunding loss, and issuance cost, and non-operating expenses, increased by $145 or 0.1% to $216,485.

Financial Highlights for Fiscal Year 2014

- Total assets of the Enterprise exceeded total liabilities by $1,180,169.
- Net position increased by $33,110 or 2.9% during the fiscal year.
- Capital assets, net of accumulated depreciation, increased by $144,716 or 8.7% to $1,805,559.
- Operating revenues, excluding interest and investment income and other non-operating revenues, increased by $7,543 or 3.0% to $260,097.
- Operating expenses, excluding interest expense and non-operating expenses, increased by $8,080 or 3.9% to $216,340.

Financial Position

In fiscal year 2015, the Enterprise adopted the provisions of Governmental Accounting Standards Board (GASB) Statement No. 68, *Accounting and Financial Reporting for Pensions*. As restatement of all prior periods is not practical, the cumulative effect of applying this statement is reported as a restatement of beginning net position as of July 1, 2014 (see *Effects of New Pronouncements* on restatement of beginning net position in Significant Accounting Policies, Note 2(u)).

The following table summarizes the Enterprise's changes in net position.

Table 1
Comparative Condensed Net Position
June 30, 2015, 2014, and 2013

	2015	2014	2013 (Restated)*	2015-2014 Change	2014-2013 Change
Total assets:					
Current and other assets	$ 263,361	303,097	448,397	(39,736)	(145,300)
Capital assets, net of accumulated depreciation	1,932,788	1,805,559	1,660,843	127,229	144,716
Total assets	2,196,149	2,108,656	2,109,240	87,493	(584)
Deferred outflows of resources:					
Unamortized loss on refunding of debt	1,255	1,698	2,075	(443)	(377)
Pensions	12,608	—	—	12,608	—
Total deferred outflows of resources	13,863	1,698	2,075	12,165	(377)
Liabilities:					
Current liabilities:					
Revenue bonds	31,115	30,895	32,805	220	(1,910)
Certificates of participation	581	557	538	24	19
Commercial paper	100,000	—	—	100,000	—
Other liabilities	58,831	63,370	61,001	(4,539)	2,369
Subtotal current liabilities	190,527	94,822	94,344	95,705	478
Long-term liabilities:					
Revenue bonds	716,070	752,906	792,016	(36,836)	(39,110)
Certificates of participation	29,742	30,393	31,032	(651)	(639)
Other liabilities	97,919	50,366	45,166	47,553	5,200
Subtotal long-term liabilities	843,731	833,665	868,214	10,066	(34,549)
Total liabilities:					
Revenue bonds	747,185	783,801	824,821	(36,616)	(41,020)
Certificates of participation	30,323	30,950	31,570	(627)	(620)
Commercial paper	100,000	—	—	100,000	—
Other liabilities	156,750	113,736	106,167	43,014	7,569
Total liabilities	1,034,258	928,487	962,558	105,771	(34,071)
Deferred inflows of resources:					
Related to pensions	33,702	—	—	33,702	—
Total deferred inflows of resources	33,702	—	—	33,702	—
Net position:					
Net investment in capital assets	1,088,552	1,066,814	1,074,526	21,738	(7,712)
Restricted for debt service	349	634	960	(285)	(326)
Restricted for capital projects	20,327	22,929	2,931	(2,602)	19,998
Unrestricted	32,824	91,490	70,340	(58,666)	21,150
Total net position	$ 1,142,052	1,181,867	1,148,757	(39,815)	33,110

* Restated per Governmental Accounting Standards Board (GASB) Statement No. 65, *Items Previously Reported as Assets and Liabilities.*

Net Position, Fiscal Year 2015

For the year ended June 30, 2015, the Enterprise's total net position decreased by $39,815 or 3.4% as a result of decreases of $58,666 in unrestricted net position, $2,602 in restricted for capital projects, and $285 in restricted for debt service, offset by an increase of $21,738 in net investment in capital assets (see Table 1).

Total assets increased by $87,493 or 4.1%. Current and other assets decreased by $39,736 or 13.1% due to decreases of $18,650 in restricted and unrestricted cash and investment with City Treasury and outside City Treasury as explained by increased spending for the Sewer System Improvement Program (SSIP) projects, and $15,162 in state and federal grants receivables from reimbursements received for the Valencia Lid Stormwater Flood, Cesar Chavez and Sunnydale Stormwater projects. Restricted receivables decreased by $3,698 due to changes in the capacity fee rate structure starting in July 2014. Charges for services receivable decreased by $2,223 mainly due to reduction in sanitary flow of 2,136 ccf or 7.8%. Other decreases included $100 in interest earnings due to lower cash balances from higher spending for the SSIP projects, and $64 in receivables due from the Department of Public Works for the Mission Bay South and Hunter's View custom work projects. The decreases were offset by increases of $88 in inventory, $49 in advances paid to the Recreation and Parks Department for the Civic Center Garage, $15 in prepayments to vendors for permits, $5 increase in custom work receivables for the Candlestick Point project, and $4 in biofuel receivables.

Capital assets, net of accumulated depreciation and amortization, increased by $127,229 or 7.0% due to the additions of facilities, improvements, machinery and equipment, and construction work in progress. The largest portion of the Enterprise's net position of $1,088,552 or 95.3% represents net investment in capital assets (see Capital Assets section of the MDA for more information). Deferred outflows of resources increased by $12,165 due to $12,608 pensions per implementation of GASB Statement No. 68, offset by $443 of amortization of refunding loss for the 2013 Series A bonds.

Total liabilities represent both current and non-current obligations, the largest portion of which included revenue bonds, certificates of participation and commercial paper, increased by $105,771 or 11.4%. As of June 30, 2015, outstanding revenue bonds payable of $747,185, commercial paper of $100,000, and certificates of participation of $30,323 represented an increase of $62,757 due to issuance of $100,000 commercial paper offset by debt repayments of $37,243. Other liabilities of $156,750, such as payables to vendors, contractors, and other government agencies for goods and services under contractual agreements, increased by $43,014 or 37.8%. The increase included $37,615 in net pension liability due to implementation of GASB Statement No. 68, $6,935 in general liability based on actuarial estimates, $4,828 in other post-employment benefits obligations for healthcare, $181 in customers' prepayments from charges for services, $20 increase in payables due to the San Francisco County Transportation Authority for the 19th Avenue project, and $9 increase from rental prepayments. The increases were offset by decreases of $3,051 in restricted and unrestricted payables primarily due to less payables at year-end, and $3,018 in employee related benefits including workers' compensation, vacation and accrued payroll primarily due to decrease in number of days of accrued payroll. Other decreases included $254 interest expense due to lower outstanding principal balances for 2013 Series A bonds, and $251 of repayment to Hetch Hetchy Power for the 525 Golden Gate Living Machine system and the Lighting Retrofit Project. In accordance with GASB Statement No. 68, $33,702 related to pensions was recognized as deferred inflows of resources in fiscal year 2015.

Net Position, Fiscal Year 2014

For the year ended June 30, 2014, the Enterprise's total net position increased by $33,110 or 2.9% as a result of increases of $21,150 in unrestricted net position and $19,998 in restricted for capital projects, offset by decreases of $7,712 in net investment in capital assets and $326 in restricted for debt service (see Table 1).

Total assets decreased by $584. Current and other assets decreased by $145,300 or 32.4% due to decrease of $139,113 in restricted and unrestricted cash and investment with City Treasury and outside City Treasury from increased SSIP spending, and $7,270 decrease in charges for services receivables resulting from more collections due to increased rates and one-time effort of transition from bi-monthly to monthly billing for residential and most commercial customers effective July 1, 2013. Other decreases included $730 of inventory obsolescence write-off, and $97 in capacity fee receivables. The decreases were offset by increases of $1,366 state grant receivables for the Valencia Lid Stormwater Flood, Cesar Chavez and Sunnydale Stormwater projects, $419 in lease prepayment to the Department of Recreation and Parks for the Civic Center Garage, $103 in interest earnings due to higher cash balance from prior year 2013 Series B bond issuance, and $22 due from the Department of Public Works for the Mission Bay South Project.

Capital assets, net of accumulated depreciation and amortization, increased by $144,716 or 8.7% primarily due to the additions of facilities, improvements, machinery and equipment, and construction work in progress. The largest portion of the Enterprise's net position of $1,066,814 or 90.3% represents net investment in capital assets (see Capital Assets section of the MDA for more information). Deferred outflows of resources decreased by $377 due to amortization of refunding loss.

Total liabilities represent both current and non-current obligations, the largest portion of which includes revenue bonds and certificates of participation, decreased by $34,071 or 3.5%. At June 30, 2014, outstanding revenue bonds payable of $783,801 and certificates of participation of $30,950, represented a decrease of $41,640 due to debt repayments of principal. Other liabilities of $113,736, such as payables to vendors, contractors, and other government agencies for goods and services under contractual agreements, increased $7,569 or 7.1%. The increases included $4,722 in restricted and unrestricted payables primarily for sewer construction project activities, $4,587 in other post-employment benefits obligations for healthcare, and $1,977 of employee-related liabilities such as accrued payroll and workers' compensation. These increases were offset by decreases of $1,846 in claims liability based on actuarial estimates, $1,379 in interest payable due to lower outstanding principal, $259 in unearned revenues relating to completion of Caesar Chavez Sewer custom work project in March 2014, and $233 due to repayments to Hetch Hetchy Power for the 525 Golden Gate living machine system and Lighting Retrofit Project.

In fiscal year 2013, the requirements of GASB Statement No. 65 restated the beginning net position in the amount of $2,511 to write-off unamortized bond issuance costs previously classified as assets. The issuance costs of $2,022 were written off and the remaining $2,075 of unamortized loss on refunding of debt were reclassified from a contra liability to a deferred outflow of resources.

Results of Operations

The following table summarizes the Enterprise's revenues, expenses, and changes in net position.

Table 2
Comparative Condensed Revenues, Expenses, and Changes in Net Position
Years ended June 30, 2015, 2014, and 2013

	2015	2014	2013 (Restated)*	2015-2014 Change	2014-2013 Change
Revenues:					
Charges for services	$ 244,604	244,705	235,479	(101)	9,226
Rents and concessions	821	828	892	(7)	(64)
Other operating revenues	10,577	14,564	16,183	(3,987)	(1,619)
Interest and investment income	1,207	2,400	524	(1,193)	1,876
Other non-operating revenues	6,564	6,882	24,886	(318)	(18,004)
Total revenues	263,773	269,379	277,964	(5,606)	(8,585)
Expenses:					
Operating expenses	216,485	216,340	208,260	145	8,080
Interest expenses	22,791	27,126	19,723	(4,335)	7,403
Amortization of premium, refunding loss, and issuance cost	(5,347)	(7,711)	(612)	2,364	(7,099)
Non-operating expenses	280	533	566	(253)	(33)
Total expenses	234,209	236,288	227,937	(2,079)	8,351
Change in net position before transfers	29,564	33,091	50,027	(3,527)	(16,936)
Transfers from (to) the City and County of San Francisco	(232)	19	888	(251)	(869)
Change in net position	29,332	33,110	50,915	(3,778)	(17,805)
Net position at beginning of year					
Beginning of year, as previously reported	1,181,867	1,148,757	1,100,353	33,110	48,404
Cumulative effect of accounting change	(69,147) **	—	(2,511)	(69,147)	2,511
Beginning of year as restated	1,112,720	1,148,757	1,097,842	(36,037)	50,915
Net position at end of year	$ 1,142,052	1,181,867	1,148,757	(39,815)	33,110

* Restated per Governmental Accounting Standards Board (GASB) Statement No. 65, *Items Previously Reported as Assets and Liabilities.*

** Cumulative effect of accounting change per GASB Statement No. 68, *Accounting and Financial Reporting for Pensions.*

Results of Operations, Fiscal Year 2015

The Enterprise's total revenues were $263,773, a decrease of $5,606 or 2.1% from prior year (see Table 2). Decrease of other operating revenues of $3,987 or 27.4% was primarily due to decreased capacity fees resulting from a rate structure change starting in July 2014. Interest and investment income decreased by $1,193 or 49.7% due to lower cash balances from higher spending for the SSIP projects and an unrealized loss from declines in market values of investments. Other non-operating revenues decreased by $318 or 4.6% due to $705 lower state grant activities as the Valencia Lid Stormwater Flood, Cesar Chavez, and Sunnydale Stormwater projects were near completion, offset by an increase of $387 in other miscellaneous revenues from $759 Pacific Gas and Electric settlement. Other decreases included charges for services of $101 due to reduction of sanitary flow of 2,136 ccf or 7.8% from single-family residential and non-residential customers, offset by an average 5% adopted rate increase. Rents and concessions decreased by $7 as a result of termination of four leases.

Total expenses decreased by $2,079 or 0.9% due to $4,335 decreased interest expenses offset by a decrease of $2,364 in amortization of premium, refunding loss and issuance cost resulting from lower outstanding revenue bonds, and $253 decrease in non-operating expenses for the summer youth program payments to the Garden Project, offset by an increase of $145 in operating expenses. The increase of $145 in operating expenses was mainly attributable to $9,836 in general and administrative expenses due to higher judgment and claims liability based on actuarial estimates, $1,852 in depreciation primarily related to increased capitalizable buildings and structures, and $959 in services provided by other departments due to $739 higher gas and electricity billings from Hetch Hetchy Power and increase of $220 from City Attorney for the 2014 Fall Storm incidents. The

increases were offset by decreases of $8,718 primarily in personnel and fringe benefits due to pension costs reduction, $2,275 in sewer repairs and replacement projects spending, $1,015 decrease in materials and supplies mainly due to prior year's inventory obsolescence write-off expense, and less purchases of equipment maintenance supplies, decrease of $473 in contractual services, and $21 decrease in billings from other City departments services.

Total transfers to City and County increased by $251 due to $200 transfer to Recreation and Parks for the 17th and Folsom Street Project and $51 from Department of Public Works for the Great Highway Project. During fiscal year 2015, net position decreased by $39,815 or 3.4%.

Results of Operations, Fiscal Year 2014

The Enterprise's total revenues of $269,379 for the year decreased by $8,585 or 3.1% over the prior year (see Table 2). Other non-operating revenues decreased by $18,004 primarily due to $17,269 reduction in revenues from two state grants for the Cesar Chavez and Sunnydale Stormwater Flood Management projects, and $411 in federal interest subsidy reduction due to sequestration. Other operating revenues decreased by $1,619 or 10.0% due to decreased capacity fee revenues from decreased permit applications, and $64 decrease in property rental due to termination of two leases. These decreases were offset by increases of $9,226 or 3.9% in charges for services as explained by a one-time effect of transition from bi-monthly to monthly billing effective July 1, 2013, increased sanitary flow of 380,116 ccf or 2.6% from residential single and non-residential customers, coupled with an average 5.0% adopted rate increase, and $1,876 or 358.0% in interest and investment income mainly due to increase in market value and interest rates, and unrealized loss recognized in prior year.

Total expenses increased by $8,351 or 3.7% due to increases of $8,080 in operating expenses and $304 in interest expense net of amortization of premium, refunding loss, and insurance costs, offset by a decrease of $33 in other non-operating expenses due to lower summer youth program payment to the Garden Project.

The increase of $8,080 in operating expenses was attributable to $6,052 in sewer improvement project costs net of write-offs, $2,055 in depreciation primarily related to building and structures, $1,133 in services provided by other departments due to $766 higher natural gas and electricity billings from Hetch Hetchy Power and two new work requests to Citybuild of $200 for employment workforce project, and $167 to Department of Parking and Traffic for the Market Street Sewer Cleaning Project. Other increases included $896 in contractual services due to increased software licensing fees of $409 and construction activities of $487, $349 in material supplies due to inventory obsolescence write-off, $324 in accrued payroll, $322 in workers' compensation and $313 in retirements benefits, offset by a decrease of $3,364 in general and administrative expenses due to lower judgment and claims liability based on actuarial estimate. During fiscal year 2014, net position increased by $33,110 or 2.9%.

Capital Assets

The following table summarizes changes in the Enterprise's capital assets.

Table 3
Capital Assets, Net of Accumulated Depreciation
As of June 30, 2015, 2014, and 2013

		2015	2014	2013	2015-2014 Change	2014-2013 Change
Facilities, improvements, machinery, and equipment	$	1,531,020	1,503,157	1,443,586	27,863	59,571
Intangible assets		3,921	4,023	4,809	(102)	(786)
Land and rights-of-way		35,737	35,737	35,737	—	—
Construction work in progress		362,110	262,642	176,711	99,468	85,931
Total	$	1,932,788	1,805,559	1,660,843	127,229	144,716

Capital Assets, Fiscal Year 2015

The Enterprise has capital assets of $1,932,788, net of accumulated depreciation and amortization, invested in a broad range of utility capital assets as of June 30, 2015 (see Table 3). This amount represents an increase of $127,229 or 7.0% over prior fiscal year. The investment in capital assets includes land, buildings, improvements, wastewater treatment plants, sewer pipes and mains, underground transport and storage boxes, pump stations, machinery, and equipment. Major additions to construction work in progress, net of capitalization and write-offs, increased by $99,468 and included various projects for assessment SSIP validation, sewer repair and replacement, and system improvements. Facilities, improvements, machinery, and equipment increased by $27,863 primarily due to the Spot Sewer Replacement project.

With the adoption of GASB Statement No. 51, *Accounting and Financial Reporting for Intangible Assets,* in fiscal year 2010, the Enterprise reclassified $4,587 in intangible assets primarily composed of sewer easements and customer care and billing computer software. The Enterprise recorded a total of $3,921 and $4,023 intangible assets as of June 30, 2015 and 2014, respectively. The decrease of $102 was due to amortization of $786, offset by an addition of $684 software system developed for two pump stations in fiscal year 2015.

Major additions to construction work in progress during the year ended June 30, 2015 include:

Biosolids and Digester Project	$	20,852
SSIP Program Management		14,009
Northshore to Channel Force Main		12,942
Wastewater Facility Reliability Improvements		12,015
Spot Sewer Replacement		10,116
Main Sewer Replacement		9,806
Oceanside Solids Handling Improvements		8,237
Channel Tunnel and Bayside Drainage		6,347
New Grit Removal and Influent Pump Station		4,735
Other project additions individually below $4,000		79,220
Total	$	178,279

Major depreciable facilities, improvements, intangible assets, machinery, and equipment placed in service, including transfers of completed projects from construction work in progress, during the year ended June 30, 2015 include:

Sewer Repair and Replacement Program	$	37,831
Spot Sewer Repair		13,930
Main Sewer Replacement		6,159
As Needed Sewer Replacement		3,685
Other project additions individually below $2,000		16,410
Total	$	78,015

See Note 4 for additional information about capital assets.

Sewer System Improvement Program

The $6.9 billion SSIP includes three phases over the span of 20 years to maintain ratepayer affordability and minimize impacts to our communities. Phase I projects were 5.6% completed as of June 2015, with 21 projects in planning phase, 14 projects in design phase, and 6 projects in construction phase. The Southeast Treatment Plant (SEP) disinfection and clarifier upgrades building design was 95% completed and the North Point Treatment

Facility (NPF) outfall system rehabilitation design was 65% completed. The SEP and NPF designs are expected to be completed by Spring 2016. Program expenditures as of June 30, 2015 totaled $241.1 million.

Capital Assets, Fiscal Year 2014

The Enterprise has capital assets of $1,805,559, net of accumulated depreciation and amortization, invested in a broad range of utility capital assets as of June 30, 2014 (see Table 3). This amount represents an increase of $144,716 or 8.7% over the prior fiscal year. The investment in capital assets includes land, buildings, improvements, wastewater treatment plants, sewer pipes and mains, underground transport and storage boxes, pump stations, machinery, and equipment. Major additions to construction work in progress, net of capitalization and write-offs, increased by $85,931 due to various projects for assessment SSIP validation, sewer repair and replacement, and system improvements. Facilities, improvements, machinery, and equipment increased by $59,571 primarily due to the Oceanside and Southeast Plant Solids Handling Improvements Project and the Spot Sewer Repair Project.

With the adoption of GASB Statement No. 51, *Accounting and Financial Reporting for Intangible Assets,* in fiscal year 2010, the Enterprise reclassified $4,587 in intangible assets primarily composed of sewer easements and customer care and billing computer software. The Enterprise recorded a total of $4,023 and $4,809 intangible assets with no addition in June 30, 2014 and 2013, respectively. The decrease in intangible assets was due to amortization of $786.

Major additions to construction work in progress during the year ended June 30, 2014 include:

Oceanside and Southeast Solids Handling Improvements	$	15,552
Northshore to Channel Force Main		14,773
SSIP Program Management		13,966
Spot Sewer Repair		13,112
Wastewater Facility Reliability Improvements		11,962
Biosolids and Digester Project		10,265
Southeast Water Pollution Control Oxygen Generation Plant		8,787
Existing Digester Facilities Reliability		7,466
Channel Tunnel and Bayside Drainage		6,440
Assessment SSIP Validation		4,430
Other projects additions individually below $4,000		87,778
Total	$	194,531

Major depreciable facilities, improvements, intangible assets, machinery, and equipment placed in service, including transfers of completed projects from construction work in progress, during the year ended June 30, 2014 include:

Sewer Repair and Replacement Program	$	32,397
Southeast and Oceanside Plant Solids Handling Improvements		20,707
Spot Sewer Repair		9,044
Wastewater Facility Security and Emergency Response		8,689
Southeast Electrical Reliability Upgrades		3,606
Southeast Community Center		3,010
Other items individually below $3,000		29,734
Total	$	107,187

See Note 4 for additional information about capital assets.

Sewer System Improvement Program

The SFPUC is underway with the initial phase of the Sewer System Improvement Program (SSIP) to upgrade the aging sewer system to provide a reliable, sustainable, and seismically safe sewer system now and for future generations. The SSIP will upgrade our existing infrastructure for operational reliability and regulatory compliance, and implement innovative green infrastructure projects, all of which will contribute towards protecting the health of our community and the environment. The $6.9 billion program includes three phases to be implemented over the span of 20 years to maintain ratepayer affordability and minimize impacts to our communities: Phase I consists of $2.7 billion in authorized funds for mission-critical repairs and construction of new facilities, Phase II consists of $3.3 billion for additional upgrades to infrastructure, construction of facilities to provide redundancy to critical system components along with grey and green infrastructure, and Phase III consists of $0.9 billion to complete seismic and reliability project upgrades to the system and ensures full implementation of green infrastructure projects.

Phase I projects are currently in planning or preliminary design phase, which includes rebuilding of the aging solids processing and energy recovery facilities at the SEP, upgrades and operational reliability improvements to existing facilities at Oceanside Treatment Plant, NPF, along with pump stations, large sewers, tunnels and combined sewer discharge structures, construction of eight green infrastructure projects, as well as planning, design, and environmental review of improvements for the Central Bayside project. The evaluation of the alternatives for the Biosolids Digester Facilities projects has been completed and is in the process for further design. Eighty-five percent of the planning and preliminary design of the green infrastructure projects are completed. The construction of the SEP oxygen generation plant and the rehabilitation of two existing digesters have been completed. Other continued efforts are the assessment analysis with the urban watershed, the condition assessment of the pump stations, and the prioritized reliability improvements for the combined sewage discharge structures and the large sewers.

Debt Administration

As of June 30, 2015, 2014 and 2013, the Enterprise's debt from revenue bonds, commercial paper and certificates of participation were $877,508, $814,751 and $856,391, respectively, as shown below in Table 4. More detailed information about the Enterprise's debt activity is presented in Notes 6, 7, and 8 to the financial statements.

Table 4
Outstanding Debt, Net of Unamortized Costs
As of June 30, 2015, 2014, and 2013

		2015	2014	2013 (Restated)*	2015-2014 Change	2014-2013 Change
Revenue bonds	$	747,185	783,801	824,821	(36,616)	(41,020)
Commercial paper		100,000	—	—	100,000	—
Certificates of participation		30,323	30,950	31,570	(627)	(620)
Total	$	877,508	814,751	856,391	62,757	(41,640)

* Restated per Governmental Accounting Standards Board (GASB) Statement No. 65, *Items Previously Reported as Assets and Liabilities.*

Total debt outstanding on June 30, 2015 consisted of $747,185 in revenue bonds, $100,000 in commercial paper and $30,323 in certificates of participation. The increase of $62,757 or 7.7% was due to $100,000 issuance of commercial paper, offset by $37,243 repayment of outstanding debt.

Credit Ratings and Bond Insurance – As of June 30, 2015 and June 30, 2014, the Enterprise carried underlying ratings of "Aa3" and "AA-" from Moody's and Standard & Poor's (S&P), respectively.

Debt Service Coverage – Pursuant to the Indenture for the Wastewater bonds, the Enterprise covenants to collect sufficient net revenues each fiscal year, together with any Enterprise funds (except Bond Reserve Funds) that are available for payment of debt service and are not budgeted to be expended, at least equal to 1.25 times annual debt service for said fiscal year. During fiscal years 2015 and 2014, the Enterprise's net revenues, together with fund balances available to pay debt service and not budgeted to be expended, were sufficient to meet the rate covenant requirements under the Indenture (see Note 8).

Debt Authorization – Pursuant to the Charter Section 8B.124, the Enterprise can incur indebtedness upon two-thirds vote of the Board of Supervisors. The Board of Supervisors has authorized under Proposition E the issuance of an additional $819,036 in revenue bonds for fiscal year 2015, totaling the authorized amount $1,639,602 with $331,585 issued against the authorization. On May 5, 2015, the Enterprise issued $100,000 in commercial paper of the $500,000 authorized.

Cost of Debt Capital – The interest rates on the Enterprise's outstanding revenue bonds ranged from 1% to 5.8%, with a blended true interest cost of 3%, after factoring in federal interest subsidy receipts on Build America Bonds at June 30, 2015. The 2009 Series C certificates of participation carried interest rates ranging from 2% to 5% and 2009 Series D certificates of participation carried interest rates from 6.4% to 6.5% in fiscal years 2015 and 2014.

Rates and Charges

Rate Setting Process

Proposition E, as approved by the voters in November 2002, amended the City Charter by adding the new Article VIIIB, entitled "Public Utilities," which established the Commission's authority to issue new revenue bonds and set wastewater rates. The Commission is required to:

- Establish rates, fees, and charges based on cost of service;
- Retain an independent rate consultant to conduct cost of service studies at least every five years;
- Consider establishing new connection fees;
- Consider conservation incentives and lifeline rates;
- Adopt a rolling five-year forecast annually; and
- Establish a Rate Fairness Board.

Pursuant to the City and County of San Francisco Charter section 8B.125, an independent rate study is performed at least once every five years. A rate study completed in April 2014 resulted in an approved four-year wastewater rate schedule by the Commission on May 13, 2014. The rates are effective July 1, 2014 through fiscal year 2018.

The following table is the Enterprise's approved average rate adjustments since July 1, 2004:

Approved Average Rate Adjustments	
Effective Date	**Rate**
July 1, 2004	11.0 %
July 1, 2005	13.0
July 1, 2006	13.0
July 1, 2007	8.0 [1]
July 1, 2008	9.0
July 1, 2009	7.0 [2]
July 1, 2010	7.0 [2]
July 1, 2011	5.0 [2]
July 1, 2012	5.0 [2]
July 1, 2013	5.0 [2]
July 1, 2014	5.0 [3]
July 1, 2015	5.0 [3]
July 1, 2016	7.0 [3]
July 1, 2017	11.0 [3]

[1] Adjustment effective July 14, 2007.

[2] Adjustment effective annually from July 1, 2009 through the fiscal year beginning July 1, 2013.

[3] Four-year rate increases adopted and effective July 1, 2014.

Request for Information

This report is designed to provide our citizens, customers, investors, and creditors with an overview of the Enterprise's finances and to demonstrate the Enterprise's accountability for the money it receives. Questions regarding any of the information provided in this report or requests for additional financial information should be addressed to San Francisco Public Utilities Commission, Chief Financial Officer, Financial Services, 525 Golden Gate Avenue, 13th Floor, San Francisco, CA 94102.

This report is available at http://www.sfwater.org/index.aspx?page=347.

SAN FRANCISCO WASTEWATER ENTERPRISE
Statements of Net Position
June 30, 2015 and 2014
(In thousands)

Assets		2015	2014
Current assets:			
Cash and investments with City Treasury	$	148,153	122,546
Cash and investments outside City Treasury		87	92
Receivables:			
Charges for services (net of allowance for doubtful accounts of $2,490 in 2015 and $2,514 in 2014)		24,031	25,499
Due from other City departments		46	110
Due from other governments		4,643	19,805
Interest		136	236
Total current receivables		28,856	45,650
Prepaid charges, advances, and other		122	97
Inventory		2,560	2,472
Restricted cash and investments outside City Treasury		12,250	24,601
Total current assets		192,028	195,458
Non-current assets:			
Restricted cash and investments with City Treasury		65,141	97,042
Restricted interest and other receivable		4,018	7,716
Charges for services, less current portion		902	1,657
Prepaid charges, advances, and other, less current portion		1,272	1,224
Capital assets, not being depreciated		400,893	301,425
Capital assets, net of accumulated depreciation		1,531,895	1,504,134
Total non-current assets		2,004,121	1,913,198
Total assets		2,196,149	2,108,656
Deferred outflows of resources			
Unamortized loss on refunding of debt		1,255	1,698
Pensions		12,608	—
Total deferred outflows of resources		13,863	1,698
Liabilities			
Current liabilities:			
Accounts payable		7,190	7,469
Accrued payroll		3,141	6,900
Accrued vacation and sick leave, current portion		3,889	3,380
Accrued workers' compensation, current portion		978	873
Due to other City departments, current portion		273	250
Damage claims liability, current portion		3,994	2,296
Unearned revenues, refunds, and other		2,092	1,902
Bond and loan interest payable		8,557	8,811
Revenue bonds, current portion		31,115	30,895
Commercial paper		100,000	—
Certificates of participation, current portion		581	557
Current liabilities payable from restricted assets		28,717	31,489
Total current liabilities		190,527	94,822
Long-term liabilities:			
Other post-employment benefits obligations		41,980	37,152
Net pension liability		37,615	—
Accrued vacation and sick leave, less current portion		2,469	2,659
Accrued workers' compensation, less current portion		4,542	4,225
Due to other City departments, less current portion		1,269	1,523
Damage claims liability, less current portion		9,473	4,236
Revenue bonds, less current portion		716,070	752,906
Certificates of participation, less current portion		29,742	30,393
Pollution remediation obligation		571	571
Total long-term liabilities		843,731	833,665
Total liabilities		1,034,258	928,487
Deferred inflows of resources			
Related to pensions		33,702	—
Total deferred inflows of resources		33,702	—
Net position			
Net investment in capital assets		1,088,552	1,066,814
Restricted for debt service		349	634
Restricted for capital projects		20,327	22,929
Unrestricted		32,824	91,490
Total net position	$	1,142,052	1,181,867

See accompanying notes to financial statements.

SAN FRANCISCO WASTEWATER ENTERPRISE

Statements of Revenues, Expenses, and Changes in Net Position

Years ended June 30, 2015 and 2014

(In thousands)

		2015	2014
Operating revenues:			
Charges for services	$	244,604	244,705
Rents and concessions		821	828
Capacity fees		6,357	9,523
Other revenues		4,220	5,041
Total operating revenues		256,002	260,097
Operating expenses:			
Personnel services		76,396	85,114
Contractual services		13,841	14,314
Materials and supplies		9,815	10,830
Depreciation		50,254	48,402
Services provided by other departments		36,212	35,274
General and administrative and other		29,967	22,406
Total operating expenses		216,485	216,340
Operating income		39,517	43,757
Non-operating revenues (expenses):			
Federal and state grants		1,075	1,780
Interest and investment income		1,207	2,400
Interest expenses		(22,791)	(27,126)
Amortization of premium, refunding loss, and issuance costs		5,347	7,711
Net gain from sale of assets		1	26
Other non-operating revenues		5,488	5,076
Other non-operating expenses		(280)	(533)
Net non-operating expenses		(9,953)	(10,666)
Change in net position before transfers		29,564	33,091
Transfers from (to) the City and County of San Francisco		(232)	19
Change in net position		29,332	33,110
Net position at beginning of year			
Beginning of year, as previously reported		1,181,867	1,148,757
Cumulative effect of accounting change		(69,147)	—
Beginning of year as restated		1,112,720	1,148,757
Net position at end of year	$	1,142,052	1,181,867

See accompanying notes to financial statements.

SAN FRANCISCO WASTEWATER ENTERPRISE
Statements of Cash Flows
Years ended June 30, 2015 and 2014
(In thousands)

		2015	2014
Cash flows from operating activities:			
Cash received from customers, including cash deposits	$	261,079	265,712
Cash received from tenants for rent		831	838
Cash paid to employees for services		(83,467)	(76,953)
Cash paid to suppliers for goods and services		(80,333)	(79,577)
Cash paid for judgments and claims		(2,897)	(2,638)
Net cash provided by operating activities		95,213	107,382
Cash flows from non-capital financing activities:			
Cash received from grants		16,237	414
Cash received from settlements		759	—
Cash received from miscellaneous revenues		750	1,195
Cash paid for rebates and program incentives		(280)	(533)
Transfers from (to) the City and County of San Francisco		(232)	19
Net cash provided by non-capital financing activities		17,234	1,095
Cash flows from capital and related financing activities:			
Proceeds from sale of capital assets		1	26
Proceeds from commercial paper borrowings		100,000	—
Principal paid on revenue bonds		(30,895)	(32,805)
Principal paid on certificates of participation		(557)	(538)
Interest paid on long-term debt		(35,369)	(37,439)
Interest paid on commercial paper		(4)	—
Issuance costs paid on long-term debt		—	(210)
Acquisition and construction of capital assets		(169,572)	(182,876)
Federal interest income subsidy for Build America Bonds		3,980	3,955
Net cash used in capital and related financing activities		(132,416)	(249,887)
Cash flows from investing activities:			
Interest income received		1,334	1,799
Proceeds from sale of investments outside City Treasury		106,977	137,958
Purchase of investments outside City Treasury		(93,825)	(148,568)
Net cash provided by (used in) investing activities		14,486	(8,811)
Decrease in cash and cash equivalents		(5,483)	(150,221)
Cash and cash equivalents:			
Beginning of year		224,448	374,669
End of year	$	218,965	224,448
Reconciliation of cash and cash equivalents to the statements of net position:			
Cash and investments with City Treasury:			
Unrestricted	$	148,153	122,546
Restricted		65,141	97,042
Cash and investments outside City Treasury:			
Unrestricted		87	92
Restricted		12,250	24,601
Less: Restricted (with maturity more than 90 days - see table in Note 3)		(6,547)	(19,699)
Less: Unrealized gain on investment		(119)	(134)
Cash and cash equivalents at the end of year on statements of cash flows	$	218,965	224,448

(Continued)

SAN FRANCISCO WASTEWATER ENTERPRISE
Statements of Cash Flows
Years ended June 30, 2015 and 2014
(In thousands)

		2015	2014
Reconciliation of operating income to net cash provided by operating activities:			
Operating income	$	39,517	43,757
Adjustment to reconcile operating income to net cash provided by operating activities:			
Depreciation		50,254	48,402
Provision for uncollectible accounts		(24)	(9)
Write-off of capital assets		1,543	2,324
Changes in operating assets and liabilities:			
Receivable charges for services, net		5,932	7,240
Inventory		(88)	730
Accounts payable		(279)	990
Due from other City departments		64	(22)
Advances and other		(64)	(342)
Accrued payroll		(3,759)	1,184
Accrued other post-employment benefits obligations		4,828	4,587
Pension obligations		(10,438)	—
Accrued vacation and sick leave		319	26
Accrued workers' compensation		422	767
Due to other City departments		(130)	(133)
Damage claims liability		6,935	(1,846)
Unearned revenues, refunds, and other liabilities		181	(273)
Total adjustments		55,696	63,625
Net cash provided by operating activities	$	95,213	107,382
Noncash transactions:			
Accrued capital asset costs	$	28,717	31,489
Interfund payable		1,542	1,773

See accompanying notes to financial statements.

(1) Description of Reporting Entity

The San Francisco Wastewater Enterprise (the Enterprise), formerly known as the San Francisco Clean Water Program (the Program), was established in 1977 following the transfer of all sewage-system-related assets and liabilities of the City and County of San Francisco (the City) to the Program.

In 1976, the electorate of the City approved a proposition authorizing the City to issue $240,000 in revenue bonds pursuant to the Revenue Bond Law of 1941 of the State of California for the purpose of acquiring, constructing, improving, and financing improvements to the City's municipal sewage treatment and disposal system. Since then, the City's Board of Supervisors has adopted resolutions (Wastewater Resolutions) providing for the issuance of various sewer revenue and refunding bond series. The Wastewater Resolutions require the City to keep separate books of records and accounts of the Enterprise.

The Enterprise was placed under the jurisdiction of the San Francisco Public Utilities Commission (the Commission) in 1996. The Commission, established in 1932, is responsible for providing operational oversight of the public utility enterprises of the City, which includes the Enterprise along with the City's power and water utilities (i.e. Hetch Hetchy Water and Power, of which the Power Enterprise is a component, and the San Francisco Water Enterprise). The Commission is responsible for determining such matters as the rates and charges for services, approval of contracts, and organizational policy.

Until August 1, 2008, the Commission consisted of five members, all appointed by the Mayor. Proposition E, a City Charter amendment approved by the voters in the June 3, 2008 election, terminated the terms of all five existing members of the Commission, changed the process for appointing new members, and set qualifications for all members. Under the amended Charter, the Mayor continues to nominate candidates to the Commission, but nominees do not take office until the Board of Supervisors votes to approve their appointments by a majority (at least six members). The amended Charter provides for staggered four-year terms for the Commission members and requires them to meet the following qualifications:

- Seat 1 must have experience in environmental policy and an understanding of environmental justice issues.
- Seat 2 must have experience in ratepayer or consumer advocacy.
- Seat 3 must have experience in project finance.
- Seat 4 must have expertise in water systems, power systems, or public utility management.
- Seat 5 is an at-large member.

The SFPUC is a department of the City, and as such, the financial operations of the Enterprise, Hetch Hetchy Water and Power, and the Water Enterprises are included in the Comprehensive Annual Financial Report of the City as enterprise funds. These financial statements are intended to present only the financial position, and the changes in financial position and cash flows of only that portion of the City that is attributable to the transactions of the Enterprise. They do not purport to, and do not, present fairly the financial position of the City as of June 30, 2015 and 2014, the changes in its financial position, or, where applicable, the cash flows for the years then ended in conformity with U.S. generally accepted accounting principles (GAAP).

(2) Significant Accounting Policies

(a) Basis of Accounting and Measurement Focus

The accounts of the Enterprise are organized on the basis of a proprietary fund type, specifically an enterprise fund of the City. The activities of this Enterprise are accounted for with a separate set of self-balancing accounts that comprise the Enterprise's assets, deferred outflows, liabilities, deferred inflows, net position, revenues, and expenses. Enterprise funds account for activities (i) that are financed with debt that is secured solely by a pledge of the net revenues from fees and charges of the activity; or (ii) that are required by laws or regulations that the activity's costs of providing services, including capital costs (such as depreciation or debt service), be recovered with fees and charges, rather than with taxes or similar revenues; or (iii) that the pricing policies of the activity establish fees and charges designed to recover its costs, including capital costs (such as depreciation or debt service).

The financial activities of the Enterprise are accounted for on a flow of economic resources measurement focus, using the accrual basis of accounting in accordance with the U.S. GAAP. Under this method, all assets and liabilities associated with operations are included on the statements of net position; revenues are recognized when earned, and expenses are recognized when liabilities are incurred. Operating revenues are defined as charges to customers, rental income, and capacity fees.

The Enterprise applies all applicable Governmental Accounting Standards Board (GASB) pronouncements.

(b) Cash and Cash Equivalents

The Enterprise considers its pooled deposits and investments held with the City Treasury to be demand deposits and, therefore, cash and cash equivalents for financial reporting. The City Treasury also holds non-pooled cash and investments for the Enterprise. Non-pooled restricted deposits and restricted deposits and investments held outside the City Treasury with original maturities of three months or less are also considered to be cash equivalents.

(c) Investments

Money market funds are carried at cost, which approximates fair value. All other investments are stated at fair value based upon quoted market prices. Changes in fair value are recognized as investment gains or losses and are recorded as a component of non-operating revenues.

(d) Inventory

Inventory consists primarily of construction materials and maintenance supplies and is valued at average cost. Inventory is expensed as it is consumed.

(e) Capital Assets

Capital assets are defined as assets with an initial individual cost of more than $5 and an estimated useful life in excess of one year. Capital assets with an original acquisition date prior to July 1, 1977 are recorded in the financial statements at estimated cost, as determined by an independent professional appraisal, or at cost, if known. All subsequent acquisitions have been recorded at cost. All donated capital assets are valued at estimated fair value at the time of donation. Depreciation and amortization are computed using the straight-line method over the estimated useful lives of the related assets, which range from 1 to 100 years. No depreciation or amortization is recorded in the year of acquisition, and depreciation or amortization is recorded in the year of disposal.

(f) Intangible Assets

Under GASB Statement No. 51, *Accounting and Financial Reporting for Intangible Assets*, intangible assets are defined as identifiable, non-financial assets capable of being separated, sold, transferred, or licensed, and include contractual or legal rights. Examples of intangible assets include rights-of-way easements, land use rights, water rights, licenses, and permits. The accounting pronouncement also provides guidance on the capitalization of internally generated intangible assets, such as the development and installation of computer software by or on behalf of the reporting entity.

According to the standard, the Enterprise is required to capitalize intangible assets with a useful life extending beyond one reporting period. The Enterprise has established a capitalization threshold of $100. GASB Statement No. 51 also requires amortization of intangible assets over the benefit period, except for certain assets having an indefinite useful life. Assets with an indefinite useful life generally provide a benefit that is not constrained by legal or contractual limitations or any other external factor, and therefore, are not amortized (see Note 4).

(g) Construction Work in Progress

The cost of acquisition and construction of major plant and equipment is recorded as construction work in progress. Costs of construction projects that are discontinued are recorded as an expense in the year in which the decision is made to discontinue such projects.

(h) Capitalization of Interest

A portion of the interest cost incurred on capital projects is capitalized on assets that require a period of time for construction or to otherwise prepare them for their intended use. Such amounts are amortized over the useful lives of the assets (see Note 4).

(i) Bond Discount, Premium, and Issuance Costs

Bond issuance costs related to prepaid insurance costs are capitalized and amortized using the effective interest method. Other bond issuance costs are expensed when incurred. Original issue bond discount or premium are offset against the related debt and are also amortized using the effective interest method.

(j) Accrued Vacation and Sick Leave

Accrued vacation pay, which may be accumulated up to 10 weeks per employee, is charged to expense as earned. Sick leave earned subsequent to December 6, 1978 is non-vesting and may be accumulated up to six months per employee.

(k) Workers' Compensation

The Enterprise is self-insured for workers' compensation claims and accrues the estimated cost of those claims, including the estimated cost of incurred but not reported claims (see Note 11(c)).

(l) General Liability

The Enterprise is self-insured for general liability and uninsurable property damage claims. Commercially uninsurable property includes assets that are underground or provide transmission and distribution. Maintained commercial coverage does not cover claims attributed to loss from earthquake, contamination, pollution remediation efforts, and other specific naturally occurring contaminants such as mold. The liability represents an estimate of the cost of all outstanding claims, including adverse loss development and estimated incurred but not reported claims (see Note 11(a)).

(m) Arbitrage Rebate Payable

Certain bonds are subject to arbitrage rebate requirements in accordance with regulations issued by the U.S. Treasury Department. The requirements generally stipulate that earnings from the investment of the tax-exempt bond proceeds that exceed related interest costs on the bonds must be remitted to the federal government on every fifth anniversary of each bond issue. No arbitrage liability is due as of June 30, 2015 and 2014.

(n) Refunding of Debt

Gains or losses occurring from refunding of debt prior to maturity are reported as deferred outflows and deferred inflows of resources from refunding of debt. Deferred outflows and deferred inflows of resources are recognized as a component of interest expense using the effective interest method over the remaining life of the old debt or the life of the new debt, whichever is shorter.

(o) Income Taxes

As a department of a government agency, the Enterprise is exempt from both federal income taxes and California state franchise taxes.

(p) Revenue Recognition

Sewer service charges are based on water usage as determined by the San Francisco Water Enterprise. Effective July 1, 2013, the majority of residential and non-residential customers are billed on a monthly basis except for building and contractor customers which are billed on a bi-monthly basis. Revenues earned but unbilled are accrued as charges for services and reflected as a receivable on the statements of net position.

(q) Use of Estimates

The preparation of financial statements in conformity with U.S. GAAP requires management to make estimates and assumptions that affect the reported amounts of assets and liabilities and the disclosure of contingent assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting period. Actual results could differ from those estimates.

(r) Reclassifications

Certain reclassifications have been made to prior year amounts to conform to current year presentation.

(s) Accounting and Financial Reporting for Pollution Remediation Obligations

According to GASB Statement No. 49, *Accounting and Financial Reporting for Pollution Remediation Obligations*, a government would have to estimate its expected outlays for pollution remediation if it knows a site is polluted and any of the following recognition triggers occur:

- Pollution poses an imminent danger to the public or environment and a government has little or no discretion to avoid fixing the problem;
- A government has violated a pollution prevention-related permit or license;
- A regulator has identified (or evidence indicates it will identify) a government as responsible (or potentially responsible) for cleaning up pollution, or for paying all or some of the cost of the cleanup;
- A government is named (or evidence indicates that it will be named) in a lawsuit to compel it to address the pollution; or

- A government begins or legally obligates itself to begin cleanup or post-cleanup activities (limited to amounts the government is legally required to complete).

As a part of ongoing operations, situations may occur requiring the removal of pollution or other hazardous material. These situations typically arise in the process of acquiring an asset, preparing an asset for its intended use, or during the design phase of projects under review by the project managers. Other times, pollution may arise during the implementation and construction of a major or minor capital project. Examples of pollution may include, but are not limited to: asbestos or lead paint removal, leaking of sewage in underground pipes or neighboring areas, chemical spills, removal and disposal of known toxic waste, harmful biological and chemical pollution of water, or contamination of surrounding soils by underground storage tanks (see Note 12(d)).

(t) Items Previously Reported as Assets and Liabilities

In March 2012, GASB issued Statement No. 65, *Items Previously Reported as Assets and Liabilities*. This Statement is a related or companion standard to GASB Statement No. 63, *Financial Reporting of Deferred Outflows of Resources, Deferred Inflows of Resources, and Net Position*. GASB Statement No. 65 establishes accounting and financial reporting standards that reclassify, as deferred outflows of resources or deferred inflows of resources, certain items that were previously reported as assets and liabilities and recognizes, as outflows of resources or inflows of resources, certain items that were previously reported as assets and liabilities. The requirements of this Statement will improve financial reporting by clarifying the appropriate use of the financial statement elements, deferred outflows of resources and deferred inflows of resources, to ensure consistency in financial reporting. The provisions of the Statement No. 65 are effective for financial statements for periods beginning after December 15, 2012. The Enterprise adopted the requirements of this Statement for fiscal year ended June 30, 2014, therefore restating the financial statements for the earliest period presented as of July 1, 2012.

As a result, the requirements of this Statement restated the beginning net position effective July 1, 2012 in the amount of $2,511 to write off unamortized bond issuance costs previously classified as assets. The requirements have also written off $2,022 of bond issuance costs included in the calculation of refunding losses, and reclassified the remaining $2,075 of unamortized loss on refunding of debt from a contra liability to a deferred outflow of resources.

(u) Effects of New Pronouncements

During fiscal year 2015, the Enterprise implemented the following accounting standards:

1) In June 2012, the GASB issued Statement No. 68, *Accounting and Financial Reporting for Pensions*, which is intended to improve accounting and financial reporting by state and local governments for pensions. It also improves information provided by state and local governmental employers about financial support for pensions that is provided by other entities. Also, in November 2013, the GASB issued Statement No. 71, *Pension Transition for Contributions Made Subsequent to the Measurement Date*, which will eliminate the source of a potential significant understatement of restated beginning net position and expense in the first year of implementation of Statement No. 68 in the accrual basis financial statements of employers and non-employer contributing entities.

   The provisions of these Statements are effective for fiscal years beginning after June 15, 2014. While restatement of all prior periods was not practical because the actuarial information was not available, the cumulative effect of applying this statement is reported as a restatement of beginning

net position as of July 1, 2014. As of July 1, 2014, the Enterprise restated its net position to record beginning net pension liability and beginning deferred outflows as follows:

As Previously Reported	Changes in Accounting Principle	As Restated
$ 1,181,867	(69,147)	1,112,720

2) GASB Statement No. 69, *Government Combinations and Disposals of Government Operations,* issued in January 2013, establishes accounting and financial reporting standards for governments that combine or dispose of their operations. The new standard is effective for periods beginning after December 15, 2013 and had no impact on the Enterprise's financial statements.

3) GASB Statement No. 70, *Accounting and Financial Reporting for Nonexchange Financial Guarantees,* issued in April 2013, establishes accounting and financial reporting standards for governments that offer or receive financial guarantees that are nonexchange transactions. The new standard is effective for periods beginning after June 15, 2013 and had no impact on the Enterprise's financial statements.

(3) Cash, Cash Equivalents, and Investments

The Enterprise's cash, cash equivalents, and investments with the City Treasury are invested in an unrated City pool pursuant to investment policy guidelines established by the City Treasurer. The objectives of the policy guidelines are, in order of priority, preservation of capital, liquidity, and yield. The policy addresses soundness of financial institutions in which the City will deposit funds, types of investment instruments as permitted by the California Government Code, and the percentage of the portfolio that may be invested in certain instruments with longer terms to maturity. The City Treasurer allocates income from the investment of pooled cash at month-end in proportion to the Enterprise's average daily cash balances. The primary objectives of the Enterprise's investment policy are consistent with the City's policy.

The restricted cash for bond reserves is held by an independent trustee outside the City investment pool. The balances as of June 30, 2015 and 2014 were $12,250 and $24,601, respectively. Funds held by the trustee established under the 2003 Indenture are invested in "Permitted Investments" as defined in the Indenture. "Permitted Investments" include money market funds registered under the Federal Investment Company Act of 1940 and whose shares are registered under the Federal Securities Act of 1933 and having a rating by S&P's of "AAAm-G", "AAAm" or "AAm" and a rating by Moody's of "Aaa", "Aa1", or "Aa2".

The following tables present the restricted cash and investments outside City Treasury as of June 30, 2015 and 2014:

Restricted Cash and Investments outside City Treasury

Investments	Credit Ratings (S&P/Moody's)	June 30, 2015 Maturities	Fair Value
U.S. Agencies	AA+/Aaa	April 18, 2016	$ 2,806
U.S. Treasury Money Market Funds	AAAm/Aaa-mf	April 18, 2016	3,741
U.S. Agencies	A-1+ / Aaa	August 20, 2015	1,868
U.S. Treasury Money Market Funds	AAAm/Aaa-mf	< 90 days	3,835
Total			$ 12,250

Restricted Cash and Investments outside City Treasury

Investments	Credit Ratings (S&P/Moody's)	June 30, 2014 Maturities	Fair Value
U.S. Agencies	AA+/Aaa	April 18, 2016	$ 6,838
U.S. Agencies	AA+/Aaa	August 20, 2015	1,868
U.S. Agencies	AA+/Aaa	March 16, 2015	4,007
Commercial Paper	A-1+/P-1	September 30, 2014	6,986
U.S. Treasury Money Market Funds	AAAm/Aaa-mf	<90 days	4,902
Total			$ 24,601

The restricted cash and investments outside City Treasury as of June 30, 2015 and 2014 included a $627 and $338 unrealized loss due to changes in fair values on U.S. Treasury Notes and U.S. Agencies, respectively.

The Enterprise's cash, cash equivalents, and investments are shown on the accompanying statements of net position as follows:

	2015	2014
Current assets:		
Cash and investments with City Treasury	$ 148,153	122,546
Cash and investments outside City Treasury	87	92
Restricted cash and investments outside City Treasury	12,250	24,601
Non-current assets:		
Restricted cash and investments with City Treasury	65,141	97,042
Total cash, cash equivalents, and investments	$ 225,631	244,281

The following table shows the percentage distribution of the City's pooled investments by maturity:

Fiscal years ending June 30	Investment maturities (in months) Under 1	1 to less than 6	6 to less than 12	12 to 60
2014	3.4%	9.6%	15.6%	71.4%
2015	12.6%	11.9%	10.5%	65.0%

(4) Capital Assets

Capital assets with a useful life of 50 years or greater include buildings and structures, sewers, wastewater treatment plants, pump stations, and other pipelines.

Capital assets as of June 30, 2015 and 2014 consisted of the following:

	2014	Increases	Decreases	2015
Capital assets not being depreciated:				
Land and rights-of-way	$ 35,737	—	—	35,737
Intangible assets	3,046	—	—	3,046
Construction work in progress	262,642	178,279	(78,811) *	362,110
Total capital assets not being depreciated	301,425	178,279	(78,811)	400,893
Capital assets being depreciated:				
Facilities and improvements	2,500,175	74,487	—	2,574,662
Intangible assets	3,931	684	—	4,615
Machinery and equipment	88,400	2,844	(43)	91,201
Total capital assets being depreciated	2,592,506	78,015 *	(43)	2,670,478
Less accumulated depreciation for:				
Facilities and improvements	(1,041,456)	(44,299)	—	(1,085,755)
Intangible assets	(2,954)	(786)	—	(3,740)
Machinery and equipment	(43,962)	(5,169)	43	(49,088)
Total accumulated depreciation	(1,088,372)	(50,254)	43	(1,138,583)
Total capital assets being depreciated, net	1,504,134	27,761	—	1,531,895
Total capital assets, net	$ 1,805,559	206,040	(78,811)	1,932,788

* Decrease in construction work in progress is greater than increase in capital assets being depreciated is explained by $1,543 in capital project write-offs, mainly related to the Electrical Arc Flash Improvements Project, the Real Time Control Project, the Oxygen Generation Plant Project, and the Collection System Condition Assessment Project.

	2013	Increases	Decreases	2014
Capital assets not being depreciated:				
Land and rights-of-way	$ 35,737	—	—	35,737
Intangible assets	3,046	—	—	3,046
Construction work in progress	176,711	194,531	(108,600) *	262,642
Total capital assets not being depreciated	215,494	194,531	(108,600)	301,425
Capital assets being depreciated:				
Facilities and improvements	2,405,198	94,977	—	2,500,175
Intangible assets	3,931	—	—	3,931
Machinery and equipment	76,697	12,210	(507)	88,400
Total capital assets being depreciated	2,485,826	107,187 *	(507)	2,592,506
Less accumulated depreciation for:				
Facilities and improvements	(998,861)	(42,595)	—	(1,041,456)
Intangible assets	(2,168)	(786)	—	(2,954)
Machinery and equipment	(39,448)	(5,021)	507	(43,962)
Total accumulated depreciation	(1,040,477)	(48,402)	507	(1,088,372)
Total capital assets being depreciated, net	1,445,349	58,785	—	1,504,134
Total capital assets, net	$ 1,660,843	253,316	(108,600)	1,805,559

* Decrease in construction work in progress is greater than increase in capital assets being depreciated is explained by $2,324 in capital project write-offs, mainly related to the Oceanside and Southeast Plant Solids Handling Improvements Project, the Facility Security and Emergency Response Project, the Existing Digester Facilities Reliability Project, and the Downtown District Sewer Repair Project.

GASB Statement No. 62, *Codification of Accounting and Financial Reporting Guidance Contained in Pre-November 30, 1989 Financial Accounting Standards Board (FASB) and American Institute of Certified Public Accountants (AICPA) Pronouncements,* requires that interest expense incurred during construction of assets be capitalized. Interest included in the construction work in progress and total interest expense incurred during the years ended June 30, 2015 and 2014 are as follows:

		2015	2014
Interest expensed	$	22,791	27,126
Interest included in construction work in progress		12,327	8,934
Total interest incurred	$	35,118	36,060

During fiscal years ended 2015 and 2014, the Enterprise expensed $1,543 and $2,324 respectively, related to demolition, maintenance, and planning costs on certain projects. The amounts of the write-offs were recognized as other operating expenses in the accompanying statements of revenues, expenses, and changes in net position.

(5) Restricted Assets

The Master Bond Resolution was discharged upon the issuance of the 2003 Series A Refunding Bonds. Pursuant to the Indenture, which became effective with the issuance of the 2003 Series A Refunding Bonds, all net revenues of the Enterprise (except amounts on deposit in the rebate fund) are irrevocably pledged to the punctual payment of debt service on the Wastewater revenue bonds. Accordingly, the net revenues of the Enterprise shall not be used for any other purpose while any of its revenue bonds are outstanding except as expressly permitted by the Indenture. Further, all net revenues shall be deposited by the City Treasurer, by instruction of the Enterprise, in special funds designated as the Revenue Fund, which must be maintained in the City Treasury. These funds, held at the City Treasury, are recorded in the statements of net position of the Enterprise as cash and investments. Deposits in the Revenue Fund, including earnings thereon, shall be appropriated, transferred, expended, or used for the following purposes and only in accordance with the following priority:

1. The payment of operation and maintenance costs of the Enterprise;
2. The payment of State revolving fund loans;
3. The payment of bonds, parity State revolving fund loans, policy costs, and amounts due as reimbursement under any letter of credit agreement; and
4. Any other lawful purpose of the Enterprise.

In accordance with the Indenture, the Enterprise maintains certain restricted cash and investment balances in trust. Restricted assets held in trust consisted of the following as of June 30, 2015 and 2014:

		2015	2014
Cash and investments with City Treasury:			
Wastewater revenue bond construction fund	$	65,141	97,042
Cash and investments outside City Treasury:			
2009 Series C Certificates of Participation - 525 Golden Gate		456	449
2009 Series D Certificates of Participation - 525 Golden Gate		1,806	2,039
2010 Series A Wastewater revenue bond fund		3,469	3,452
2010 Series B Wastewater revenue bond fund		6,489	6,468
2013 Series B Wastewater revenue bond fund		30	12,193
Total cash and investments outside City Treasury		12,250	24,601
Interest and other receivables:			
Wastewater revenue bond construction fund including capacity fee receivables		4,018	7,716
Total restricted assets	$	81,409	129,359

Restricted assets listed above as cash and investments with City Treasury are held in subfunds accounts within the Sewer Revenue Fund of the City Treasury.

(6) Short-Term Debt

Under the voter-approved 2002 Proposition E, the Commission and Board of Supervisors authorized the issuance of up to $500,000 in commercial paper for the purpose of reconstructing, expanding, repairing, or improving the Enterprise's facilities. The Enterprise had $100,000 and $0 commercial paper outstanding as of June 30, 2015 and 2014, respectively.

(7) Changes in Long-Term Liabilities

Long-term liability activities for the years ended June 30, 2015 and 2014 are as follows:

	Interest rate *	Maturity (Calendar Year)	2014	Additions	Reductions	2015	Due within one year
Revenue Bonds:							
2010 Series A	4.00% - 5.00%	2021	$ 47,050	—	—	47,050	—
2010 Series B (Build America)	4.65 - 5.82	2040	192,515	—	—	192,515	—
2013 Series A	1.00 - 5.00	2025	160,595	—	(30,895)	129,700	31,115
2013 Series B	4.00 - 5.00	2042	331,585	—	—	331,585	—
For issuance premiums			52,056	—	(5,721)	46,335	—
Total revenue bonds payable			783,801	—	(36,616)	747,185	31,115
2009 Series C Certificates of Participation (COPs)	2.00 - 5.00	2022	6,138	—	(557)	5,581	581
2009 Series C COPs issuance premiums			354	—	(70)	284	—
2009 Series D COPs (Build America)	6.36 - 6.49	2041	24,458	—	—	24,458	—
Other post-employment benefits obligations			37,152	7,599	(2,771)	41,980	—
Net pension liability			—	37,615	—	37,615	—
Accrued vacation and sick leave			6,039	3,435	(3,116)	6,358	3,889
Accrued workers' compensation			5,098	1,979	(1,557)	5,520	978
Damage claims liability			6,532	13,541	(6,606)	13,467	3,994
Pollution remediation obligation			571	—	—	571	—
Total			$ 870,143	64,169	(51,293)	883,019	40,557

* After adjusting for federal interest subsidy, the true interest costs for revenue bond 2010 Series B and 2009 Series D certificates of participation, both issued as Build America Bonds, are 3.7% and 4.3%, respectively.

	Interest rate *	Maturity (Calendar Year)	2013#	Additions	Reductions	2014	Due within one year
Revenue Bonds:							
2010 Series A	4.00% - 5.00%	2021	$ 47,050	—	—	47,050	—
2010 Series B (Build America)	4.65 - 5.82	2040	192,515	—	—	192,515	—
2013 Series A	1.00 - 5.00	2025	193,400	—	(32,805)	160,595	30,895
2013 Series B	4.00 - 5.00	2042	331,585	—	—	331,585	—
For issuance premiums			60,271	—	(8,215)	52,056	—
Total revenue bonds payable			824,821	—	(41,020)	783,801	30,895
2009 Series C Certificates of Participation (COPs)	2.00 - 5.00	2022	6,676	—	(538)	6,138	557
2009 Series C COPs issuance premiums			436	—	(82)	354	—
2009 Series D COPs (Build America)	6.36 - 6.49	2041	24,458	—	—	24,458	—
Other post-employment benefits obligations			32,565	7,265	(2,678)	37,152	—
Accrued vacation and sick leave			6,013	3,072	(3,046)	6,039	3,380
Accrued workers' compensation			4,331	2,364	(1,597)	5,098	873
Damage claims liability			8,378	—	(1,846)	6,532	2,296
Pollution remediation obligation			571	—	—	571	—
Total			$ 908,249	12,701	(50,807)	870,143	38,001

*After adjusting for federal interest subsidy, the true interest costs for revenue bond 2010 Series B and 2009 Series D certificates of participation, both issued as Build America Bonds, are 3.7% and 4.3%, respectively.
#The June 30, 2013 balance was restated to reflect the impact of GASB Statement No. 65 implementation. For fiscal year 2013, the unamortized refunding loss of $2,075 was reclassified to deferred outflows of resources.

The payments of principal and interest amounts on various bonds are secured by net revenues of the Enterprise.

(a) Wastewater Revenue Bonds 2010 Series A

During fiscal year 2010, the Enterprise issued revenue bonds 2010 Series A in the amount of $47,050 with interest rates ranging from 4.0% to 5.0%. Proceeds from the bonds were used to redeem $50,000 in outstanding commercial paper notes, fund a cash debt service reserve fund, and pay the costs of issuing the bonds. The bonds were rated "Aa3" and "AA-" by Moody's and S&P, respectively. Bonds mature through October 1, 2021. The true interest cost is 2.9%. As of June 30, 2015 and 2014, the 2010 Series A bonds' principal amount outstanding was $47,050.

(b) Wastewater Revenue Bonds 2010 Series B

During fiscal year 2010, the Enterprise issued revenue bonds 2010 Series B (Federally Taxable – Build America Bonds – Direct Payment) in the amount of $192,515 with interest rates ranging from 4.7% to 5.8%. Proceeds from the bonds were used to redeem $53,500 in outstanding commercial paper notes, provide funding for capital projects in the amount of $112,429, fund a cash debt service reserve fund, and pay financing costs for the bonds. The bonds were rated "Aa3" and "AA-" by Moody's and S&P, respectively. Bonds mature through October 1, 2040. The true interest cost is 3.7%. As of June 30, 2015 and 2014, the 2010 Series B bonds' principal amount outstanding was $192,515.

(c) Wastewater Revenue Bonds 2013 Series A

In January 2013, the Enterprise issued tax-exempt revenue bonds 2013 Series A in the amount of $193,400 for the purpose of refunding the remaining portion of the outstanding 2003 Series A bonds maturing on and after October 1, 2013. The bonds carried "Aa3" and "AA-" ratings from Moody's and S&P, respectively. The 2013 Series A refunding bonds include serial bonds with interest rates varying from 1.0% to 5.0% and have a final maturity in October 2025. The Series A bonds have a true interest cost of 1.2%. The 2013 Series A bonds also refunded the remaining portion of the outstanding State revolving fund loans. The refunding resulted in the recognition of a deferred accounting loss of $2,986, gross debt service savings of $35,107 over the next 13 years, and an economic gain of $32,783 or 15.4% of the refunded principal. As of June 30, 2015 and 2014, the principal amount outstanding of the 2013 Series A bonds was $129,700 and $160,595, respectively.

(d) Wastewater Revenue Bonds 2013 Series B

In February 2013, the Enterprise issued revenue bonds 2013 Series B in the amount of $331,585 with interest rates ranging from 4.0% to 5.0%. Proceeds from the bonds were used for Wastewater capital projects, pay off all outstanding Wastewater commercial paper notes, and pay the costs of issuing the bonds. The bonds were rated "Aa3" and "AA-" by Moody's and S&P, respectively. Bonds mature through October 1, 2042. The true interest cost is 3.6%. As of June 30, 2015 and 2014, the principal amount outstanding of the 2013 Series B bonds was $331,585.

(e) Future Annual Debt Services of Revenue and Refunding Bonds

The following table presents the future annual debt service relating to the revenue and refunding bonds outstanding as of June 30, 2015. The interest before subsidy amounts include the interest for the 2010 Series A and B and 2013 Series A and B bonds. The federal interest subsidy amounts represent 32.5% of the interest for the 2010 Series B revenue bonds.

		Principal	Interest before subsidy	Federal interest subsidy*	Interest net of subsidy
Fiscal years ending June 30:					
2016	$	31,115	32,383	(3,467)	28,916
2017		20,870	31,384	(3,467)	27,917
2018		20,015	30,481	(3,467)	27,014
2019		21,010	29,544	(3,467)	26,077
2020		22,085	28,555	(3,467)	25,088
2021-2025		101,040	127,272	(16,825)	110,447
2026-2030		100,395	103,643	(13,830)	89,813
2031-2035		124,870	74,610	(9,605)	65,005
2036-2040		152,500	41,276	(4,319)	36,957
2041-2043		106,950	6,657	(130)	6,527
		700,850	505,805	(62,044)	443,761
Less: Current portion		(31,115)			
Add: Unamortized bond premiums		46,335			
Long-term portion as of June 30, 2015	$	716,070			

* The SFPUC received IRS notice dated October 14, 2014 that the federal interest subsidy on the 2010 Series B bonds is reduced by 7.3% or a total reduction of $4,886 due to sequestration over the remaining life of the bonds.

As defined in the Indenture, the principal and interest of the Enterprise's refunding bonds are payable from its corresponding revenue as well as monies deposited in certain funds and accounts pledged thereto (see Note 5).

(f) Certificates of Participation Issued for the 525 Golden Gate Avenue Headquarters Building

In October 2009, the City issued $167,670 in certificates of participation to fund the headquarters building of the SFPUC at 525 Golden Gate Avenue. The 2009 Series C were issued for $38,120 and 2009 Series D for $129,550 as "Build America Bonds" on a taxable basis under the 2009 American Recovery and Reinvestment Act. The 2009 Series C certificates carry interest rates ranging from 2.0% to 5.0% and mature on November 1, 2022. The 2009 Series D certificates carry interest rates ranging from 6.4% to 6.5% and mature on November 1, 2041, after adjusting for the federal interest subsidy, the true interest cost averages 3.4% and 4.3% for Series C and Series D, respectively.

Under the terms of a memorandum of understanding (MOU) between the City and the SFPUC dated October 1, 2009, the City conveyed the real property to the Trustee, the Bank of New York Mellon Trust Company, N.A., which was replaced by U.S. Bank in March 2014 under a property lease in exchange for the proceeds of the sale of the certificates. The Trustee has leased the property back to the City for the City's use under a Project Lease. The City is obligated under the Project Lease to pay base rental payments and other payments to the Trustee each year during the 32-year term of the Project Lease. The Commission makes annual base rental payments to the City for the building equal to annual debt service on the certificates. It is anticipated these lease costs will be offset with reductions in costs associated with current office rental expense.

Each of the three Enterprises has an ownership interest in the building equal to their projected usage of space as follows: Water (73%), Wastewater (15%), and Power (12%). Similarly, each Enterprise is responsible for a portion of the annual Base Rental Payment based on their ownership percentages less contributed equity. The percentage share of Base Rental Payments for the Enterprises is as follows: Water (71.4%), Wastewater (18.9%), and Power (9.7%).

The future annual debt services relating to the certificates of participation 2009 Series C outstanding as of June 30, 2015 are as follows:

Certificates of Participation 2009 Series C (Tax Exempt)	Principal	Interest	Total
Fiscal years ending June 30:			
2016	$ 581	265	846
2017	612	235	847
2018	643	203	846
2019	676	170	846
2020	711	136	847
2021-2023	2,358	180	2,538
	5,581	1,189	6,770
Less: Current portion	(581)		
Add: Unamortized bond premiums	284		
Long-term portion as of June 30, 2015	$ 5,284		

The future annual debt services relating to the certificates of participation 2009 Series D outstanding as of June 30, 2015 are as follows:

Certificates of Participation 2009 Series D (Taxable)	Principal	Interest before subsidy	Federal interest subsidy*	Interest net of subsidy
Fiscal years ending June 30:				
2016	$ —	1,578	(512)	1,066
2017	—	1,578	(512)	1,066
2018	—	1,578	(512)	1,066
2019	—	1,578	(512)	1,066
2020	—	1,578	(512)	1,066
2021-2025	1,763	7,779	(2,524)	5,255
2026-2030	5,099	6,545	(2,123)	4,422
2031-2035	6,276	4,723	(1,532)	3,191
2036-2040	7,740	2,458	(798)	1,660
2041-2042	3,580	236	(76)	160
Total		29,631	(9,613)	20,018
Long-term portion as of June 30, 2015	$ 24,458			

* The SFPUC received IRS notice dated October 14, 2014 that the federal interest subsidy on the 2009 Series D bonds is reduced by 7.3% or a total reduction of $757 due to sequestration over the remaining life of the bonds.

(8) Revenue Pledge

The Enterprise has pledged future revenues to repay various revenue bonds. Proceeds from the revenue bonds provided financing for various capital construction projects, and to refund previously issued bonds. The bonds are payable through fiscal year 2043 and are solely from revenues of the Enterprise.

The original amount of revenue bonds issued, total principal and interest remaining, principal and interest paid during fiscal years 2015 and 2014, applicable net revenues, and funds available for debt service are as follows:

	2015	2014
Bonds issued with revenue pledge	$ 764,550	764,550
Principal and interest remaining due at the end of the year	1,206,655	1,271,023
Principal and interest paid during the year	60,901	64,852
Net revenues for the year ended June 30	96,547	109,181
Funds available for debt service	230,960	218,988

(9) Employee Benefits

(a) Pension Plan

The Enterprise participates in a cost-sharing multiple-employer defined benefit pension plan (the Plan). The Plan is administered by the San Francisco Employees' Retirement System (SFERS). For purposes of measuring the net pension liability, deferred outflows and inflows of resources related to pensions, pension expense, information about the fiduciary net position of the SFERS plans, and additions to/deductions from the fiduciary plan's net position have been determined on the same basis as they are reported by Cheiron for SFERS Plan. Benefit payments (including refunds of employee contributions) are recognized when currently due and payable in accordance with the benefit terms. Investments are reported at fair value.

GASB Statement No. 68 requires that the reported results must pertain to liability and asset information within certain defined timeframes. For this report, the following timeframes are used:

San Francisco Employees' Retirement System – Cost Sharing	
Valuation Date	June 30, 2013 updated to June 30, 2014
Measurement Date	June 30, 2014
Measurement Period	July 1, 2013 to June 30, 2014

The City is an employer of the Plan with a proportionate share of 93.78% as of June 30, 2014. The Enterprise's allocation percentage was determined based on its contributions divided by the City's total employer contributions for fiscal year 2014. The Enterprise's net pension liability, deferred outflows and inflows of resources related to pensions, amortization of deferred outflows and inflows and pension expense to each department is based on its allocated percentage. The Enterprise's allocation of the City's proportionate share was 2.27% as of the measurement date.

Plan Description – The Plan provides basic service retirement, disability, and death benefits based on specified percentages of defined final average monthly salary and provides annual cost of living adjustments (COLA) after retirement. The Plan also provides pension continuation benefits to qualified survivors. The City Charter and the Administrative Code are the authorities which establish and amend the benefit provisions and employer obligations of the Plan. The Retirement System issues a publicly available financial report that includes financial statements and required supplementary information for the Plan. That report may be obtained by writing to the San Francisco Employees' Retirement System, 1145 Market Street, 5th Floor, San Francisco, CA 94103, or by calling (415) 487-7000.

Benefits – Benefits and refunds are recognized when due and payable in accordance with the terms of the Plan. The Retirement System pays benefits according to the category of employment and the type of benefit coverage provided by the City. The four main categories of Plan members are:

a) Miscellaneous Non-Safety Members – staff, operational, supervisory, and all other eligible employees who are not in special membership categories.

b) Sheriff's Department and Miscellaneous Safety Members – sheriffs assuming office on and after January 7, 2012, and undersheriffs, deputized personnel of the Sheriff's Department, and miscellaneous safety employees hired on and after January 7, 2012.

c) Firefighter Members – firefighters and other employees whose principal duties are in fire prevention and suppression work or who occupy positions designated by law as firefighter member positions.

d) Police Members – police officers and other employees whose principal duties are in active law enforcement or who occupy positions designated by law as police member positions.

The membership groups and the related service retirement benefits are included in the Notes to the Basic Financial Statements of San Francisco Employees' Retirement System.

All members are eligible to apply for a disability retirement benefit, regardless of age, when they have 10 or more years of credited service and they sustain an injury or illness that prevents them from performing their duties. Safety members are eligible to apply for an industrial disability retirement benefit from their first day on the job if their disability is caused by an illness or injury that they receive while performing their duties.

All retired members receive a benefit adjustment each July 1, which is the Basic COLA. The majority of adjustments are determined by changes in Consumer Price Index (CPI) with increases capped at 2%. Effective July 1, 2012, the Plan provides for a Supplemental COLA in years when there are sufficient "excess" investment earnings in the Plan and the Plan is fully funded on a market value of assets basis. The maximum benefit adjustment is 3.5% including that Basic COLA. For members hired on or after January 7, 2012, Supplemental COLAs will not be permanent adjustments to retirement benefits.

Funding and Contribution Policy – Contributions are made to the basic plan by both the City and the participating employees. Employee contributions are mandatory as required by the Charter. Employee contribution rates for fiscal years 2015 and 2014 varied from 7.5% to 13.0% as a percentage of gross covered salary. For fiscal year ended June 30, 2015, most employee groups agreed through collective bargaining for employees to contribute the full amount of the employee contributions on a pretax basis. The Enterprise is required to contribute at an actuarially determined rate. Based on the July 1, 2013 actuarial report, the Enterprise's required employer contribution rates for fiscal years 2015 and 2014 were 26.76% and 24.82%, respectively.

Employer contributions and employee contributions made by the employer to the Plan are recognized when due and the employer has made a formal commitment to provide the contributions. The City's proportionate share of employer contributions recognized by the Retirement System in fiscal year ended June 30, 2014 was $499,752. The Enterprise's allocation of employer contributions for fiscal years 2015 and 2014 were $12,288 and $11,290, respectively.

Pension Liabilities, Pension Expenses, Deferred Outflows and Inflows of Resources Related to Pensions – As of June 30, 2015, the City reported net pension liabilities for its proportionate share of the pension liability of the Plan of $1,660,365. The City's net pension liability for the Plan was measured as the proportionate share of the net pension liability. The net pension liability of the Plan was measured as of June 30, 2014, and the total pension liability for the Plan used to calculate the net pension liability was determined by an actuarial valuation as of June 30, 2013 rolled forward to June 30, 2014 using standard update procedures. The City's proportion of the net pension liability was based on a projection of the City's long-term share of contributions to the pension plan relative to the projected contributions of all participating employers, actuarially determined. The Enterprise's allocation of the City's proportionate share of the net pension liability for the Plan as of June 30, 2015 was $37,615.

For the year ended June 30, 2015, the City's recognized pension expense was $95,710 including amortization of deferred outflows and inflows related pension items. The Enterprise's allocation of pension expense including amortization of deferred outflows and inflows related pension items was $2,169. As of June 30, 2015, the City's reported deferred outflows of resources and deferred inflows of resources related to pensions from the following sources and will be amortized annually and recognized as pension expense.

Schedule of Deferred Outflows and Inflows of Resources

	Deferred Outflows of Resources	Deferred Inflows of Resources
Employer contribution subsequent to the measurement date	$ 12,608	—
Changes of assumptions	—	1,247
Net difference between projected and actual earnings on pension plan investments	—	32,223
Change in proportionate share	—	232
Total	**$ 12,608**	**33,702**

Amounts reported as deferred outflows, exclusive of contributions made after the measurement date, and deferred inflows of resources will be recognized in pension expense as follows:

Fiscal years	Deferred Outflows/(Inflows) of Resources
2016	$ (8,425)
2017	(8,425)
2018	(8,425)
2019	(8,427)

Actuarial Assumptions – A summary of the actuarial assumptions and methods used to calculate the total pension liability as of June 30, 2014 is provided below, including any assumptions that differ from those used in the July 1, 2013 actuarial valuation. Refer to the July 1, 2013 actuarial valuation report for a complete description of all other assumptions, which can be found on the Retirement System's website.

Key Actuarial Assumptions				
Valuation Date	June 30, 2013 updated to June 30, 2014			
Measurement Date	June 30, 2014			
Actuarial Cost Method	Entry-Age Normal Cost Method			
Expected Rate of Return	7.58%			
Municipal Bond Yield	4.39% as of June 30, 2013			
	4.31% as of June 30, 2014			
	Bond Buyer 20-Bond GO Index, July 3, 2013 and July 2, 2014			
Discount Rate	7.52% as of June 30, 2013			
	7.58% as of June 30, 2014			
Administrative Expenses	0.45% of payroll			
	Old Miscellaneous and all New Plans	Old Police & Fire, pre 7/1/75 Retirements	Old Police & Fire, Charters A8.595 and A8.596	Old Police & Fire, Charters A8.559 and A8.585
Basic COLA	2.00%	3.00%	4.00%	5.00%

Mortality rates for active members were based upon the RP-2000 Employee Tables for Males and Females projected using Scale AA to 2030 for females and to 2005 for males. Mortality rates for healthy annuitants were based upon the RP-2000 Healthy Annuitant Tables for Males and Females projected using Scale AA to 2020.

Discount Rate – The beginning and end of year measurements are based on different assumptions and contribution methods that result in different discount rates. The discount rate was 7.52% as of June 30, 2013 and 7.58% as of June 30, 2014.

The discount rate used to measure the total pension liability as of June 30, 2014 was 7.58%. The projection of cash flows used to determine the discount rate assumed that plan member contributions will continue to be made at the rates specified in the Charter. Employer contributions were assumed to be made in accordance with the contribution policy in effect for July 1, 2014 actuarial valuation. That policy includes contributions equal to the employer portion of the entry age normal costs for members as of the valuation data plus an amortization payment on the unfunded actuarial liability. The amortization payment is based on 15-year closed amortization of Charter amendments as a level percentage of payment and closed 20-year amortization as a level percentage of payroll of experience gains and losses and assumption changes. Supplemental COLAs are amortized over a closed 5-year period from the date they are granted. The unfunded actuarial liability is based on an actuarial value of assets that smooths investment gains and losses over five years and a measurement of the actuarial liability that excludes the value of any future Supplemental COLAs.

While the contributions and measure of actuarial liability in the valuation do not anticipate any Supplemental COLAs, the projected contributions for the determination of the discount rate include the anticipated future amortization payments on future Supplemental COLA's for current members when they are expected to be granted. For a Supplemental COLA to be granted, the market value of assets must exceed the actuarial liability at the beginning of the year and the actual investment earnings during the year must exceed the expected investment earnings on the actuarial value of assets. When a Supplemental COLA is granted, the amount depends on the amount of excess earnings and the basic COLA amount for each membership group. In most cases, the large majority of members receive a 1.50% Supplemental COLA.

Because the probability of a Supplemental COLA depends on the current funded level of the System, we developed an assumption as of June 30, 2014 of the probability and amount of Supplemental COLA for each future year. The table below shows the net assumed Supplemental COLAs for member with a 2.00% basic COLAs for sample years.

Assumed Supplemental COLA for Members with a 2.0% Basic COLA

Fiscal years	Assumption
2015	0.00 %
2020	0.36
2025	0.38
2030	0.38
2035+	0.38

The projection of benefit payments to current members for determining the discount rate includes the payment of anticipated future Supplemental COLAs.

Based on these assumptions, the Retirement System's fiduciary net position was projected to be available to make projected future benefit payments for current members until fiscal year end 2083 when only a portion of the projected benefit payments can be made from the projected fiduciary net position. Projected benefit payments are discounted at the long-term expected return on assets of 7.58% to the extent the fiduciary net position is available to make the payments and at the municipal bond rate of 4.31% to the extent they are not available. Since the payments discounted at the municipal bond rate are relatively few and far in the future, the municipal bond rate does not affect the single equivalent rate when rounded to two decimal places. Consequently, the single equivalent rate used to determine the total pension liability as of June 30, 2014 is 7.58%.

The long-term expected rate of return on pension plan investments was 7.58%. It was set by the Retirement Board after consideration of both expected future returns and historical returns experienced by the Retirement System. Expected future returns were determined by using a building-block method in which best-estimate ranges of expected future real rates of return were developed for each major asset class. These ranges were combined to produce the long-term expected rate of return by weighting the expected future real rates of return by the target asset allocation percentage and by adding expected inflation. Target allocation and best estimates of geometric long-term expected real rates of return (net of pension plan investment expense and inflation) for each major asset class are summarized in the following table.

Long-Term Expected Real Rates of Return

Asset Class	Target Allocation	Long-Term Expected Real Rate of Return
Global Equity	47.00 %	5.30 %
Fixed Income	25.00	1.80
Private Equity	16.00	8.80
Real Asset	12.00	5.80

Sensitivity of Proportionate Share of the Net Pension Liability to Changes in the Discount Rate – The following presents the Enterprise's allocation of the employer's proportionate share of the net pension liability for the Plan, calculated using the discount rate, as well as what the Enterprise's allocation of the employer's proportionate share of the net pension liability would be if it were calculated using a discount rate that is 1% lower or 1% higher than the current rate.

Employer	Proportionate Share	1% Decrease share of NPL @ 6.58%	Share of NPL @ 7.58%	1% Increase Share of NPL @ 8.58%
Wastewater	2.27%	$ 93,172	37,615	(9,040)

(b) Healthcare Benefits

Healthcare benefits for the Enterprise employees, retired employees, and surviving spouses are financed by beneficiaries and by the City through the City and County of San Francisco Health Service System (the Health Service System). The Enterprise's annual contribution for both active and retired employees was $9,921 and $10,091 in 2015 and 2014, respectively. Included in these amounts are $2,771 and $2,678 in 2015 and 2014, respectively, to provide post-retirement benefits for the Enterprise's retired employees, on a pay-as-you-go basis.

The City has determined a citywide Annual Required Contribution (ARC), interest on net other post-employment benefits (OPEB) other than pensions obligations, ARC adjustment, and OPEB cost based upon an actuarial valuation performed in accordance with GASB Statement No. 45, *Accounting and Financial Reporting by Employers for Postemployment Benefits Other Than Pensions,* by the City's actuaries. The ARC represents a level of funding that, if paid on an ongoing basis, is projected to cover the normal cost of each year and any unfunded actuarial liabilities (or funding excess) amortized over 30 years.

The following table shows the components of the City's annual OPEB allocations for the Enterprise based upon its percentage of citywide payroll costs for the years ended June 30, 2015 and 2014, for the amount contributed to the plan, and changes in the City's net OPEB obligations:

		2015	2014
Annual required contribution	$	7,323	7,020
Interest on net OPEB obligations		1,666	1,469
Adjustment to ARC		(1,390)	(1,224)
Annual OPEB cost		7,599	7,265
Contribution made		(2,771)	(2,678)
Increase in net OPEB obligations		4,828	4,587
Net OPEB obligations - beginning of year		37,152	32,565
Net OPEB obligations - end of year	$	41,980	37,152

The City issues a publicly available financial report at a citywide level that includes the complete note disclosures and Required Supplementary Information (RSI) related to the City's post-retirement healthcare obligations. The report may be obtained by writing to the City and County of San Francisco, Office of the Controller, 1 Dr. Carlton B. Goodlett Place, Room 316, San Francisco, CA 94102, or by calling (415) 554-7500.

(10) Related Parties

Various common costs incurred by the Commission are allocated proratably among the Enterprise, Hetch Hetchy Water and Power, and the Water Enterprise. The allocations are based on the Commission management's best estimate and may change from year to year depending on the activities incurred by each Enterprise and the information available. The administrative costs of $25,408 or 31.3% and $24,281 or 31.6% are allocated to the Enterprise for the years ended June 30, 2015 and 2014, respectively.

The City performs certain administrative services such as maintenance of accounting records and investment of cash for all fund groups within the City. The various funds are charged for these services based on the City's indirect cost allocation plan. The overhead allocation paid to the General Fund of the City by the Enterprise is $0 and $170 for the years ended June 30, 2015 and 2014, respectively, and is

included in other operating expenses in the accompanying financial statements. The fiscal years 2015 and 2014 reflected the true-up adjustment between projection and actual.

The Enterprise purchases electricity from Hetch Hetchy Power at market rates. This amount, totaling $9,893 and $9,154 for the years ended June 30, 2015 and 2014, respectively, has been included in services provided by other departments in the accompanying financial statements.

The Enterprise provides sewer services to other City departments at non-residential rates established by the Commission, and through the Customer Services Department, bills and collects sewer service charges on behalf of the Enterprise.

The City's Department of Public Works provides certain engineering and other services to the Enterprise and charges amounts designed to recover its costs. These services are primarily related to street cleaning, engineering, building repair, and sewer repair. This amount totaling approximately $14,418 and $14,587 for the years ended June 30, 2015 and 2014, respectively, has been included in services provided by other departments in the accompanying financial statements.

A variety of other City departments provide services such as purchasing, legal, data processing, telecommunications, and human resources to the Enterprise and charge amounts designed to recover those departments' costs. These charges totaling approximately $11,901 and $11,532 for the years ended June 30, 2015 and 2014, respectively, have been included in services provided by other departments in the accompanying financial statements.

As of June 30, 2015, the Enterprise has payables in the amount of $1,542, of which $1,372 is associated with the SFPUC Headquarters Living Machine system, and $150 for the Lighting Retrofit Projects to Hetch Hetchy Power, and $20 for the 19th Avenue project to San Francisco County Transportation Authority.

SFPUC's 75-year lease agreement with the San Francisco Recreation and Parks Department, for the use of parking spaces for its fleet of vehicles at the Civic Center Garage, commenced on February 1, 2011. Total payment under this agreement is $6,274, which was fully made as of fiscal year 2015. The expenses and prepayments among the three SFPUC Enterprises are based on 525 Golden Gate occupancy. As of June 30, 2015, the Enterprise's allocable shares of expenses and prepayment were $22 and $1,294, respectively, and as of June 30, 2014 were $20 and $1,244, respectively.

(11) Risk Management

The Enterprise's Risk Management program includes both self-insured (i.e. self-retention) and insured exposures at risk. Risk assessments and purchasing of insurance coverage are collaboratively coordinated by SFPUC Enterprise Risk Management and the City's Office of Risk Management. With certain exceptions, the City and the Enterprise's general approach is to first evaluate the exposure at risk for self-insurance. Based on this analysis, internal mitigation strategies and financing through a self-retention mechanism is generally more economical as the SFPUC in coordination with the City Attorney's Office administers, adjusts, settles, defends, and pays claims from budgeted resources (i.e. pay-as-you-go fund). When economically more viable or when required by debt financing covenants, the Enterprise obtains commercial insurance. At least annually, the City actuarially determines general liability and workers' compensation risk exposures. The Enterprise does not maintain commercial earthquake coverage, with certain minor exceptions, such as a sub-limit for fire-sprinkler leakage due to earthquake under the Property Insurance program.

Primary Risks	Typical Coverage Approach
General liability	Self-Insured
Property	Purchased Insurance and Self-Insured
Electronic data processing	Purchased Insurance and Self-Insured
Workers' compensation	Self-Insured through Citywide Pool

Other Risks	Typical Coverage Approach
Surety bonds	Purchased and Contractually Transferred
Errors and omissions	Combination of Self-Insured and Contractual Risk Transfer
Professional liability	Combination of Self-Insured and Contractual Risk Transfer
Public officials liability	Purchased Insurance
Employment practices liability	Purchased Insurance
Builders' risk	Contractually Transferred
Crime	Purchased Insurance

(a) General Liability

Through coordination with the Controller and the City Attorney's Office, the general liability risk exposure is actuarially determined and is addressed through pay-as-you-go funding as part of the budgetary process. Associated costs and estimates are booked as expenses as required under GAAP for financial statement purposes for both the Enterprise and the City and County of San Francisco's Comprehensive Annual Financial Report. The claim expense allocations are determined based on actuarially determined anticipated claim payments and the projected timing of disbursement.

The changes for the general liability (damage claims) for the years ended June 30, 2015 and 2014 are as follows:

Fiscal years	Beginning of year	Claims and changes in estimates	Claims paid	End of year
2014	$ 8,378	(798)	(1,048)	6,532
2015	6,532	13,541	(6,606)	13,467

(b) Property and Electronic Data Processing

The Enterprise's property risk management approach varies depending on whether the facility is currently under construction, the property is part of revenue-generating operations, the property is of high value, or is mission-critical in nature. During the course of construction, the Enterprise requires each contractor to provide its own insurance, while ensuring the full scope of work be covered with satisfactory levels to limit the Enterprise's risk exposure. Once construction is complete, the Enterprise performs an assessment to determine whether liability/loss coverage will be obtained through the commercial property policy or self-insurance. The majority of property scheduled in the insurance program is for either: (1) revenue generating facilities, (2) debt financed facilities, (3) mandated coverage to meet statutory requirements for bonding of various public officials, or (4) high-value, mission-critical property or equipment. The Electronic Data Processing policy protects selected high-value electronic property in case of damage or loss.

(c) Workers' Compensation

The City actuarially determines and allocates workers' compensation costs to the Enterprise according to a formula based on the following: (i) the dollar amount of claims; (ii) yearly projections of payments based on historical experience; and (iii) the size of the Enterprise's payroll. The administration of workers' compensation claims and payouts are handled by the Workers' Compensation Division of the City's Department of Human Resources. Statewide workers' compensation reforms have resulted in

budgetary savings in recent years. The City continues to develop and implement improved programs, such as return-to-work programs, to lower or mitigate the growth of workers' compensation costs. Programs include accident prevention, investigation, and duty modification for injured employees with medical restrictions so return to work can occur as soon as possible.

The changes in the liabilities for workers' compensation for the years ended June 30, 2015 and 2014 are as follows:

Fiscal years	Beginning of year	Claims and changes in estimates	Claims paid	End of year
2014	$ 4,331	2,364	(1,597)	5,098
2015	5,098	1,979	(1,557)	5,520

(d) Surety Bonds

Bonds are required in most phases of the public utilities construction contracting process for such phases as bid, performance, and payment or maintenance. Additionally, bonds may be required in other contracts where goods or services are provided to ensure compliance with applicable terms and conditions such as warranty.

(e) Errors and Omissions, Professional Liability

Errors and omissions and professional liability are commonly transferred through contract to the contracted professional, or retained through self-insurance on a case-by-case basis depending on the size, complexity, or scope of construction or professional service contracts. Examples of such contracts are inclusive of services provided by engineers, architects, design professionals, and other licensed or certified professional service providers.

(f) Public Officials Liability, Employment Practices Liability

All Enterprise public officials with financial oversight responsibilities are provided coverage through a commercial Public Officials Liability policy. An Employment Practices Liability policy is retained to protect against employment-related claims and liabilities.

(g) Builders' Risk

Builders' risk policies of insurance are required to be provided by the contractor on all construction projects for the full value of construction.

(h) Crime

The Enterprise also retains a commercial crime policy, in lieu of bonding its employees, to provide coverage against liabilities or losses due to third-party crime or employee fraud.

(12) Commitments and Litigation

(a) Commitments

As of June 30, 2015 and 2014, the Enterprise has outstanding commitments with third parties of $124,657 and $116,287, respectively, for various capital projects and other purchase agreements for materials and services.

(b) Grants

Grants that the Enterprise received are subject to audit and final acceptance by the granting agency. Current and prior year costs of such grants are subject to adjustment upon audit.

(c) Litigation

The Enterprise is a defendant in various legal actions and claims that arise during the normal course of business. The final disposition of these legal actions and claims is not determinable. However, in the opinion of management, the outcome of any litigation of these matters will not have a material effect on the financial position or changes in net position of the Enterprise.

(d) Environmental Issue

The Enterprise has been listed as potentially responsible parties in the clean-up effort of Yosemite Creek. Yosemite Creek has been identified as having toxic sediments, primarily polychlorinated biphenyls, in the drainage areas to the creek; contaminated flows emanating from a local industrial discharger as the likely responsible source of the contamination. The pollution remediation obligation reported in the accompanying statements of net position is based on estimated contractual costs. The liability balance remained at $571 as of June 30, 2015 and 2014.

(13) Subsequent Event

Commercial Paper

The Enterprise is authorized to issue up to $500,000 in commercial paper, with $100,000 issued as of June 30, 2015. On September 22, 2015, an additional $35,000 of commercial paper was issued for the SSIP projects, totaling $135,000 issued against the authorization.

APPENDIX C

PROPOSED FORM OF OPINION OF CO-BOND COUNSEL

Upon issuance of the Bonds, Stradling, Yocca, Carlson & Rauth, a Professional Corporation, and Amira Jackmon, Attorney at Law, Co-Bond Counsel, propose to render their final approving opinions in substantially the following form:

[Closing Date]

Public Utilities Commission of the City and County of San Francisco
525 Golden Gate Avenue
San Francisco, California 94102

$240,580,000 Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2016 Series A (Green Bonds)	$67,820,000 Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2016 Series B

Members of the Commission:

We have acted as co-bond counsel to the Public Utilities Commission of the City and County of San Francisco (the "Commission"), a commission duly constituted under the Charter (the "Charter") of the City and County of San Francisco (the "City"), in connection with the issuance of Commission's Wastewater Revenue Bonds, 2016 Series A (Green Bonds) (the "2016 Series A Bonds"), in the aggregate principal amount of $240,580,000 and its Wastewater Revenue Bonds, 2016 Series B (the "2016 Series B Bonds"), in the aggregate principal amount of $67,820,000 (the 2016 Series A Bonds and the 2016 Series B Bonds being collectively referred to as the "Bonds"). The Bonds are being issued pursuant to authority granted by the Charter of the City, and an Indenture, dated as of January 1, 2003, by and between the Commission and U.S. Bank, N.A., as trustee (the "Trustee"), as amended and supplemented (the "Original Indenture"), and as further supplemented by a Fourth Supplemental Indenture, dated as of May 1, 2016 (the "Fourth Supplemental Indenture" and together with the Original Indenture, the "Indenture") by and between the Commission and the Trustee. Capitalized terms not otherwise defined herein shall have the respective meanings ascribed thereto in the Indenture.

In our capacity as co-bond counsel, we have reviewed the Indenture, ordinances and resolutions of the Commission and the City, certifications of the Commission, the Trustee and others, opinions of counsel to the Commission, the City and the Trustee, and such other documents, opinions and instruments as we deemed necessary to render the opinions set forth herein. We have assumed, but have not verified, that the signatures on all documents, certificates and opinions that we have reviewed are genuine. In our examination, we have assumed, but have not verified, the legal capacity of all natural persons, the authenticity of all documents submitted to us as originals, the conformity to original documents of all documents submitted to us as certified or photostatic copies or by facsimile

or other means of electronic transmission or which we obtained from sites on the internet, and the authenticity of the originals of such latter documents. As to facts and certain other matters and the consequences thereof relevant to the opinions expressed herein and the other statements made herein, we have relied without investigation or verification upon, and assumed the accuracy and completeness of certificates, letters (including opinion letters), and oral and written statements and representations of public officials, officers and other representatives of the Commission and the City, counsel for the Commission and the City, and others.

Our services as co-bond counsel were limited to such examination and to rendering the opinions set forth below. Furthermore, we have assumed compliance with all covenants and agreements compliance with which is necessary to assure that future actions, omissions or events will not cause the interest on the Bonds to be included in gross income for federal income tax purposes. With respect to the opinions expressed herein, we call attention to the fact that the enforceability of the rights and obligations under the Bonds and the Indenture are subject to bankruptcy, insolvency, reorganization, moratorium, fraudulent conveyance and other similar laws affecting creditors' rights, to the application of equitable principles if equitable remedies are sought, to the exercise of judicial discretion in appropriate cases and to limitations on legal remedies against cities and counties in the State of California. In addition, the imposition of certain fees and charges by the Commission relating to the Enterprise is subject to the provisions of the Charter and Articles XIII C and XIII D of the California Constitution.

Based upon the foregoing and subject to the limitations and qualifications herein specified we are of the opinion, as of the date hereof and under existing law, that:

(1) The Bonds have been duly authorized, issued by, and constitute the valid and binding limited obligations of, the Commission.

(2) The Fourth Supplemental Indenture has been duly authorized, executed and delivered by the Commission, and the Indenture constitutes the valid and binding obligation of the Commission. The Indenture creates a valid pledge of the Net Revenues of the Enterprise to secure the payment of the principal of and interest on the Bonds, as and to the extent set forth in the Indenture and subject to the provisions of the Indenture permitting the application thereof for the purposes and on the terms and conditions set forth therein.

(3) The Bonds are limited obligations of the Commission and are payable exclusively from and are secured by a pledge of the Net Revenues of the Enterprise. The general fund of the City is not liable and the credit or taxing power of the City is not pledged for the payment of the Bonds or the interest thereon. The Commission has no taxing power. The Bonds are not secured by a legal or equitable pledge of, or charge, lien or encumbrance upon, any of the property of the City or of the Commission or any of their income or receipts, except the Net Revenues.

(4) Under existing statutes, regulations, rulings and judicial decisions, the interest on the Bonds is excluded from gross income for federal income tax purposes and is not an item of tax preference for purposes of calculating the federal alternative minimum tax imposed on individuals and corporations. With respect to corporations, the interest on the Bonds may be included as an adjustment in the calculation of alternative minimum taxable income, which may affect the alternative minimum tax liability of corporations.

(5) The interest on the Bonds is exempt from State of California personal income tax.

(6) The amount by which a Bond owner's original basis for determining loss on sale or exchange in the applicable Bonds (generally, the purchase price) exceeds the amount payable on maturity (or on an earlier call date) constitutes amortizable premium, which must be amortized under Section 171 of the Code; such amortizable premium reduces the Bond owner's basis in the applicable Bonds (and the amount of tax-exempt interest received), and is not deductible for federal income tax purposes. The basis reduction as a result of the amortization of Bond premium may result in a Bond owner realizing a taxable gain when a Bond is sold by the owner for an amount equal to or less (under certain circumstances) than the original cost of the Bond to the owner.

The opinions expressed herein as to the exclusion from gross income of interest on the Bonds are based upon certain representations of fact and certifications made by the Commission and others and are subject to the condition that the Commission complies with all requirements of the Internal Revenue Code of 1986, as amended (the "Code"), that must be satisfied subsequent to the issuance of the Bonds to assure that such interest will not become includable in gross income for federal income tax purposes. Failure to comply with such requirements of the Code might cause the interest to be included in gross income for federal income tax purposes retroactive to the date of issuance of the Bonds. The Commission has covenanted to comply with all such requirements.

The opinions expressed herein may be affected by actions taken (or not taken) or events occurring (or not occurring) after the date hereof. We have not undertaken to determine, or to inform any person, whether any such actions or events are taken or do occur. The Indenture and the Tax Certificate permit certain actions to be taken or to be omitted if a favorable opinion of Co-Bond Counsel is provided with respect thereto. No opinion is expressed herein as to the effect on the exclusion from gross income of interest for federal income tax purposes with respect to any Bonds if any such action is taken or omitted based upon the opinion or advice of counsel other than ourselves. Other than expressly stated herein, we express no other opinion regarding tax consequences with respect to the Bonds.

The opinions expressed and the statements made herein are based on an analysis of existing laws, regulations, rulings and judicial decisions and cover certain matters not directly addressed by such authorities. Such opinions and statements may be adversely affected by actions taken or events occurring, including a change in law, regulation or ruling (or in the application or official interpretation of any law, regulation or ruling) after the date hereof. We have not undertaken to determine, or to inform any person, whether such actions are taken or such events occur, and we have no obligation to update this letter in light of such actions or events or for any other reason.

No opinion is expressed herein on the accuracy, completeness or sufficiency of the official statement relating to the Bonds or other offering material relating to the Bonds and we expressly disclaim any duty to advise the owners of the Bonds with respect to matters contained in such official statement or other offering material.

This opinion is limited to the laws of the State of California and the federal laws of the United States. We assume no responsibility with respect to the applicability or the effect of the laws of any other jurisdiction. The opinions in this letter are expressed solely as of the date hereof for your benefit and may not be relied upon in any manner for any purposes by any other person.

Respectfully submitted,

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APPENDIX D

FORM OF CONTINUING DISCLOSURE CERTIFICATE

$308,400,000
Public Utilities Commission Of The City And County Of San Francisco
Wastewater Revenue Bonds

$240,580,000	**$67,820,000**
2016 Series A (Green Bonds)	**2016 Series B**

This Continuing Disclosure Certificate (the "Disclosure Certificate") is executed and delivered by the Public Utilities Commission of the City and County of San Francisco (the "SFPUC") in connection with the issuance of the wastewater revenue bonds captioned above (collectively, the "2016 Series AB Bonds"). The 2016 Series AB Bonds are being issued pursuant to an Indenture dated as of January 1, 2003, between the SFPUC and U.S. Bank National Association, as trustee (the "Trustee"), as amended and supplemented, including as amended and supplemented by a Fourth Supplemental Indenture dated as of May 1, 2016, by and between the SFPUC and the Trustee (collectively, the "Indenture").

The SFPUC covenants and agrees as follows:

SECTION 1. Purpose of the Disclosure Certificate. This Disclosure Certificate is being executed and delivered by the SFPUC for the benefit of the Holders and Beneficial Owners of the 2016 Series AB Bonds and in order to assist the Participating Underwriters in complying with Securities and Exchange Commission (the "S.E.C.") Rule 15c2-12(b)(5).

SECTION 2. Definitions. In addition to the definitions set forth in the Indenture, which apply to any capitalized term used in this Disclosure Certificate unless otherwise defined in this Section 2, the following capitalized terms shall have the following meanings:

"Annual Report" shall mean any Annual Report provided by the SFPUC pursuant to, and as described in, Sections 3 and 4 of this Disclosure Certificate.

"Beneficial Owner" shall mean any person that: (a) has or shares the power, directly or indirectly, to make investment decisions concerning ownership of any 2016 Series AB Bonds (including persons holding 2016 Series AB Bonds through nominees, depositories or other intermediaries) including, but not limited to, the power to vote or consent with respect to any 2016 Series AB Bonds or to dispose of ownership of any 2016 Series AB Bonds; or (b) is treated as the owner of any 2016 Series AB Bonds for federal income tax purposes.

"Dissemination Agent" shall mean the SFPUC, acting in its capacity as Dissemination Agent under this Disclosure Certificate, or any successor Dissemination Agent designated in writing by the SFPUC and which has filed with the SFPUC a written acceptance of such designation.

"Holder" shall mean either the registered owners of the 2016 Series AB Bonds, or, if the 2016 Series AB Bonds are registered in the name of The Depository Trust Company or another recognized depository, any applicable participant in such depository system.

"Listed Events" shall mean any of the events listed in Section 5(a) of this Disclosure Certificate.

"MSRB" shall mean the Municipal Securities Rulemaking Board or any other entity designated or authorized by the Securities and Exchange Commission to receive continuing disclosure filings pursuant to the Rule. Until otherwise designated by the MSRB or the Securities and Exchange Commission, filings with the MSRB are to be made through the Electronic Municipal Market Access (EMMA) website of the MSRB currently located at *http://emma.msrb.org*.

"Official Statement" shall mean the final Official Statement, dated May 10, 2016, prepared in connection with the sale and offering of the 2016 Series AB Bonds.

"Participating Underwriter" shall mean any of the original underwriters or purchasers of the 2016 Series AB Bonds required to comply with the Rule in connection with the offering of the 2016 Series AB Bonds.

"Rule" shall mean Rule 15c2-12(b)(5) adopted by the S.E.C. under the Securities Exchange Act of 1934, as the same may be amended from time to time.

SECTION 3. Provision of Annual Reports.

(a) The SFPUC shall, or shall cause the Dissemination Agent to, not later than 9 months after the end of the SFPUC's Fiscal Year (which currently ends June 30), commencing March 31, 2013, with the report for the 2011-12 Fiscal Year, provide to the MSRB an Annual Report which is consistent with the requirements of Section 4 of this Disclosure Certificate. If the Dissemination Agent is not the SFPUC, the SFPUC shall provide the Annual Report to the Dissemination Agent not later than 15 days prior to said date. The Annual Report must be submitted in electronic format and accompanied by such identifying information as is prescribed by the MSRB, and may cross-reference other information as provided in Section 4 of this Disclosure Certificate. However, if the audited financial statements of the SFPUC are not available by the date required above for the filing of the Annual Report, the SFPUC shall submit unaudited financial statements and submit the audited financial statements as soon as they are available. If the SFPUC's Fiscal Year changes, it shall give notice of such change in the same manner as for a Listed Event under Section 5(c).

(b) If the SFPUC is unable to provide to the MSRB an Annual Report by the date required in subsection (a), the SFPUC shall send a notice to the MSRB in substantially the form attached as Exhibit A.

(c) The Dissemination Agent shall (if the Dissemination Agent is other than the SFPUC), file a report with the SFPUC certifying the date that the Annual Report was provided to the MSRB pursuant to this Disclosure Certificate.

SECTION 4. Content of Annual Reports. SFPUC's Annual Report shall contain or incorporate by reference the following information:

(a) Audited Financial Statements of the municipal water supply, storage and distribution system of the SFPUC for the prior Fiscal Year, prepared in accordance with generally accepted accounting principles applicable to the SFPUC from time to time. If the SFPUC's audited financial statements are not available by the date the Annual Report is required to be filed, the Annual Report shall contain unaudited financial statements in a format similar to the financial statements contained in the final Official Statements, and the audited financial statements shall be filed in the same manner as the Annual Report when they become available;

(b) An update for the prior Fiscal Year of the table in the Official Statement entitled "TABLE 6, SUMMARY OF ACTIVE SEWER ACCOUNTS BY USER TYPE;"

(c) An update for the prior Fiscal Year of the table in the Official Statement entitled "TABLE 12, HISTORICAL AND ADOPTED SINGLE-FAMILY RESIDENTIAL USER SEWER RATES";

(d) An update for the prior Fiscal Year of the table in the Official Statement entitled "TABLE 13, HISTORICAL AND ADOPTED MULTI-FAMILY RESIDENTIAL USER SEWER RATES";

(e) An update for the prior Fiscal Year of the table in the Official Statement entitled "TABLE 14, HISTORICAL AND ADOPTED NON-RESIDENTIAL USER DISCHARGE RATES";

(f) An update for the prior Fiscal Year of the table in the Official Statement in the section entitled "OBLIGATIONS PAYABLE FROM REVENUES – Outstanding Parity Revenue Bonds" showing all bonds and other obligations of the SFPUC secured by Net Revenues; and

(g) An update for the prior Fiscal Year of the table in the Official Statement entitled "TABLE 20, HISTORICAL REVENUE, OPERATING & MAINTENANCE EXPENSES AND DEBT SERVICE COVERAGE."

Any or all of the items listed above may be set forth in a document or set of documents, or may be included by specific reference to other documents, including official statements of debt issues of the SFPUC or related public entities, which are available to the public on the MSRB website. If the document included by reference is a final official statement, it must be available from the MSRB. The SFPUC shall clearly identify each such other document so included by reference.

SECTION 5. Reporting of Significant Events.

(a) The SFPUC shall give, or cause to be given, notice of the occurrence of any of the following Listed Events with respect to the 2016 Series AB Bonds:

1. Principal and interest payment delinquencies.
2. Non payment related defaults, if material.
3. Unscheduled draws on debt service reserves reflecting financial difficulties.
4. Unscheduled draws on credit enhancements reflecting financial difficulties.
5. Substitution of credit or liquidity providers, or their failure to perform.
6. Adverse tax opinions, the issuance by the Internal Revenue Service of proposed or final determinations of taxability, Notices of Proposed Issue (IRS Form 5701-TEB) or other material notices or determinations with respect to the tax status of the 2016 Series AB Bonds, or other material events affecting the tax exempt status of the 2016 Series AB Bonds.
7. Modifications to rights of security holders, if material.
8. Bond calls, if material, and tender offers.
9. Defeasances.
10. Release, substitution, or sale of property securing repayment of the 2016 Series AB Bonds, if material.
11. Rating changes.
12. Bankruptcy, insolvency, receivership or similar event of the SFPUC.
13. The consummation of a merger, consolidation, or acquisition involving the SFPUC or the sale of all or substantially all of the assets of the SFPUC, other than in the ordinary course of business, the entry into a definitive agreement to undertake such an action or the termination of a definitive agreement relating to any such actions, other than pursuant to its terms, if material.
14. Appointment of a successor or additional trustee or the change of name of a trustee, if material.

(b) Whenever the SFPUC obtains knowledge of the occurrence of a Listed Event, and, if the Listed Event is described in subsections (a)(2), (a)(7), (a)(8) (if the event is a bond call), (a)(10), (a)(13) or (a)(14) above, the SFPUC determines that knowledge of the occurrence of that Listed Event would be material under applicable federal securities law, the SFPUC shall, or shall cause the Dissemination Agent (if not the SFPUC) to, file a notice of such occurrence with the MSRB, in an electronic format as prescribed by the MSRB, in a timely manner not in excess of 10 business days after the occurrence of the Listed Event. Notwithstanding the foregoing, notice of Listed Events described in subsections (a)(8) and (9) above need not be given under this subsection any earlier than the notice (if any) of the underlying event is given to holders of affected 2016 Series AB Bonds under the Indenture.

(c) For purposes of this Certificate, any event described in paragraph (a)(12) above is considered to occur when any of the following occur: the appointment of a receiver, fiscal agent, or similar officer for the SFPUC in a proceeding under the United States Bankruptcy Code or in any other proceeding under state or federal law in which a court or governmental authority has assumed jurisdiction over substantially all of the assets or business of the SFPUC, or if such jurisdiction has been assumed by leaving the existing governing body and officials or officers in possession but subject to the supervision and orders of a court or governmental authority, or the entry of an order confirming a plan of reorganization, arrangement, or liquidation by a court or governmental authority having supervision or jurisdiction over substantially all of the assets or business of said party.

(d) The SFPUC shall give, or cause to be given, notice of the occurrence of the receipt by the SFPUC of a written statement by the Climate Standards Board to the effect that the Public Utilities Commission of the City and County of San Francisco Wastewater Revenue Bonds, 2016 Series A (Green Bonds) are no longer "Climate Bond Certified". The SFPUC shall, or shall cause the Dissemination Agent (if not the SFPUC) to, file a notice of such occurrence with the MSRB, in electronic format as prescribed by the MSRB, in a timely manner not in excess of 30 days after the receipt by the SFPUC of such written statement.

SECTION 6. Termination of Reporting Obligation. The SFPUC's obligations under this Disclosure Certificate shall terminate upon the legal defeasance, prior redemption or payment in full of all of the 2016 Series AB Bonds. If such termination occurs prior to the final maturity of the 2016 Series AB Bonds, the SFPUC shall give notice of such termination in the same manner as for a Listed Event under Section 5(c).

SECTION 7. Dissemination Agent. The SFPUC may, from time to time, appoint or engage a Dissemination Agent to assist it in carrying out its obligations under this Disclosure Certificate, and may discharge any such Dissemination Agent, with or without appointing a successor Dissemination Agent. The Dissemination Agent shall have only such duties as are specifically set forth in this Disclosure Certificate.

SECTION 8. Amendment; Waiver. Notwithstanding any other provision of this Disclosure Certificate, the SFPUC may amend or waive this Disclosure Certificate or any provision of this Disclosure Certificate, provided that the following conditions are satisfied:

(a) If the amendment or waiver relates to the provisions of Sections 3(a), 3(b), 4 or 5(a), it may only be made in connection with a change in circumstances that arises from a change in legal requirements, change in law, or change in the identity, nature or status of an obligated person with respect to the 2016 Series AB Bonds or the type of business conducted;

(b) The undertaking, as amended or taking into account such waiver, would, in the opinion of the City Attorney or nationally recognized bond counsel, have complied with the requirements of the Rule at the time of the original issuance of the 2016 Series AB Bonds, after taking into account any amendments or interpretations of the Rule, as well as any change in circumstances; and

(c) The amendment or waiver either (i) is approved by the owners of a majority in aggregate principal amount of the 2016 Series AB Bonds or (ii) does not, in the opinion of the City Attorney or nationally recognized bond counsel, materially impair the interests of the Holders.

In the event of any amendment or waiver of a provision of this Disclosure Certificate, the SFPUC shall describe such amendment in the next Annual Report, and shall include, as applicable, a narrative explanation of the reason for the amendment or waiver and its impact on the type (or in the case of a change of accounting principles, on the presentation) of financial information or operating data being presented by the SFPUC. In addition, if the amendment relates to the accounting principles to be followed in preparing financial statements: (i) notice of such change shall be given in the same manner as for a Listed Event under Section 5; and (ii) the Annual Report for the year in which the change is made should present a comparison (in narrative form and also, if feasible, in quantitative form) between the financial statements as prepared on the basis of the new accounting principles and those prepared on the basis of the former accounting principles.

SECTION 9. Additional Information. Nothing in this Disclosure Certificate shall be deemed to prevent the SFPUC from disseminating any other information, using the means of dissemination set forth in this Disclosure Certificate or any other means of communication, or including any other information in any Annual Report or notice of occurrence of a Listed Event, in addition to that which is required by this Disclosure Certificate. If the SFPUC chooses to include any information in any Annual Report or notice of occurrence of a Listed Event in addition to that which is specifically required by this Disclosure Certificate, the SFPUC shall have no obligation under this Disclosure Certificate to update such information or include it in any future Annual Report or notice of occurrence of a Listed Event.

SECTION 10. Default. In the event of a failure of the SFPUC to comply with any provision of this Disclosure Certificate, any Participating Underwriter, Holder or Beneficial Owner of the 2016 Series AB Bonds may take such actions as may be necessary and appropriate, including seeking mandate or specific performance by court order, to cause the SFPUC to comply with its obligations under this Disclosure Certificate; provided that any such action may be instituted only in a federal or State court located in the City and County of San Francisco, State of California. Failure by the SFPUC to comply with any provision of this Disclosure Certificate shall not be deemed an Event of Default under the Indenture and the sole remedy under this Disclosure Certificate in the event of any failure of the SFPUC to comply with this Disclosure Certificate shall be an action to compel performance.

SECTION 11. Beneficiaries. This Disclosure Certificate shall inure solely to the benefit of the SFPUC, the Dissemination Agent, the Participating Underwriters and Holders and Beneficial Owners from time to time of the 2016 Series AB Bonds, and shall create no rights in any other person or entity.

Date: May 24, 2016.

PUBLIC UTILITIES COMMISSION OF THE
CITY AND COUNTY OF SAN FRANCISCO

By_____
Harlan L. Kelly, Jr.
General Manager

Approved as to Form:

DENNIS J. HERRERA
CITY ATTORNEY

By:_____
Deputy City Attorney

CONTINUING DISCLOSURE CERTIFICATE EXHIBIT A

FORM OF NOTICE TO THE
MUNICIPAL SECURITIES RULEMAKING BOARD
OF FAILURE TO FILE ANNUAL REPORT

Name of Issuer: PUBLIC UTILITIES COMMISSION OF THE CITY AND COUNTY OF SAN FRANCISCO

Name of Issue: WASTEWATER REVENUE BONDS, [2016 SERIES A (Green Bonds) / 2016 SERIES B]

Date of Issuance: May 24, 2016

NOTICE IS HEREBY GIVEN that the SFPUC has not provided an Annual Report with respect to the above-named Bonds as required by Section 3 of the Continuing Disclosure Certificate of the Public Utilities Commission of the City and County of San Francisco, dated May __, 2016. The SFPUC anticipates that the Annual Report will be filed by _____.

Dated:_____

PUBLIC UTILITIES COMMISSION OF THE CITY AND COUNTY OF SAN FRANCISCO

By:_____[to be signed only if filed]_____
Title _____

APPENDIX E

SECURITIES DEPOSITORY AND BOOK-ENTRY SYSTEM

The information in this Appendix has been provided by DTC for use in securities offering documents, and the SFPUC takes no responsibility for the accuracy or completeness thereof. The SFPUC cannot and does not give any assurances that DTC, DTC Participants or Indirect Participants will distribute to the beneficial owners either (a) payments of interest, principal or premium, if any, with respect to the Bonds or (b) certificates representing ownership interest in or other confirmation of ownership interest in the Bonds, or that they will so do on a timely basis or that DTC, DTC Participants or DTC Indirect Participants will act in the manner described in this Official Statement. The current "Rules" applicable to DTC are on file with the Securities and Exchange Commission and the current "Procedures" of DTC to be followed in dealing with DTC Participants are on file with DTC.

As used in this Appendix, "Securities" means the 2016 Series AB Bonds, "Issuer" means the SFPUC, and "Agent" means the Trustee.

1. The Depository Trust Company ("DTC"), New York, NY, will act as securities depository for the securities (the "Securities"). The Securities will be issued as fully-registered securities registered in the name of Cede & Co. (DTC's partnership nominee) or such other name as may be requested by an authorized representative of DTC. One fully-registered Security certificate will be issued for each maturity of the Securities, each in the aggregate principal amount of such maturity, and will be deposited with DTC. If, however, the aggregate principal amount of any maturity exceeds $500 million, one certificate will be issued with respect to each $500 million of principal amount, and an additional certificate will be issued with respect to any remaining principal amount of such maturity.

2. DTC, one of the world's largest securities depositories, is a limited-purpose trust company organized under the New York Banking Law, a "banking organization" within the meaning of the New York Banking Law, a member of the Federal Reserve System, a "clearing corporation" within the meaning of the New York Uniform Commercial Code, and a "clearing agency" registered pursuant to the provisions of Section 17A of the Securities Exchange Act of 1934. DTC holds and provides asset servicing for over 3.5 million issues of U.S. and non-U.S. equity issues, corporate and municipal debt issues, and money market instruments (from over 100 countries) that DTC's participants ("Direct Participants") deposit with DTC. DTC also facilitates the post-trade settlement among Direct Participants of sales and other securities transactions in deposited securities, through electronic computerized book-entry transfers and pledges between Direct Participants' accounts. This eliminates the need for physical movement of securities certificates. Direct Participants include both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, clearing corporations, and certain other organizations. DTC is a wholly-owned subsidiary of The Depository Trust & Clearing Corporation ("DTCC"). DTCC is the holding company for DTC, National Securities Clearing Corporation and Fixed Income Clearing Corporation, all of which are registered clearing agencies. DTCC is owned by the users of its regulated subsidiaries. Access to the DTC system is also available to others such as both U.S. and non-U.S. securities brokers and dealers, banks, trust companies, and clearing corporations that clear through or maintain a custodial relationship with a Direct Participant, either directly or indirectly ("Indirect Participants"). DTC has a Standard & Poor's rating of AA+. The DTC Rules applicable to its Participants are on file with the Securities and Exchange Commission. More information about DTC can be found at www.dtcc.com. The information contained on this Internet site is not incorporated herein by reference.

3. Purchases of Securities under the DTC system must be made by or through Direct Participants, which will receive a credit for the Securities on DTC's records. The ownership interest of each actual purchaser of each Security ("Beneficial Owner") is in turn to be recorded on the Direct and Indirect Participants' records. Beneficial Owners will not receive written confirmation from DTC of their purchase. Beneficial Owners are, however, expected to receive written confirmations providing details of the transaction, as well as periodic statements of their holdings, from the Direct or Indirect Participant through which the Beneficial Owner entered into the transaction. Transfers of ownership interests in the Securities are to be accomplished by entries made on the books of Direct and Indirect Participants acting on behalf of Beneficial Owners. Beneficial Owners will not receive certificates representing their ownership interests in Securities, except in the event that use of the book-entry system for the Securities is discontinued.

4. To facilitate subsequent transfers, all Securities deposited by Direct Participants with DTC are registered in the name of DTC's partnership nominee, Cede & Co., or such other name as may be requested by an authorized representative of DTC. The deposit of Securities with DTC and their registration in the name of Cede & Co. or such other DTC nominee do not affect any change in beneficial ownership. DTC has no knowledge of the actual Beneficial Owners of the Securities; DTC's records reflect only the identity of the Direct Participants to whose accounts such Securities are credited, which may or may not be the Beneficial Owners. The Direct and Indirect Participants will remain responsible for keeping account of their holdings on behalf of their customers.

5. Conveyance of notices and other communications by DTC to Direct Participants, by Direct Participants to Indirect Participants, and by Direct Participants and Indirect Participants to Beneficial Owners will be governed by arrangements among them, subject to any statutory or regulatory requirements as may be in effect from time to time. Beneficial Owners of Securities may wish to take certain steps to augment the transmission to them of notices of significant events with respect to the Securities, such as redemptions, tenders, defaults, and proposed amendments to the Security documents. For example, Beneficial Owners of Securities may wish to ascertain that the nominee holding the Securities for their benefit has agreed to obtain and transmit notices to Beneficial Owners. In the alternative, Beneficial Owners may wish to provide their names and addresses to the registrar and request that copies of notices be provided directly to them.

6. Redemption notices shall be sent to DTC. If less than all of the Securities within an issue are being redeemed, DTC's practice is to determine by lot the amount of the interest of each Direct Participant in such issue to be redeemed.

7. Neither DTC nor Cede & Co. (nor any other DTC nominee) will consent or vote with respect to Securities unless authorized by a Direct Participant in accordance with DTC's MMI Procedures. Under its usual procedures, DTC mails an Omnibus Proxy to Issuer as soon as possible after the record date. The Omnibus Proxy assigns Cede & Co.'s consenting or voting rights to those Direct Participants to whose accounts Securities are credited on the record date (identified in a listing attached to the Omnibus Proxy).

8. Redemption proceeds, distributions, and dividend payments on the Securities will be made to Cede & Co., or such other nominee as may be requested by an authorized representative of DTC. DTC's practice is to credit Direct Participants' accounts upon DTC's receipt of funds and corresponding detail information from Issuer or Agent, on payable date in accordance with their respective holdings shown on DTC's records. Payments by Participants to Beneficial Owners will be governed by standing instructions and customary practices, as is the case with securities held for the accounts of customers in bearer form or registered in "street name," and will be the responsibility of such Participant and not of DTC, Agent, or Issuer, subject to any statutory or regulatory requirements as may be in effect from time to time. Payment of redemption proceeds, distributions, and dividend payments to Cede & Co. (or such other nominee as may be requested by an authorized representative of DTC) is the responsibility of Issuer or Agent, disbursement of such payments to Direct Participants will be the responsibility of DTC, and disbursement of such payments to the Beneficial Owners will be the responsibility of Direct and Indirect Participants.

9. DTC may discontinue providing its services as depository with respect to the Securities at any time by giving reasonable notice to Issuer or Agent. Under such circumstances, in the event that a successor depository is not obtained, Security certificates are required to be printed and delivered.

10. Issuer may decide to discontinue use of the system of book-entry-only transfers through DTC (or a successor securities depository). In that event, Security certificates will be printed and delivered to DTC.

11. The information in this section concerning DTC and DTC's book-entry system has been obtained from sources that Issuer believes to be reliable, but Issuer takes no responsibility for the accuracy thereof.

APPENDIX F

ANTICIPATED 2016 SERIES A GREEN BOND PROJECTS

The SFPUC anticipates financing and refinancing a portion of the following SSIP projects using proceeds of the 2016 Series A Bonds:

1. Collection System Improvements
2. Central Bayside System Improvements
3. SSIP Biosolids/Digester Project
4. Stormwater Management
5. Northshore to Channel Force Main
6. Southeast Treatment Plant Improvements
7. Other Treatment Plant Improvements
8. Urban Watershed Assessment Project
9. Biofuel Alternative Energy Project
10. SSIP Program-wide Management

The SFPUC expects to post information respecting the expenditure of the proceeds of the 2016 Series A Bonds on its website. See "PLAN OF FINANCE," "ESTIMATED SOURCES AND USES OF FUNDS," "GREEN BONDS DESIGNATION AND CLIMATE CERTIFICATION" and "SEWER SYSTEM IMPROVEMENT PROGRAM" in the main body of this Official Statement.

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San Francisco
Water
Power
Sewer

Services of the San Francisco Public Utilities Commission

Mixed Sources
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